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Gender differences in faculty rank among academic physicians: A systematic review and meta-analysis

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Gender differences in faculty rank among academic physicians: A systematic review and meta-analysis

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ABSTRACT

Objective: To analyze the difference in the proportion of men versus women with full professorship among academic physicians.

Design: Systematic review and meta-analysis.

Data sources: MEDLINE, Embase, Cochrane CENTRAL, ERIC, and PsycINFO were searched from inception to July 3, 2020.

Study selection: All original studies reporting faculty rank stratified by gender were included.

Data extraction and synthesis: Study screening, data extraction, and quality assessment were performed by two independent reviewers, with a third author resolving discrepancies. Meta-analysis was conducted using random-effects models.

Results: Our search yielded 5,897 articles. 218 studies were included with 991,207 academic physician data points. Men were 2.77 times more likely to be full professors (182,271/643,790 men vs. 30,349/251,501 women, OR 2.77, 95% CI 2.57-2.98). Although men practiced for longer (median 18 vs. 12 years, $p < .00002$), the gender gap remained after pooling 7 studies that adjusted for factors including time in practice, specialty, publications, h-index, additional PhD, and institution (adjusted OR 1.83, 95% CI 1.04-3.20). Meta-regression by data collection year demonstrated improvement over time ($p = .0011$), however, subgroup analysis showed that gender disparities remain significant in the 2010-2020 decade (OR 2.63, 95% CI 2.48-2.80). The gender gap was present across all specialties and both within and outside of North America. Men published more papers (mean difference 17.2, 95% CI 14.7-19.7), earned higher salaries (mean

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difference $33,256, 95% CI $25,969-$40,542), and were more likely to be departmental chairs (OR 2.61, 95% CI 2.19-3.12).

Conclusions: Gender inequity in academic medicine exists across all specialties, geographic regions, and multiple measures of success, including academic rank, publications, salary, and leadership. Men are more likely than women to be full professors after controlling for experience, academic productivity, and specialty. Although there has been some improvement over time, the gender disparity in faculty rank persists.

PROSPERO registration number: CRD42020197414.

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- This systematic review and meta-analysis comprehensively synthesizes the literature on gender inequity in academic medicine with full professorship as the primary outcome.
- We analyzed trends over time, across specialties, and between geographic regions.
- Both unadjusted and adjusted data were pooled.
- There was high heterogeneity between studies as data from multiple sources, specialties, and countries were pooled.
- Publication bias was present in 3 of 5 secondary outcomes.

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INTRODUCTION

Gender equity remains an important issue in academic medicine. In the United States, women make up more than half of medical school graduates (1) and 37% of full-time faculty (2). However, only 24% of full professors (3), 18% of department chairs (4), and 17% of deans are women (5). Women are under-represented on editorial boards, have fewer publications, and receive less research funding (6,7). Due to these inequities, women are more likely to leave academia (8).

Many studies have analyzed gender bias in academic medicine; however, no comprehensive synthesis of the literature has been performed. Single studies are limited by the analysis of one institution (9), specialty (10), or country (11–14) and small sample sizes (15). Previous groups have studied gender differences in academic rank among radiologists (16), anesthesiologists (17), urologists (18), and neurosurgeons (19). Some have examined gender disparities in Australia (20), Germany (21), Japan (22), and Nordic countries (23). We conducted a systematic review and meta-analysis to synthesize these studies, with gender differences in full professorship as the primary outcome.

METHODS

Protocol and registration

We conducted a systematic review and meta-analysis according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement guidelines (24). Our study protocol (CRD42020197414) was registered with the International Prospective Register of Systematic Reviews (PROSPERO) (25).

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Information sources and search strategy

MEDLINE, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), PsycINFO, and Education Resources Information Center (ERIC) were searched from inception to July 3, 2020 for studies analyzing gender differences in faculty rank among academic physicians. A combination of Medical Subject Heading (MeSH) terms and keywords for gender AND physician AND faculty AND rank were used to maximize sensitivity. Our search did not apply language limitations and Google Translate was used for non-English studies (26). The search strategy is detailed in eTable 1.

Study selection and data collection

Title and abstract screening, full-text review, data collection, and study quality assessment were conducted by two independent reviewers (BL and JJB), with a third author resolving discrepancies (FD). We included all original studies reporting faculty rank stratified by gender among academic physicians. Case reports, reviews, and studies reporting data on only one gender were excluded. We hand-searched the reference lists of included studies for additional articles.

A standardized form was used to collect data for included studies. Variables obtained were study authors, publication year, geographic region, data source, specialty, data collection year, sample size, outcomes reported, baseline demographics, and effect size measurements. Authors were contacted through email for baseline demographics and effect size measurements if they were not reported in the original publication. When studies reported duplicate data (same data source, year of data collection, and outcome), the most comprehensive article was included (largest sample size and/or most outcomes reported). In studies reporting data from multiple

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years, results from the most recent year were included. Study quality was assessed using the Newcastle-Ottawa Scale (NOS) modified for cross-sectional studies (27,28).

Statistical analysis

We calculated a κ statistic to assess inter-rater agreement at the title/abstract screening and full-text review stages, with a threshold of ≥ 0.8 indicating strong agreement (29). Baseline covariates for men vs. women were compared by calculating median values across included studies and applying unpaired t test.

We conducted a meta-analysis of the proportion of men versus women with full professorship. Unadjusted estimates were reported as odds ratios (OR) with 95% confidence intervals (CI). Adjusted estimates were calculated using the inverse variance method and reported as adjusted OR with 95% CI (30). Sensitivity analyses were conducted by excluding low quality studies (NOS score ≤ 5) (31). We performed meta-regression and subgroup analysis based on data collection year to assess for trends over time and gender disparities in the modern era. We also conducted subgroup analyses for studies within vs. outside of North America, surgical vs. non-surgical specialties, and across specialties. For subgroup analysis by specialty, data was included from studies that reported specialty-specific results either as their primary or subgroup analysis. Subgroup differences were analyzed by calculating between-subgroup variance as described by Borenstein and Higgins (2013) (32,33). Secondary outcomes were publications, h-index, salary, federal research funding, and leadership (departmental chair). Continuous outcome estimates were reported as mean differences with 95% CI. Heterogeneity was quantified using the I^2 statistic and publication bias was assessed using funnel plots.

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Meta-analyses were conducted using Review Manager 5.4.1 (The Cochrane Collaboration, Copenhagen, Denmark) based on random-effects models. Meta-regression was performed using R version 4.0.3 (R Project for Statistical Computing) (34) with meta (35) and metaphor (36) packages. Statistical tests were 2-sided with a significance threshold of $p < .05$.

Patient and public involvement

Patients or the public were not involved in the design, conduct, reporting, or dissemination plans of our research.

RESULTS

Study screening and selection

We identified 8,050 articles through our search of MEDLINE (n = 3,383), Embase (n = 2,810), PsycINFO (n = 907), ERIC (n = 633), and CENTRAL (n = 317). 5,897 articles remained after duplicates were removed, all of which underwent title and abstract screening. 4,827 records were excluded and 1,070 underwent full-text review. 852 were excluded, most commonly because there was no relevant outcome (n = 606), no stratification by gender (n = 115), or they contained duplicate data of an included study (n = 54). Hand-search of reference lists of included studies identified no additional articles. 218 studies were included in the final systematic review and meta-analysis. Inter-rater agreement was strong at title/abstract screening ($\kappa = 0.88$) and full-text review ($\kappa = 0.89$). Our search results are summarized in the PRISMA study flow diagram in Figure 1.

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Study characteristics

We included 218 cross-sectional studies published between 1979 and 2020. In total, 991,207 academic physician data points were included (699,010 men vs. 292,197 women). 170 studies reported academic rank (7 adjusted), 42 reported publications (1 adjusted), 41 reported h-index (1 adjusted), 20 reported salary (2 adjusted), 21 reported federal research funding (1 adjusted), and 42 reported departmental chair data (0 adjusted). An increasing number of studies were published in the past decade (2010-2020: 164 studies [75%] vs. before 2010: 54 studies [25%]). Outcomes were reported on 24 specialties across 18 countries. Data sources included surveys (n = 89, 40%), departmental websites (n = 71, 33%), and faculty databases (n = 58, 27%). The characteristics of included studies are summarized in eTable 2.

Baseline demographic data

90 studies reported baseline demographic data. Median values of covariates across included studies were assessed. Men were older (median age 47.7 vs. 43.5, $p < .00001$) and practiced for longer (median 18 vs. 12 years, $p < .00002$). Most academic physicians were white (median 74% [men] and 75% [women], $p = .56$). Men were more likely to be married (median 91% vs. 75%, $p < .0002$) and have children (median 84% vs. 71%, $p < .0007$). There were no significant differences in number of hours worked (median 58.0 [men] vs. 54.7 [women], $p = .056$), percentage of time spent on research (median 27.1% [men] vs. 25.2% [women], $p = 0.80$), or proportion of faculty holding an additional PhD (14.8% [men] vs. 14.6% [women], $p = .57$). Baseline demographic data for included studies are presented in eTable 3.

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Study quality

Of the 218 cross-sectional studies, 127 were good quality (NOS score ≥ 8), 57 were fair quality (NOS score 6-7), and 34 were poor quality (NOS score ≤ 5). The most common reason for diminished study quality was poor comparability of cohorts due to inadequate statistical adjustment for confounders, with 131 studies receiving a score of 0/2 and 24 studies receiving a score of 1/2. Another contributor to poor quality was insufficient characterization of non-respondents in survey-based articles, with 81 studies receiving a score of 0/1. A summary of study quality assessment is presented in eTable 4.

Meta-analysis of full professorship

For full professorship, 170 studies with 895,291 academic physician data points (643,790 men vs. 251,501 women) were included. Meta-analysis demonstrated that men were 2.77 times more likely to be full professors (182,271 [28.3%] vs. 30,349 [12.1%], OR 2.77, 95% CI 2.57 – 2.98, $p < .00001$, $I^2 = 94\%$, Figure 2). Funnel plot showed no evidence of publication bias (eFigure 1). Sensitivity analysis removing 27 poor quality studies (NOS score ≤ 5) continued to demonstrate that men were more likely to be full professors (OR 2.81, 95% CI 2.60 – 3.04, $p < .00001$, $I^2 = 95\%$, eFigure 2).

Seven studies adjusted for potential confounders, including years in practice (Dossani (37), Gawad (38), Jena (12), Orhurhu (39), Riaz (40), Smith (41), Sperling (42)), specialty (Jena), publications (Dossani, Gawad, Jena, Riaz, Sperling), h-index (Dossani, Gawad, Orhurhu, Riaz, Sperling), additional PhD (Dossani, Gawad, Sperling), and institution (Dossani, Jena, Orhurhu, Riaz, Sperling). Pooling the adjusted effect estimates comparing 64,469 men to 32,063

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women demonstrated that men were 1.83 times more likely to be full professors (adjusted OR 1.83, 95% CI 1.04 – 3.20, p = .03, I^2 = 91%, Figure 3).

To assess trends over time, meta-regression was performed based on data collection year, which demonstrated a reduction in the gender gap between 1979 and 2020 (p = .0011, Figure 4). To evaluate gender disparities in the modern era, subgroup analysis was performed by decade (Figure 5). Between 2010-2020, men were 2.63 times more likely to be full professors (OR 2.63, 95% CI 2.48 – 2.80, p < .00001, I^2 = 63%). Between 2000-2009, men were 2.85 times more likely to be full professors (OR 2.85, 95% CI 2.36 – 3.42, p < .00001, I^2 = 97%). Before 2000, men were 3.62 times more likely to be full professors (OR 3.62, 95% CI 3.27 – 4.01, p < .00001, I^2 = 85%).

Differences between geographic regions were analyzed (eFigure 3). Within North America, men were 2.87 times more likely to be full professors (OR 2.87, 95% CI 2.65 – 3.10, p < .00001, I^2 = 95%). Outside of North America, men were 2.34 times more likely to be full professors (OR 2.34, 95% CI 1.80 – 3.04, p < .00001, I^2 = 82%). There was no significant difference between countries in North America vs. outside of North America (Chi^2 = 2.12, p = .15).

Surgical specialties were compared to non-surgical specialties (eFigure 4). For surgical specialties, men were 2.99 times more likely to be full professors (OR 2.99, 95% CI 2.64 – 3.40, p < .00001, I^2 = 72%). For non-surgical specialties, men were 2.72 times more likely to be full professors (OR 2.72, 95% CI 2.48 to 2.98, p < .00001, I^2 = 84%). There was no significant difference between surgical and non-surgical specialties (Chi^2 = 1.43, p = .23).

Analysis of individual specialties demonstrated that men were more likely to be full professors across all specialties (eFigure 5). The specialty with the smallest gender difference in

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full professorship was neurosurgery (5 studies, OR 2.14, 95% CI 1.30 – 3.52, p = .003, I^2 = 59%). The specialty with the greatest gender difference in full professorship was obstetrics and gynecology (8 studies, OR 4.52, 95% CI 3.51 – 5.81, p < .00001, I^2 = 70%).

Meta-analysis of secondary outcomes

For total publications, 42 studies were included comparing 96,483 men to 47,910 women. Over their careers, men published 17.20 more papers (mean difference 17.20, 95% CI 14.68 – 19.72, p < .00001, I^2 = 99%, eFigure 6). Funnel plot demonstrated publication bias skewed towards a higher mean difference (eFigure 7). Sensitivity analysis removing 5 poor quality studies continued to show that men published more papers (mean difference 17.66, 95% CI 14.97 – 20.35, p < .00001, I^2 = 99%, eFigure 8). One study demonstrated that women continued to have fewer publications after adjusting for career duration (adjusted relative rate 0.46, 95% CI 0.39 – 0.55) (43).

For h-index, 41 studies were included comparing 29,923 men to 12,286 women. Men had a higher h-index than women (mean difference 5.97, 95% CI 4.77 – 7.16, p < .00001, I^2 = 98%, eFigure 9). Funnel plot demonstrated no evidence of publication bias (eFigure 10). Sensitivity analysis removing 1 poor quality study continued to show that men had a higher h-index (mean difference 5.98, 95% CI 4.77 – 7.19, p < .00001, I^2 = 98%, eFigure 11). One study demonstrated that women continued to have a lower h-index after adjusting for career duration (adjusted relative rate 0.62, 95% CI 0.54 – 0.72) (43).

For federal research funding, 21 studies were included comparing 87,205 men to 39,638 women. Men were 1.71 times more likely to hold a federal research grant (OR 1.71, 95% CI 1.41 – 2.08, p < .00001, I^2 = 88%, eFigure 12). Funnel plot demonstrated evidence of publication bias

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skewed towards a higher OR (eFigure 13). Sensitivity analysis removing 2 poor quality studies continued to show that men were more likely to hold a federal research grant (OR 1.55, 95% CI 1.30 – 1.84, $p < .00001$, $I^2 = 75\%$, eFigure 14). One study demonstrated no difference in the proportion of men vs. women receiving federal research funding after controlling for time in practice, additional degrees, publications, and h-index (adjusted OR 1.23, 95% CI 0.79 – 1.92) (42).

For salary, 20 studies were included comparing 15,155 men to 7,837 women. Men's salary was $32,520 higher (mean difference $32,520, 95% CI $25,790 - $39,260, $p < .00001$, $I^2 = 94\%$, eFigure 15). Funnel plot showed evidence of publication bias skewed towards a higher mean difference (eFigure 16). Sensitivity analysis removing 1 poor quality study continued to show that men received a higher salary than women (mean difference $34,930, 95% CI $25,920 - $43,950, $p < .00001$, $I^2 = 95\%$, eFigure 17). Two studies reported adjusted estimates for salary. Jena et al. (2016) demonstrated that men earned more after controlling for time in practice, research funding, publications, and clinical trial participation (adjusted absolute difference $19,878, 95% CI $15,261 - $24,495) (44). Gambhir et al. (2020) also showed that men received a higher salary after adjusting for academic rank and specialty (adjusted absolute difference $45,904, 95% CI $13,264 - $103,137) (45).

For leadership, 42 studies were included comparing 48,613 men to 26,958 women. Men were 2.61 times more likely to be a departmental chair (OR 2.61, 95% CI 2.19 – 3.12, $p < .00001$, $I^2 = 55\%$, eFigure 18). Funnel plot demonstrated no evidence of publication bias (eFigure 19). Sensitivity analysis removing 6 poor quality studies continued to show that men were more likely to be departmental chairs (OR 2.60, 95% CI 2.16 – 3.13, $p < .00001$, $I^2 = 59\%$, eFigure 20). No studies reported adjusted estimates for this outcome.

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A meta-analysis of adjusted estimates for secondary outcomes was not performed due to a paucity of studies.

DISCUSSION

Summary of findings

This systematic review and meta-analysis of 218 studies including 991,207 academic physician data points demonstrated that men were 2.77 times more likely to be full professors than women. This was consistent across specialties and geographic regions. There has been some improvement over time, but in the 2010-2020 decade, men were 2.63 times more likely to be full professors. Although men practiced 6 years longer than women, men continued to be 1.83 times more likely to be full professors after pooling results from 7 studies that controlled for number of years in practice. Furthermore, men published 17.2 more papers, had a h-index that was 5.97 higher, received a salary that was $32,520 greater, and were 2.61 and 1.71 times more likely to be departmental chairs and receive federal research funding, respectively. Among single studies, men continued to have more publications, higher h-indices, and greater salaries after adjusting for factors including time in practice and specialty (43–45).

Comparison to existing literature

This was the first systematic review and meta-analysis of gender differences in faculty rank among academic physicians. Our findings are consistent with most of the literature demonstrating significant gender disparities in full professorship.

Jena et al. (2015) performed the largest single study on this topic, analyzing 91,073 US academic physicians. They showed that 28.6% of men were full professors compared to 11.9%

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of women, for an absolute difference of 16.7% (12). In their study, men were older, more likely to hold federal research grants, and published more papers. However, after adjusting for experience, research productivity, and specialty, women continued to be less likely than men to be full professors, with an absolute difference of 4% (12). We demonstrated similar findings across specialties, over time, and between geographic regions in a larger pooled cohort.

Many studies have analyzed gender inequity in a single specialty or country. Qamar and colleagues (2020) showed that men were more likely to be full professors than women in academic radiology in Canada (46). Others have demonstrated similar gender gaps in anesthesiology and surgery in Spain and Japan, respectively (47,48). Our analysis of 24 specialties across 18 countries confirmed that the gender disparity in academic rank is pervasive across specialties globally.

Explanation of findings

In our study, men practiced for 6 years longer than women, which may have contributed to men being 2.77 times more likely to be full professors. However, after pooling 7 studies that adjusted for number of years in practice, men continued to be 1.83 times more likely to attain full professorship. Therefore, differences in career duration does not fully explain the gender gap. This is consistent with previous studies showing that men were more likely to hold higher academic positions after adjusting for time in practice (37–39).

Several studies have explored reasons for gender inequity in academic medicine. In a survey of 4,285 US pediatricians, women were less likely to report that they received adequate mentorship (49). Other studies have demonstrated gender-based discrimination (50), unequal allocation of institutional resources (51), and lack of promotion tracks for women who take time

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off for childcare (52). In addition to inequity at work, inequity at home may also influence progress in academic medicine. Jolly and colleagues (2014) surveyed 1,049 US clinician-scientists and found that women spent 8.5 more hours per week on household activities than men (53). Due to these inequities, women are more likely to leave academia (54). These reasons could explain the gender differences in academic rank, research productivity, salary, and leadership.

Notably, our study demonstrated that a similar gender gap in full professorship exists for countries within and outside of North America. Despite geographic and cultural differences, gender inequity is a global phenomenon. This systemic problem exists not only in health care, but also in education (55), income (56), and political power (57).

Our analysis showed that men were more likely than women to be full professors across every specialty. Interestingly, 39% of faculty in obstetrics and gynecology were women, yet men were 4 times more likely to be full professors. Rayburn and colleagues (2012) analyzed US faculty in obstetrics and gynecology and found an increase in the proportion of female entry-level faculty, but they were primarily in the non-tenure track (58). They showed that promotion rates dropped from 35% in 1980-1989 to 26% in 2000-2009 (58). Therefore, in specialties where there is an increasing number of women entering but few opportunities for advancement, the gender gap may be widening.

We showed minimal improvement in gender disparities over time, with men being 2.63 times more likely to be full professors in 2010-2020. As demonstrated in the example of obstetrics and gynecology, waiting for more women to enter the academic pipeline will be insufficient to close the gender gap. One method of actively improving gender equity is implementing programs that directly support women. For example, Stanford University's School of Medicine established an Office of Diversity and Leadership in 2004 (59). They developed

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research awards, mentorship programs, and networking events for women (59). Over the next 6 years, the number of female full professors increased from 52 to 91 (59). This rise was 60% greater than the national promotion rate to full professor for women (59). Similar career development programs at Johns Hopkins University (60) and the University of Michigan (61) have demonstrated benefit to female academic physicians. However, Carr and colleagues (2017) demonstrated that many institutions do not have programs to support gender equity (62). This may explain the ongoing gender gap, which can be further closed by expanding interventions that support women to a national and global scale.

Limitations

This study was limited by high heterogeneity in most of the analyses conducted. This was expected as we pooled effect estimates across multiple specialties, data sources, and countries. Subgroup analysis by specialty demonstrated a reduction in heterogeneity. Therefore, data variability across specialties may be a source of heterogeneity in our study.

Funnel plots for secondary outcomes including publications, federal research funding, and salary demonstrated evidence of publication bias, which may have contributed to overestimation of the gender gap. However, there was no evidence of publication bias for the primary outcome of academic rank.

There may be selection bias due to the inclusion of cross-sectional studies that did not statistically adjusting for confounding. However, pooling of studies that controlled for potential confounders demonstrated that men continued to be more likely than women to be full professors. Sensitivity analysis were also conducted to remove poor quality studies, which did not have a significant impact on the pooled estimate for any outcome.

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CONCLUSIONS

This systematic review and meta-analysis provides a comprehensive synthesis of the literature on gender differences in faculty rank among academic physicians. We demonstrated that men were more likely to be full professors than women after adjusting for experience, academic productivity, and specialty. The gender gap exists across all specialties and geographic regions. Gender inequity persists even in the most contemporaneous studies, indicating that gender disparities in faculty rank will remain unless actively addressed. Greater efforts are needed to support women in academic medicine.

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None.

CONTRIBUTORS

All authors were involved in designing the study. TK and BL developed the database search. BL, JJB, and FH screened citations for inclusion, performed risk of bias analysis, and extracted data. BL performed statistical analysis. All authors were involved in interpreting the data. BL drafted the manuscript. All authors were involved in critical revision of the manuscript for important intellectual content. BL and MAO obtained funding. NNB and MAO provided supervision. The study guarantor is BL, who had full access to the data and takes responsibility for the integrity of data collection and accuracy of data analysis. All authors reviewed the final manuscript and agreed to be accountable for all aspects of the work and approved the final manuscript for submission. The corresponding author attests that all listed authors meet authorship criteria and no others meeting the criteria have been omitted.

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COMPETING INTERESTS

All authors have completed the *Unified Competing Interest form* (available on request from the corresponding author) and declare: Ben Li had financial support from the Canadian Society for Vascular Surgery National Student Research Award for the submitted work; no financial relationships with any organisations that might have an interest in the submitted work in the previous three years; no other relationships or activities that could appear to have influenced the submitted work.

PATIENT CONSENT FOR PUBLICATION

Not required.

ETHICS APPROVAL

This systematic review and meta-analysis was based on published data. The authors did not access any information that could lead to identification of a specific patient. Therefore, ethics approval was not required.

PROVENANCE AND PEER REVIEW

Not commissioned.

DATA SHARING STATEMENT

All data relevant to the study are included in the article or uploaded as supplementary information. Additional data are available upon reasonable request.

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TRANSPARENCY STATEMENT

BL, the study guarantor, affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that there were no discrepancies from the study as originally planned.

DISSEMINATION DECLARATION

Dissemination of results to study participants or patient organisations is not applicable for this systematic review and meta-analysis.

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FIGURE LEGENDS

Figure 1. PRISMA study flow diagram.

Figure 2. Forest plot of full professorship for men vs. women. M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

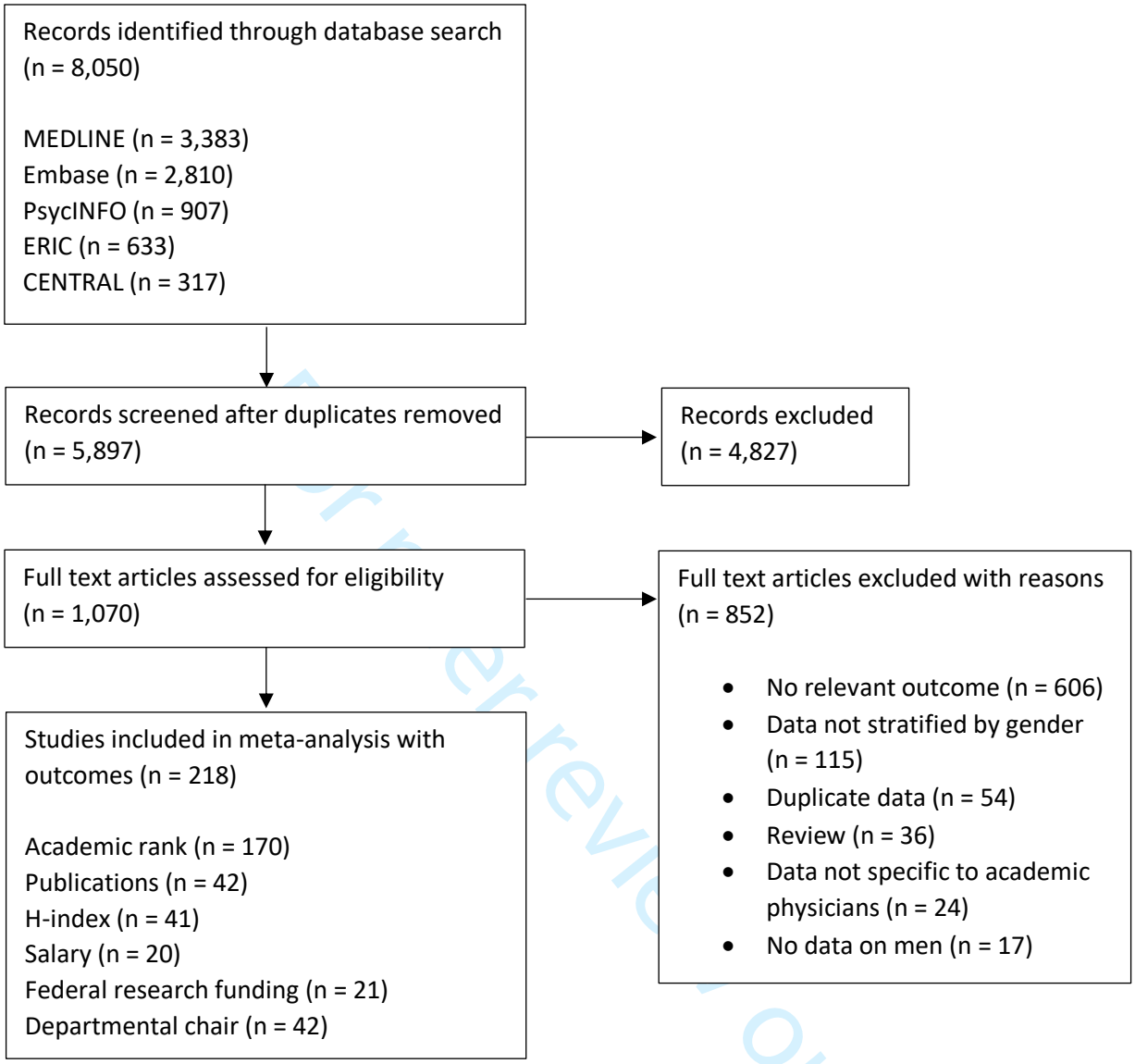
Figure 3. Forest plot of full professorship adjusted for years in practice (Dossani, Gawad, Jena, Orhurhu, Riaz, Smith, Sperling), specialty (Jena), publications (Dossani, Gawad, Jena, Riaz, Sperling), h-index (Dossani, Gawad, Orhurhu, Riaz, Sperling), additional PhD (Dossani, Gawad, Sperling), and institution (Dossani, Jena, Orhurhu, Riaz, Sperling). SE (standard error), Total (sample size), IV (inverse variance), Random (random effects), CI (confidence interval).

Figure 4. Meta-regression of full professorship based on data collection year. Each circle represents a unique study, the size of the circle indicates weight, and the blue line shows the trend over time.

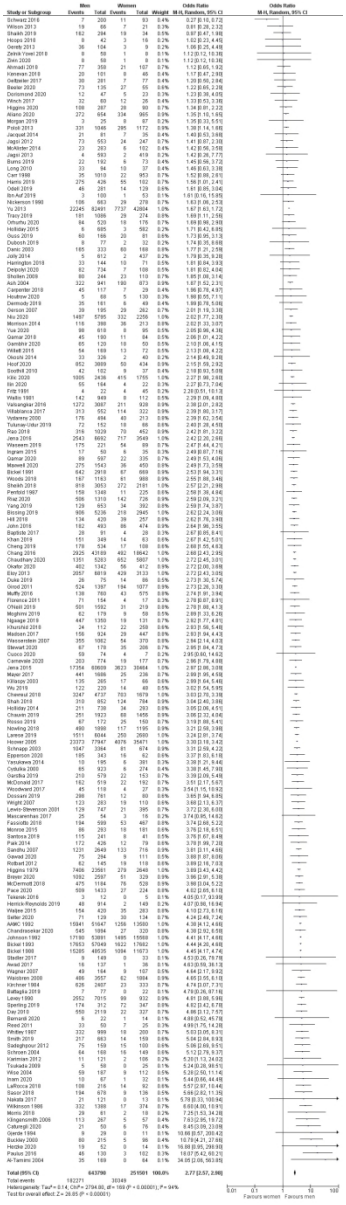
Figure 5. Subgroup analysis of full professorship by decade based on data collection year (Before 2000, 2000-2009, 2010-2020). M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

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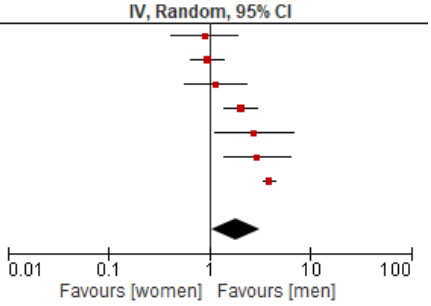
Forest plot of full professorship for men vs. women. M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

286x1004mm (72 x 72 DPI)

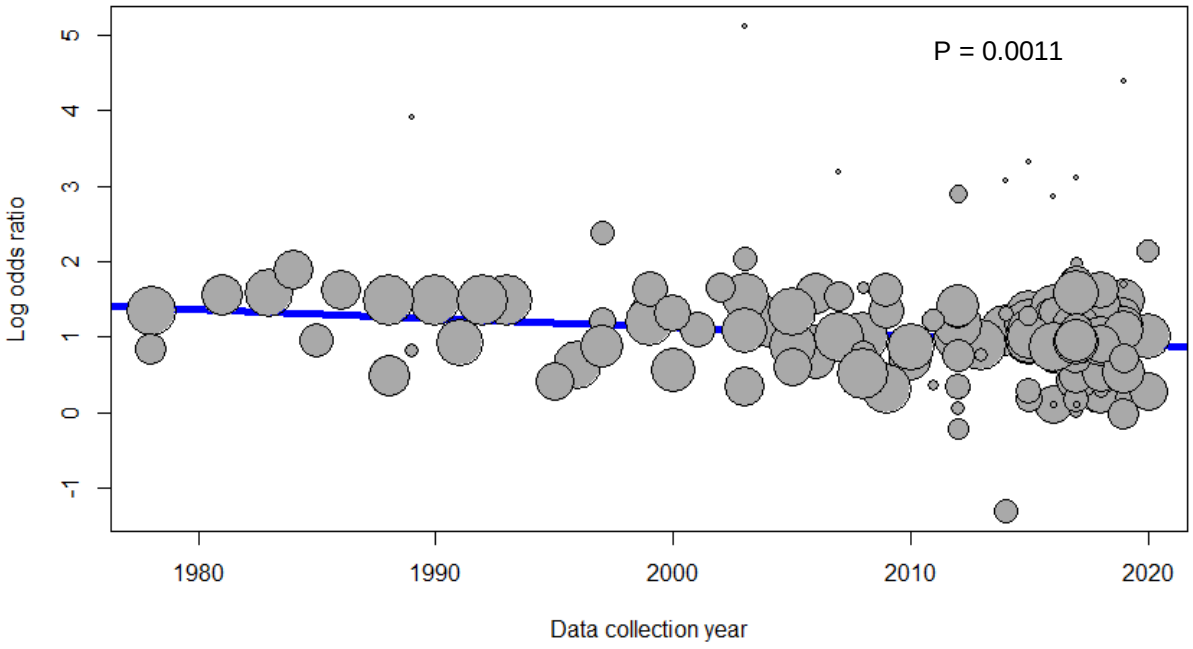
Study or Subgroup	log[Odds Ratio]	SE	Men Total	Women Total	Weight	Odds Ratio IV, Random, 95% CI
Dossani 2019	-0.1054	0.3889	761	80	13.0%	0.90 [0.42, 1.93]
Riaz 2020	-0.0513	0.2015	1310	726	15.8%	0.95 [0.64, 1.41]
Orhurhu 2020	0.1398	0.3671	520	176	13.4%	1.15 [0.56, 2.36]
Sperling 2019	0.7129	0.1957	312	347	15.9%	2.04 [1.39, 2.99]
Gawad 2020	1.026	0.4611	294	111	11.9%	2.79 [1.13, 6.89]
Smith 2019	1.0784	0.3896	663	159	13.0%	2.94 [1.37, 6.31]
Jena 2015	1.3552	0.0727	60609	30464	17.0%	3.88 [3.36, 4.47]
Total (95% CI)			**64469**	**32063**	**100.0%**	**1.83 [1.04, 3.20]**

Heterogeneity: Tau² = 0.47; Chi² = 65.23, df = 6 (P < 0.00001); I² = 91%

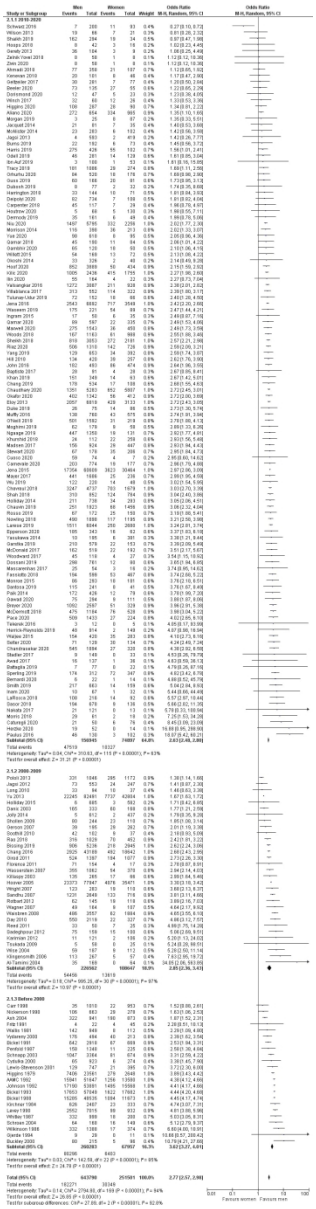
Test for overall effect: Z = 2.11 (P = 0.03)



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Subgroup analysis of full professorship by decade based on data collection year (Before 2000, 2000-2009, 2010-2020). M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

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Supplementary online content

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eTable 1. Search strategies for A) MEDLINE, B) Embase, C) Cochrane CENTRAL, PsycINFO, and ERIC

A) MEDLINE

1	Sex Characteristics/ (54058)
2	sex distribution/ (55145)
3	Sex Factors/ (263100)
4	Sexism/ (1967)
5	(sex or sexes or sexism).tw,kf. (559154)
6	gender*.tw,kf. (331420)
7	wom?n*.tw,kf. (1170111)
8	female*.tw,kf. (959904)
9	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 (2625210)
10	exp Physicians/ (140699)
11	physician*.tw,kf. (392562)
12	(medicine or medical).tw,kf. (1578309)
13	(surgery or surgical or surgeon*).tw,kf. (1912383)
14	(doctor or doctors).tw,kf. (122123)
15	(resident or residents).tw,kf. (156159)
16	(intern or interne or interns or internes).tw,kf. (5853)
17	(trainee or trainees).tw,kf. (25596)
18	clinician*.tw,kf. (224670)
19	10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 (3873810)
20	Faculty, Medical/ (13263)
21	exp Academic Medical Centers/ (93396)
22	(faculty or faculties).tw,kf. (48393)
23	(academic or academics).tw,kf. (133076)
24	20 or 21 or 22 or 23 (253893)
25	Achievement/ (15369)
26	career mobility/ (11441)
27	achievement*.tw,kf. (61273)
28	(rank or ranks).tw,kf. (110562)
29	professor*.tw,kf. (21283)
30	(career or careers).tw,kf. (42059)
31	(job or jobs).tw,kf. (61905)
32	(ladder or ladders).tw,kf. (10401)
33	advancement*.tw,kf. (54164)
34	promot*.tw,kf. (1017322)
35	25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (1364135)
36	9 and 19 and 24 and 35 (3465)
37	limit 36 to (case reports or comment or editorial or letter or meta analysis or "review" or "systematic review") (207)
38	36 not 37 (3383)

B) Embase

1	exp sex difference/ (400581)
2	sex ratio/ (72668)
3	sexism/ or gender bias/ (6087)
4	(sex or sexes or sexism).tw,kw. (794517)
5	gender*.tw,kw. (535055)

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6	wom?n*.tw,kw. (1669994)
7	female*.tw,kw. (1510495)
8	1 or 2 or 3 or 4 or 5 or 6 or 7 (3876952)
9	exp physician/ (790780)
10	physician*.tw,kw. (571261)
11	(medicine or medical).tw,kw. (2386371)
12	(surgery or surgical or surgeon*).tw,kw. (2693032)
13	(doctor or doctors).tw,kw. (182709)
14	(resident or residents).tw,kw. (207836)
15	(intern or interne or interns or internes).tw,kw. (9879)
16	(trainee or trainees).tw,kw. (42786)
17	clinician*.tw,kw. (322509)
18	9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 (5742530)
19	medical school/ (63562)
20	exp teaching hospital/ (173020)
21	(faculty or faculties).tw,kw. (70075)
22	(academic or academics).tw,kw. (223412)
23	19 or 20 or 21 or 22 (478892)
24	achievement/ (37218)
25	job performance/ (16941)
26	career mobility/ (10384)
27	achievement*.tw,kw. (81730)
28	(rank or ranks).tw,kw. (188284)
29	professor*.tw,kw. (23538)
30	(career or careers).tw,kw. (52124)
31	(job or jobs).tw,kw. (80710)
32	(ladder or ladders).tw,kw. (11600)
33	advancement*.tw,kw. (70446)
34	promot*.tw,kw. (1278758)
35	24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (1777343)
36	8 and 18 and 23 and 35 (6798)
37	limit 36 to embase (3085)
38	limit 37 to (editorial or letter or "review") (313)
39	37 not 38 (2810)

C) Cochrane CENTRAL

1	Sex Characteristics/ (1241)
2	sex distribution/ (746)
3	Sex Factors/ (5904)
4	Sexism/ (26)
5	(sex or sexes or sexism).tw. (43624)
6	gender*.tw. (32982)
7	wom?n*.tw. (151619)
8	female*.tw. (90532)
9	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 (281682)
10	exp Physicians/ (1794)
11	physician*.tw. (41471)
12	(medicine or medical).tw. (141009)
13	(surgery or surgical or surgeon*).tw. (197021)
14	(doctor or doctors).tw. (10694)
15	(resident or residents).tw. (10895)

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16	(intern or interne or interns or internes).tw. (631)
17	(trainee or trainees).tw. (2553)
18	clinician*.tw. (22755)
19	10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 (368556)
20	Faculty, Medical/ (106)
21	exp Academic Medical Centers/ (1757)
22	(faculty or faculties).tw. (3115)
23	(academic or academics).tw. (12131)
24	20 or 21 or 22 or 23 (16538)
25	Achievement/ (481)
26	career mobility/ (11)
27	achievement*.tw. (6235)
28	(rank or ranks).tw. (15788)
29	professor*.tw. (570)
30	(career or careers).tw. (763)
31	(job or jobs).tw. (2806)
32	(ladder or ladders).tw. (317)
33	advancement*.tw. (2232)
34	promot*.tw. (42698)
35	25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (70085)
36	9 and 19 and 24 and 35 (317)

D) PsycINFO

1	human sex differences/ (112551)
2	exp sex role attitudes/ (12493)
3	sex discrimination/ (2293)
4	(sex or sexes or sexism).ti,ab,id. (210417)
5	gender*.ti,ab,id. (221794)
6	wom?n*.ti,ab,id. (307718)
7	female*.ti,ab,id. (365668)
8	1 or 2 or 3 or 4 or 5 or 6 or 7 (800890)
9	exp Physicians/ (43420)
10	physician*.ti,ab,id. (64370)
11	(medicine or medical).ti,ab,id. (238071)
12	(surgery or surgical or surgeon*).ti,ab,id. (43436)
13	(doctor or doctors).ti,ab,id. (27901)
14	(resident or residents).ti,ab,id. (47303)
15	(intern or interne or interns or internes).ti,ab,id. (3474)
16	(trainee or trainees).ti,ab,id. (16486)
17	clinician*.ti,ab,id. (90454)
18	9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 (455043)
19	educational personnel/ (11565)
20	(faculty or faculties).ti,ab,id. (35800)
21	(academic or academics).ti,ab,id. (149957)
22	19 or 20 or 21 (182798)
23	achievement/ (9479)
24	occupational success/ (2867)
25	occupational mobility/ (1101)
26	exp occupational tenure/ (1223)
27	achievement*.ti,ab,id. (111516)
28	(rank or ranks).ti,ab,id. (23580)

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29	professor*.ti,ab,id. (15563)
30	(career or careers).ti,ab,id. (64031)
31	(job or jobs).ti,ab,id. (91586)
32	(ladder or ladders).ti,ab,id. (1646)
33	advancement*.ti,ab,id. (13461)
34	promot*.ti,ab,id. (191176)
35	23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (476581)
36	8 and 18 and 22 and 35 (952)
37	limit 36 to ("comment/reply" or editorial or letter or reviews) (46)
38	36 not 37 (907)

E) ERIC

NOFT(sex or sexes or sexism or gender* or wom?n* or female*) AND NOFT(physician* or medicine or medical or surgery or surgical or surgeon* or doctor or doctors or resident or residents or intern or interns or interne or internes or trainee or trainees or clinician*) AND NOFT(faculty or faculties or academic or academics) AND NOFT(achievement* or rank or ranks or professor* or career or careers or job or jobs or ladder or ladders or advancement* or promot*)

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eTable 2. Summary of included studies

Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
AAMC[1]	1992	US	Database	Multiple	1990	51647	13580	Academic rank
Abdellatif[2]	2019	North America, Asia, Oceania, Europe	Departmental websites	Multiple	2016	976	328	Publications, h-index
Ahmadi[3]	2018	US, Canada	Departmental websites	Radiology	2016	358	107	Academic rank
Akabas[4]	2019	US	Survey	Multiple	2015	2546	1105	Federal research funding
Aliano[5]	2020	Spain	Survey	Anesthesia	2018	654	965	Academic rank, departmental chair
Al-Tamimi[6]	2004	Saudi Arabia	Survey	Multiple	2003	169	64	Academic rank
Amering[7]	2011	Global	Database	Psychiatry	2007	3775	3775	Publications
Ash[8]	2004	US	Survey	Multiple	1996	941	873	Academic rank, salary
Awad[9]	2017	US	Departmental websites	Urology	2015	137	36	Academic rank, publications, h-index, federal research funding
Baptiste[10]	2017	US	Survey	General surgery	2015	91	28	Academic rank, tenure
Battaglia[11]	2019	US, Canada	Departmental websites	Radiology	2017	77	22	Academic rank, h-index
Beeler[12]	2020	Global	Survey	Radiation oncology	2018	135	55	Academic rank, publications, h-index
Bernardi[13]	2020	US	Survey	General surgery	2019	22	14	Academic rank
Bickel[14]	1988	US	Database	Multiple	1988	48535	11673	Academic rank

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Bickel[15]	1993	US	Database	Multiple	1993	57049	17682	Academic rank, tenure
Bickel[16]	1991	US	Database	Multiple	1991	2918	669	Academic rank
Bissing[17]	2019	US	Database	Anesthesia	2015	5236	2945	Academic rank
Breyer[18]	2020	US	Survey	Urology	2017	2597	329	Academic rank
Bruggmann[19]	2017	Germany	Database	Obstetrics and gynecology, otolaryngology	2013	12591	12857	Departmental chair
Buckley[20]	2000	US	Survey	Multiple	1997	215	96	Academic rank, tenure
Burns[21]	2019	Canada	Survey	Anesthesia	2018	192	73	Academic rank, department chair
Camacci[22]	2020	US	Departmental websites	Ophthalmology	2018	158	60	Publications, h-index
Carnevale[23]	2020	US	Departmental websites	Vascular surgery	2019	774	177	Academic rank, publications, h-index, federal research funding, departmental chair
Carpenter[24]	2018	US	Departmental websites	General surgery	2017	117	29	Academic rank
Carr[25]	1992	US	Survey	Internal medicine	1986	435	145	Tenure, department chair
Carr[26]	1998	US	Survey	Multiple	1995	1010	953	Publications
Caturegli[27]	2020	US	Database	General surgery	2020	50	76	Academic rank, publications, h-index, federal research funding
Chaiyachati[28]	2019	US	Survey	Internal medicine, pediatrics, family medicine	2016	58	104	Publications

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Chandrasekar [29]	2020	US, Canada	Departmental websites	Urology	2019	1894	320	Academic rank
Chang[30]	2016	US	Database	Multiple	2008	43189	18642	Academic rank
Chaudhary[31]	2020	US	Database	Psychiatry	2018	5283	5807	Academic rank
Chauvin[32]	2019	Canada	Departmental websites	Psychiatry	2016	1923	1456	Academic rank, publications
Cheng[33]	2019	US	Departmental websites	Vascular surgery	2018	534	108	Academic rank
Cheng[34]	2019	US	Survey	Multiple	2018	37	35	Salary
Chevreul[35]	2018	France	Database	Multiple	2015	4737	1679	Academic rank
Chokshi[36]	2009	US	Survey	General surgery	2007	275	36	Federal research funding
Colletti[37]	2000	US	Survey	General surgery	2000	45	9	Tenure
Cuoco[38]	2020	US	Database	Neurosurgery	2018	74	7	Academic rank, salary
Cydulka[39]	2000	US	Survey	Emergency medicine	1997	923	274	Academic rank
Danic[40]	2003	Croatia	Database	Multiple	2000	333	168	Academic rank
Day[41]	2010	US	Database	Orthopedic surgery	2006	2119	327	Academic rank
Deipolyi[42]	2020	US	Departmental websites	Interventional radiology	2019	734	108	Academic rank, publications, h-index
Dennis[43]	1990	US	Survey	Multiple	1987	85	51	Salary
Dermody[44]	2019	US	Database	Otolaryngology	2016	161	49	Academic rank, h-index, salary
Desai[45]	2018	US	Departmental websites	General surgery	2017	325	105	H-index
Diehl[46]	2008	US	Survey	Family medicine	2006	276	49	Salary

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Dorismond[47]	2020	US	Departmental websites	Otolaryngology	2019	47	23	Academic rank, h-index, federal research funding
Dossani[48]	2019	US	Database	Neurosurgery	2019	761	80	Academic rank
Dubosh[49]	2019	US	Departmental websites	Emergency medicine	2018	77	32	Academic rank, federal research funding
Duke[50]	2019	US	Survey	Multiple	2020	75	86	Academic rank
Eloy[51]	2013	US	Departmental websites	Multiple	2012	6819	3133	Academic rank, h-index, departmental chair
Eloy[52]	2013	US	Departmental websites	Otolaryngology	2012	149	55	Federal research funding
Epperson[53]	2020	US, Canada	Departmental websites	Otolaryngology	2018	343	62	Academic rank, h-index, departmental chair
Fassiotto[54]	2018	US	Database	Multiple	2015	599	467	Academic rank
Florence[55]	2011	US	Survey	General surgery	2006	154	17	Academic rank, departmental chair
Fritz[56]	1991	US	Survey	Pediatrics	1989	22	45	Academic rank
Gambhir[57]	2020	US	Database	General surgery	2016	120	50	Academic rank, salary
Garstka[58]	2019	US	Departmental websites	Otolaryngology	2017	579	153	Academic rank, h-index, departmental chair
Gauthier[59]	2013	Germany, Switzerland, Austria	Survey	Multiple	2011	67	62	Department chair
Gawad[60]	2020	Canada	Departmental websites	General surgery	2018	294	111	Academic rank, publications
Geltzeiler[61]	2017	US	Departmental websites	General surgery	2015	281	77	Academic rank, publications

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Gerety[62]	2013	US, Canada, Asia, South America, Europe	Survey	Plastic surgery	2012	104	9	Academic rank
Gerson[63]	2007	US	Survey	Gastroenterology	2006	195	262	Academic rank, departmental chair
Girod[64]	2011	US	Survey	Multiple	2007	1397	1077	Academic rank, publications
Gjerde[65]	1994	US	Survey	Family medicine	1989	29	11	Academic rank
Grimm[66]	2016	US	Departmental websites	Radiology	2015	2502	1262	Departmental chairs
Guss[67]	2019	US	Departmental websites	Radiation oncology	2017	166	81	Academic rank, federal research funding, salary, departmental chair
Hafeez[68]	2019	Global	Departmental websites	Psychiatry	2017	579	257	Publications, h-index
Harrington[69]	2018	US	Departmental websites	Radiology	2017	144	71	Academic rank
Harris[70]	2019	Global	Departmental websites	General surgery	2017	426	102	Academic rank, h-index
Herrick-Reynolds[71]	2019	US	Database	General surgery	2017	914	149	Academic rank
Herzke[72]	2020	US	Survey	Family medicine	2019	52	14	Academic rank
Higgins[73]	2020	US	Survey	Orthopedic surgery	2020	287	90	Academic rank
Higgins[74]	1979	US	Database	Multiple	1978	23561	2648	Academic rank
Hill[75]	2018	US	Survey	Obstetrics and gynecology	2015	420	257	Academic rank
Holliday[76]	2015	US	Survey	Multiple	2009	685	582	Academic rank

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Holliday[77]	2014	US	Departmental websites	Radiation oncology	2012	738	293	Academic rank, federal research funding
Hoof[78]	2020	US	Departmental websites	Orthopedic surgery	2018	3889	434	Academic rank
Hoops[79]	2018	US	Database	General surgery	2017	42	16	Academic rank
Hoover[80]	2005	US	Database	Multiple	2004	77947	35471	Academic rank
Houtrow[81]	2020	US	Survey	Physical medicine and rehabilitation	2017	68	130	Academic rank, salary, departmental chair
Houtrow[82]	2011	US	Survey	Physical medicine and rehabilitation	2009	25	61	Salary
Hurley[83]	2014	US	Database	Anesthesia	2008	4360	1783	Publications
Ibn Auf[84]	2019	Sudan	Survey	Multiple	2017	100	53	Academic rank
Ilin[85]	2020	Canada	Departmental websites	Urology	2019	164	22	Academic rank
Inam[86]	2020	Pakistan	Survey	General surgery	2019	67	32	Academic rank
Ingram[87]	2015	US	Departmental websites	Respirology	2015	50	35	Academic rank, h-index
Jacquet[88]	2014	US	Survey	Emergency medicine	2012	81	35	Academic rank
Jagsi[89]	2013	US	Survey	Multiple	2011	593	419	Academic rank, salary
Jagsi[90]	2012	US	Survey	Multiple	2003	553	247	Academic rank, publications, salary
Jaque[91]	2020	Germany	Database	Urology	2015	1944	499	Departmental chair
Jena[92]	2016	US	Database	Multiple	2013	6692	3549	Academic rank, publications, federal research funding, salary

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Jena[93]	2015	US	Database	Multiple	2014	60609	30464	Academic rank, publications, federal research funding
John[94]	2016	US	Departmental websites	Dermatology	2015	493	474	Academic rank, h-index, departmental chair
Johnson[95]	1992	US	Database	Multiple	1992	53891	15568	Academic rank
Jolly[96]	2014	US	Survey	Multiple	2009	612	437	Academic rank
Kaderli[97]	2015	Switzerland	Survey	General surgery	2011	448	64	Department chair
Karimian[98]	2012	Iran	Survey	Multiple	2008	121	106	Academic rank
Kenevan[99]	2018	US	Departmental websites	Anesthesia	2017	101	46	Academic rank, department chair
Khan[100]	2019	US	Departmental websites	Cardiology	2017	349	63	Academic rank, publications, h-index, federal research funding
Khan[101]	2014	US	Departmental websites	Neurosurgery	2012	1144	81	H-index
Khurshid[102]	2018	US, Canada	Departmental websites	Radiology	2017	112	258	Academic rank
Kilic[103]	2020	US	Survey	Pathology	2018	2436	1755	Academic rank
Killaspy[104]	2003	UK	Survey	Psychiatry	2001	265	66	Academic rank
Kim[105]	2019	US	Departmental websites	Urology	2017	327	118	Publications
Kirchner[106]	1984	US	Survey	Radiology	1981	2407	333	Academic rank
Klingensmith[107]	2006	US	Survey	General surgery	2003	267	57	Academic rank
Kloosterboer[108]	2020	US	Departmental websites	Ophthalmology	2019	84	32	Publications, h-index

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Komlenac[109]	2019	Sweden, Netherlands, Austria	Survey	Multiple	2012	468	550	Departmental chair
Larese[110]	2019	Italy	Database	Multiple	2015	6044	2680	Academic rank
LaRocca[111]	2018	US	Departmental websites	General surgery	2017	216	92	Academic rank
Levey[112]	1990	US	Survey	Internal medicine	1983	7015	932	Academic rank, federal research funding
Lewis-Stevenson[113]	2001	US	Survey	Family medicine	1999	747	395	Academic rank
Long[114]	2010	US	Departmental websites	Anesthesia	2008	94	37	Academic rank, department chair
Madsen[115]	2017	US	Survey	Emergency medicine	2015	924	447	Academic rank, salary, departmental chair
Mascarenhas[116]	2017	Canada	Database	Multiple	2014	54	16	Academic rank
Maxwell[117]	2020	US	Database	General surgery	2016	1543	450	Academic rank, h-index, salary
Mayer[118]	2017	US	Departmental websites	Urology	2016	1686	236	Academic rank, departmental chair, federal research funding
McAlister[119]	2014	Canada	Survey	Ophthalmology	2012	283	102	Academic rank
McDermott[120]	2018	US	Departmental websites	Neurology	2016	1184	528	Academic rank, publications, h-index
McDonald[121]	2017	US	Departmental websites	Radiology	2016	519	192	Academic rank, h-index
Mendoza[122]	2015	US	Survey	Pediatrics	2015	5104	6064	Departmental chair
Modena[123]	1999	Italy	Survey	Cardiology	1996	1407	294	Publications

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Moghimi[124]	2019	US, Canada	Departmental websites	Radiology	2016	179	58	Academic rank
Monroe[125]	2015	US	Survey	Multiple	2012	293	181	Academic rank, department chair
Morgan[126]	2019	US	Survey	Obstetrics and gynecology	2018	25	87	Academic rank
Morris[127]	2018	US	Database	General surgery	2017	61	18	Academic rank
Morrison[128]	2014	Canada	Database	Internal medicine	2010	398	213	Academic rank
Mueller[129]	2017	US	Departmental websites	Surgery	2017	160	52	Publications, h-index
Muffly[130]	2016	US	Database	Anesthesia	2015	760	575	Academic rank
Nakata[131]	2017	Japan	Database	General surgery	2015	121	13	Academic rank
Ngaage[132]	2019	US	Departmental websites	Neurosurgery	2018	1350	131	Academic rank, h-index
Nguyen[133]	2018	US	Database	General surgery	2017	331	213	H-index
Nickerson[134]	1990	US	Database	Multiple	1988	663	278	Academic rank
Niu[135]	2020	US	Database	Radiology	2017	5795	2256	Academic rank, departmental chair
Nowling[136]	2018	US	Database	Multiple	2012	1898	1195	Academic rank
Odell[137]	2019	US, Canada	Departmental websites	Neurosurgery	2017	1592	219	Academic rank
Okafor[138]	2020	US	Departmental websites	Otolaryngology	2020	1342	412	Academic rank
Okoshi[139]	2014	Japan	Database	General surgery	2013	326	40	Academic rank
O'Neill[140]	2019	US, Canada	Departmental websites	Radiology	2018	281	129	Academic rank
Orhurhu[141]	2020	US	Departmental websites	Anesthesia	2018	520	176	Academic rank

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Pace[142]	2020	US	Departmental websites	Urology	2018	1433	224	Academic rank, publications, h-index
Pagel[143]	2015	US	Database	Anesthesia	2015	315	82	Publications, h-index, federal research funding
Paik[144]	2014	US	Departmental websites	Plastic surgery	2012	426	79	Academic rank, departmental chair
Paulus[145]	2016	US	Survey	Multiple	2012	130	102	Academic rank
Penfold[146]	1987	Canada	Survey	Psychiatry	1985	1348	225	Academic rank, departmental chair
Pfeiffer[147]	2016	Germany	Database	Multiple	2013	385	539	Publications
Pololi[148]	2013	US	Survey	Multiple	2009	1046	1172	Academic rank
Qamar[149]	2020	Canada	Departmental websites	Radiology	2017	597	335	Academic rank
Qamar[150]	2018	US, Canada	Departmental websites	Radiology	2016	190	84	Academic rank, departmental chair
Rao[151]	2018	US	Database	Multiple	2005	1029	452	Academic rank
Raptis[152]	2012	North America, Europe	Survey	General surgery	2011	365	74	Publications
Rayburn[153]	2012	US	Database	Multiple	2009	2181	2077	Tenure
Reed[154]	2011	US	Database	Multiple	2007	50	25	Academic rank, publications
Reiser[155]	1993	US	Survey	Psychiatry	1984	235	45	Publications
Riaz[156]	2020	US	Database	Hematology	2017	1310	726	Academic rank, publications, h-index, federal research funding, departmental chair

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Robboy[157]	2017	US	Database	Pathology	2016	35	24	Publications, tenure
Rosso[158]	2019	France	Database	Multiple	2019	172	150	Academic rank
Rotbart[159]	2012	US	Database	Pediatrics	2009	145	118	Academic rank, departmental chair, tenure
Sadeghpour[160]	2012	US	Survey	Dermatology	2009	159	100	Academic rank, salary, tenure
Sandhu[161]	2007	UK	Survey	Multiple	2005	2649	716	Academic rank
Santosa[162]	2019	US	Departmental websites	Plastic surgery	2016	241	41	Academic rank, publications
Sasor[163]	2018	US	Departmental websites	Plastic surgery	2017	678	136	Academic rank, federal research funding
Schnapp[164]	2003	US	Survey	Respirology	1999	3364	674	Academic rank
Schroen[165]	2004	US	Survey	General surgery	1999	168	149	Academic rank, publications, tenure
Schwarz[166]	2016	Germany	Survey	General surgery	2014	200	93	Academic rank, department chair
Selter[167]	2020	US	Departmental websites	Obstetrics and gynecology	2018	129	134	Academic rank, departmental chair
Shah[168]	2018	US, Canada	Departmental websites	Dermatology	2017	852	784	Academic rank
Shaikh[169]	2019	Global	Database	Neurosurgery	2019	294	34	Academic rank
Sheikh[170]	2018	US	Departmental websites	Psychiatry	2018	3053	2181	Academic rank
Shiber[171]	2020	US	Survey	Obstetrics and gynecology	2019	41	88	Salary
Shollen[172]	2009	US	Survey	Multiple	2005	244	110	Academic rank
Singh[173]	2008	US	Survey	Gastroenterology	2015	168	25	Salary

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Smith[174]	2019	US	Departmental websites	Plastics	2018	663	159	Academic rank, departmental chair
Soothill[175]	2010	UK	Survey	Psychiatry	2008	102	37	Academic rank
Speck[176]	2012	US	Database	Multiple	2007	566	335	Tenure
Spencer[177]	2016	US	Survey	Urology	2017	660	73	Salary
Sperling[178]	2019	US	Departmental websites	Obstetrics and gynecology	2017	312	347	Academic rank, federal research funding
Stadler[179]	2017	Global	Survey	Multiple	2014	149	33	Academic rank
Steger[180]	2009	Austria	Survey	Multiple	2005	220	173	Publications, departmental chair
Stewart[181]	2020	US	Departmental websites	Dermatology	2019	178	206	Academic rank, publications, federal research funding
Svider[182]	2014	US	Database	Ophthalmology	2013	433	157	H-index, federal research funding
Tekerek[183]	2016	Turkey	Survey	Multiple	2016	12	5	Academic rank
Therattil[184]	2016	US	Departmental websites	Plastic surgery	2013	491	101	H-index
Tracy[185]	2019	US	Departmental websites	Otolaryngology	2019	1086	274	Academic rank
Tsukada[186]	2009	Japan	Survey	Cardiology	2007	58	25	Academic rank
Tulunay-Ugur[187]	2019	US	Survey	Otolaryngology	2018	152	66	Academic rank
Turaga[188]	2013	US	Departmental websites	General surgery	2012	87	3	Publications, h-index
Valsangkar[189]	2016	US	Departmental websites	General surgery	2016	3087	928	Academic rank, publications, federal research

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
								funding, departmental chair
Villablanca[190]	2017	US	Survey	Multiple	2010	552	322	Academic rank
Vydareny[191]	2000	US	Survey	Radiology	1997	494	213	Academic rank, tenure
Wagner[192]	2007	US	Survey	Physical rehabilitation and medicine	2007	164	107	Academic rank, federal research funding, tenure
Waisbren[193]	2008	US	Database	Multiple	2003	3557	1884	Academic rank
Waljee[194]	2015	US	Survey	Multiple	2012	420	283	Academic rank
Wallis[195]	1981	US	Survey	Multiple	1978	949	112	Academic rank
Warner[196]	2017	US	Database	Multiple	2015	3010	2435	Publications, h-index
Waseem[197]	2019	Global	Departmental websites	Internal medicine	2018	221	89	Academic rank
Wasserstein[198]	2007	US	Survey	Multiple	2003	1062	370	Academic rank
Weaver[199]	2015	US	Survey	Family medicine	2010	513	263	Salary
Whitley[200]	1987	US	Survey	Radiology	1986	999	200	Academic rank, publications
Wiler[201]	2019	US	Survey	Emergency medicine	2017	1430	797	Salary
Wilkes[202]	2015	UK, Ireland	Departmental websites	Neurosurgery	2014	290	25	H-index
Wilkinson[203]	1986	US	Database	Anesthesia	1984	1388	374	Academic rank
Willett[204]	2015	US	Survey	Internal medicine	2012	169	72	Academic rank
Wilson[205]	2013	US	Departmental websites	Radiation oncology	2012	66	21	Academic rank
Winch[206]	2017	UK	Survey	Pediatrics	2015	60	26	Academic rank

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Wise[207]	2004	Canada	Survey	Obstetrics and gynecology	2002	187	112	Academic rank
Woods[208]	2018	US	Database	Multiple	2017	1163	988	Academic rank
Woodward[209]	2017	US	Departmental websites	Gastroenterology	2015	118	27	Academic rank
Wright[210]	2007	US	Database	Multiple	2000	283	110	Academic rank
Wu[211]	2019	North America, Europe, Oceania	Departmental websites	General surgery	2017	220	48	Academic rank
Xierali[212]	2020	US	Database	Family medicine	2017	2887	2865	Tenure
Yang[213]	2019	US, Canada	Departmental websites	Physical medicine and rehabilitation	2017	653	392	Academic rank
Yasukawa[214]	2014	Japan	Survey	Multiple	2011	195	381	Academic rank
Yu[215]	2013	US	Database	Multiple	2008	82491	42804	Academic rank
Yue[216]	2020	Canada	Departmental websites	Orthopedic surgery	2019	618	95	Academic rank
Zelnik[217]	2018	Israel	Survey	Internal medicine	2016	58	8	Academic rank
Zivin[218]	2020	US	Survey	Multiple	2017	683	627	Academic rank

* AAMC (American Association of Medical Colleges), US (United States), UK (United Kingdom)

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eTable 3. Demographic data reported by 90 studies

Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Aliano 2020	Men	-	-	-	-	-	71.5%	82.2%	21.3%
	Women	-	-	-	-	-	62.2%	63.0%	14.5%
Ash 2004	Men	47.0	80.0%	-	58.0	28.0%	-	-	-
	Women	45.0	82.0%	-	56.0	29.0%	-	-	-
Awad 2017	Men	-	-	9.8	-	-	-	-	-
	Women	-	-	9.5	-	-	-	-	-
Baptiste 2017	Men	-	-	-	-	-	92.3%	94.1%	-
	Women	-	-	-	-	-	71.4%	90.0%	-
Beeler 2020	Men	-	65.9%	-	-	21.5%	-	-	-
	Women	-	63.6%	-	-	12.7%	-	-	-
Bernardi 2020	Men	-	-	-	-	-	91.0%	77.0%	-
	Women	-	-	-	-	-	71.0%	71.0%	-
Buckley 2000	Men	48.0	-	12.0	-	-	-	-	-
	Women	41.0	-	7.0	-	-	-	-	-
Burns 2019	Men	45.4	-	12.9	-	-	89.1%	57.1%	-
	Women	43.6	-	11.1	-	-	76.4%	43.8%	-
Camacci 2020	Men	-	-	-	-	-	-	-	7.0%
	Women	-	-	-	-	-	-	-	15.0%

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Carr 1992	Men	40.0	97.0%	-	58.0	18.0%	-	-	-
	Women	37.0	93.0%	-	55.0	14.0%	-	-	-
Carr 1998	Men	47.0	87.0%	13.0	59.0	28.0%	88.0%	84.0%	-
	Women	45.0	91.0%	11.0	55.0	28.1%	74.0%	69.0%	-
Chaiyachati 2019	Men	-	91.0%	-	-	-	-	-	-
	Women	-	89.0%	-	-	-	-	-	-
Chauvin 2019	Men	-	-	20.0	-	-	-	-	-
	Women	-	-	15.3	-	-	-	-	-
Colletti 2000	Men	-	-	-	-	-	100.0%	99.0%	11.0%
	Women	-	-	-	-	-	89.0%	44.0%	0.0%
Cydulka 2000	Men	40.7	86.0%	-	-	-	-	-	-
	Women	38.5	84.0%	-	-	-	-	-	-
Deipolyi 2020	Men	-	-	20.0	-	-	-	-	-
	Women	-	-	16.0	-	-	-	-	-
Dennis 1990	Men	-	-	-	-	-	65.0%	-	-
	Women	-	-	-	-	-	86.0%	-	-
Dermody 2019	Men	-	-	24.6	-	-	-	-	-
	Women	-	-	18.4	-	-	-	-	-
Dorismond 2020	Men	-	-	-	-	-	-	-	23.4%
	Women	-	-	-	-	-	-	-	26.1%

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Dossani 2019	Men	-	-	-	-	-	-	-	13.9%
	Women	-	-	-	-	-	-	-	18.6%
Epperson 2020	Men	-	-	20.3	-	-	-	-	-
	Women	-	-	13.9	-	-	-	-	-
Fassiotto 2018	Men	-	63.6%	-	-	-	-	-	-
	Women	-	54.2%	-	-	-	-	-	-
Florence 2011	Men	48.7	-	15.8	-	-	88.0%	90.5%	-
	Women	43.7	-	6.8	-	-	88.0%	64.7%	-
Fritz 1991	Men	-	-	-	52.0	25.0%	95.0%	-	-
	Women	-	-	-	48.0	15.0%	70.0%	-	-
Gauthier 2013	Men	45.6	-	-	-	-	68.6%	65.2%	-
	Women	36.6	-	-	-	-	77.4%	35.5%	-
Gawad 2020	Men	-	-	18.5	-	-	-	-	9.5%
	Women	-	-	13.5	-	-	-	-	12.6%
Geltzeiler 2017	Men	-	-	18.0	-	-	-	-	-
	Women	-	-	11.0	-	-	-	-	-
Gerety 2013	Men	53.6	-	-	-	-	-	-	-
	Women	45.1	-	-	-	-	-	-	-
Gerson 2007	Men	46.0	66.0%	8.6	-	-	-	-	-
	Women	42.0	62.0%	6.5	-	-	-	-	-

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Girod 2011	Men	49.9	96.6%	21.3	-	-	-	-	-
	Women	47.2	92.9%	16.8	-	-	-	-	-
Guss 2019	Men	-	-	12.0	-	-	-	-	26.5%
	Women	-	-	10.0	-	-	-	-	18.5%
Harris 2019	Men	-	-	23.0	-	-	-	-	21.9%
	Women	-	-	17.0	-	-	-	-	36.1%
Higgins 2020	Men	-	-	-	-	-	92.7%	94.4%	-
	Women	-	-	-	-	-	68.9%	71.1%	-
Higgins 1979	Men	-	-	-	-	-	-	-	7.8%
	Women	-	-	-	-	-	-	-	3.9%
Hill 2018	Men	-	89.0%	-	-	-	93.0%	91.0%	-
	Women	-	78.0%	-	-	-	87.0%	72.0%	-
Holliday 2015	Men	40.0	70.4%	-	-	-	92.2%	82.0%	30.4%
	Women	40.5	70.6%	-	-	-	87.4%	76.6%	15.5%
Holliday 2014	Men	-	-	17.0	-	-	-	-	17.2%
	Women	-	-	12.0	-	-	-	-	9.2%
Houtrow 2020	Men	-	-	-	-	-	-	-	23.9%
	Women	-	-	-	-	-	-	-	14.6%
Ibn 2019	Men	42.6	-	9.1	-	-	86.9%	-	10.2%
	Women	38.8	-	6.2	-	-	64.2%	-	11.3%

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Inam 2020	Men	-	-	-	-	-	74.6%	-	-
	Women	-	-	-	-	-	43.8%	-	-
Jagsi 2013	Men	40.5	68.8%	4.6	59.4	65.7%	92.9%	83.0%	34.4%
	Women	39.9	68.8%	4.7	54.0	65.4%	89.0%	79.5%	21.0%
Jagsi 2012	Men	45.0	75.1%	-	63.2	58.0%	91.0%	89.3%	49.2%
	Women	45.3	76.5%	-	58.1	57.0%	87.0%	84.2%	48.6%
Jena 2016	Men	51.5	-	19.4	-	-	-	-	-
	Women	46.4	-	14.6	-	-	-	-	-
Jena 2015	Men	52.3	-	20.8	-	-	-	-	-
	Women	46.8	-	15.1	-	-	-	-	-
Jolly 2014	Men	40.6	68.3%	-	59.0	-	92.5%	83.2%	-
	Women	40.0	66.4%	-	54.0	-	88.8%	79.2%	-
Kaderli 2015	Men	50.0	-	-	-	-	83.3%	84.8%	-
	Women	43.0	-	-	-	-	76.6%	42.2%	-
Kenevan 2018	Men	52.0	-	-	-	-	-	-	-
	Women	52.5	-	-	-	-	-	-	-
Khan 2019	Men	-	-	24.0	-	-	-	-	8.6%
	Women	-	-	21.0	-	-	-	-	7.9%
Klingensmith 2006	Men	49.9	-	-	-	-	-	-	-
	Women	43.5	-	-	-	-	-	-	-

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Komlenac 2019	Men	-	-	-	54.0	-	-	64.0%	-
	Women	-	-	-	43.0	-	-	39.0%	-
Long 2010	Men	53.0	-	-	-	-	-	-	-
	Women	51.0	-	-	-	-	-	-	-
Madsen 2017	Men	-	77.8%	12.0	-	-	-	-	-
	Women	-	77.8%	8.4	-	-	-	-	-
Maxwell 2020	Men	-	-	26.3	-	-	-	-	-
	Women	-	-	18.1	-	-	-	-	-
Mayer 2017	Men	-	-	22.0	-	-	-	-	2.8%
	Women	-	-	13.0	-	-	-	-	0.3%
McAlister 2014	Men	-	-	-	-	-	-	90.0%	-
	Women	-	-	-	-	-	-	81.0%	-
Modena 1999	Men	-	-	-	39.0	-	87.0%	-	-
	Women	-	-	-	37.7	-	66.0%	-	-
Monroe 2015	Men	49.0	-	-	-	-	-	-	-
	Women	43.0	-	-	-	-	-	-	-
Mueller 2017	Men	-	-	14.3	-	-	-	-	-
	Women	-	-	12.6	-	-	-	-	-
Muffly 2016	Men	50.9	-	-	-	-	-	-	-
	Women	47.5	-	-	-	-	-	-	-

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Ngaage 2019	Men	-	-	10.0	-	-	-	-	-
	Women	-	-	4.0	-	-	-	-	-
Pagel 2015	Men	-	-	20.0	-	-	-	-	-
	Women	-	-	16.0	-	-	-	-	-
Paulus 2016	Men	51.0	-	19.0	-	-	-	-	-
	Women	45.0	-	14.0	-	-	-	-	-
Reed 2011	Men	-	-	26.6	-	-	-	-	-
	Women	-	-	26.6	-	-	-	-	-
Riaz 2020	Men	-	-	20.0	-	-	-	-	-
	Women	-	-	14.0	-	-	-	-	-
Robboy 2017	Men	-	71.4%	-	-	-	-	-	-
	Women	-	87.5%	-	-	-	-	-	-
Sadeghpour 2012	Men	49.9	-	-	52.9	-	-	-	-
	Women	42.6	-	-	50.2	-	-	-	-
Santosa 2019	Men	-	-	-	-	-	-	-	11.6%
	Women	-	-	-	-	-	-	-	29.3%
Sasor 2018	Men	-	-	16.4	-	-	-	-	4.3%
	Women	-	-	9.9	-	-	-	-	4.4%
Schnapp 2003	Men	47.0	-	-	-	-	-	-	-
	Women	42.0	-	-	-	-	-	-	-

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Schroen 2004	Men	48.1	-	15.5	-	-	96.0%	90.0%	-
	Women	42.7	-	9.3	-	-	77.0%	62.0%	-
Schwarz 2016	Men	47.1	-	-	65.5	9.1%	93.0%	86.2%	-
	Women	47.5	-	-	59.7	15.1%	74.6%	77.4%	-
Shaikh 2019	Men	-	-	-	-	-	-	-	20.4%
	Women	-	-	-	-	-	-	-	17.6%
Shollen 2009	Men	-	-	-	57.0	42.0%	-	-	-
	Women	-	-	-	57.0	42.0%	-	-	-
Singh 2008	Men	45.0	-	-	58.0	-	95.0%	97.0%	-
	Women	46.0	-	-	53.0	-	92.0%	87.0%	-
Smith 2019	Men	51.6	75.7%	14.0	-	-	-	-	4.3%
	Women	44.6	73.4%	9.0	-	-	-	-	3.8%
Speck 2012	Men	37.3	71.9%	-	-	-	54.4%	-	15.7%
	Women	36.9	66.0%	-	-	-	44.5%	-	7.2%
Spencer 2016	Men	50.0	-	19.0	60.0	-	-	-	-
	Women	42.0	-	10.0	55.0	-	-	-	-
Stadler 2017	Men	-	-	9.3	-	-	-	-	-
	Women	-	-	9.2	-	-	-	-	-
Stewart 2020	Men	-	70.2%	22.4	-	-	-	-	27.0%
	Women	-	61.7%	16.1	-	-	-	-	22.8%

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Tracy 2019	Men	-	-	17.2	-	-	-	-	-
	Women	-	-	10.8	-	-	-	-	-
Tsukada 2009	Men	-	-	18.1	-	-	85.4%	69.2%	-
	Women	-	-	10.8	-	-	66.7%	50.0%	-
Tulunay-Ugur 2019	Men	-	72.0%	-	-	-	96.8%	-	-
	Women	-	64.2%	-	-	-	75.0%	-	-
Valsangkar 2016	Men	-	-	-	-	-	-	-	4.9%
	Women	-	-	-	-	-	-	-	11.3%
Wagner 2007	Men	47.3	81.3%	15.2	59.8	26.1%	-	-	9.8%
	Women	45.1	79.0%	12.4	54.3	22.2%	-	-	0.9%
Warner 2017	Men	-	71.7%	-	-	-	-	-	9.3%
	Women	-	71.8%	-	-	-	-	-	4.5%
Weaver 2015	Men	45.0	-	6.0	-	-	86.0%	46.0%	-
	Women	42.0	-	5.0	-	-	80.0%	45.0%	-
Wise 2004	Men	52.8	-	-	-	-	96.0%	-	18.0%
	Women	43.4	-	-	-	-	74.0%	-	28.0%
Yasukawa 2014	Men	-	-	-	-	-	90.0%	83.0%	53.0%
	Women	-	-	-	-	-	72.0%	78.0%	40.0%
Median across studies	Men	47.4	74.0%	18.0	58.0	27.1%	91.0%	84.0%	14.8%
	Women	43.5	75.0%	12.0	54.7	25.2%	75.0%	71.0%	14.6%

A dashed line (-) represents values that were not reported, could not be calculated from reported data, and authors were not able to provide.

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eTable 4. Quality assessment of included studies

Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
AAMC 1992	1	1	1	2	0	2	0	7
Abdellatif 2019	0	1	1	2	1	2	1	8
Ahmadi 2018	1	1	1	2	2	2	1	10
Akabas 2019	0	1	0	2	0	1	1	5
Aliano 2020	1	1	0	2	1	1	1	7
Al-Tamimi 2004	1	1	0	2	0	1	1	6
Amering 2011	1	1	1	2	0	2	1	8
Ash 2004	1	1	0	2	2	1	1	8
Awad 2017	1	1	1	2	2	2	1	10
Baptiste 2017	1	1	0	2	0	1	1	6
Battaglia 2019	1	1	1	2	0	2	1	8
Beeler 2020	0	1	0	2	0	1	1	5
Bernardi 2020	0	0	0	2	0	1	0	3
Bickel 1988	1	1	1	2	0	2	1	8
Bickel 1991	1	1	1	2	0	2	0	7
Bickel 1993	1	1	1	2	0	2	0	7
Bissing 2019	1	1	1	2	0	2	1	8
Breyer 2020	1	1	1	2	0	1	1	7

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Bruggmann 2017	1	1	1	2	0	2	0	7
Buckley 2000	1	1	0	2	0	1	1	6
Burns 2019	0	1	0	2	0	1	1	5
Camacci 2020	0	1	1	2	0	2	1	7
Carnevale 2020	1	1	1	2	0	2	1	8
Carpenter 2018	0	1	1	2	0	2	1	7
Carr 1992	1	1	0	2	2	1	1	8
Carr 1998	1	1	1	2	2	1	1	9
Caturegli 2020	1	1	1	2	2	2	1	10
Chaiyachati 2019	1	1	0	2	2	1	1	8
Chandrasekar 2020	1	1	1	2	2	2	1	10
Chang 2016	1	1	1	2	0	2	1	8
Chaudhary 2020	1	1	1	2	0	2	1	8
Chauvin 2019	1	1	1	2	1	2	1	9
Cheng 2019	1	1	0	2	0	1	1	6
Cheng 2019	1	1	1	2	2	2	1	10
Chevreul 2018	1	1	1	2	2	2	1	10
Chokshi 2009	0	1	0	2	2	1	1	7
Colletti 2000	1	1	0	2	0	1	1	6

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Cuoco 2020	0	1	1	2	0	2	1	7
Cydulka 2000	1	1	0	2	2	1	1	8
Danic 2003	1	1	1	2	0	2	1	8
Day 2010	1	1	1	2	0	2	1	8
Deipolyi 2020	1	1	1	2	2	2	1	10
Dennis 1990	0	1	0	2	0	1	1	5
Dermody 2019	1	1	1	2	2	2	1	10
Desai 2018	1	1	1	2	1	2	1	9
Diehl 2008	0	1	0	2	2	1	1	7
Dorismond 2020	0	1	1	2	0	2	1	7
Dossani 2019	1	1	1	2	2	2	1	10
Dubosh 2019	1	1	1	2	0	2	1	8
Duke 2019	1	1	1	2	0	2	1	8
Eloy 2013	1	1	1	2	1	2	1	9
Eloy 2013	0	1	1	2	2	2	1	9
Epperson 2020	1	1	1	2	2	2	1	10
Fassiotto 2018	1	1	1	2	2	2	1	10
Florence 2011	0	1	0	2	0	1	1	5
Fritz 1991	1	1	0	2	0	1	1	6
Gambhir 2020	1	1	1	2	2	2	1	10

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Garstka 2019	1	1	1	2	1	2	1	9
Gauthier 2013	1	1	0	2	0	1	1	6
Gawad 2020	1	1	1	2	2	2	1	10
Geltzeiler 2017	1	1	1	2	1	2	1	9
Gerety 2013	0	1	0	2	0	1	1	5
Gerson 2007	0	1	0	2	0	1	1	5
Girod 2011	0	1	0	2	2	1	1	7
Gjerde 1994	0	1	0	2	0	1	1	5
Grimm 2016	1	1	1	2	0	2	0	7
Guss 2019	1	1	1	2	2	2	1	10
Hafeez 2019	0	1	1	2	0	2	1	7
Harrington 2018	0	1	1	2	0	2	1	7
Harris 2019	0	1	1	2	2	2	1	9
Herrick-Reynolds 2019	1	1	1	2	0	2	1	8
Herzke 2020	0	1	0	2	0	1	1	5
Higgins 1979	1	1	1	2	1	2	0	8
Higgins 2020	0	1	0	2	0	1	1	5
Hill 2018	0	1	0	2	0	1	1	5
Holliday 2014	1	1	1	2	2	2	1	10
Holliday 2015	0	1	0	2	0	1	1	5

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Hoof 2020	1	1	1	2	0	2	1	8
Hoops 2018	1	1	1	2	0	2	1	8
Hoover 2005	1	1	1	2	0	2	1	8
Houtrow 2011	1	1	0	2	0	1	1	6
Houtrow 2020	1	1	1	2	2	1	1	9
Hurley 2014	1	1	1	2	1	2	1	9
Ibn 2019	1	1	0	2	0	1	1	6
Ilin 2020	1	1	1	2	0	2	1	8
Inam 2020	1	1	0	2	0	1	1	6
Ingram 2015	1	1	1	2	0	2	1	8
Jacquet 2014	0	1	0	2	0	1	1	5
Jagsi 2012	0	1	1	2	2	1	1	8
Jagsi 2013	0	1	1	2	2	2	1	9
Jaque 2020	1	1	1	2	0	2	1	8
Jena 2015	1	1	1	2	2	2	1	10
Jena 2016	1	1	1	2	2	2	1	10
John 2016	1	1	1	2	1	2	1	9
Johnson 1992	1	1	1	2	0	2	0	7
Jolly 2014	0	1	0	2	1	1	1	6
Kaderli 2015	1	1	0	2	0	1	1	6

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Karimian 2012	1	1	0	2	0	1	1	6
Kenevan 2018	0	1	1	2	0	2	1	7
Khan 2014	1	1	1	2	1	2	1	9
Khan 2019	1	1	1	2	1	2	1	9
Khurshid 2018	1	1	1	2	0	2	1	8
Kilic 2020	1	1	1	2	0	2	1	8
Killaspy 2003	1	1	0	2	0	1	1	6
Kim 2019	1	1	1	2	2	2	1	10
Kirchner 1984	0	1	0	2	0	1	0	4
Klingensmith 2006	1	1	0	2	0	1	1	6
Kloosterboer 2020	0	1	1	2	0	2	1	7
Komlenac 2019	0	1	0	2	0	1	1	5
Larese 2019	1	1	1	2	0	2	1	8
LaRocca 2018	1	1	1	2	2	2	1	10
Levey 1990	0	1	0	2	0	1	1	5
Lewis-Stevenson 2001	1	1	0	2	0	1	1	6
Long 2010	1	1	1	2	0	2	1	8
Madsen 2017	1	1	0	2	2	1	1	8
Mascarenhas 2017	1	1	1	2	0	2	0	7

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Maxwell 2020	1	1	1	2	2	2	1	10
Mayer 2017	1	1	1	2	0	2	1	8
McAlister 2014	0	1	0	2	0	1	1	5
McDermott 2018	1	1	1	2	1	2	1	9
McDonald 2017	1	1	1	2	0	2	1	8
Mendoza 2015	0	1	0	2	0	1	1	5
Modena 1999	0	1	0	2	0	1	1	5
Moghimi 2019	1	1	1	2	0	2	1	8
Monroe 2015	1	1	0	2	0	1	1	6
Morgan 2019	0	1	0	2	0	1	1	5
Morris 2018	1	1	1	2	0	2	1	8
Mueller 2017	1	1	1	2	1	2	1	9
Muffly 2016	1	1	1	2	0	2	1	8
Nakata 2017	1	1	1	2	2	2	1	10
Ngaage 2019	1	1	1	2	2	2	1	10
Nguyen 2018	1	1	1	2	2	2	1	10
Nickerson 1990	1	1	1	2	0	2	1	8
Niu 2020	1	1	1	2	0	2	1	8
Nowling 2018	1	1	1	2	0	2	1	8

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Odell 2019	1	1	1	2	0	2	1	8
Okafor 2020	1	1	1	2	2	2	1	10
Okoshi 2014	1	1	1	2	0	2	1	8
O'Neill 2019	1	1	1	2	0	2	1	8
Orhurhu 2020	1	1	1	2	2	2	1	10
Pace 2020	1	1	1	2	2	2	1	10
Pagel 2015	0	1	1	2	1	2	1	8
Paik 2014	1	1	1	2	1	2	1	9
Paulus 2016	1	1	0	2	0	1	1	6
Penfold 1987	0	1	0	2	0	1	0	4
Pfeiffer 2016	1	1	1	2	0	2	1	8
Pololi 2013	1	1	0	2	0	1	1	6
Qamar 2018	1	1	1	2	1	2	1	9
Qamar 2020	1	1	1	2	2	2	1	10
Rao 2018	1	1	1	2	0	2	1	8
Raptis 2012	1	1	0	2	0	1	1	6
Rayburn 2012	1	1	1	2	1	2	1	9
Reed 2011	1	1	1	2	2	2	1	10
Reiser 1993	0	1	0	2	0	1	1	5
Riaz 2020	1	1	1	2	2	2	1	10

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Robboy 2017	1	1	1	2	0	2	1	8
Rose 2015	1	1	1	2	2	2	1	10
Rosso 2019	1	1	1	2	0	2	1	8
Rotbart 2012	1	1	1	2	1	2	1	9
Sadeghpour 2012	0	1	0	2	2	1	1	7
Sandhu 2007	1	1	1	2	0	1	1	7
Santosa 2019	0	1	1	2	2	2	1	9
Sasor 2018	1	1	1	2	0	2	1	8
Schnapp 2003	1	1	0	2	0	1	1	6
Schroen 2004	0	1	0	2	0	1	1	5
Schwarz 2016	1	1	0	2	0	2	1	7
Selter 2020	1	1	1	2	0	2	1	8
Shah 2018	1	1	1	2	0	2	1	8
Shaikh 2019	1	1	1	2	0	2	1	8
Sheikh 2018	1	1	1	2	0	2	1	8
Shiber 2020	1	1	0	2	2	1	1	8
Shollen 2009	1	1	0	2	0	1	1	6
Singh 2008	0	1	0	2	2	1	1	7
Smith 2019	1	1	1	2	1	2	1	9
Soothill 2010	0	1	0	2	0	1	1	5

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Speck 2012	1	1	1	2	0	2	1	8
Spencer 2016	0	1	0	2	2	1	1	7
Sperling 2019	1	1	1	2	2	2	1	10
Stadler 2017	1	1	0	2	0	1	1	6
Steger 2009	0	1	0	2	0	1	1	5
Stewart 2020	0	1	1	2	2	2	1	9
Svider 2014	0	1	1	2	2	2	1	9
Tekerek 2016	0	0	0	2	0	1	1	4
Therattil 2016	1	1	1	2	2	2	1	10
Tracy 2019	1	1	1	2	0	2	1	8
Tsukada 2009	0	1	0	2	0	1	1	5
Tulunay-Ugur 2019	0	0	0	2	0	1	1	4
Turaga 2013	0	1	1	2	1	2	1	8
Valsangkar 2016	1	1	1	2	1	2	1	9
Villablanca 2017	1	1	0	2	0	1	1	6
Vydareny 2000	0	1	0	2	0	1	1	5
Wagner 2007	0	1	0	2	2	1	1	7
Waisbren 2008	1	1	1	2	2	2	1	10
Waljee 2015	1	1	0	2	0	1	1	6

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Wallis 1981	0	1	0	2	0	1	0	4
Warner 2017	1	1	1	2	2	2	1	10
Waseem 2019	1	1	1	2	0	1	1	7
Wasserstein 2007	1	1	0	2	0	1	1	6
Weaver 2015	0	1	0	2	2	1	1	7
Whitley 1987	0	1	0	2	0	1	1	5
Wiler 2019	1	1	0	2	2	1	1	8
Wilkes 2015	1	1	1	2	0	2	1	8
Wilkinson 1986	1	1	1	2	0	2	1	8
Willett 2015	1	1	0	2	2	1	1	8
Wilson 2013	0	1	1	2	0	2	1	7
Winch 2017	0	1	0	2	0	1	1	5
Wise 2004	1	1	0	2	0	1	1	6
Woods 2018	1	1	1	2	0	2	1	8
Woodward 2017	1	1	1	2	0	2	1	8
Wright 2007	1	1	1	2	2	2	1	10
Wu 2019	1	1	1	2	0	1	1	7
Xierali 2020	1	1	1	2	0	2	1	8
Yang 2019	1	1	1	2	0	2	1	8

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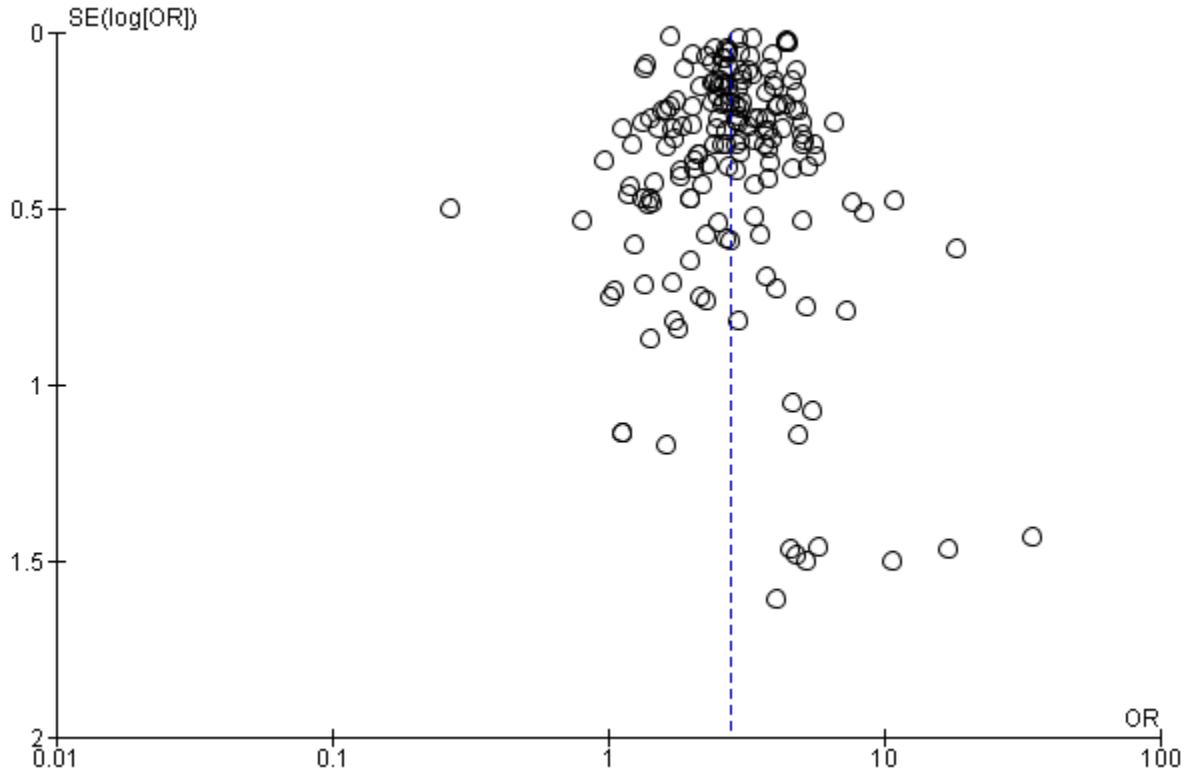
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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Yasukawa 2014	1	1	0	2	2	1	1	8
Yu 2013	1	1	1	2	1	2	1	9
Yue 2020	1	1	1	2	0	2	1	8
Zelnik 2018	0	1	0	2	0	1	1	5
Zivin 2020	1	1	0	2	2	1	1	8

*Quality assessment was performed using Newcastle-Ottawa Scale (NOS) for cross-sectional studies. A maximum of one point is awarded for representativeness of cohort, adequacy of sample size, characterization of non-respondents, and statistical tests. A maximum of two points is awarded for characterization of non-respondents, ascertainment of exposure, and comparability of groups. An overall maximum score of 10 can be awarded per study. NOS score ≥ 8 is low risk, 6 – 7 is moderate risk, and ≤ 5 is high risk.

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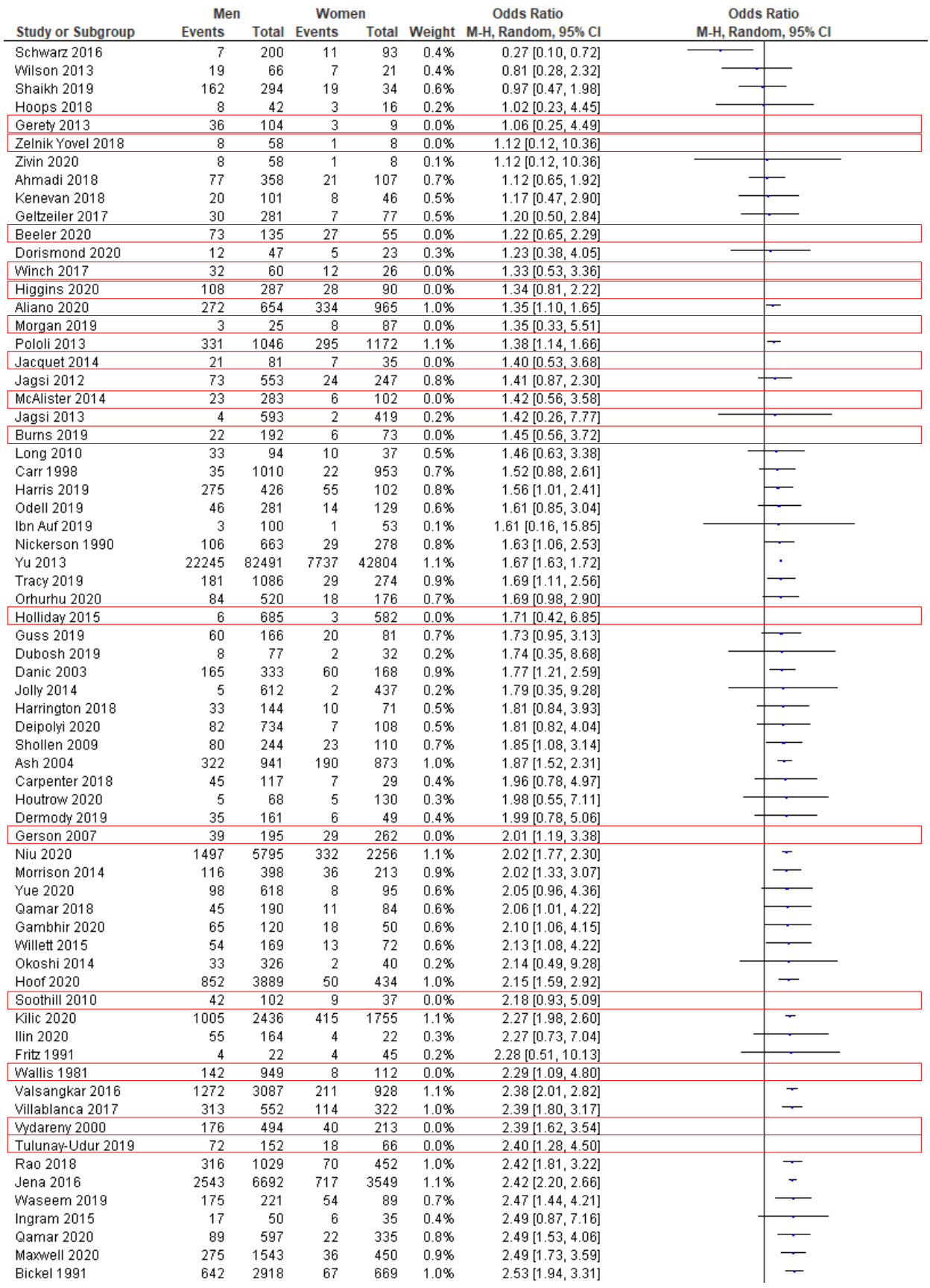
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eFigure 1. Funnel plot of full professorship for men vs. women. SE (standard error), OR (odds ratio).

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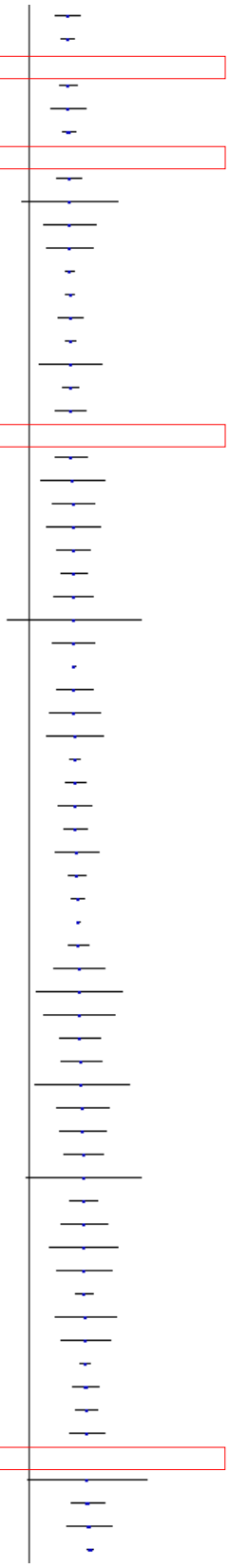
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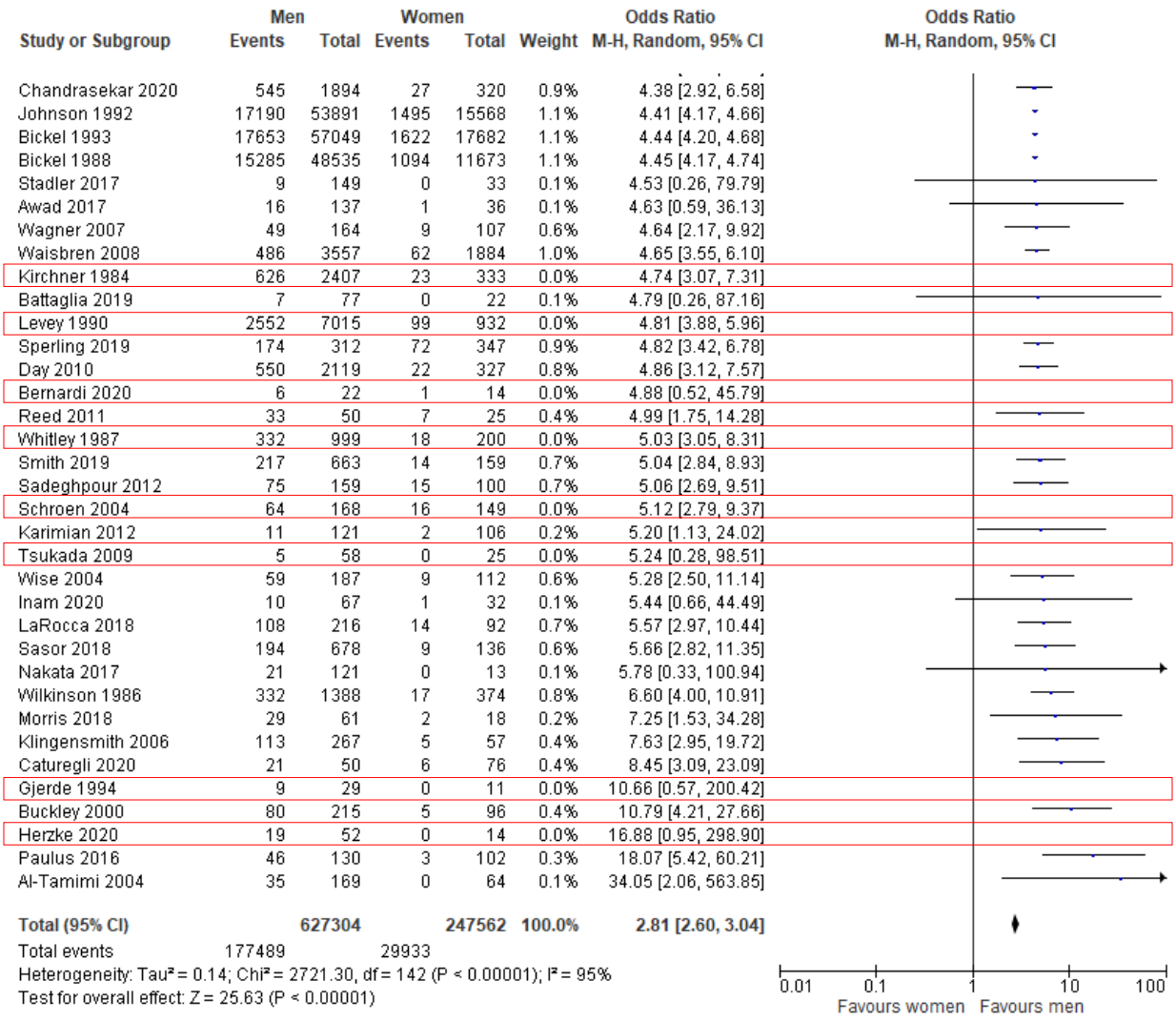
Study or Subgroup	Men Events	Men Total	Women Events	Women Total	Weight	Odds Ratio M-H, Random, 95% CI
Schwarz 2016	7	200	11	93	0.4%	0.27 [0.10, 0.72]
Wilson 2013	19	66	7	21	0.4%	0.81 [0.28, 2.32]
Shaikh 2019	162	294	19	34	0.6%	0.97 [0.47, 1.98]
Hoops 2018	8	42	3	16	0.2%	1.02 [0.23, 4.45]
Gerety 2013	36	104	3	9	0.0%	1.06 [0.25, 4.49]
Zelnik Yovel 2018	8	58	1	8	0.0%	1.12 [0.12, 10.36]
Zivin 2020	8	58	1	8	0.1%	1.12 [0.12, 10.36]
Ahmadi 2018	77	358	21	107	0.7%	1.12 [0.65, 1.92]
Kenevan 2018	20	101	8	46	0.5%	1.17 [0.47, 2.90]
Geltzeiler 2017	30	281	7	77	0.5%	1.20 [0.50, 2.84]
Beeler 2020	73	135	27	55	0.0%	1.22 [0.65, 2.29]
Dorismond 2020	12	47	5	23	0.3%	1.23 [0.38, 4.05]
Winch 2017	32	60	12	26	0.0%	1.33 [0.53, 3.36]
Higgins 2020	108	287	28	90	0.0%	1.34 [0.81, 2.22]
Aliano 2020	272	654	334	965	1.0%	1.35 [1.10, 1.65]
Morgan 2019	3	25	8	87	0.0%	1.35 [0.33, 5.51]
Pololi 2013	331	1046	295	1172	1.1%	1.38 [1.14, 1.66]
Jacquet 2014	21	81	7	35	0.0%	1.40 [0.53, 3.68]
Jagsi 2012	73	553	24	247	0.8%	1.41 [0.87, 2.30]
McAlister 2014	23	283	6	102	0.0%	1.42 [0.56, 3.58]
Jagsi 2013	4	593	2	419	0.2%	1.42 [0.26, 7.77]
Burns 2019	22	192	6	73	0.0%	1.45 [0.56, 3.72]
Long 2010	33	94	10	37	0.5%	1.46 [0.63, 3.38]
Carr 1998	35	1010	22	953	0.7%	1.52 [0.88, 2.61]
Harris 2019	275	426	55	102	0.8%	1.56 [1.01, 2.41]
Odell 2019	46	281	14	129	0.6%	1.61 [0.85, 3.04]
Ibn Auf 2019	3	100	1	53	0.1%	1.61 [0.16, 15.85]
Nickerson 1990	106	663	29	278	0.8%	1.63 [1.06, 2.53]
Yu 2013	22245	82491	7737	42804	1.1%	1.67 [1.63, 1.72]
Tracy 2019	181	1086	29	274	0.9%	1.69 [1.11, 2.56]
Orhurhu 2020	84	520	18	176	0.7%	1.69 [0.98, 2.90]
Holliday 2015	6	685	3	582	0.0%	1.71 [0.42, 6.85]
Guss 2019	60	166	20	81	0.7%	1.73 [0.95, 3.13]
Dubosh 2019	8	77	2	32	0.2%	1.74 [0.35, 8.68]
Danic 2003	165	333	60	168	0.9%	1.77 [1.21, 2.59]
Jolly 2014	5	612	2	437	0.2%	1.79 [0.35, 9.28]
Harrington 2018	33	144	10	71	0.5%	1.81 [0.84, 3.93]
Deipolyi 2020	82	734	7	108	0.5%	1.81 [0.82, 4.04]
Shollen 2009	80	244	23	110	0.7%	1.85 [1.08, 3.14]
Ash 2004	322	941	190	873	1.0%	1.87 [1.52, 2.31]
Carpenter 2018	45	117	7	29	0.4%	1.96 [0.78, 4.97]
Houtrow 2020	5	68	5	130	0.3%	1.98 [0.55, 7.11]
Dermody 2019	35	161	6	49	0.4%	1.99 [0.78, 5.06]
Gerson 2007	39	195	29	262	0.0%	2.01 [1.19, 3.38]
Niu 2020	1497	5795	332	2256	1.1%	2.02 [1.77, 2.30]
Morrison 2014	116	398	36	213	0.9%	2.02 [1.33, 3.07]
Yue 2020	98	618	8	95	0.6%	2.05 [0.96, 4.36]
Qamar 2018	45	190	11	84	0.6%	2.06 [1.01, 4.22]
Gambhir 2020	65	120	18	50	0.6%	2.10 [1.06, 4.15]
Willett 2015	54	169	13	72	0.6%	2.13 [1.08, 4.22]
Okoshi 2014	33	326	2	40	0.2%	2.14 [0.49, 9.28]
Hoof 2020	852	3889	50	434	1.0%	2.15 [1.59, 2.92]
Soothill 2010	42	102	9	37	0.0%	2.18 [0.93, 5.09]
Kilic 2020	1005	2436	415	1755	1.1%	2.27 [1.98, 2.60]
Ilin 2020	55	164	4	22	0.3%	2.27 [0.73, 7.04]
Fritz 1991	4	22	4	45	0.2%	2.28 [0.51, 10.13]
Wallis 1981	142	949	8	112	0.0%	2.29 [1.09, 4.80]
Valsangkar 2016	1272	3087	211	928	1.1%	2.38 [2.01, 2.82]
Villablanca 2017	313	552	114	322	1.0%	2.39 [1.80, 3.17]
Vydareny 2000	176	494	40	213	0.0%	2.39 [1.62, 3.54]
Tulunay-Udur 2019	72	152	18	66	0.0%	2.40 [1.28, 4.50]
Rao 2018	316	1029	70	452	1.0%	2.42 [1.81, 3.22]
Jena 2016	2543	6692	717	3549	1.1%	2.42 [2.20, 2.66]
Waseem 2019	175	221	54	89	0.7%	2.47 [1.44, 4.21]
Ingram 2015	17	50	6	35	0.4%	2.49 [0.87, 7.16]
Qamar 2020	89	597	22	335	0.8%	2.49 [1.53, 4.06]
Maxwell 2020	275	1543	36	450	0.9%	2.49 [1.73, 3.59]
Bickel 1991	642	2918	67	669	1.0%	2.53 [1.94, 3.31]

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Study or Subgroup	Men Events	Men Total	Women Events	Women Total	Weight	Odds Ratio M-H, Random, 95% CI
Woods 2018	167	1163	61	988	1.0%	2.55 [1.88, 3.46]
Sheikh 2018	818	3053	272	2181	1.1%	2.57 [2.21, 2.98]
Penfold 1987	158	1348	11	225	0.0%	2.58 [1.38, 4.84]
Riaz 2020	506	1310	142	726	1.0%	2.59 [2.09, 3.21]
Yang 2019	129	653	34	392	0.9%	2.59 [1.74, 3.87]
Bissing 2019	906	5236	218	2945	1.1%	2.62 [2.24, 3.06]
Hill 2018	134	420	39	257	0.0%	2.62 [1.76, 3.90]
John 2016	182	493	86	474	1.0%	2.64 [1.96, 3.55]
Baptiste 2017	28	91	4	28	0.3%	2.67 [0.85, 8.41]
Khan 2019	151	349	14	63	0.7%	2.67 [1.42, 5.01]
Cheng 2019	178	534	17	108	0.7%	2.68 [1.55, 4.63]
Chang 2016	2925	43189	492	18642	1.1%	2.68 [2.43, 2.95]
Chaudhary 2020	1351	5283	652	5807	1.1%	2.72 [2.45, 3.01]
Okafor 2020	402	1342	56	412	1.0%	2.72 [2.00, 3.69]
Eloy 2013	2057	6819	429	3133	1.1%	2.72 [2.43, 3.05]
Duke 2019	26	75	14	86	0.6%	2.73 [1.30, 5.74]
Girod 2011	524	1397	194	1077	1.1%	2.73 [2.26, 3.30]
Muffly 2016	138	760	43	575	0.9%	2.74 [1.91, 3.94]
Florence 2011	71	154	4	17	0.0%	2.78 [0.87, 8.91]
O'Neill 2019	501	1592	31	219	0.9%	2.78 [1.88, 4.13]
Moghimi 2019	62	179	9	58	0.5%	2.89 [1.33, 6.26]
Ngaage 2019	447	1350	19	131	0.8%	2.92 [1.77, 4.81]
Khurshid 2018	24	112	22	258	0.7%	2.93 [1.56, 5.48]
Madsen 2017	156	924	29	447	0.9%	2.93 [1.94, 4.43]
Wasserstein 2007	355	1062	54	370	1.0%	2.94 [2.14, 4.03]
Stewart 2020	67	178	35	206	0.8%	2.95 [1.84, 4.73]
Cuoco 2020	59	74	4	7	0.2%	2.95 [0.60, 14.62]
Carnevale 2020	203	774	19	177	0.8%	2.96 [1.79, 4.88]
Jena 2015	17354	60609	3623	30464	1.1%	2.97 [2.86, 3.09]
Mayer 2017	441	1686	25	236	0.8%	2.99 [1.95, 4.59]
Killaspy 2003	135	265	17	66	0.7%	2.99 [1.64, 5.46]
Wu 2019	122	220	14	48	0.6%	3.02 [1.54, 5.95]
Chevreul 2018	3247	4737	703	1679	1.1%	3.03 [2.70, 3.39]
Shah 2018	310	852	124	784	1.0%	3.04 [2.40, 3.86]
Holliday 2014	211	738	34	293	0.9%	3.05 [2.06, 4.51]
Chauvin 2019	251	1923	68	1456	1.0%	3.06 [2.32, 4.04]
Rosso 2019	67	172	25	150	0.8%	3.19 [1.88, 5.41]
Nowling 2018	490	1898	117	1195	1.0%	3.21 [2.58, 3.98]
Larese 2019	1511	6044	250	2680	1.1%	3.24 [2.81, 3.74]
Hoover 2005	23373	77947	4076	35471	1.1%	3.30 [3.18, 3.42]
Schnapp 2003	1047	3364	81	674	1.0%	3.31 [2.59, 4.22]
Epperson 2020	185	343	16	62	0.7%	3.37 [1.83, 6.18]
Yasukawa 2014	10	195	6	381	0.4%	3.38 [1.21, 9.44]
Cydulka 2000	65	923	6	274	0.5%	3.38 [1.45, 7.90]
Garstka 2019	210	579	22	153	0.8%	3.39 [2.09, 5.49]
McDonald 2017	162	519	22	192	0.8%	3.51 [2.17, 5.67]
Woodward 2017	45	118	4	27	0.3%	3.54 [1.15, 10.92]
Dossani 2019	298	761	12	80	0.7%	3.65 [1.94, 6.85]
Wright 2007	123	283	19	110	0.7%	3.68 [2.13, 6.37]
Lewis-Stevenson 2001	129	747	21	395	0.8%	3.72 [2.30, 6.00]
Mascarenhas 2017	25	54	3	16	0.3%	3.74 [0.95, 14.62]
Fassiotto 2018	194	599	53	467	0.9%	3.74 [2.68, 5.22]
Monroe 2015	86	293	18	181	0.7%	3.76 [2.18, 6.51]
Santosa 2019	115	241	8	41	0.5%	3.76 [1.67, 8.49]
Paik 2014	172	426	12	79	0.6%	3.78 [1.99, 7.20]
Sandhu 2007	1231	2649	133	716	1.0%	3.81 [3.11, 4.66]
Gawad 2020	75	294	9	111	0.6%	3.88 [1.87, 8.06]
Rotbart 2012	62	145	19	118	0.7%	3.89 [2.16, 7.03]
Higgins 1979	7406	23561	279	2648	1.1%	3.89 [3.43, 4.42]
Breyer 2020	1092	2597	51	329	1.0%	3.96 [2.91, 5.38]
McDermott 2018	475	1184	76	528	1.0%	3.98 [3.04, 5.22]
Pace 2020	509	1433	27	224	0.9%	4.02 [2.65, 6.10]
Tekerek 2016	3	12	0	5	0.0%	4.05 [0.17, 93.99]
Herrick-Reynolds 2019	48	914	2	149	0.2%	4.07 [0.98, 16.94]
Waljee 2015	154	420	35	283	0.9%	4.10 [2.73, 6.16]
Selter 2020	71	129	30	134	0.7%	4.24 [2.49, 7.24]
AAMC 1992	15941	51647	1256	13580	1.1%	4.38 [4.12, 4.66]

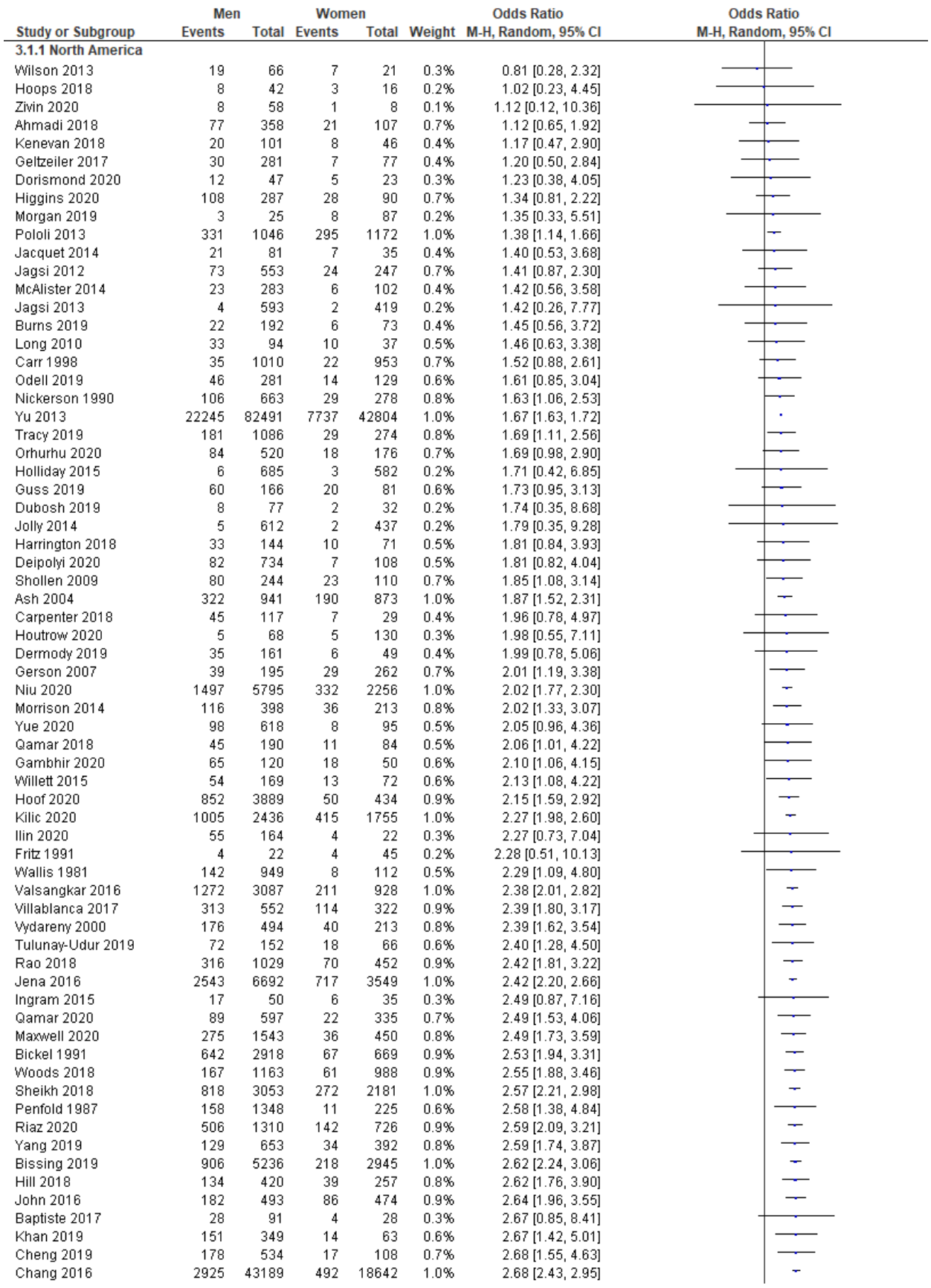


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eFigure 2. Sensitivity analysis of full professorship removing 27 poor quality studies (red box, NOS score ≤ 5). M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

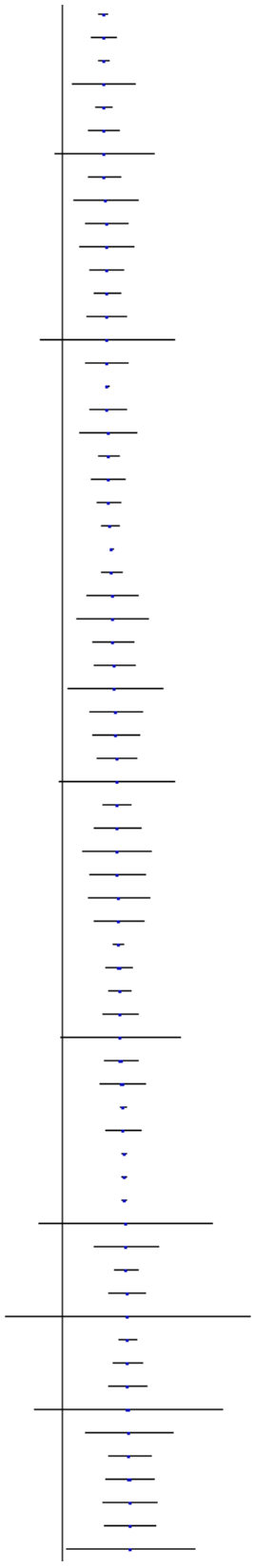
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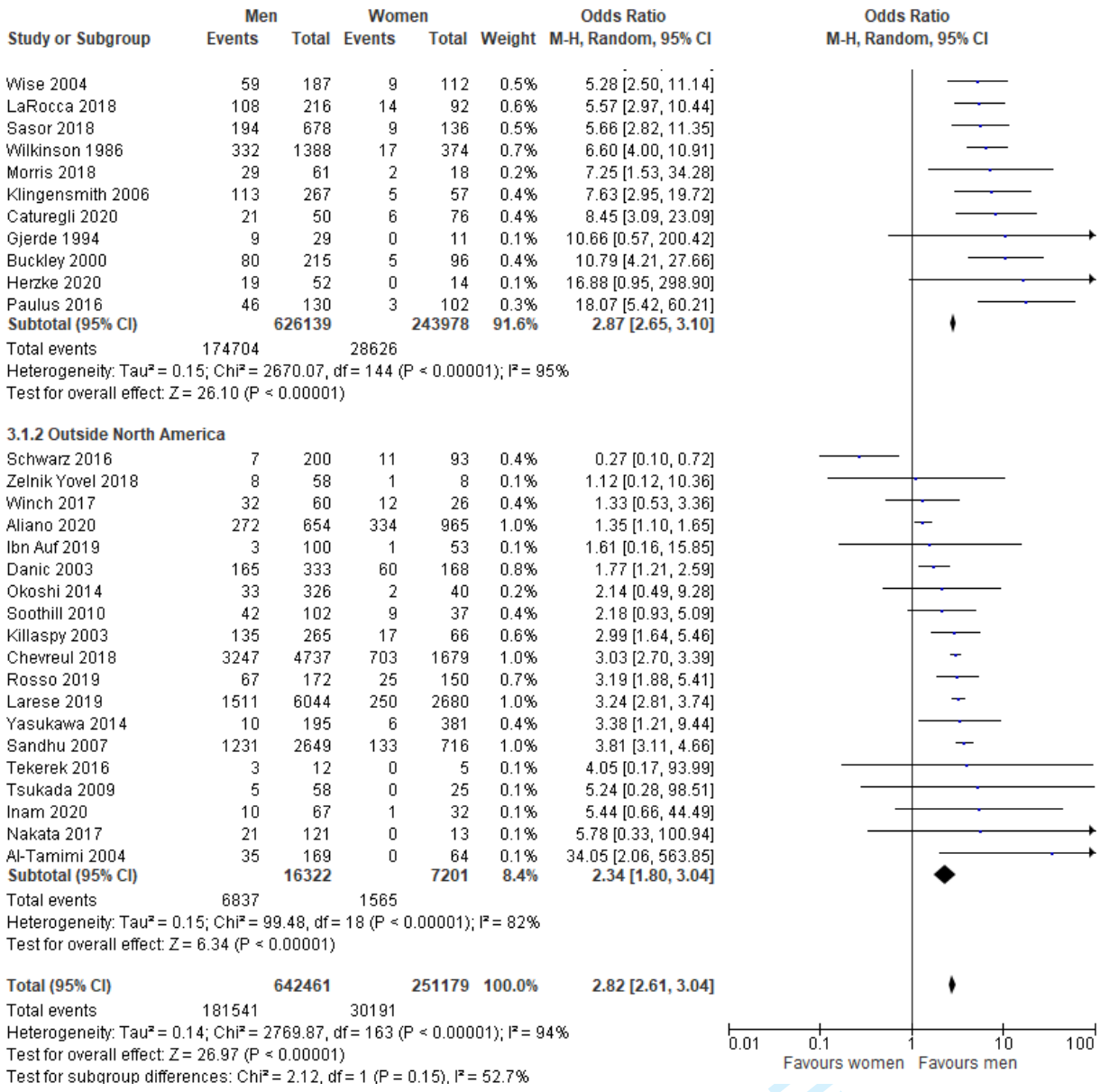
Study or Subgroup	Men Events	Men Total	Women Events	Women Total	Weight	Odds Ratio M-H, Random, 95% CI
3.1.1 North America						
Wilson 2013	19	66	7	21	0.3%	0.81 [0.28, 2.32]
Hoops 2018	8	42	3	16	0.2%	1.02 [0.23, 4.45]
Zivin 2020	8	58	1	8	0.1%	1.12 [0.12, 10.36]
Ahmadi 2018	77	358	21	107	0.7%	1.12 [0.65, 1.92]
Kenevan 2018	20	101	8	46	0.4%	1.17 [0.47, 2.90]
Geltzeiler 2017	30	281	7	77	0.4%	1.20 [0.50, 2.84]
Dorismond 2020	12	47	5	23	0.3%	1.23 [0.38, 4.05]
Higgins 2020	108	287	28	90	0.7%	1.34 [0.81, 2.22]
Morgan 2019	3	25	8	87	0.2%	1.35 [0.33, 5.51]
Pololi 2013	331	1046	295	1172	1.0%	1.38 [1.14, 1.66]
Jacquet 2014	21	81	7	35	0.4%	1.40 [0.53, 3.68]
Jagsi 2012	73	553	24	247	0.7%	1.41 [0.87, 2.30]
McAlister 2014	23	283	6	102	0.4%	1.42 [0.56, 3.58]
Jagsi 2013	4	593	2	419	0.2%	1.42 [0.26, 7.77]
Burns 2019	22	192	6	73	0.4%	1.45 [0.56, 3.72]
Long 2010	33	94	10	37	0.5%	1.46 [0.63, 3.38]
Carr 1998	35	1010	22	953	0.7%	1.52 [0.88, 2.61]
Odell 2019	46	281	14	129	0.6%	1.61 [0.85, 3.04]
Nickerson 1990	106	663	29	278	0.8%	1.63 [1.06, 2.53]
Yu 2013	22245	82491	7737	42804	1.0%	1.67 [1.63, 1.72]
Tracy 2019	181	1086	29	274	0.8%	1.69 [1.11, 2.56]
Orhurhu 2020	84	520	18	176	0.7%	1.69 [0.98, 2.90]
Holliday 2015	6	685	3	582	0.2%	1.71 [0.42, 6.85]
Guss 2019	60	166	20	81	0.6%	1.73 [0.95, 3.13]
Dubosh 2019	8	77	2	32	0.2%	1.74 [0.35, 8.68]
Jolly 2014	5	612	2	437	0.2%	1.79 [0.35, 9.28]
Harrington 2018	33	144	10	71	0.5%	1.81 [0.84, 3.93]
Deipolyi 2020	82	734	7	108	0.5%	1.81 [0.82, 4.04]
Shollen 2009	80	244	23	110	0.7%	1.85 [1.08, 3.14]
Ash 2004	322	941	190	873	1.0%	1.87 [1.52, 2.31]
Carpenter 2018	45	117	7	29	0.4%	1.96 [0.78, 4.97]
Houtrow 2020	5	68	5	130	0.3%	1.98 [0.55, 7.11]
Dermody 2019	35	161	6	49	0.4%	1.99 [0.78, 5.06]
Gerson 2007	39	195	29	262	0.7%	2.01 [1.19, 3.38]
Niu 2020	1497	5795	332	2256	1.0%	2.02 [1.77, 2.30]
Morrison 2014	116	398	36	213	0.8%	2.02 [1.33, 3.07]
Yue 2020	98	618	8	95	0.5%	2.05 [0.96, 4.36]
Qamar 2018	45	190	11	84	0.5%	2.06 [1.01, 4.22]
Gambhir 2020	65	120	18	50	0.6%	2.10 [1.06, 4.15]
Willett 2015	54	169	13	72	0.6%	2.13 [1.08, 4.22]
Hoof 2020	852	3889	50	434	0.9%	2.15 [1.59, 2.92]
Kilic 2020	1005	2436	415	1755	1.0%	2.27 [1.98, 2.60]
Ilin 2020	55	164	4	22	0.3%	2.27 [0.73, 7.04]
Fritz 1991	4	22	4	45	0.2%	2.28 [0.51, 10.13]
Wallis 1981	142	949	8	112	0.5%	2.29 [1.09, 4.80]
Valsangkar 2016	1272	3087	211	928	1.0%	2.38 [2.01, 2.82]
Villablanca 2017	313	552	114	322	0.9%	2.39 [1.80, 3.17]
Vydareny 2000	176	494	40	213	0.8%	2.39 [1.62, 3.54]
Tulunay-Udur 2019	72	152	18	66	0.6%	2.40 [1.28, 4.50]
Rao 2018	316	1029	70	452	0.9%	2.42 [1.81, 3.22]
Jena 2016	2543	6692	717	3549	1.0%	2.42 [2.20, 2.66]
Ingram 2015	17	50	6	35	0.3%	2.49 [0.87, 7.16]
Qamar 2020	89	597	22	335	0.7%	2.49 [1.53, 4.06]
Maxwell 2020	275	1543	36	450	0.8%	2.49 [1.73, 3.59]
Bickel 1991	642	2918	67	669	0.9%	2.53 [1.94, 3.31]
Woods 2018	167	1163	61	988	0.9%	2.55 [1.88, 3.46]
Sheikh 2018	818	3053	272	2181	1.0%	2.57 [2.21, 2.98]
Penfold 1987	158	1348	11	225	0.6%	2.58 [1.38, 4.84]
Riaz 2020	506	1310	142	726	0.9%	2.59 [2.09, 3.21]
Yang 2019	129	653	34	392	0.8%	2.59 [1.74, 3.87]
Bissing 2019	906	5236	218	2945	1.0%	2.62 [2.24, 3.06]
Hill 2018	134	420	39	257	0.8%	2.62 [1.76, 3.90]
John 2016	182	493	86	474	0.9%	2.64 [1.96, 3.55]
Baptiste 2017	28	91	4	28	0.3%	2.67 [0.85, 8.41]
Khan 2019	151	349	14	63	0.6%	2.67 [1.42, 5.01]
Cheng 2019	178	534	17	108	0.7%	2.68 [1.55, 4.63]
Chang 2016	2925	43189	492	18642	1.0%	2.68 [2.43, 2.95]

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Study or Subgroup	Men Events	Men Total	Women Events	Women Total	Weight	Odds Ratio M-H, Random, 95% CI
Chaudhary 2020	1351	5283	652	5807	1.0%	2.72 [2.45, 3.01]
Okafor 2020	402	1342	56	412	0.9%	2.72 [2.00, 3.69]
Eloy 2013	2057	6819	429	3133	1.0%	2.72 [2.43, 3.05]
Duke 2019	26	75	14	86	0.5%	2.73 [1.30, 5.74]
Girod 2011	524	1397	194	1077	1.0%	2.73 [2.26, 3.30]
Muffly 2016	138	760	43	575	0.8%	2.74 [1.91, 3.94]
Florence 2011	71	154	4	17	0.3%	2.78 [0.87, 8.91]
O'Neill 2019	501	1592	31	219	0.8%	2.78 [1.88, 4.13]
Moghimi 2019	62	179	9	58	0.5%	2.89 [1.33, 6.26]
Ngaage 2019	447	1350	19	131	0.7%	2.92 [1.77, 4.81]
Khurshid 2018	24	112	22	258	0.6%	2.93 [1.56, 5.48]
Madsen 2017	156	924	29	447	0.8%	2.93 [1.94, 4.43]
Wasserstein 2007	355	1062	54	370	0.9%	2.94 [2.14, 4.03]
Stewart 2020	67	178	35	206	0.7%	2.95 [1.84, 4.73]
Cuoco 2020	59	74	4	7	0.2%	2.95 [0.60, 14.62]
Carnevale 2020	203	774	19	177	0.7%	2.96 [1.79, 4.88]
Jena 2015	17354	60609	3623	30464	1.0%	2.97 [2.86, 3.09]
Mayer 2017	441	1686	25	236	0.8%	2.99 [1.95, 4.59]
Wu 2019	122	220	14	48	0.6%	3.02 [1.54, 5.95]
Shah 2018	310	852	124	784	0.9%	3.04 [2.40, 3.86]
Holliday 2014	211	738	34	293	0.8%	3.05 [2.06, 4.51]
Chauvin 2019	251	1923	68	1456	0.9%	3.06 [2.32, 4.04]
Nowling 2018	490	1898	117	1195	0.9%	3.21 [2.58, 3.98]
Hoover 2005	23373	77947	4076	35471	1.0%	3.30 [3.18, 3.42]
Schnapp 2003	1047	3364	81	674	0.9%	3.31 [2.59, 4.22]
Epperson 2020	185	343	16	62	0.6%	3.37 [1.83, 6.18]
Cydulka 2000	65	923	6	274	0.4%	3.38 [1.45, 7.90]
Garstka 2019	210	579	22	153	0.7%	3.39 [2.09, 5.49]
McDonald 2017	162	519	22	192	0.7%	3.51 [2.17, 5.67]
Woodward 2017	45	118	4	27	0.3%	3.54 [1.15, 10.92]
Dossani 2019	298	761	12	80	0.6%	3.65 [1.94, 6.85]
Wright 2007	123	283	19	110	0.7%	3.68 [2.13, 6.37]
Lewis-Stevenson 2001	129	747	21	395	0.7%	3.72 [2.30, 6.00]
Mascarenhas 2017	25	54	3	16	0.2%	3.74 [0.95, 14.62]
Fassiotto 2018	194	599	53	467	0.9%	3.74 [2.68, 5.22]
Monroe 2015	86	293	18	181	0.7%	3.76 [2.18, 6.51]
Santosa 2019	115	241	8	41	0.5%	3.76 [1.67, 8.49]
Paik 2014	172	426	12	79	0.6%	3.78 [1.99, 7.20]
Gawad 2020	75	294	9	111	0.5%	3.88 [1.87, 8.06]
Rotbart 2012	62	145	19	118	0.6%	3.89 [2.16, 7.03]
Higgins 1979	7406	23561	279	2648	1.0%	3.89 [3.43, 4.42]
Breyer 2020	1092	2597	51	329	0.9%	3.96 [2.91, 5.38]
McDermott 2018	475	1184	76	528	0.9%	3.98 [3.04, 5.22]
Pace 2020	509	1433	27	224	0.8%	4.02 [2.65, 6.10]
Herrick-Reynolds 2019	48	914	2	149	0.2%	4.07 [0.98, 16.94]
Waljee 2015	154	420	35	283	0.8%	4.10 [2.73, 6.16]
Selter 2020	71	129	30	134	0.7%	4.24 [2.49, 7.24]
AAMC 1992	15941	51647	1256	13580	1.0%	4.38 [4.12, 4.66]
Chandrasekar 2020	545	1894	27	320	0.8%	4.38 [2.92, 6.58]
Johnson 1992	17190	53891	1495	15568	1.0%	4.41 [4.17, 4.66]
Bickel 1993	17653	57049	1622	17682	1.0%	4.44 [4.20, 4.68]
Bickel 1988	15285	48535	1094	11673	1.0%	4.45 [4.17, 4.74]
Awad 2017	16	137	1	36	0.1%	4.63 [0.59, 36.13]
Wagner 2007	49	164	9	107	0.5%	4.64 [2.17, 9.92]
Waisbren 2008	486	3557	62	1884	0.9%	4.65 [3.55, 6.10]
Kirchner 1984	626	2407	23	333	0.8%	4.74 [3.07, 7.31]
Battaglia 2019	7	77	0	22	0.1%	4.79 [0.26, 87.16]
Levey 1990	2552	7015	99	932	0.9%	4.81 [3.88, 5.96]
Sperling 2019	174	312	72	347	0.8%	4.82 [3.42, 6.78]
Day 2010	550	2119	22	327	0.8%	4.86 [3.12, 7.57]
Bernardi 2020	6	22	1	14	0.1%	4.88 [0.52, 45.79]
Reed 2011	33	50	7	25	0.3%	4.99 [1.75, 14.28]
Whitley 1987	332	999	18	200	0.7%	5.03 [3.05, 8.31]
Smith 2019	217	663	14	159	0.6%	5.04 [2.84, 8.93]
Sadeghpour 2012	75	159	15	100	0.6%	5.06 [2.69, 9.51]
Schroen 2004	64	168	16	149	0.6%	5.12 [2.79, 9.37]
Karimian 2012	11	121	2	106	0.2%	5.20 [1.13, 24.02]

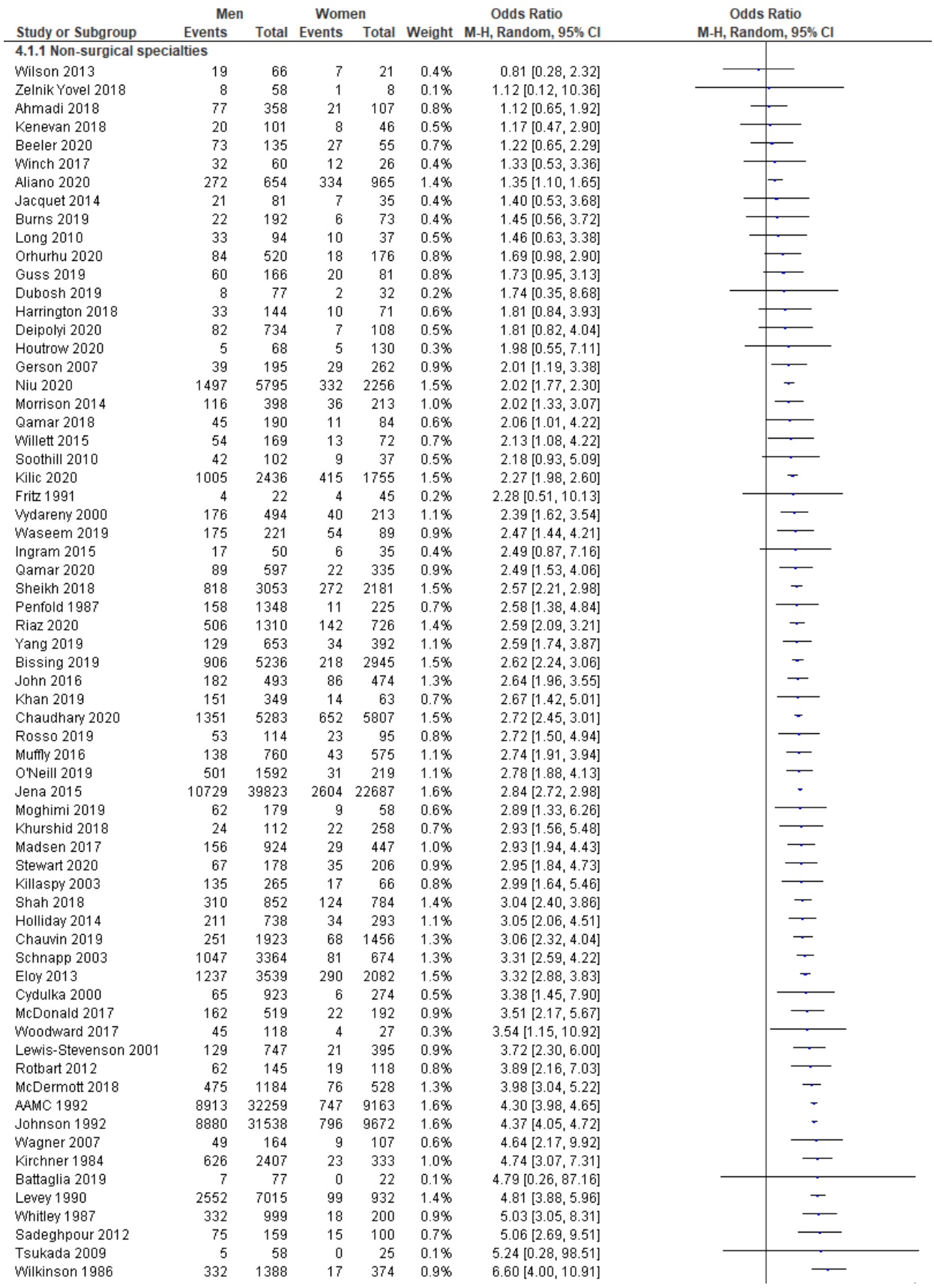


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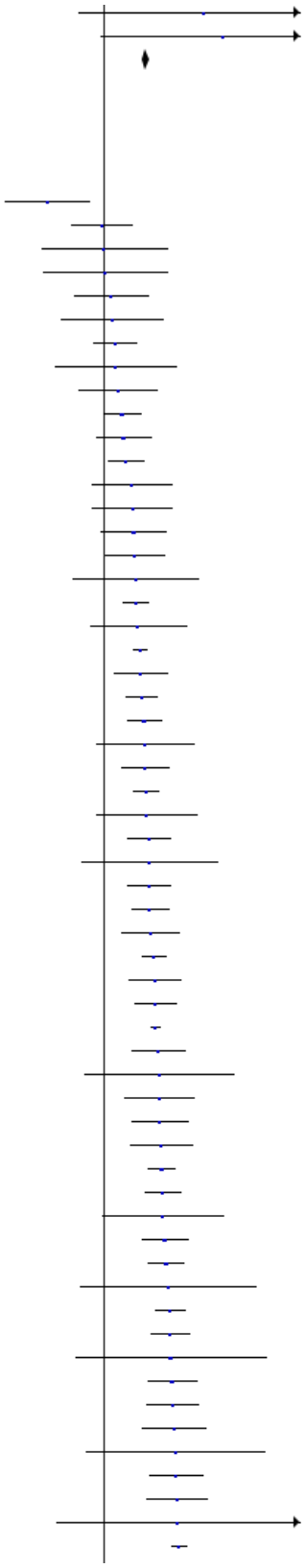
eFigure 3. Subgroup analysis of full professorship by geographic region (within North America versus outside of North America). M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

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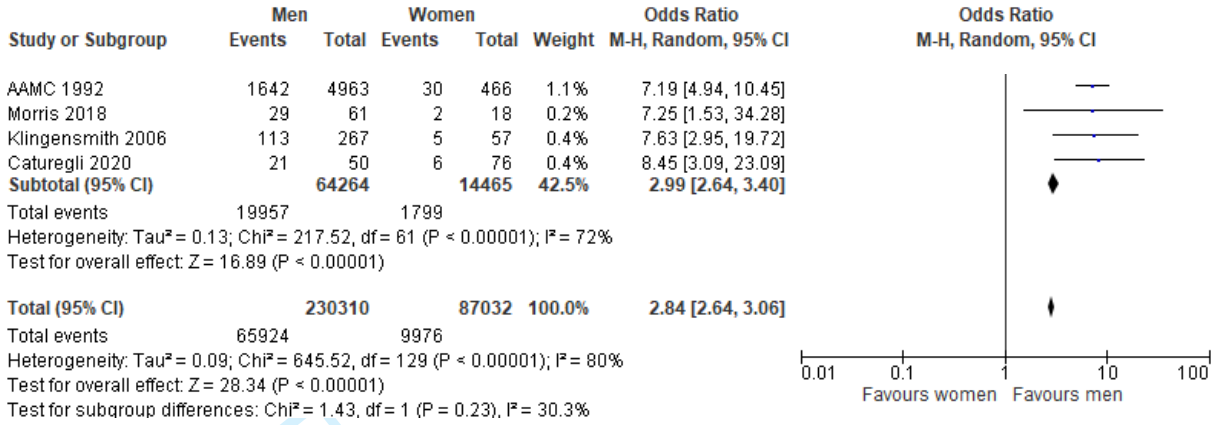


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Study or Subgroup	Men Events	Men Total	Women Events	Women Total	Weight	Odds Ratio M-H, Random, 95% CI
Gjerde 1994	9	29	0	11	0.1%	10.66 [0.57, 200.42]
Herzke 2020	19	52	0	14	0.1%	16.88 [0.95, 298.90]
Subtotal (95% CI)		**166046**		**72567**	**57.5%**	**2.72 [2.48, 2.98]**
Total events	45967		8177			
Heterogeneity: Tau² = 0.08; Chi² = 423.11, df = 67 (P < 0.00001); I² = 84%						
Test for overall effect: Z = 21.33 (P < 0.00001)						
4.1.2 Surgical specialties						
Schwarz 2016	7	200	11	93	0.4%	0.27 [0.10, 0.72]
Shaikh 2019	162	294	19	34	0.6%	0.97 [0.47, 1.98]
Hoops 2018	8	42	3	16	0.2%	1.02 [0.23, 4.45]
Gerety 2013	36	104	3	9	0.2%	1.06 [0.25, 4.49]
Geltzeiler 2017	30	281	7	77	0.5%	1.20 [0.50, 2.84]
Dorismond 2020	12	47	5	23	0.3%	1.23 [0.38, 4.05]
Higgins 2020	108	287	28	90	0.9%	1.34 [0.81, 2.22]
Morgan 2019	3	25	8	87	0.2%	1.35 [0.33, 5.51]
McAlister 2014	23	283	6	102	0.4%	1.42 [0.56, 3.58]
Harris 2019	275	426	55	102	1.0%	1.56 [1.01, 2.41]
Odell 2019	46	281	14	129	0.7%	1.61 [0.85, 3.04]
Tracy 2019	181	1086	29	274	1.0%	1.69 [1.11, 2.56]
Carpenter 2018	45	117	7	29	0.4%	1.96 [0.78, 4.97]
Dermody 2019	35	161	6	49	0.4%	1.99 [0.78, 5.06]
Yue 2020	98	618	8	95	0.6%	2.05 [0.96, 4.36]
Gambhir 2020	65	120	18	50	0.7%	2.10 [1.06, 4.15]
Okoshi 2014	33	326	2	40	0.2%	2.14 [0.49, 9.28]
Hoof 2020	852	3889	50	434	1.2%	2.15 [1.59, 2.92]
Ilin 2020	55	164	4	22	0.3%	2.27 [0.73, 7.04]
Valsangkar 2016	1272	3087	211	928	1.5%	2.38 [2.01, 2.82]
Tulunay-Udur 2019	72	152	18	66	0.7%	2.40 [1.28, 4.50]
Maxwell 2020	275	1543	36	450	1.1%	2.49 [1.73, 3.59]
Hill 2018	134	420	39	257	1.1%	2.62 [1.76, 3.90]
Baptiste 2017	28	91	4	28	0.3%	2.67 [0.85, 8.41]
Cheng 2019	178	534	17	108	0.8%	2.68 [1.55, 4.63]
Okafor 2020	402	1342	56	412	1.2%	2.72 [2.00, 3.69]
Florence 2011	71	154	4	17	0.3%	2.78 [0.87, 8.91]
Ngaage 2019	447	1350	19	131	0.9%	2.92 [1.77, 4.81]
Cuoco 2020	59	74	4	7	0.2%	2.95 [0.60, 14.62]
Carnevale 2020	203	774	19	177	0.9%	2.96 [1.79, 4.88]
Mayer 2017	441	1686	25	236	1.0%	2.99 [1.95, 4.59]
Wu 2019	122	220	14	48	0.7%	3.02 [1.54, 5.95]
Eloy 2013	553	1740	66	534	1.3%	3.30 [2.50, 4.36]
Epperson 2020	185	343	16	62	0.7%	3.37 [1.83, 6.18]
Garstka 2019	210	579	22	153	0.9%	3.39 [2.09, 5.49]
Jena 2015	3925	13010	436	3873	1.5%	3.41 [3.06, 3.79]
Dossani 2019	298	761	12	80	0.7%	3.65 [1.94, 6.85]
Rosso 2019	10	22	2	11	0.2%	3.75 [0.65, 21.51]
Santosa 2019	115	241	8	41	0.5%	3.76 [1.67, 8.49]
Paik 2014	172	426	12	79	0.7%	3.78 [1.99, 7.20]
Gawad 2020	75	294	9	111	0.6%	3.88 [1.87, 8.06]
Breyer 2020	1092	2597	51	329	1.2%	3.96 [2.91, 5.38]
Pace 2020	509	1433	27	224	1.0%	4.02 [2.65, 6.10]
Herrick-Reynolds 2019	48	914	2	149	0.2%	4.07 [0.98, 16.94]
Selter 2020	71	129	30	134	0.9%	4.24 [2.49, 7.24]
Chandrasekar 2020	545	1894	27	320	1.1%	4.38 [2.92, 6.58]
Awad 2017	16	137	1	36	0.1%	4.63 [0.59, 36.13]
Sperling 2019	174	312	72	347	1.2%	4.82 [3.42, 6.78]
Day 2010	550	2119	22	327	1.0%	4.86 [3.12, 7.57]
Bernardi 2020	6	22	1	14	0.1%	4.88 [0.52, 45.79]
Smith 2019	217	663	14	159	0.8%	5.04 [2.84, 8.93]
Schroen 2004	64	168	16	149	0.8%	5.12 [2.79, 9.37]
Wise 2004	59	187	9	112	0.6%	5.28 [2.50, 11.14]
Inam 2020	10	67	1	32	0.1%	5.44 [0.66, 44.49]
LaRocca 2018	108	216	14	92	0.7%	5.57 [2.97, 10.44]
Sasor 2018	194	678	9	136	0.6%	5.66 [2.82, 11.35]
Nakata 2017	21	121	0	13	0.1%	5.78 [0.33, 100.94]
Johnson 1992	3147	9672	128	1711	1.4%	5.96 [4.96, 7.18]



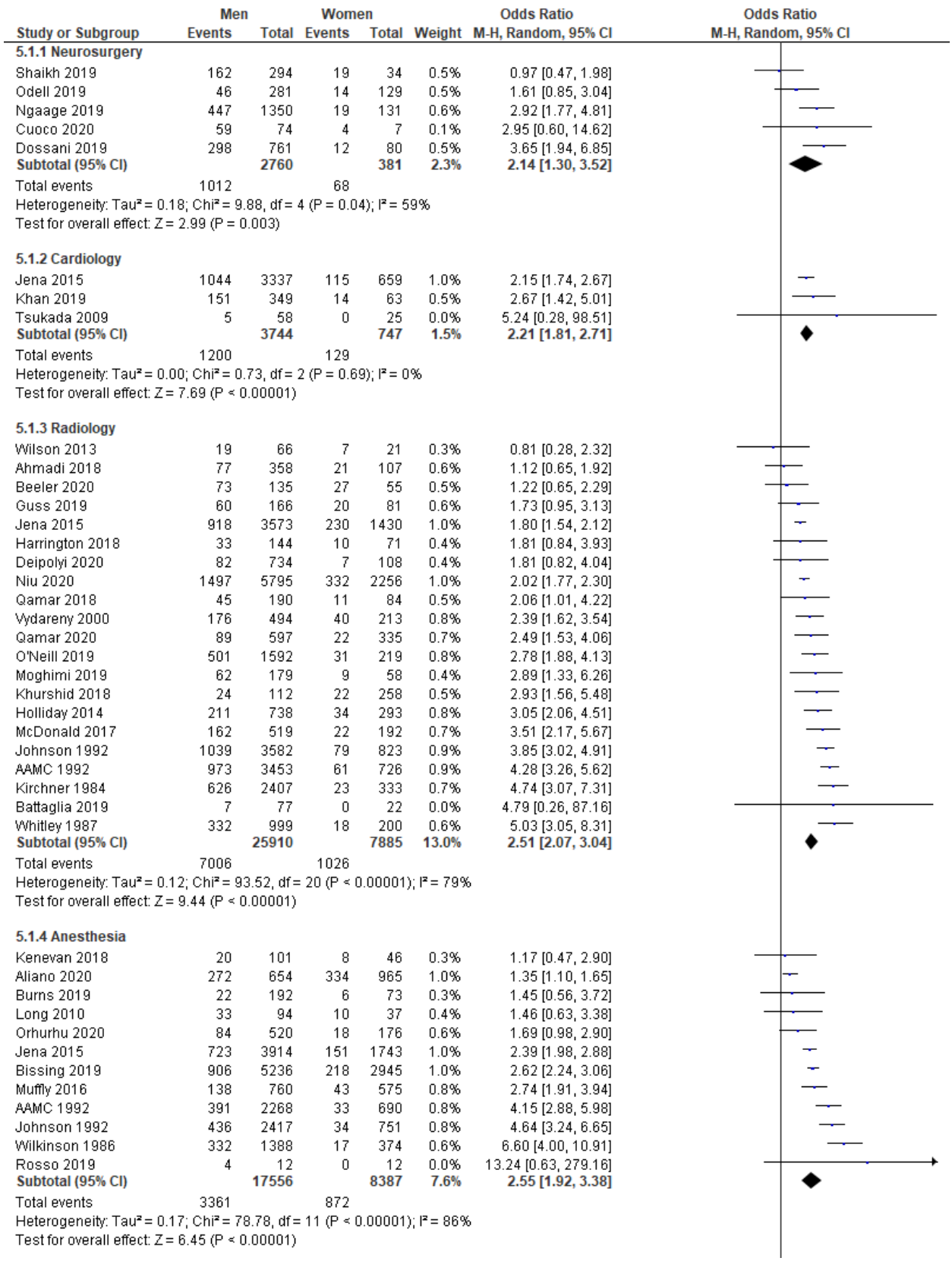
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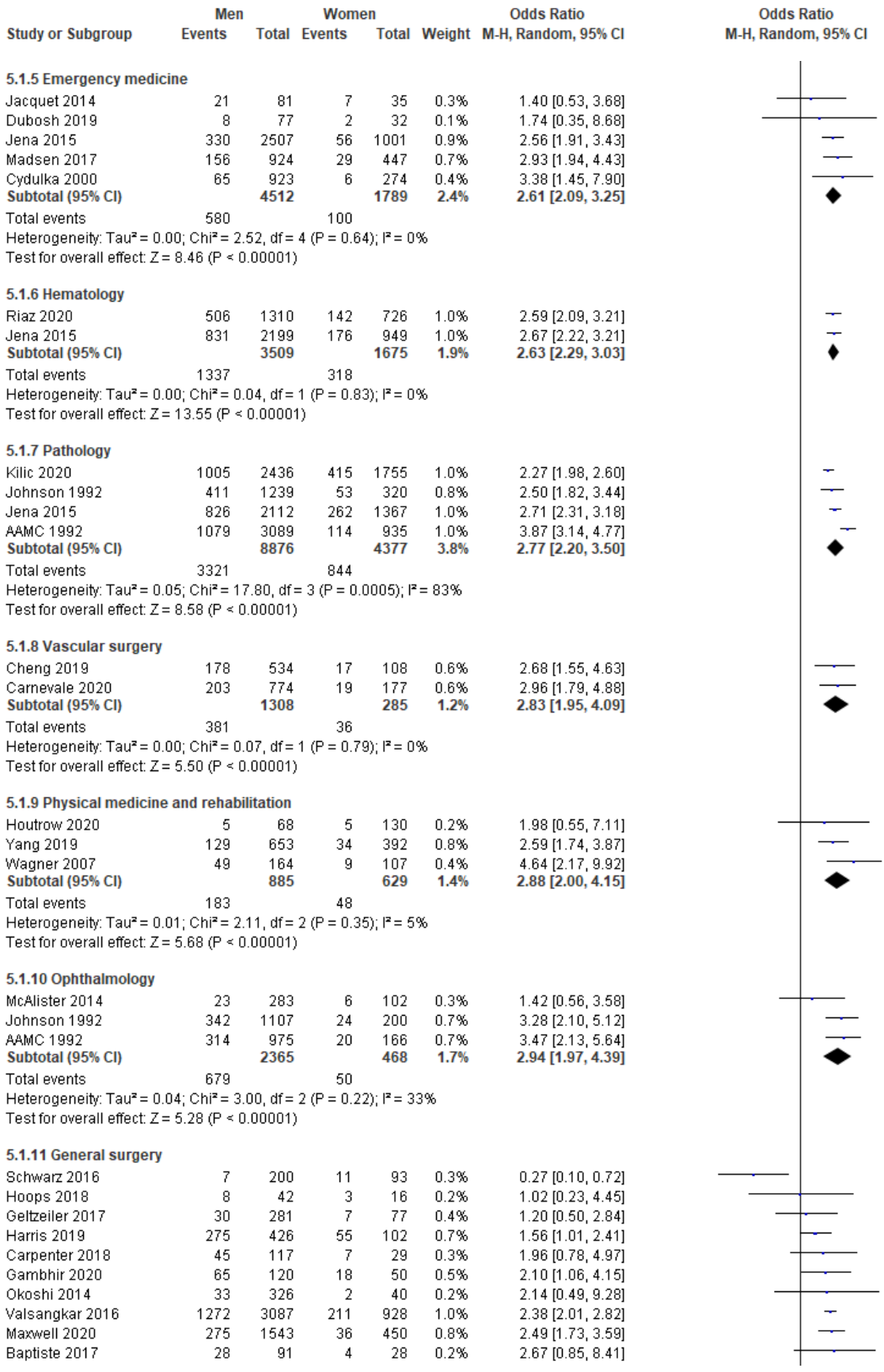
eFigure 4. Subgroup analysis of full professorship for surgical versus non-surgical specialties. M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

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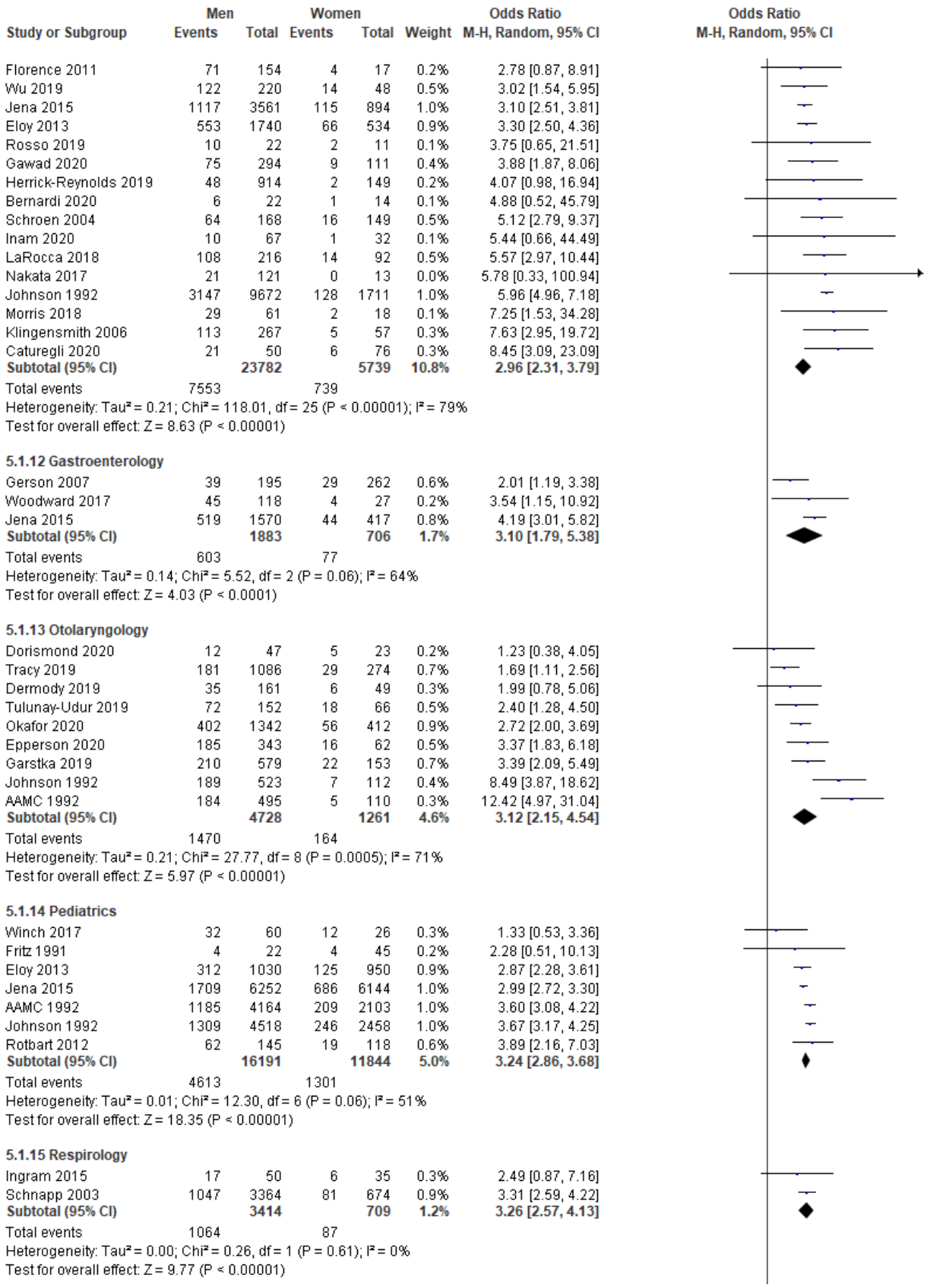
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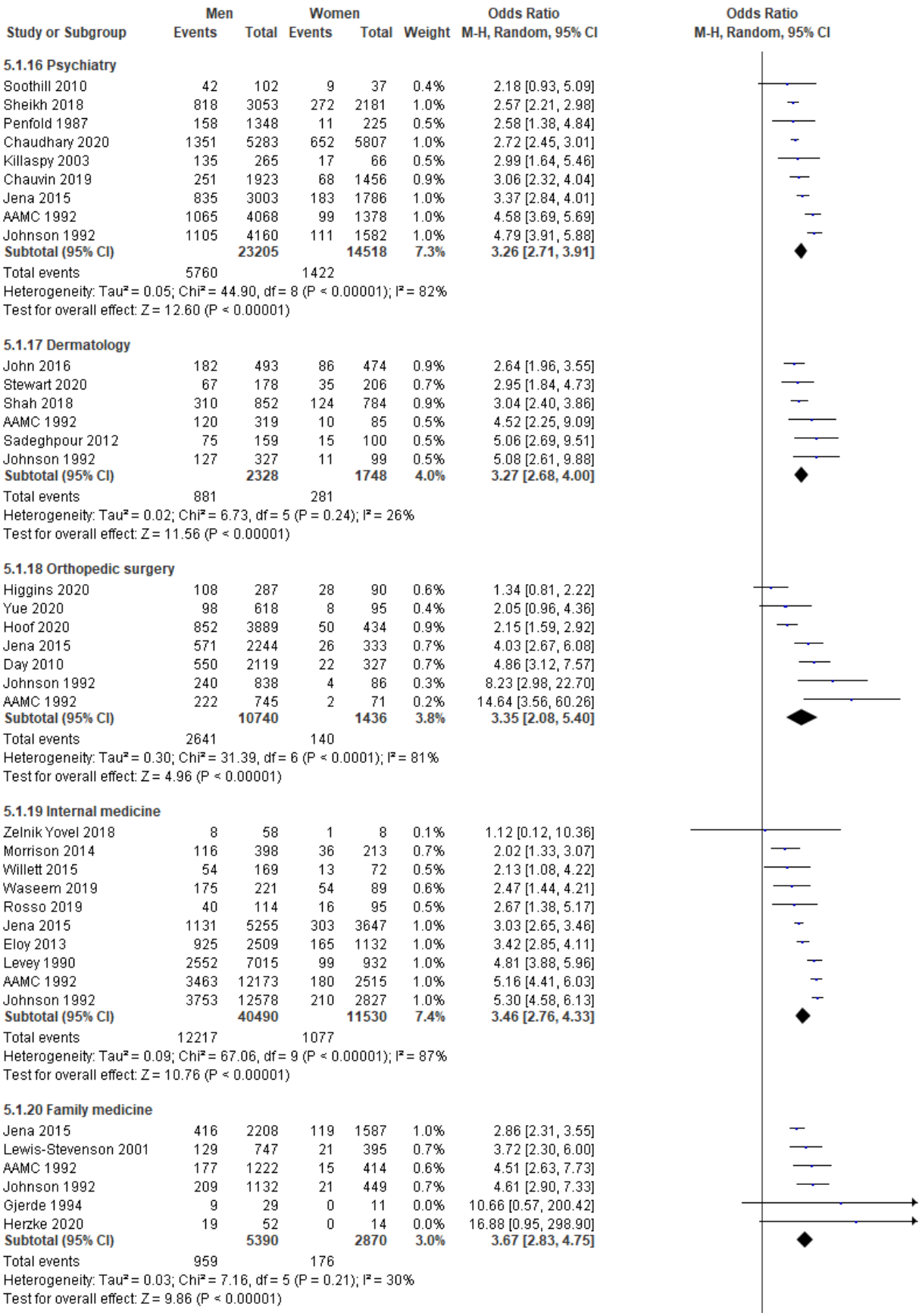
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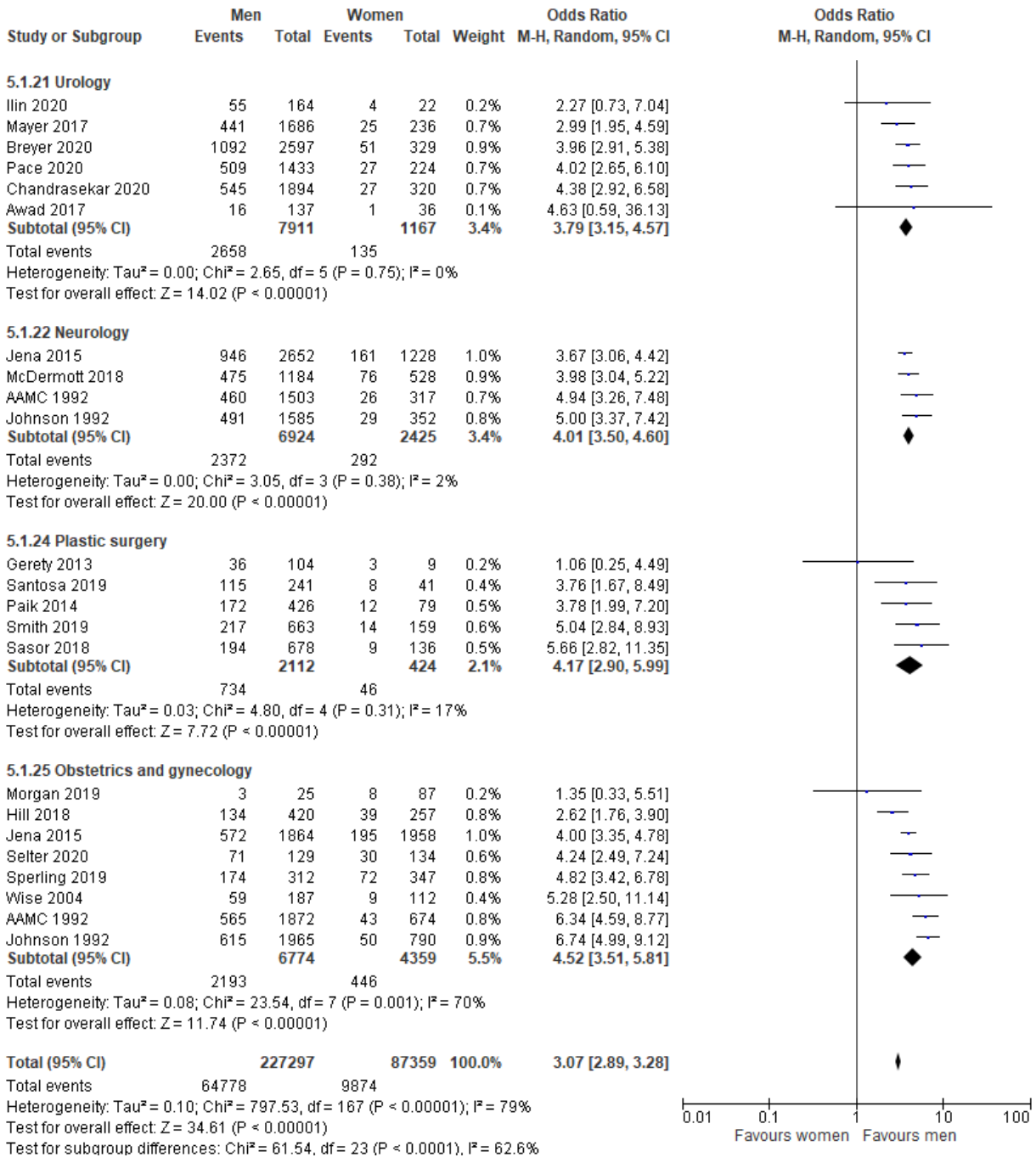
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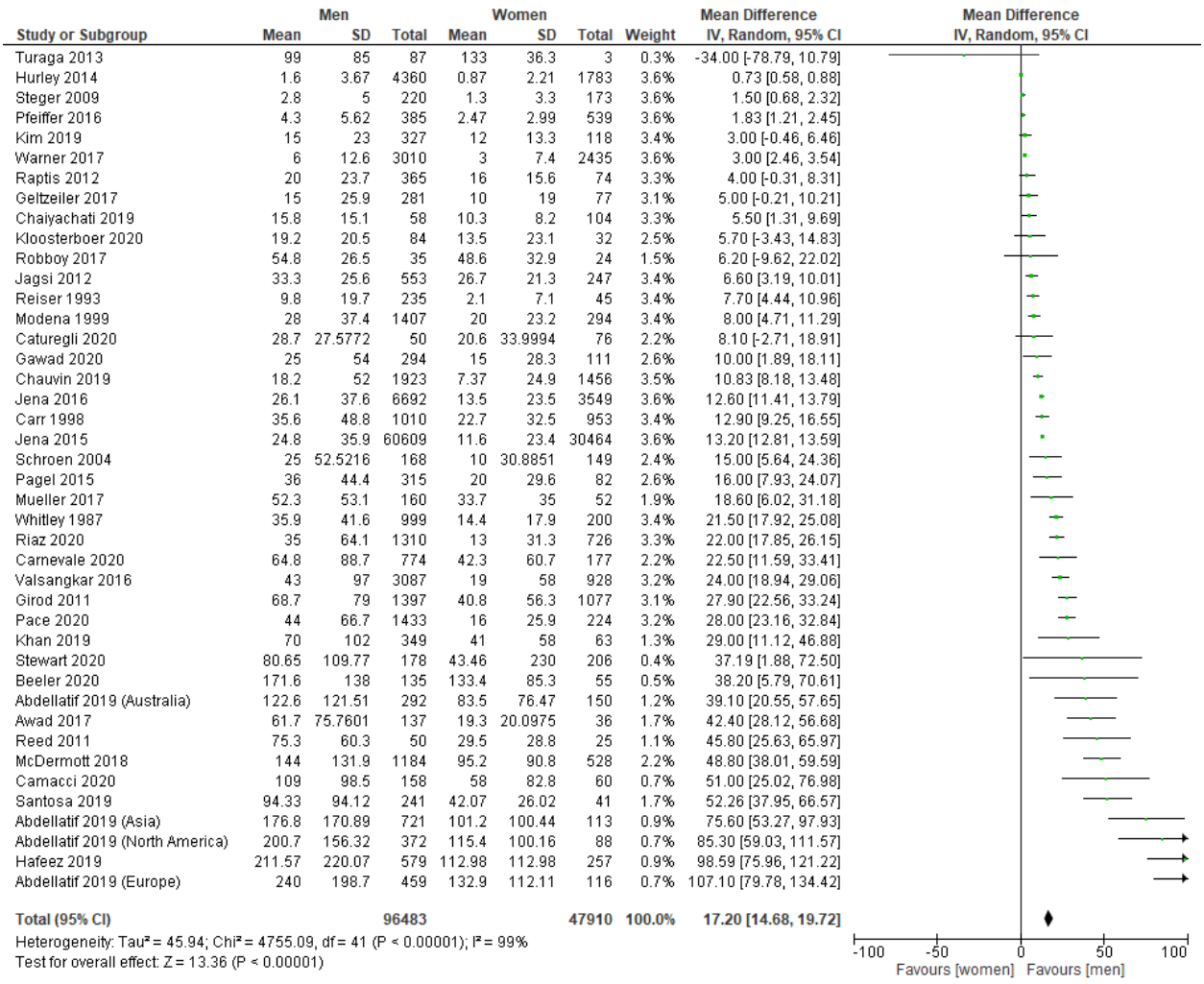


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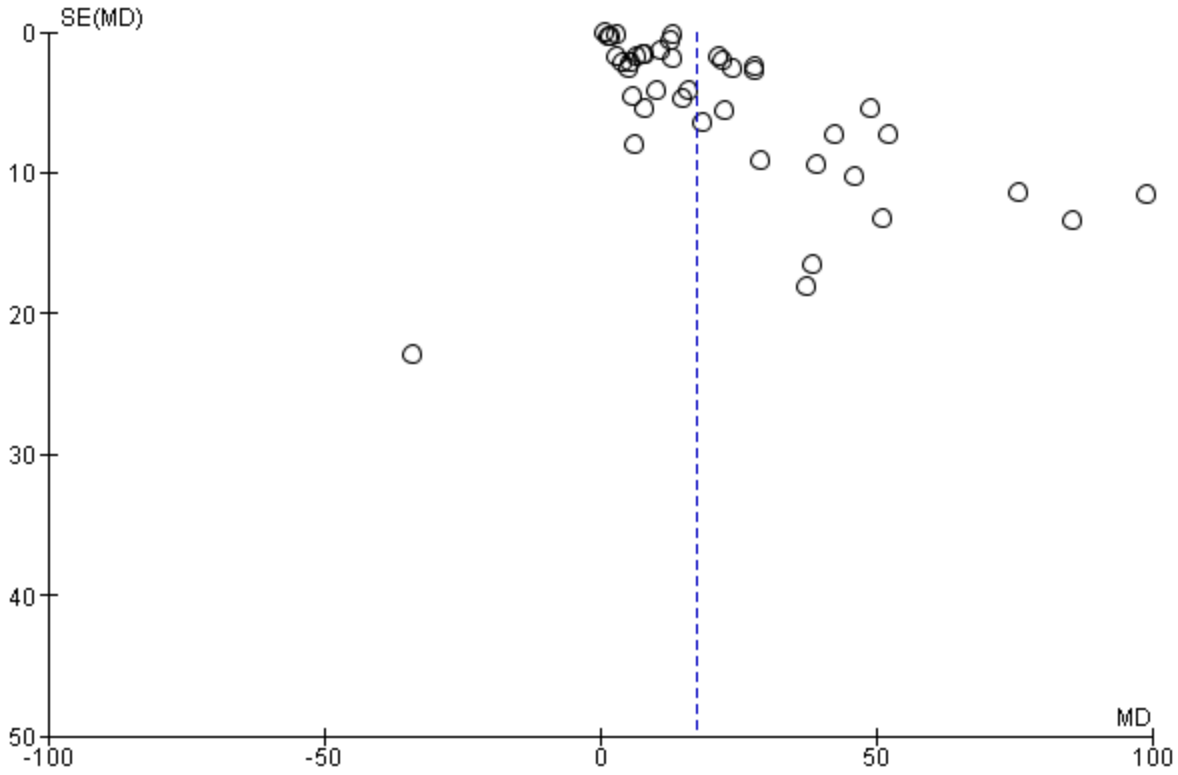
eFigure 5. Subgroup analysis of full professorship by specialty. M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

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eFigure 6. Forest plot of total publications for men vs. women. SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).

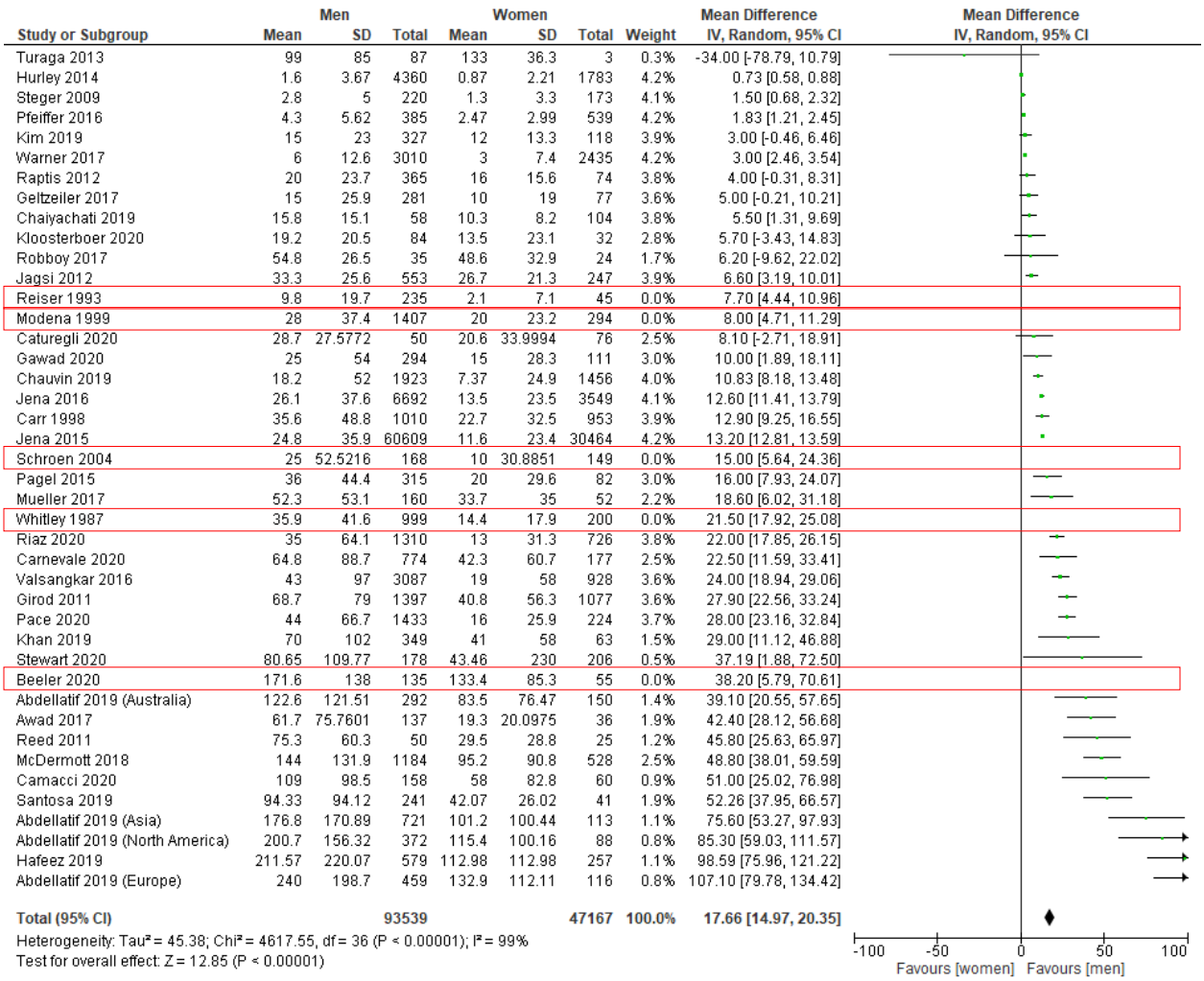
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eFigure 7. Funnel plot of total publications for men vs. women. SE (standard error), MD (mean difference).

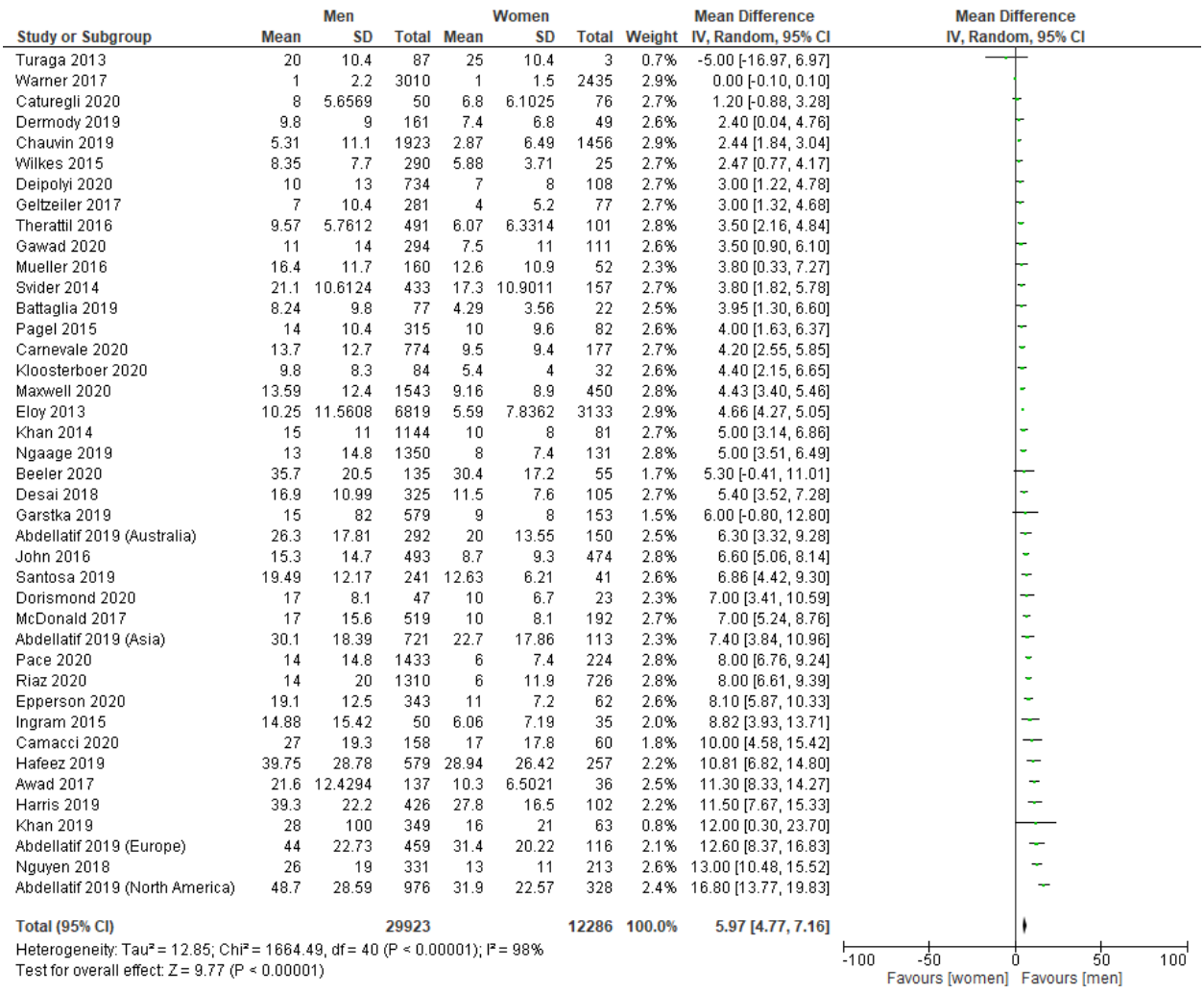
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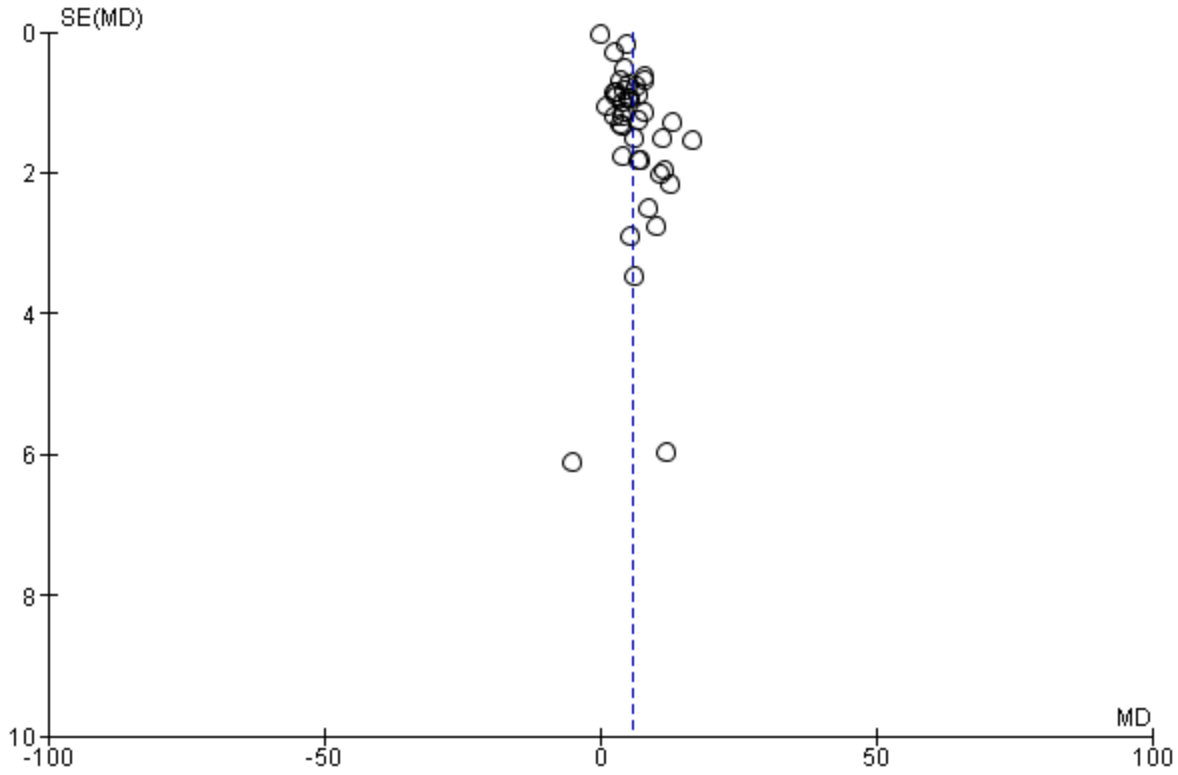
eFigure 8. Sensitivity analysis of total publications removing 5 poor quality studies (red box, NOS score ≤ 5). SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).

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eFigure 9. Forest plot of h-index for men vs. women. SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).

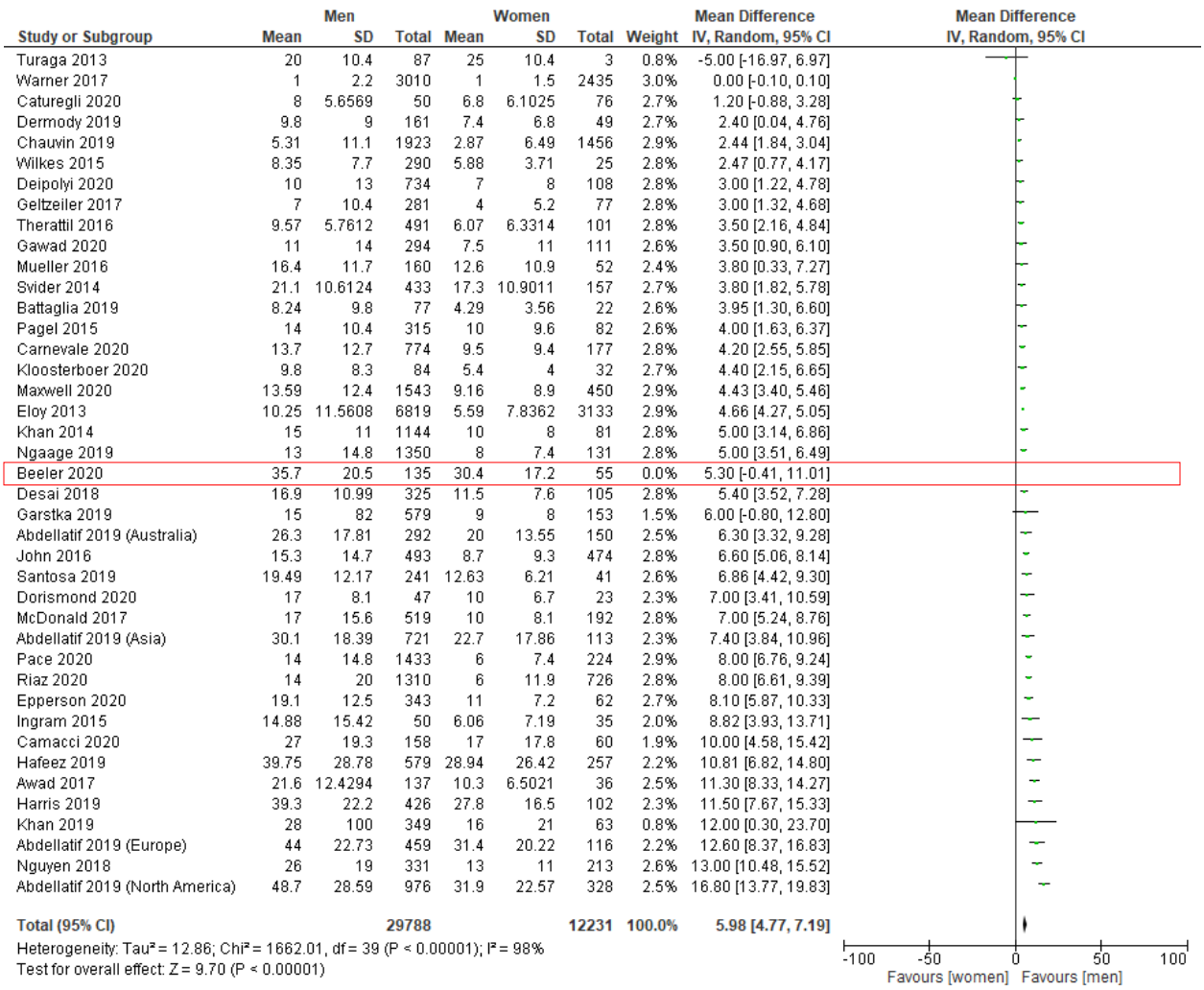
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eFigure 10. Funnel plot of h-index for men vs. women. SE (standard error), MD (mean difference).

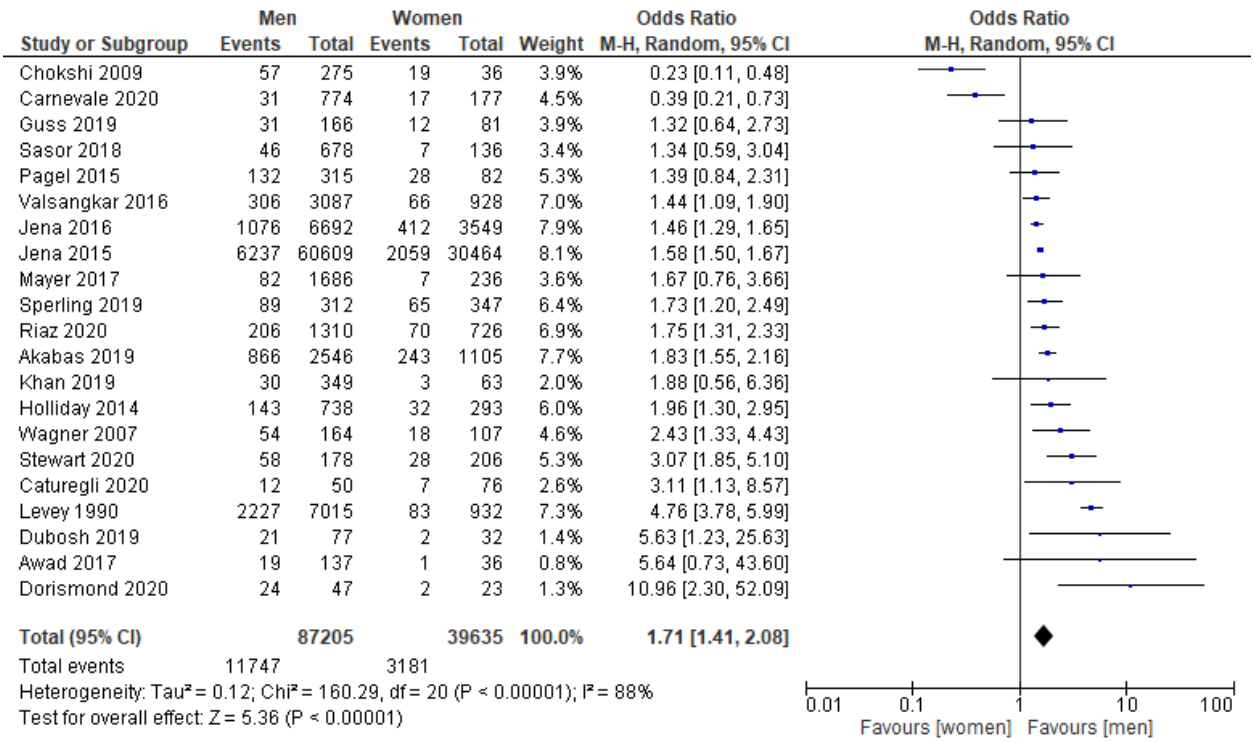
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eFigure 11. Sensitivity analysis of h-index removing 1 poor quality study (red box, NOS score ≤ 5). SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).

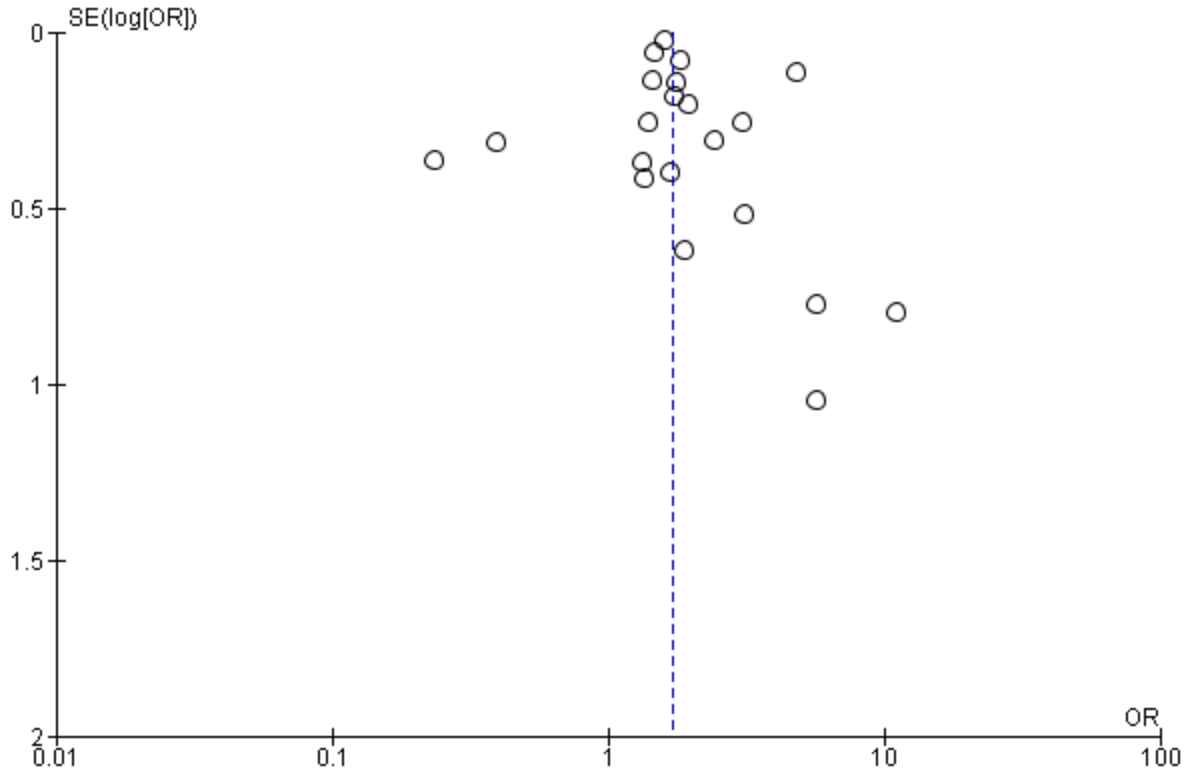
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eFigure 12. Forest plot of federal research funding for men vs. women. M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).

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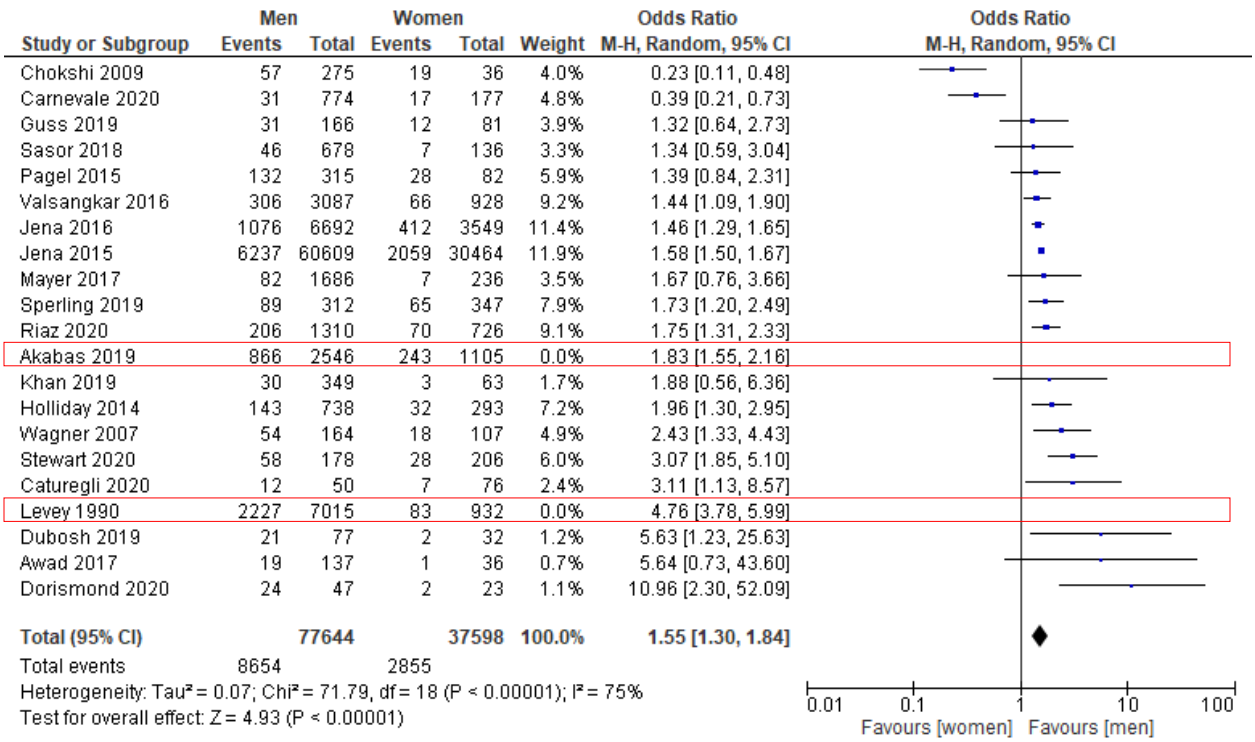
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eFigure 13. Funnel plot of federal research funding for men vs. women. SE (standard error), OR (odds ratio).

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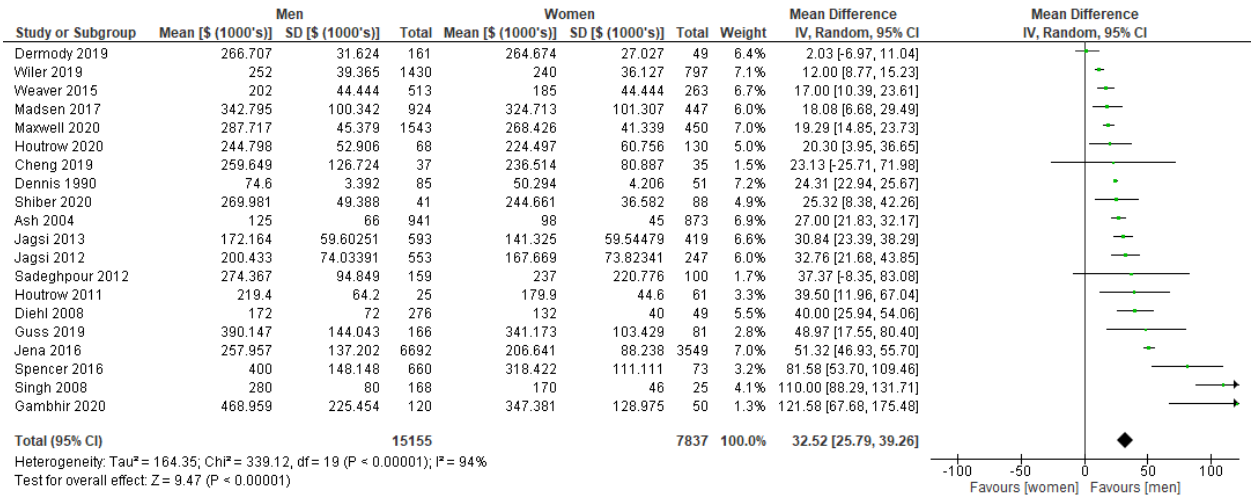
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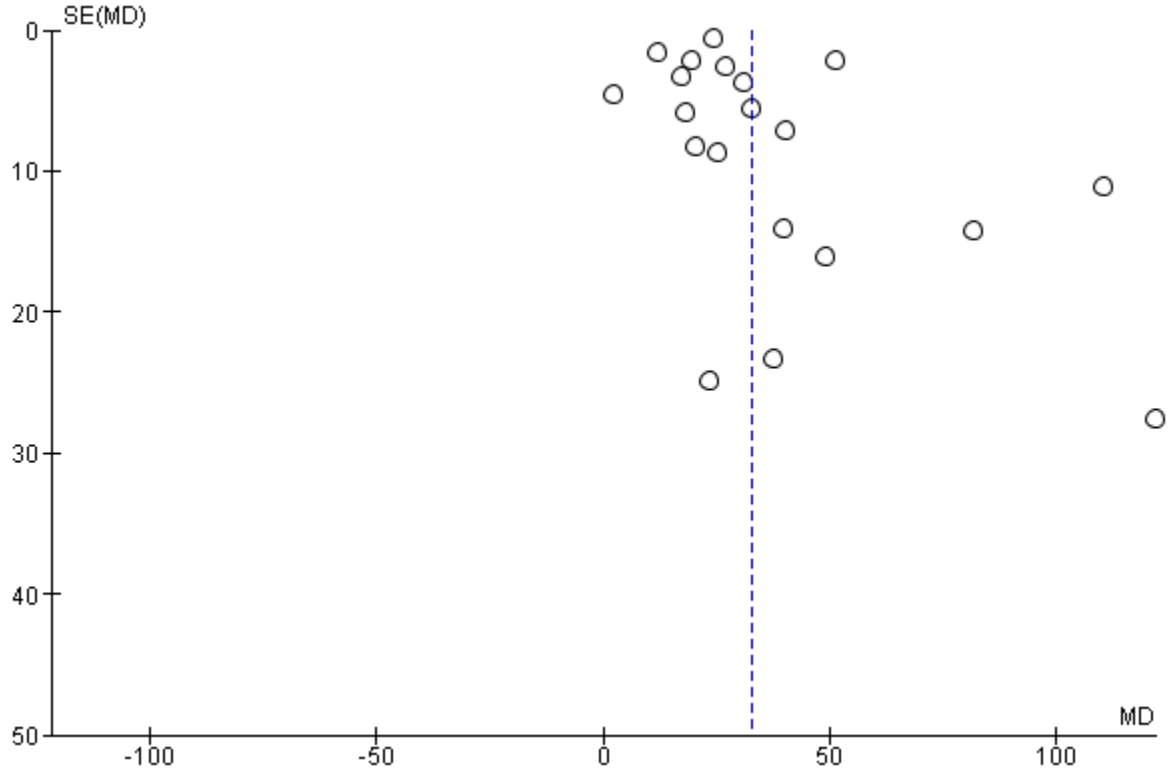
eFigure 14. Sensitivity analysis of federal research funding removing 2 poor quality studies (red box, NOS score ≤ 5). M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).

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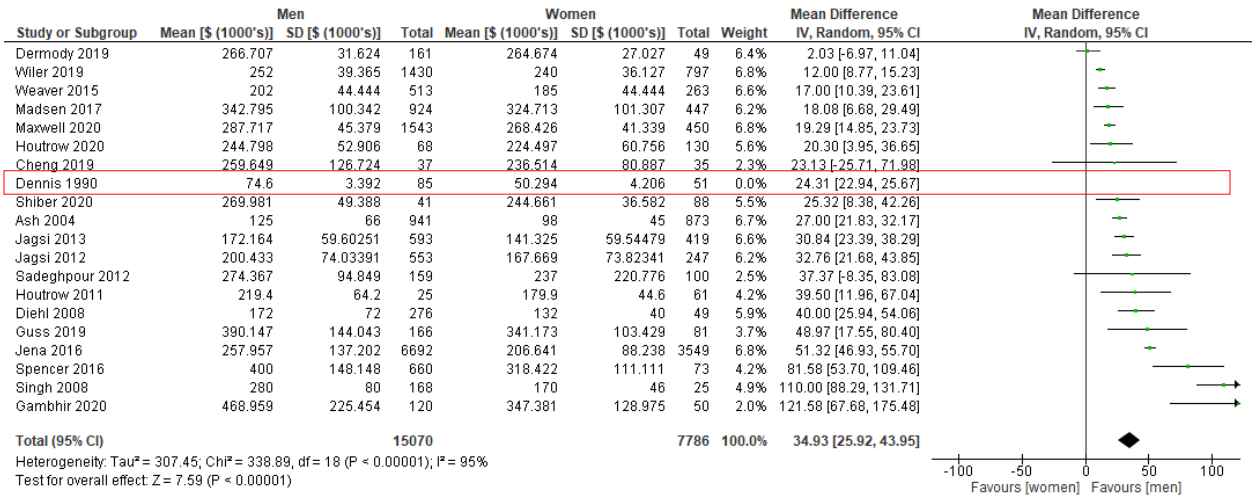


eFigure 15. Forest plot of salary for men vs. women. Means and standard deviations (SD) are presented in thousands of dollars. Total (sample size), IV (inverse variance), Random (random effects), CI (confidence interval).



eFigure 16. Funnel plot of salary for men vs. women. SE (standard error), MD (mean difference).

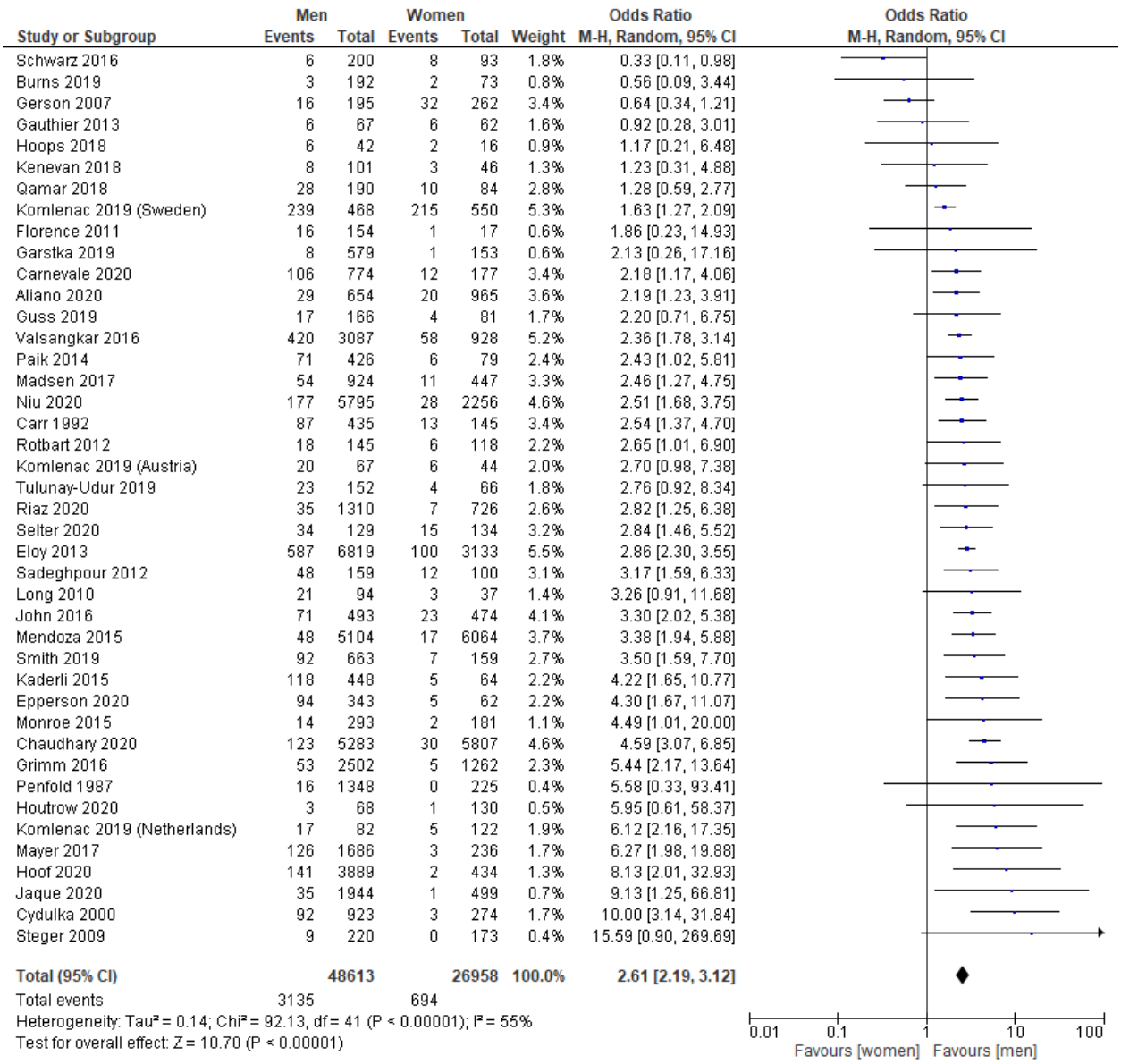
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eFigure 17. Sensitivity analysis of salary removing 1 poor quality study (red box, NOS score ≤ 5). Means and standard deviations (SD) are presented in thousands of dollars. Total (sample size), IV (inverse variance), Random (random effects), CI (confidence interval).

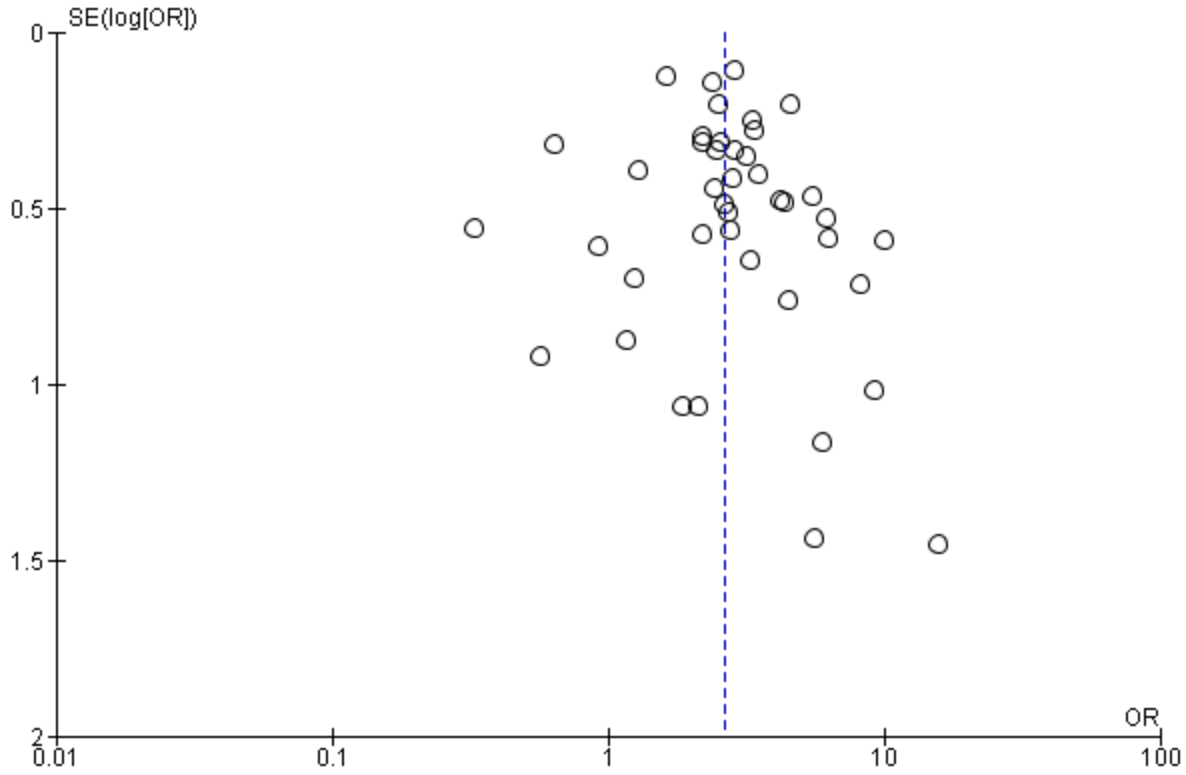
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eFigure 18. Forest plot of leadership (departmental chair) for men vs. women. M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).

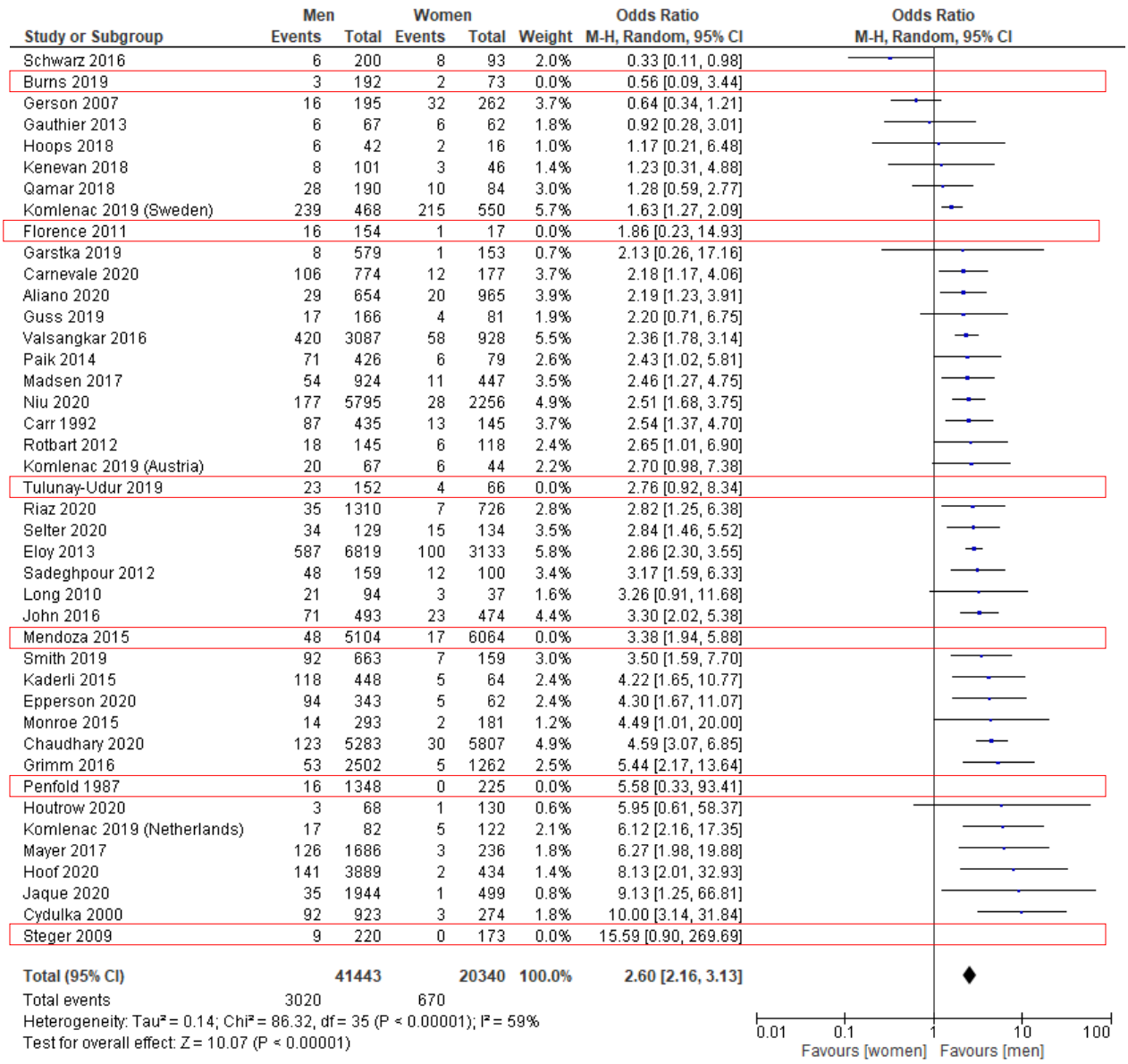
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eFigure 19. Funnel plot of leadership (departmental chair) for men vs. women. M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).

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eFigure 20. Sensitivity analysis of leadership (departmental chair) removing 6 poor quality studies (red box, NOS score ≤ 5). M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	eTable 1 (Supplement pages 2-5)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6-7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6-7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7-8

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Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7-8

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6-7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7-8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9, eTable 2 (Supplement pages 6-19)
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10, eTable 4 (Supplement pages 29-40)
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-14
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10-14
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	10-14
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10-14
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	18
FUNDING			

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Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	19

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Gender differences in faculty rank among academic physicians: A systematic review and meta-analysis

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ABSTRACT

Objective: Many studies have analyzed gender bias in academic medicine; however, no comprehensive synthesis of the literature has been performed. We conducted a pooled analysis of the difference in the proportion of men versus women with full professorship among academic physicians.

Design: Systematic review and meta-analysis.

Data sources: MEDLINE, Embase, Cochrane CENTRAL, ERIC, and PsycINFO were searched from inception to July 3, 2020.

Study selection: All original studies reporting faculty rank stratified by gender worldwide were included.

Data extraction and synthesis: Study screening, data extraction, and quality assessment were performed by two independent reviewers, with a third author resolving discrepancies. Meta-analysis was conducted using random-effects models.

Results: Our search yielded 5,897 articles. 218 studies were included with 991,207 academic physician data points. Men were 2.77 times more likely to be full professors (182,271/643,790 men vs. 30,349/251,501 women, OR 2.77, 95% CI 2.57-2.98). Although men practiced for longer (median 18 vs. 12 years, $p < .00002$), the gender gap remained after pooling 7 studies that adjusted for factors including time in practice, specialty, publications, h-index, additional PhD, and institution (adjusted OR 1.83, 95% CI 1.04-3.20). Meta-regression by data collection year demonstrated improvement over time ($p = .0011$), however, subgroup analysis showed that gender disparities remain significant in the 2010-2020 decade (OR 2.63, 95% CI 2.48-2.80). The

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gender gap was present across all specialties and both within and outside of North America. Men published more papers (mean difference 17.2, 95% CI 14.7-19.7), earned higher salaries (mean difference $33,256, 95% CI $25,969-$40,542), and were more likely to be departmental chairs (OR 2.61, 95% CI 2.19-3.12).

Conclusions: Gender inequity in academic medicine exists across all specialties, geographic regions, and multiple measures of success, including academic rank, publications, salary, and leadership. Men are more likely than women to be full professors after controlling for experience, academic productivity, and specialty. Although there has been some improvement over time, the gender disparity in faculty rank persists.

PROSPERO registration number: CRD42020197414.

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- This systematic review and meta-analysis comprehensively synthesizes the literature on gender inequity in academic medicine with full professorship as the primary outcome.
- Five major databases were searched including studies globally with no language limitations.
- We analyzed trends over time, across specialties, and between geographic regions.
- Both unadjusted and adjusted data were pooled.
- Individual participant-level data was not available; a study-level meta-analysis was conducted.

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INTRODUCTION

Gender equity remains an important issue in academic medicine. In the United States, women make up more than half of medical school graduates (1) and 37% of full-time faculty (2). However, only 24% of full professors (3), 18% of department chairs (4), and 17% of deans are women (5). Women are under-represented on editorial boards, have fewer publications, and receive less research funding (6,7). Due to these inequities, women are more likely to leave academia (8). Previous studies have demonstrated the positive impact of health care workforce diversity on patient outcomes, costs, and innovation (9). Therefore, gender imbalances in academic medicine, particularly in leadership positions, have broad negative implications (10).

An important indicator of gender inequity in academic medicine is the difference in the proportion of women vs. men with full professorship (11). The decision to appoint a physician to full professor is usually made by a medical school committee and generally takes into consideration a candidate's clinical and academic contributions (12). However, this varies across settings based on institutional values, specialty, and conscious or unconscious bias of promotions committee members (13–15). Importantly, several studies have demonstrated that women are less likely than men to be full professors even after adjusting for clinical and academic productivity (16–18).

The current literature on gender bias in academic medicine is limited by single studies that assess one institution (19,20), specialty (21–24), or country (25–29). Furthermore, the cross-sectional nature of most analyses precludes an investigation of trends over time (30–32). We conducted a systematic review and meta-analysis to calculate a pooled measurement of gender differences in full professorship worldwide and assessed variations across specialties, between geographic regions, and over time.

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METHODS

Protocol and registration

We conducted a systematic review and meta-analysis and reported our findings according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement guidelines (33). Our study protocol (CRD42020197414) was registered with the International Prospective Register of Systematic Reviews (PROSPERO) (34).

Information sources and search strategy

MEDLINE, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), PsycINFO, and Education Resources Information Center (ERIC) were searched from inception to July 3, 2020 for studies analyzing gender differences in faculty rank among academic physicians. A combination of Medical Subject Heading (MeSH) terms and keywords for gender AND physician AND faculty AND rank were used to maximize sensitivity. Our search did not apply language limitations and Google Translate was used for non-English studies (35). The search strategy is detailed in eTable 1.

Study selection and data collection

Title and abstract screening, full-text review, data collection, and study quality assessment were conducted by two independent reviewers (BL and JJB), with a third author resolving discrepancies (FD). We included all original studies reporting faculty rank stratified by gender among academic physicians. Included articles did not need to be specifically designed to assess gender differences in faculty rank. As long as studies reported the proportion of full professors in a pool of academic physicians for women and men, they were included. This

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approach allowed us to capture articles broadly and sensitively. Case reports, reviews, and studies reporting data on only one gender were excluded. We hand-searched the reference lists of included studies for additional articles.

A standardized form was used to collect data for included studies. Variables obtained were study authors, publication year, geographic region, data source, specialty, data collection year, sample size, outcomes reported, baseline demographics, and effect size measurements. Authors were contacted through email for baseline demographics and effect size measurements if they were not reported in the original publication. When studies reported duplicate data (same data source, year of data collection, and outcome), the most comprehensive article was included (largest sample size and/or most outcomes reported). In studies reporting data from multiple years, results from the most recent year were included. Study quality was assessed using the Newcastle-Ottawa Scale (NOS) modified for cross-sectional studies (36,37).

Outcomes

The primary outcome was the difference in the proportion of female vs. male academic physicians with full professorship. An academic physician was defined as a physician with a medical school appointment (e.g., instructors and assistant/associate/full professors) (38). This primary outcome was chosen because full professorship is generally considered the highest academic rank and an important indicator of the overall status of an academic physician (11).

The secondary outcomes were total publications, h-index, salary, proportion receiving federal research funding, and leadership (proportion who are departmental chairs). Total publications, h-index, and federal research funding are markers of academic productivity (39). Federal research funding was defined as any funding received by academic physicians from a

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national government (40). Salary and departmental chair positions are other indicators of physician status (41,42).

Statistical analysis

We calculated a κ statistic to assess inter-rater agreement at the title/abstract screening and full-text review stages, with a threshold of ≥ 0.8 indicating strong agreement (43). Baseline covariates for men vs. women were compared by calculating median values across included studies and applying unpaired t test.

We conducted a meta-analysis of the proportion of men versus women with full professorship. Unadjusted estimates were reported as odds ratios (OR) with 95% confidence intervals (CI). Adjusted estimates were calculated using the inverse variance method and reported as adjusted OR with 95% CI (44). Sensitivity analyses were conducted by excluding low quality studies (NOS score ≤ 5) (45). We performed meta-regression and subgroup analysis based on data collection year to assess for trends over time and gender disparities in the modern era. We also conducted subgroup analyses for studies within vs. outside of North America, surgical vs. non-surgical specialties, and across specialties. For subgroup analysis by specialty, data was included from studies that reported specialty-specific results either as their primary or subgroup analysis. Subgroup differences were analyzed by calculating between-subgroup variance as described by Borenstein and Higgins (2013) (46,47). Continuous outcome estimates for publications, h-index, and salary were reported as mean differences with 95% CI. Monetary values for salary were converted to US dollars using a standardized currency converter (48). Heterogeneity was quantified using the I^2 statistic and publication bias was assessed using funnel plots.

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Meta-analyses were conducted using Review Manager 5.4.1 (The Cochrane Collaboration, Copenhagen, Denmark) based on random-effects models (49). Meta-regression was performed using R version 4.0.3 (R Project for Statistical Computing) (50) with meta (51) and metaphor (52) packages. Statistical tests were 2-sided with a significance threshold of $p < .05$.

Patient and public involvement

Patients or the public were not involved in the design, conduct, reporting, or dissemination plans of our research.

RESULTS

Study screening and selection

We identified 8,050 articles through our search of MEDLINE (n = 3,383), Embase (n = 2,810), PsycINFO (n = 907), ERIC (n = 633), and CENTRAL (n = 317). 5,897 articles remained after duplicates were removed, all of which underwent title and abstract screening. 4,827 records were excluded and 1,070 underwent full-text review. 852 were excluded, most commonly because there was no relevant outcome (n = 606), no stratification by gender (n = 115), or they contained duplicate data of an included study (n = 54). Hand-search of reference lists of included studies identified no additional articles. 218 studies were included in the final systematic review and meta-analysis. Inter-rater agreement was strong at title/abstract screening ($\kappa = 0.88$) and full-text review ($\kappa = 0.89$). Our search results are summarized in the PRISMA study flow diagram in Figure 1.

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Study characteristics

We included 218 cross-sectional studies published between 1979 and 2020. In total, 991,207 academic physician data points were included (699,010 men vs. 292,197 women). The population included instructors and assistant/associate/full professors. 170 studies reported academic rank (7 adjusted), 42 reported publications (1 adjusted), 41 reported h-index (1 adjusted), 20 reported salary (2 adjusted), 21 reported federal research funding (1 adjusted), and 42 reported departmental chair data (0 adjusted). An increasing number of studies were published in the past decade (2010-2020: 164 studies [75%] vs. before 2010: 54 studies [25%]). Outcomes were reported on 24 specialties across 16 regions, including countries from Europe (7), Asia (7), and North America (2). Data sources included surveys (n = 89, 40%), departmental websites (n = 71, 33%), and faculty databases (n = 58, 27%). The characteristics of included studies are summarized in eTable 2a and the number of included studies by country are presented in eTable 2b.

Baseline demographic data

90 studies reported baseline demographic data. Median values of covariates across included studies were assessed. Men were older (median age 47.7 vs. 43.5, $p < .00001$) and practiced for longer (median 18 vs. 12 years, $p < .00002$). Most academic physicians were white (median 74% [men] and 75% [women], $p = .56$). Men were more likely to be married (median 91% vs. 75%, $p < .0002$) and have children (median 84% vs. 71%, $p < .0007$). There were no significant differences in number of hours worked (median 58.0 [men] vs. 54.7 [women], $p = .056$), percentage of time spent on research (median 27.1% [men] vs. 25.2% [women], $p = 0.80$),

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or proportion of faculty holding an additional PhD (14.8% [men] vs. 14.6% [women], p = .57). Baseline demographic data for included studies are presented in eTable 3.

Study quality

Of the 218 cross-sectional studies, 127 were good quality (NOS score ≥ 8), 57 were fair quality (NOS score 6-7), and 34 were poor quality (NOS score ≤ 5). The most common reason for diminished study quality was poor comparability of cohorts due to inadequate statistical adjustment for confounders, with 131 studies receiving a score of 0/2 and 24 studies receiving a score of 1/2. Another contributor to poor quality was insufficient characterization of non-respondents in survey-based articles, with 81 studies receiving a score of 0/1. A summary of study quality assessment is presented in eTable 4.

Meta-analysis of full professorship

For full professorship, 170 studies with 895,291 academic physician data points (643,790 men vs. 251,501 women) were included. Meta-analysis demonstrated that men were 2.77 times more likely to be full professors (182,271 [28.3%] vs. 30,349 [12.1%], OR 2.77, 95% CI 2.57 – 2.98, p < .00001, I^2 = 94%, Figure 2). Funnel plot showed no evidence of publication bias (eFigure 1). Sensitivity analysis removing 27 poor quality studies (NOS score ≤ 5) continued to demonstrate that men were more likely to be full professors (OR 2.81, 95% CI 2.60 – 3.04, p < .00001, I^2 = 95%, eFigure 2).

Seven studies adjusted for potential confounders, including years in practice (Dossani (17), Gawad (18), Jena (16), Orhurhu (53), Riaz (54), Smith (55), Sperling (56)), specialty (Jena), publications (Dossani, Gawad, Jena, Riaz, Sperling), h-index (Dossani, Gawad, Orhurhu,

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Riaz, Sperling), additional PhD (Dossani, Gawad, Sperling), and institution (Dossani, Jena, Orhurhu, Riaz, Sperling). Pooling the adjusted effect estimates comparing 64,469 men to 32,063 women demonstrated that men were 1.83 times more likely to be full professors (adjusted OR 1.83, 95% CI 1.04 – 3.20, p = .03, I^2 = 91%, Figure 3).

To assess trends over time, meta-regression was performed based on data collection year, which demonstrated a reduction in the gender gap between 1979 and 2020 (p = .0011, Figure 4). Of note, 1979 was the data collection year for the first study on gender differences in faculty rank of academic physicians based on our search. To evaluate gender disparities in the modern era, subgroup analysis was performed by decade (Figure 5). Before 2000, men were 3.62 times more likely to be full professors (OR 3.62, 95% CI 3.27 – 4.01, p < .00001, I^2 = 85%). Between 2000-2009, men were 2.85 times more likely to be full professors (OR 2.85, 95% CI 2.36 – 3.42, p < .00001, I^2 = 97%). Between 2010-2020, men were 2.63 times more likely to be full professors (OR 2.63, 95% CI 2.48 – 2.80, p < .00001, I^2 = 63%).

Differences between geographic regions were analyzed (eFigure 3). Given that the majority of included studies were published in North America (182/218), we compared countries within North America to countries outside of North America. Within North America, men were 2.87 times more likely to be full professors (OR 2.87, 95% CI 2.65 – 3.10, p < .00001, I^2 = 95%). Outside of North America, men were 2.34 times more likely to be full professors (OR 2.34, 95% CI 1.80 – 3.04, p < .00001, I^2 = 82%). There was no significant difference between countries in North America vs. outside of North America (Chi^2 = 2.12, p = .15).

Surgical specialties were compared to non-surgical specialties (eFigure 4). For surgical specialties, men were 2.99 times more likely to be full professors (OR 2.99, 95% CI 2.64 – 3.40, p < .00001, I^2 = 72%). For non-surgical specialties, men were 2.72 times more likely to be full

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professors (OR 2.72, 95% CI 2.48 to 2.98, p < .00001, I^2 = 84%). There was no significant difference between surgical and non-surgical specialties (Chi^2 = 1.43, p = .23).

Analysis of individual specialties demonstrated that men were more likely to be full professors across all specialties (eFigure 5). The specialty with the smallest gender difference in full professorship was neurosurgery (5 studies, OR 2.14, 95% CI 1.30 – 3.52, p = .003, I^2 = 59%). The specialty with the greatest gender difference in full professorship was obstetrics and gynecology (8 studies, OR 4.52, 95% CI 3.51 – 5.81, p < .00001, I^2 = 70%).

Meta-analysis of secondary outcomes

For total publications, 42 studies were included comparing 96,483 men to 47,910 women. Over their careers, men published 17.20 more papers (mean difference 17.20, 95% CI 14.68 – 19.72, p < .00001, I^2 = 99%, eFigure 6). Funnel plot demonstrated publication bias skewed towards a higher mean difference (eFigure 7). Sensitivity analysis removing 5 poor quality studies continued to show that men published more papers (mean difference 17.66, 95% CI 14.97 – 20.35, p < .00001, I^2 = 99%, eFigure 8). One study demonstrated that women continued to have fewer publications after adjusting for career duration (adjusted relative rate 0.46, 95% CI 0.39 – 0.55) (57).

For h-index, 41 studies were included comparing 29,923 men to 12,286 women. Men had a higher h-index than women (mean difference 5.97, 95% CI 4.77 – 7.16, p < .00001, I^2 = 98%, eFigure 9). Funnel plot demonstrated no evidence of publication bias (eFigure 10). Sensitivity analysis removing 1 poor quality study continued to show that men had a higher h-index (mean difference 5.98, 95% CI 4.77 – 7.19, p < .00001, I^2 = 98%, eFigure 11). One study demonstrated

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that women continued to have a lower h-index after adjusting for career duration (adjusted relative rate 0.62, 95% CI 0.54 – 0.72) (57).

For federal research funding, 21 studies were included comparing 87,205 men to 39,638 women. Men were 1.71 times more likely to hold a federal research grant (OR 1.71, 95% CI 1.41 – 2.08, $p < .00001$, $I^2 = 88\%$, eFigure 12). Funnel plot demonstrated evidence of publication bias skewed towards a higher OR (eFigure 13). Sensitivity analysis removing 2 poor quality studies continued to show that men were more likely to hold a federal research grant (OR 1.55, 95% CI 1.30 – 1.84, $p < .00001$, $I^2 = 75\%$, eFigure 14). One study demonstrated no difference in the proportion of men vs. women receiving federal research funding after controlling for time in practice, additional degrees, publications, and h-index (adjusted OR 1.23, 95% CI 0.79 – 1.92) (56).

For salary, 20 studies were included comparing 15,155 men to 7,837 women. Men's salary was $32,520 higher (mean difference $32,520, 95% CI $25,790 - $39,260, $p < .00001$, $I^2 = 94\%$, eFigure 15). Funnel plot showed evidence of publication bias skewed towards a higher mean difference (eFigure 16). Sensitivity analysis removing 1 poor quality study continued to show that men received a higher salary than women (mean difference $34,930, 95% CI $25,920 - $43,950, $p < .00001$, $I^2 = 95\%$, eFigure 17). Two studies reported adjusted estimates for salary. Jena et al. (2016) demonstrated that men earned more after controlling for time in practice, research funding, publications, and clinical trial participation (adjusted absolute difference $19,878, 95% CI $15,261 - $24,495) (41). Gambhir et al. (2020) also showed that men received a higher salary after adjusting for academic rank and specialty (adjusted absolute difference $45,904, 95% CI $13,264 - $103,137) (58).

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For leadership, 42 studies were included comparing 48,613 men to 26,958 women. Men were 2.61 times more likely to be a departmental chair (OR 2.61, 95% CI 2.19 – 3.12, p < .00001, I^2 = 55%, eFigure 18). Funnel plot demonstrated no evidence of publication bias (eFigure 19). Sensitivity analysis removing 6 poor quality studies continued to show that men were more likely to be departmental chairs (OR 2.60, 95% CI 2.16 – 3.13, p < .00001, I^2 = 59%, eFigure 20). No studies reported adjusted estimates for this outcome.

A meta-analysis of adjusted estimates for secondary outcomes was not performed due to a paucity of studies.

DISCUSSION

Summary of findings

This systematic review and meta-analysis of 218 studies including 991,207 academic physician data points demonstrated that men were 2.77 times more likely to be full professors than women. This was consistent across specialties and geographic regions. There has been some improvement over time, but in the 2010-2020 decade, men were 2.63 times more likely to be full professors. Although men practiced 6 years longer than women, men continued to be 1.83 times more likely to be full professors after pooling results from 7 studies that controlled for number of years in practice. Furthermore, men published 17.2 more papers, had a h-index that was 5.97 higher, received a salary that was $32,520 greater, and were 2.61 and 1.71 times more likely to be departmental chairs and receive federal research funding, respectively. Among single studies, men continued to have more publications, higher h-indices, and greater salaries after adjusting for factors including time in practice and specialty (41,57,58).

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Comparison to existing literature

This was the first systematic review and meta-analysis of gender differences in faculty rank among academic physicians. Our findings are consistent with most of the single studies in the literature demonstrating significant gender disparities in academic rank. Specifically, our pooled analysis of 24 specialties across 16 countries confirmed that the gender disparity in full professorship is pervasive across specialties globally.

Jena et al. (2015) performed the largest single study on this topic, analyzing 91,073 US academic physicians. They showed that 28.6% of men were full professors compared to 11.9% of women, for an absolute difference of 16.7% (16). In their study, men were older, more likely to hold federal research grants, and published more papers. However, after adjusting for experience, research productivity, and specialty, women continued to be less likely than men to be full professors, with an absolute difference of 4% (16). We demonstrated similar findings across specialties, over time, and between geographic regions in a larger pooled cohort.

Explanation of findings

In our study, men practiced for 6 years longer than women, which may have contributed to men being 2.77 times more likely to be full professors. However, after pooling 7 studies that adjusted for number of years in practice, men continued to be 1.83 times more likely to attain full professorship. Therefore, differences in career duration does not fully explain the gender gap. This is consistent with previous studies showing that men were more likely to hold higher academic positions after adjusting for time in practice (17,18,53).

Several studies have explored reasons for gender inequity in academic medicine. In a survey of 4,285 US pediatricians, women were less likely to report that they received adequate

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mentorship (59). Other studies have demonstrated gender-based discrimination (60), unequal allocation of institutional resources (61), and lack of promotion tracks for women who take time off for childcare (62). In addition to inequity at work, inequity at home may also influence progress in academic medicine. Jolly and colleagues (2014) surveyed 1,049 US clinician-scientists and found that women spent 8.5 more hours per week on household activities than men (63). Due to these inequities, women are more likely to leave academia (64). These reasons could explain the gender differences in academic rank, research productivity, salary, and leadership.

Notably, our study demonstrated that a similar gender gap in full professorship exists for countries within and outside of North America. Despite geographic and cultural differences, gender inequity is a global phenomenon. This systemic problem exists not only in health care, but also in education (65), income (66), and political power (67).

Our analysis showed that men were more likely than women to be full professors across every specialty. Interestingly, 39% of faculty in obstetrics and gynecology were women, yet men were 4 times more likely to be full professors. Rayburn and colleagues (2012) analyzed US faculty in obstetrics and gynecology and found an increase in the proportion of female entry-level faculty, but they were primarily in the non-tenure track (68). They showed that promotion rates dropped from 35% in 1980-1989 to 26% in 2000-2009 (68). Therefore, in specialties where there is an increasing number of women entering but few opportunities for advancement, the gender gap may be widening.

We showed minimal improvement in gender disparities over time, with men being 2.63 times more likely to be full professors in 2010-2020. As demonstrated in the example of obstetrics and gynecology, waiting for more women to enter the academic pipeline will be insufficient to close the gender gap. One method of actively improving gender equity is

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implementing programs that directly support women. For example, Stanford University's School of Medicine established an Office of Diversity and Leadership in 2004 (69). They developed research awards, mentorship programs, and networking events for women (69). Over the next 6 years, the number of female full professors increased from 52 to 91 (69). This rise was 60% greater than the national promotion rate to full professor for women (69). Similar career development programs at Johns Hopkins University (70) and the University of Michigan (71) have demonstrated benefit to female academic physicians. In Europe, many medical schools have instituted Athena SWAN (Scientific Women's Academic Network) action plans, which are a set of multi-level interventions that address structural, cultural, and institutional factors that create barriers for women (72). This broad-based strategy has significantly improved female representation in leadership positions (73). However, Carr and colleagues (2017) demonstrated that many institutions do not have programs to support gender equity (74). This may explain the ongoing gender gap, which can be further closed by expanding interventions that support women to a national and global scale.

Limitations

This study was limited by high heterogeneity in most of the analyses conducted. This was expected as we pooled effect estimates across multiple specialties, data sources, and countries. Subgroup analysis by specialty demonstrated a reduction in heterogeneity. Therefore, data variability across specialties may be a source of heterogeneity in our study.

Funnel plots for secondary outcomes including publications, federal research funding, and salary demonstrated evidence of publication bias, which may have contributed to

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overestimation of the gender gap. However, there was no evidence of publication bias for the primary outcome of academic rank.

There may be selection bias due to the inclusion of cross-sectional studies that did not statistically adjust for confounding. However, pooling of studies that controlled for potential confounders demonstrated that men continued to be more likely than women to be full professors. Sensitivity analysis was also conducted to remove poor quality studies, which did not have a significant impact on the pooled estimate for any outcome.

We intended to reduce the risk of counting a study participant more than once by including only the most comprehensive article (largest sample size and/or most outcomes reported) when multiple studies reported duplicate data (same data source, year of data collection, and outcome). However, this risk could not be completely mitigated without participant-level data, which was not available in our study-level meta-analysis.

In our cohort, we did not have information on the academic ranks of all the participants (e.g., instructors and assistant/associate professors) as many studies only reported the number of full professors out of a pool of academic physicians. We recommend that future studies on this topic report the academic ranks of all their participants.

CONCLUSIONS

This systematic review and meta-analysis provides a comprehensive synthesis of the literature on gender differences in faculty rank among academic physicians. We demonstrated that men were more likely to be full professors than women after adjusting for experience, academic productivity, and specialty. The gender gap exists across all specialties and geographic regions. Gender inequity persists even in the most contemporaneous studies, indicating that

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gender disparities in faculty rank will remain unless actively addressed. Greater efforts are needed to support women in academic medicine.

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CONTRIBUTORS

All authors (BL, JJB, FD, KS, TK, EG, NNB, MAO) were involved in designing the study. TK and BL developed the database search. BL, JJB, and FD screened citations for inclusion, performed risk of bias analysis, and extracted data. BL performed statistical analysis. All authors (BL, JJB, FD, KS, TK, EG, NNB, MAO) were involved in interpreting the data. BL drafted the manuscript. All authors (BL, JJB, FD, KS, TK, EG, NNB, MAO) were involved in critical revision of the manuscript for important intellectual content. BL and MAO obtained funding. NNB and MAO provided supervision. The study guarantor is BL, who had full access to the data and takes responsibility for the integrity of data collection and accuracy of data analysis. All authors reviewed the final manuscript and agreed to be accountable for all aspects of the work and approved the final manuscript for submission. The corresponding author attests that all listed authors meet authorship criteria and no others meeting the criteria have been omitted.

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and do not reflect the official policy or position of the funding source. All researchers were independent of the funder.

COMPETING INTERESTS

All authors have completed the *Unified Competing Interest form* (available on request from the corresponding author) and declare: Ben Li had financial support from the Canadian Society for Vascular Surgery National Student Research Award for the submitted work; no financial relationships with any organisations that might have an interest in the submitted work in the previous three years; no other relationships or activities that could appear to have influenced the submitted work.

PATIENT CONSENT FOR PUBLICATION

Not required.

ETHICS APPROVAL

This systematic review and meta-analysis was based on published data. The authors did not access any information that could lead to identification of a specific patient. Therefore, ethics approval was not required.

PROVENANCE AND PEER REVIEW

Not commissioned.

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DATA SHARING STATEMENT

All data relevant to the study are included in the article or uploaded as supplementary information. Additional data are available upon reasonable request.

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TRANSPARENCY STATEMENT

BL, the study guarantor, affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that there were no discrepancies from the study as originally planned.

DISSEMINATION DECLARATION

Dissemination of results to study participants or patient organisations is not applicable for this systematic review and meta-analysis.

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FIGURE LEGENDS

Figure 1. PRISMA study flow diagram.

Figure 2. Forest plot of full professorship for men vs. women. M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

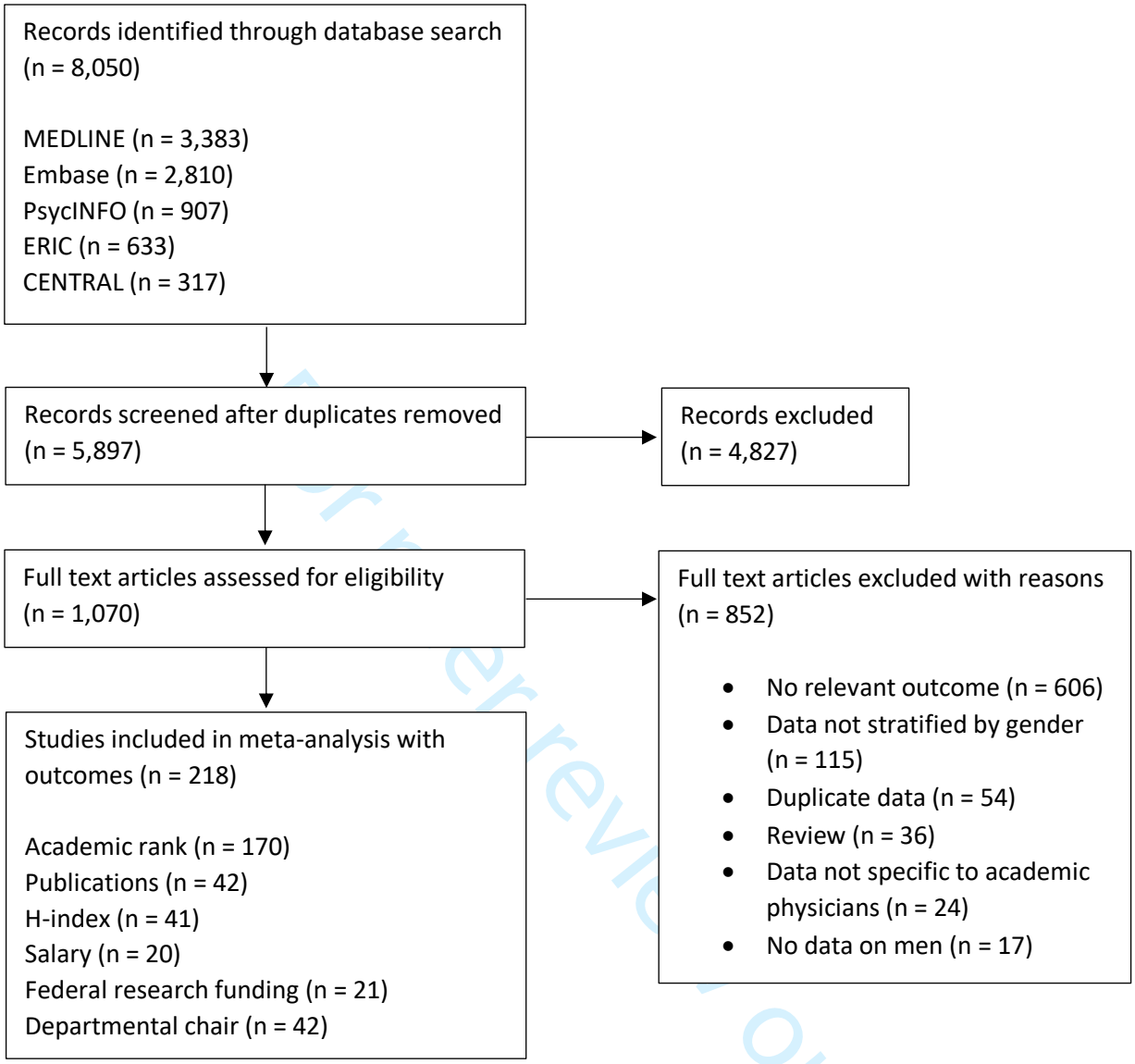
Figure 3. Forest plot of full professorship adjusted for years in practice (Dossani, Gawad, Jena, Orhurhu, Riaz, Smith, Sperling), specialty (Jena), publications (Dossani, Gawad, Jena, Riaz, Sperling), h-index (Dossani, Gawad, Orhurhu, Riaz, Sperling), additional PhD (Dossani, Gawad, Sperling), and institution (Dossani, Jena, Orhurhu, Riaz, Sperling). SE (standard error), Total (sample size), IV (inverse variance), Random (random effects), CI (confidence interval).

Figure 4. Meta-regression of full professorship based on data collection year. Each circle represents a unique study, the size of the circle indicates weight, and the blue line shows the trend over time. The Y axis represents the log odds ratio of the proportion of men vs. women with full professorship.

Figure 5. Subgroup analysis of full professorship by decade based on data collection year (Before 2000, 2000-2009, 2010-2020). M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

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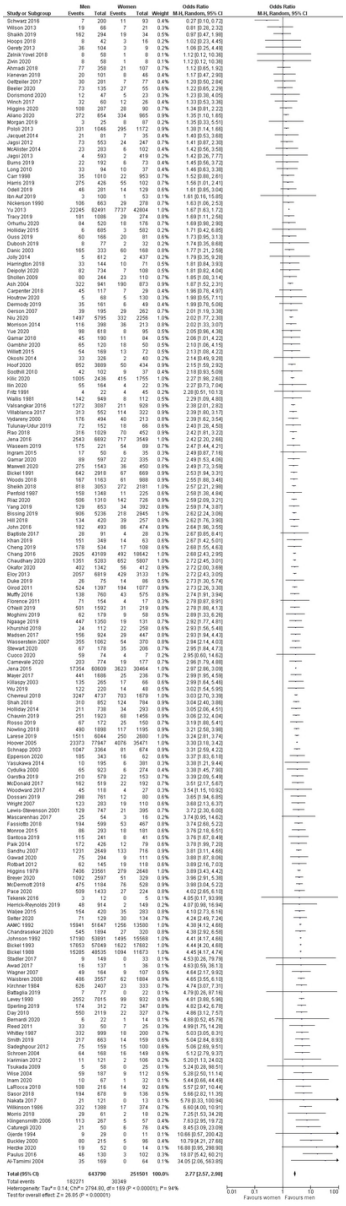
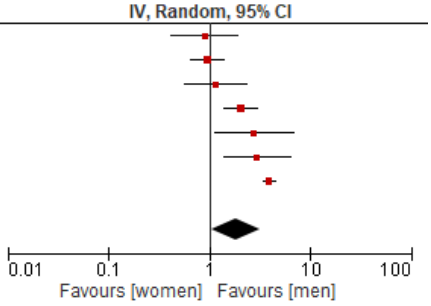


Figure 2. Forest plot of full professorship for men vs. women. M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

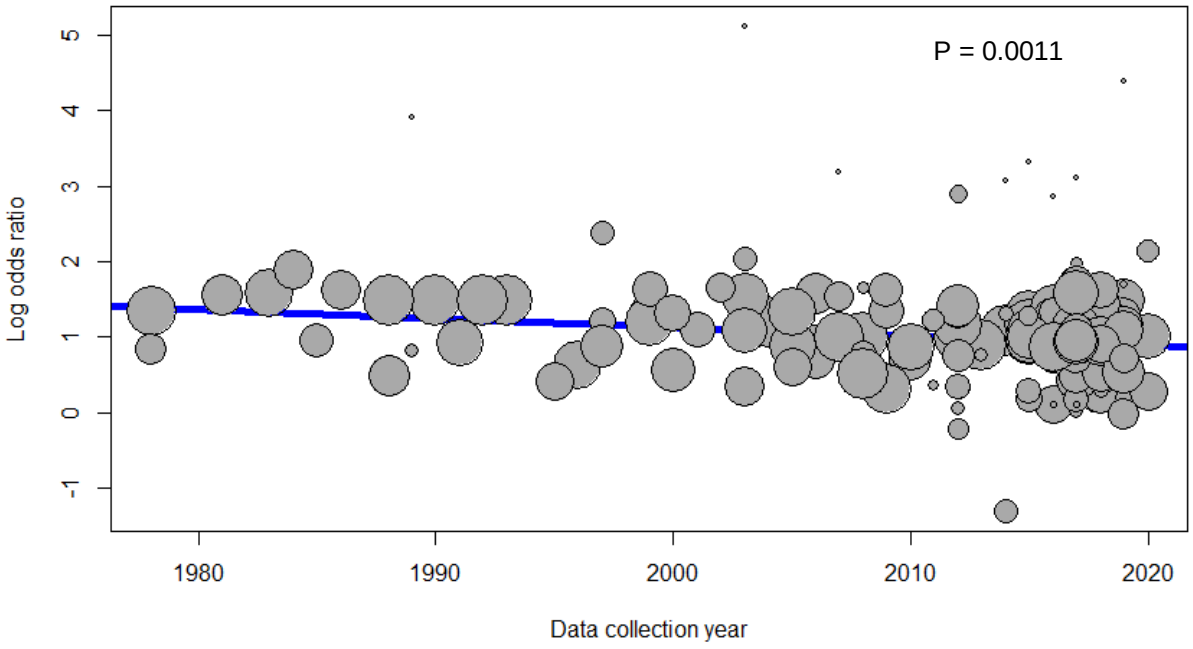
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Study or Subgroup	log[Odds Ratio]	SE	Men Total	Women Total	Weight	Odds Ratio IV, Random, 95% CI
Dossani 2019	-0.1054	0.3889	761	80	13.0%	0.90 [0.42, 1.93]
Riaz 2020	-0.0513	0.2015	1310	726	15.8%	0.95 [0.64, 1.41]
Orhurhu 2020	0.1398	0.3671	520	176	13.4%	1.15 [0.56, 2.36]
Sperling 2019	0.7129	0.1957	312	347	15.9%	2.04 [1.39, 2.99]
Gawad 2020	1.026	0.4611	294	111	11.9%	2.79 [1.13, 6.89]
Smith 2019	1.0784	0.3896	663	159	13.0%	2.94 [1.37, 6.31]
Jena 2015	1.3552	0.0727	60609	30464	17.0%	3.88 [3.36, 4.47]
Total (95% CI)			**64469**	**32063**	**100.0%**	**1.83 [1.04, 3.20]**

Heterogeneity: Tau² = 0.47; Chi² = 65.23, df = 6 (P < 0.00001); I² = 91%
Test for overall effect: Z = 2.11 (P = 0.03)



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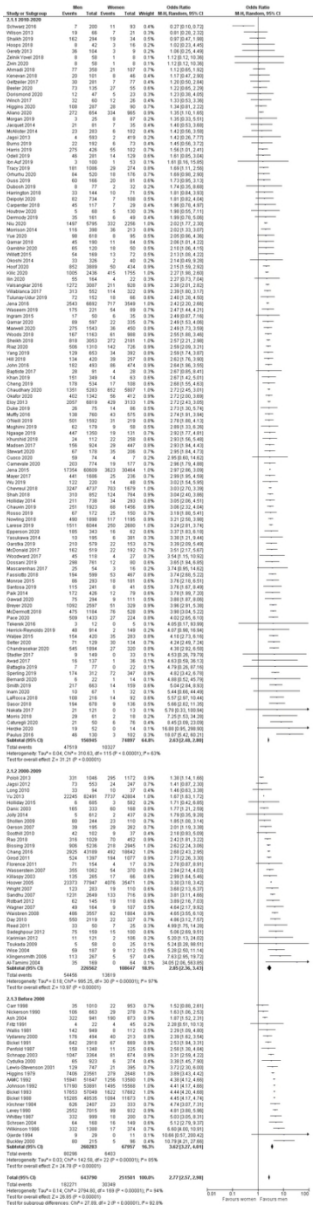


Figure 5. Subgroup analysis of full professorship by decade based on data collection year (Before 2000, 2000-2009, 2010-2020). M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

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Supplementary online content

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eTable 1. Search strategies for A) MEDLINE, B) Embase, C) Cochrane CENTRAL, PsycINFO, and ERIC

A) MEDLINE

1	Sex Characteristics/ (54058)
2	sex distribution/ (55145)
3	Sex Factors/ (263100)
4	Sexism/ (1967)
5	(sex or sexes or sexism).tw,kf. (559154)
6	gender*.tw,kf. (331420)
7	wom?n*.tw,kf. (1170111)
8	female*.tw,kf. (959904)
9	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 (2625210)
10	exp Physicians/ (140699)
11	physician*.tw,kf. (392562)
12	(medicine or medical).tw,kf. (1578309)
13	(surgery or surgical or surgeon*).tw,kf. (1912383)
14	(doctor or doctors).tw,kf. (122123)
15	(resident or residents).tw,kf. (156159)
16	(intern or interne or interns or internes).tw,kf. (5853)
17	(trainee or trainees).tw,kf. (25596)
18	clinician*.tw,kf. (224670)
19	10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 (3873810)
20	Faculty, Medical/ (13263)
21	exp Academic Medical Centers/ (93396)
22	(faculty or faculties).tw,kf. (48393)
23	(academic or academics).tw,kf. (133076)
24	20 or 21 or 22 or 23 (253893)
25	Achievement/ (15369)
26	career mobility/ (11441)
27	achievement*.tw,kf. (61273)
28	(rank or ranks).tw,kf. (110562)
29	professor*.tw,kf. (21283)
30	(career or careers).tw,kf. (42059)
31	(job or jobs).tw,kf. (61905)
32	(ladder or ladders).tw,kf. (10401)
33	advancement*.tw,kf. (54164)
34	promot*.tw,kf. (1017322)
35	25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (1364135)
36	9 and 19 and 24 and 35 (3465)
37	limit 36 to (case reports or comment or editorial or letter or meta analysis or "review" or "systematic review") (207)
38	36 not 37 (3383)

B) Embase

1	exp sex difference/ (400581)
2	sex ratio/ (72668)
3	sexism/ or gender bias/ (6087)
4	(sex or sexes or sexism).tw,kw. (794517)
5	gender*.tw,kw. (535055)

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6	wom?n*.tw,kw. (1669994)
7	female*.tw,kw. (1510495)
8	1 or 2 or 3 or 4 or 5 or 6 or 7 (3876952)
9	exp physician/ (790780)
10	physician*.tw,kw. (571261)
11	(medicine or medical).tw,kw. (2386371)
12	(surgery or surgical or surgeon*).tw,kw. (2693032)
13	(doctor or doctors).tw,kw. (182709)
14	(resident or residents).tw,kw. (207836)
15	(intern or interne or interns or internes).tw,kw. (9879)
16	(trainee or trainees).tw,kw. (42786)
17	clinician*.tw,kw. (322509)
18	9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 (5742530)
19	medical school/ (63562)
20	exp teaching hospital/ (173020)
21	(faculty or faculties).tw,kw. (70075)
22	(academic or academics).tw,kw. (223412)
23	19 or 20 or 21 or 22 (478892)
24	achievement/ (37218)
25	job performance/ (16941)
26	career mobility/ (10384)
27	achievement*.tw,kw. (81730)
28	(rank or ranks).tw,kw. (188284)
29	professor*.tw,kw. (23538)
30	(career or careers).tw,kw. (52124)
31	(job or jobs).tw,kw. (80710)
32	(ladder or ladders).tw,kw. (11600)
33	advancement*.tw,kw. (70446)
34	promot*.tw,kw. (1278758)
35	24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (1777343)
36	8 and 18 and 23 and 35 (6798)
37	limit 36 to embase (3085)
38	limit 37 to (editorial or letter or "review") (313)
39	37 not 38 (2810)

C) Cochrane CENTRAL

1	Sex Characteristics/ (1241)
2	sex distribution/ (746)
3	Sex Factors/ (5904)
4	Sexism/ (26)
5	(sex or sexes or sexism).tw. (43624)
6	gender*.tw. (32982)
7	wom?n*.tw. (151619)
8	female*.tw. (90532)
9	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 (281682)
10	exp Physicians/ (1794)
11	physician*.tw. (41471)
12	(medicine or medical).tw. (141009)
13	(surgery or surgical or surgeon*).tw. (197021)
14	(doctor or doctors).tw. (10694)
15	(resident or residents).tw. (10895)

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16	(intern or interne or interns or internes).tw. (631)
17	(trainee or trainees).tw. (2553)
18	clinician*.tw. (22755)
19	10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 (368556)
20	Faculty, Medical/ (106)
21	exp Academic Medical Centers/ (1757)
22	(faculty or faculties).tw. (3115)
23	(academic or academics).tw. (12131)
24	20 or 21 or 22 or 23 (16538)
25	Achievement/ (481)
26	career mobility/ (11)
27	achievement*.tw. (6235)
28	(rank or ranks).tw. (15788)
29	professor*.tw. (570)
30	(career or careers).tw. (763)
31	(job or jobs).tw. (2806)
32	(ladder or ladders).tw. (317)
33	advancement*.tw. (2232)
34	promot*.tw. (42698)
35	25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (70085)
36	9 and 19 and 24 and 35 (317)

D) PsycINFO

1	human sex differences/ (112551)
2	exp sex role attitudes/ (12493)
3	sex discrimination/ (2293)
4	(sex or sexes or sexism).ti,ab,id. (210417)
5	gender*.ti,ab,id. (221794)
6	wom?n*.ti,ab,id. (307718)
7	female*.ti,ab,id. (365668)
8	1 or 2 or 3 or 4 or 5 or 6 or 7 (800890)
9	exp Physicians/ (43420)
10	physician*.ti,ab,id. (64370)
11	(medicine or medical).ti,ab,id. (238071)
12	(surgery or surgical or surgeon*).ti,ab,id. (43436)
13	(doctor or doctors).ti,ab,id. (27901)
14	(resident or residents).ti,ab,id. (47303)
15	(intern or interne or interns or internes).ti,ab,id. (3474)
16	(trainee or trainees).ti,ab,id. (16486)
17	clinician*.ti,ab,id. (90454)
18	9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 (455043)
19	educational personnel/ (11565)
20	(faculty or faculties).ti,ab,id. (35800)
21	(academic or academics).ti,ab,id. (149957)
22	19 or 20 or 21 (182798)
23	achievement/ (9479)
24	occupational success/ (2867)
25	occupational mobility/ (1101)
26	exp occupational tenure/ (1223)
27	achievement*.ti,ab,id. (111516)
28	(rank or ranks).ti,ab,id. (23580)

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29	professor*.ti,ab,id. (15563)
30	(career or careers).ti,ab,id. (64031)
31	(job or jobs).ti,ab,id. (91586)
32	(ladder or ladders).ti,ab,id. (1646)
33	advancement*.ti,ab,id. (13461)
34	promot*.ti,ab,id. (191176)
35	23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 (476581)
36	8 and 18 and 22 and 35 (952)
37	limit 36 to ("comment/reply" or editorial or letter or reviews) (46)
38	36 not 37 (907)

E) ERIC

NOFT(sex or sexes or sexism or gender* or wom?n* or female*) AND NOFT(physician* or medicine or medical or surgery or surgical or surgeon* or doctor or doctors or resident or residents or intern or interns or interne or internes or trainee or trainees or clinician*) AND NOFT(faculty or faculties or academic or academics) AND NOFT(achievement* or rank or ranks or professor* or career or careers or job or jobs or ladder or ladders or advancement* or promot*)

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eTable 2a. Summary of included studies

Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
AAMC[1]	1992	US	Database	Multiple	1990	51647	13580	Academic rank
Abdellatif[2]	2019	North America, Asia, Oceania, Europe	Departmental websites	Multiple	2016	976	328	Publications, h-index
Ahmadi[3]	2018	US, Canada	Departmental websites	Radiology	2016	358	107	Academic rank
Akabas[4]	2019	US	Survey	Multiple	2015	2546	1105	Federal research funding
Aliano[5]	2020	Spain	Survey	Anesthesia	2018	654	965	Academic rank, departmental chair
Al-Tamimi[6]	2004	Saudi Arabia	Survey	Multiple	2003	169	64	Academic rank
Amering[7]	2011	Global	Database	Psychiatry	2007	3775	3775	Publications
Ash[8]	2004	US	Survey	Multiple	1996	941	873	Academic rank, salary
Awad[9]	2017	US	Departmental websites	Urology	2015	137	36	Academic rank, publications, h-index, federal research funding
Baptiste[10]	2017	US	Survey	General surgery	2015	91	28	Academic rank, tenure
Battaglia[11]	2019	US, Canada	Departmental websites	Radiology	2017	77	22	Academic rank, h-index
Beeler[12]	2020	Global	Survey	Radiation oncology	2018	135	55	Academic rank, publications, h-index
Bernardi[13]	2020	US	Survey	General surgery	2019	22	14	Academic rank
Bickel[14]	1988	US	Database	Multiple	1988	48535	11673	Academic rank

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Bickel[15]	1993	US	Database	Multiple	1993	57049	17682	Academic rank, tenure
Bickel[16]	1991	US	Database	Multiple	1991	2918	669	Academic rank
Bissing[17]	2019	US	Database	Anesthesia	2015	5236	2945	Academic rank
Breyer[18]	2020	US	Survey	Urology	2017	2597	329	Academic rank
Bruggmann[19]	2017	Germany	Database	Obstetrics and gynecology, otolaryngology	2013	12591	12857	Departmental chair
Buckley[20]	2000	US	Survey	Multiple	1997	215	96	Academic rank, tenure
Burns[21]	2019	Canada	Survey	Anesthesia	2018	192	73	Academic rank, department chair
Camacci[22]	2020	US	Departmental websites	Ophthalmology	2018	158	60	Publications, h-index
Carnevale[23]	2020	US	Departmental websites	Vascular surgery	2019	774	177	Academic rank, publications, h-index, federal research funding, departmental chair
Carpenter[24]	2018	US	Departmental websites	General surgery	2017	117	29	Academic rank
Carr[25]	1992	US	Survey	Internal medicine	1986	435	145	Tenure, department chair
Carr[26]	1998	US	Survey	Multiple	1995	1010	953	Publications
Caturegli[27]	2020	US	Database	General surgery	2020	50	76	Academic rank, publications, h-index, federal research funding
Chaiyachati[28]	2019	US	Survey	Internal medicine, pediatrics, family medicine	2016	58	104	Publications

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Chandrasekar [29]	2020	US, Canada	Departmental websites	Urology	2019	1894	320	Academic rank
Chang[30]	2016	US	Database	Multiple	2008	43189	18642	Academic rank
Chaudhary[31]	2020	US	Database	Psychiatry	2018	5283	5807	Academic rank
Chauvin[32]	2019	Canada	Departmental websites	Psychiatry	2016	1923	1456	Academic rank, publications
Cheng[33]	2019	US	Departmental websites	Vascular surgery	2018	534	108	Academic rank
Cheng[34]	2019	US	Survey	Multiple	2018	37	35	Salary
Chevreul[35]	2018	France	Database	Multiple	2015	4737	1679	Academic rank
Chokshi[36]	2009	US	Survey	General surgery	2007	275	36	Federal research funding
Colletti[37]	2000	US	Survey	General surgery	2000	45	9	Tenure
Cuoco[38]	2020	US	Database	Neurosurgery	2018	74	7	Academic rank, salary
Cydulka[39]	2000	US	Survey	Emergency medicine	1997	923	274	Academic rank
Danic[40]	2003	Croatia	Database	Multiple	2000	333	168	Academic rank
Day[41]	2010	US	Database	Orthopedic surgery	2006	2119	327	Academic rank
Deipolyi[42]	2020	US	Departmental websites	Interventional radiology	2019	734	108	Academic rank, publications, h-index
Dennis[43]	1990	US	Survey	Multiple	1987	85	51	Salary
Dermody[44]	2019	US	Database	Otolaryngology	2016	161	49	Academic rank, h-index, salary
Desai[45]	2018	US	Departmental websites	General surgery	2017	325	105	H-index
Diehl[46]	2008	US	Survey	Family medicine	2006	276	49	Salary

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Dorismond[47]	2020	US	Departmental websites	Otolaryngology	2019	47	23	Academic rank, h-index, federal research funding
Dossani[48]	2019	US	Database	Neurosurgery	2019	761	80	Academic rank
Dubosh[49]	2019	US	Departmental websites	Emergency medicine	2018	77	32	Academic rank, federal research funding
Duke[50]	2019	US	Survey	Multiple	2020	75	86	Academic rank
Eloy[51]	2013	US	Departmental websites	Multiple	2012	6819	3133	Academic rank, h-index, departmental chair
Eloy[52]	2013	US	Departmental websites	Otolaryngology	2012	149	55	Federal research funding
Epperson[53]	2020	US, Canada	Departmental websites	Otolaryngology	2018	343	62	Academic rank, h-index, departmental chair
Fassiotto[54]	2018	US	Database	Multiple	2015	599	467	Academic rank
Florence[55]	2011	US	Survey	General surgery	2006	154	17	Academic rank, departmental chair
Fritz[56]	1991	US	Survey	Pediatrics	1989	22	45	Academic rank
Gambhir[57]	2020	US	Database	General surgery	2016	120	50	Academic rank, salary
Garstka[58]	2019	US	Departmental websites	Otolaryngology	2017	579	153	Academic rank, h-index, departmental chair
Gauthier[59]	2013	Germany, Switzerland, Austria	Survey	Multiple	2011	67	62	Department chair
Gawad[60]	2020	Canada	Departmental websites	General surgery	2018	294	111	Academic rank, publications
Geltzeiler[61]	2017	US	Departmental websites	General surgery	2015	281	77	Academic rank, publications

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Gerety[62]	2013	US, Canada, Asia, South America, Europe	Survey	Plastic surgery	2012	104	9	Academic rank
Gerson[63]	2007	US	Survey	Gastroenterology	2006	195	262	Academic rank, departmental chair
Girod[64]	2011	US	Survey	Multiple	2007	1397	1077	Academic rank, publications
Gjerde[65]	1994	US	Survey	Family medicine	1989	29	11	Academic rank
Grimm[66]	2016	US	Departmental websites	Radiology	2015	2502	1262	Departmental chairs
Guss[67]	2019	US	Departmental websites	Radiation oncology	2017	166	81	Academic rank, federal research funding, salary, departmental chair
Hafeez[68]	2019	Global	Departmental websites	Psychiatry	2017	579	257	Publications, h-index
Harrington[69]	2018	US	Departmental websites	Radiology	2017	144	71	Academic rank
Harris[70]	2019	Global	Departmental websites	General surgery	2017	426	102	Academic rank, h-index
Herrick-Reynolds[71]	2019	US	Database	General surgery	2017	914	149	Academic rank
Herzke[72]	2020	US	Survey	Family medicine	2019	52	14	Academic rank
Higgins[73]	2020	US	Survey	Orthopedic surgery	2020	287	90	Academic rank
Higgins[74]	1979	US	Database	Multiple	1978	23561	2648	Academic rank
Hill[75]	2018	US	Survey	Obstetrics and gynecology	2015	420	257	Academic rank
Holliday[76]	2015	US	Survey	Multiple	2009	685	582	Academic rank

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Holliday[77]	2014	US	Departmental websites	Radiation oncology	2012	738	293	Academic rank, federal research funding
Hoof[78]	2020	US	Departmental websites	Orthopedic surgery	2018	3889	434	Academic rank
Hoops[79]	2018	US	Database	General surgery	2017	42	16	Academic rank
Hoover[80]	2005	US	Database	Multiple	2004	77947	35471	Academic rank
Houtrow[81]	2020	US	Survey	Physical medicine and rehabilitation	2017	68	130	Academic rank, salary, departmental chair
Houtrow[82]	2011	US	Survey	Physical medicine and rehabilitation	2009	25	61	Salary
Hurley[83]	2014	US	Database	Anesthesia	2008	4360	1783	Publications
Ibn Auf[84]	2019	Sudan	Survey	Multiple	2017	100	53	Academic rank
Ilin[85]	2020	Canada	Departmental websites	Urology	2019	164	22	Academic rank
Inam[86]	2020	Pakistan	Survey	General surgery	2019	67	32	Academic rank
Ingram[87]	2015	US	Departmental websites	Respirology	2015	50	35	Academic rank, h-index
Jacquet[88]	2014	US	Survey	Emergency medicine	2012	81	35	Academic rank
Jagsi[89]	2013	US	Survey	Multiple	2011	593	419	Academic rank, salary
Jagsi[90]	2012	US	Survey	Multiple	2003	553	247	Academic rank, publications, salary
Jaque[91]	2020	Germany	Database	Urology	2015	1944	499	Departmental chair
Jena[92]	2016	US	Database	Multiple	2013	6692	3549	Academic rank, publications, federal research funding, salary

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Jena[93]	2015	US	Database	Multiple	2014	60609	30464	Academic rank, publications, federal research funding
John[94]	2016	US	Departmental websites	Dermatology	2015	493	474	Academic rank, h-index, departmental chair
Johnson[95]	1992	US	Database	Multiple	1992	53891	15568	Academic rank
Jolly[96]	2014	US	Survey	Multiple	2009	612	437	Academic rank
Kaderli[97]	2015	Switzerland	Survey	General surgery	2011	448	64	Department chair
Karimian[98]	2012	Iran	Survey	Multiple	2008	121	106	Academic rank
Kenevan[99]	2018	US	Departmental websites	Anesthesia	2017	101	46	Academic rank, department chair
Khan[100]	2019	US	Departmental websites	Cardiology	2017	349	63	Academic rank, publications, h-index, federal research funding
Khan[101]	2014	US	Departmental websites	Neurosurgery	2012	1144	81	H-index
Khurshid[102]	2018	US, Canada	Departmental websites	Radiology	2017	112	258	Academic rank
Kilic[103]	2020	US	Survey	Pathology	2018	2436	1755	Academic rank
Killaspy[104]	2003	UK	Survey	Psychiatry	2001	265	66	Academic rank
Kim[105]	2019	US	Departmental websites	Urology	2017	327	118	Publications
Kirchner[106]	1984	US	Survey	Radiology	1981	2407	333	Academic rank
Klingensmith[107]	2006	US	Survey	General surgery	2003	267	57	Academic rank
Kloosterboer[108]	2020	US	Departmental websites	Ophthalmology	2019	84	32	Publications, h-index

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Komlenac[109]	2019	Sweden, Netherlands, Austria	Survey	Multiple	2012	468	550	Departmental chair
Larese[110]	2019	Italy	Database	Multiple	2015	6044	2680	Academic rank
LaRocca[111]	2018	US	Departmental websites	General surgery	2017	216	92	Academic rank
Levey[112]	1990	US	Survey	Internal medicine	1983	7015	932	Academic rank, federal research funding
Lewis-Stevenson[113]	2001	US	Survey	Family medicine	1999	747	395	Academic rank
Long[114]	2010	US	Departmental websites	Anesthesia	2008	94	37	Academic rank, department chair
Madsen[115]	2017	US	Survey	Emergency medicine	2015	924	447	Academic rank, salary, departmental chair
Mascarenhas[116]	2017	Canada	Database	Multiple	2014	54	16	Academic rank
Maxwell[117]	2020	US	Database	General surgery	2016	1543	450	Academic rank, h-index, salary
Mayer[118]	2017	US	Departmental websites	Urology	2016	1686	236	Academic rank, departmental chair, federal research funding
McAlister[119]	2014	Canada	Survey	Ophthalmology	2012	283	102	Academic rank
McDermott[120]	2018	US	Departmental websites	Neurology	2016	1184	528	Academic rank, publications, h-index
McDonald[121]	2017	US	Departmental websites	Radiology	2016	519	192	Academic rank, h-index
Mendoza[122]	2015	US	Survey	Pediatrics	2015	5104	6064	Departmental chair
Modena[123]	1999	Italy	Survey	Cardiology	1996	1407	294	Publications

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Moghimi[124]	2019	US, Canada	Departmental websites	Radiology	2016	179	58	Academic rank
Monroe[125]	2015	US	Survey	Multiple	2012	293	181	Academic rank, department chair
Morgan[126]	2019	US	Survey	Obstetrics and gynecology	2018	25	87	Academic rank
Morris[127]	2018	US	Database	General surgery	2017	61	18	Academic rank
Morrison[128]	2014	Canada	Database	Internal medicine	2010	398	213	Academic rank
Mueller[129]	2017	US	Departmental websites	Surgery	2017	160	52	Publications, h-index
Muffly[130]	2016	US	Database	Anesthesia	2015	760	575	Academic rank
Nakata[131]	2017	Japan	Database	General surgery	2015	121	13	Academic rank
Ngaage[132]	2019	US	Departmental websites	Neurosurgery	2018	1350	131	Academic rank, h-index
Nguyen[133]	2018	US	Database	General surgery	2017	331	213	H-index
Nickerson[134]	1990	US	Database	Multiple	1988	663	278	Academic rank
Niu[135]	2020	US	Database	Radiology	2017	5795	2256	Academic rank, departmental chair
Nowling[136]	2018	US	Database	Multiple	2012	1898	1195	Academic rank
Odell[137]	2019	US, Canada	Departmental websites	Neurosurgery	2017	1592	219	Academic rank
Okafor[138]	2020	US	Departmental websites	Otolaryngology	2020	1342	412	Academic rank
Okoshi[139]	2014	Japan	Database	General surgery	2013	326	40	Academic rank
O'Neill[140]	2019	US, Canada	Departmental websites	Radiology	2018	281	129	Academic rank
Orhurhu[141]	2020	US	Departmental websites	Anesthesia	2018	520	176	Academic rank

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Pace[142]	2020	US	Departmental websites	Urology	2018	1433	224	Academic rank, publications, h-index
Pagel[143]	2015	US	Database	Anesthesia	2015	315	82	Publications, h-index, federal research funding
Paik[144]	2014	US	Departmental websites	Plastic surgery	2012	426	79	Academic rank, departmental chair
Paulus[145]	2016	US	Survey	Multiple	2012	130	102	Academic rank
Penfold[146]	1987	Canada	Survey	Psychiatry	1985	1348	225	Academic rank, departmental chair
Pfeiffer[147]	2016	Germany	Database	Multiple	2013	385	539	Publications
Pololi[148]	2013	US	Survey	Multiple	2009	1046	1172	Academic rank
Qamar[149]	2020	Canada	Departmental websites	Radiology	2017	597	335	Academic rank
Qamar[150]	2018	US, Canada	Departmental websites	Radiology	2016	190	84	Academic rank, departmental chair
Rao[151]	2018	US	Database	Multiple	2005	1029	452	Academic rank
Raptis[152]	2012	North America, Europe	Survey	General surgery	2011	365	74	Publications
Rayburn[153]	2012	US	Database	Multiple	2009	2181	2077	Tenure
Reed[154]	2011	US	Database	Multiple	2007	50	25	Academic rank, publications
Reiser[155]	1993	US	Survey	Psychiatry	1984	235	45	Publications
Riaz[156]	2020	US	Database	Hematology	2017	1310	726	Academic rank, publications, h-index, federal research funding, departmental chair

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Robboy[157]	2017	US	Database	Pathology	2016	35	24	Publications, tenure
Rosso[158]	2019	France	Database	Multiple	2019	172	150	Academic rank
Rotbart[159]	2012	US	Database	Pediatrics	2009	145	118	Academic rank, departmental chair, tenure
Sadeghpour[160]	2012	US	Survey	Dermatology	2009	159	100	Academic rank, salary, tenure
Sandhu[161]	2007	UK	Survey	Multiple	2005	2649	716	Academic rank
Santosa[162]	2019	US	Departmental websites	Plastic surgery	2016	241	41	Academic rank, publications
Sasor[163]	2018	US	Departmental websites	Plastic surgery	2017	678	136	Academic rank, federal research funding
Schnapp[164]	2003	US	Survey	Respirology	1999	3364	674	Academic rank
Schroen[165]	2004	US	Survey	General surgery	1999	168	149	Academic rank, publications, tenure
Schwarz[166]	2016	Germany	Survey	General surgery	2014	200	93	Academic rank, department chair
Selter[167]	2020	US	Departmental websites	Obstetrics and gynecology	2018	129	134	Academic rank, departmental chair
Shah[168]	2018	US, Canada	Departmental websites	Dermatology	2017	852	784	Academic rank
Shaikh[169]	2019	Global	Database	Neurosurgery	2019	294	34	Academic rank
Sheikh[170]	2018	US	Departmental websites	Psychiatry	2018	3053	2181	Academic rank
Shiber[171]	2020	US	Survey	Obstetrics and gynecology	2019	41	88	Salary
Shollen[172]	2009	US	Survey	Multiple	2005	244	110	Academic rank
Singh[173]	2008	US	Survey	Gastroenterology	2015	168	25	Salary

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Smith[174]	2019	US	Departmental websites	Plastics	2018	663	159	Academic rank, departmental chair
Soothill[175]	2010	UK	Survey	Psychiatry	2008	102	37	Academic rank
Speck[176]	2012	US	Database	Multiple	2007	566	335	Tenure
Spencer[177]	2016	US	Survey	Urology	2017	660	73	Salary
Sperling[178]	2019	US	Departmental websites	Obstetrics and gynecology	2017	312	347	Academic rank, federal research funding
Stadler[179]	2017	Global	Survey	Multiple	2014	149	33	Academic rank
Steger[180]	2009	Austria	Survey	Multiple	2005	220	173	Publications, departmental chair
Stewart[181]	2020	US	Departmental websites	Dermatology	2019	178	206	Academic rank, publications, federal research funding
Svider[182]	2014	US	Database	Ophthalmology	2013	433	157	H-index, federal research funding
Tekerek[183]	2016	Turkey	Survey	Multiple	2016	12	5	Academic rank
Therattil[184]	2016	US	Departmental websites	Plastic surgery	2013	491	101	H-index
Tracy[185]	2019	US	Departmental websites	Otolaryngology	2019	1086	274	Academic rank
Tsukada[186]	2009	Japan	Survey	Cardiology	2007	58	25	Academic rank
Tulunay-Ugur[187]	2019	US	Survey	Otolaryngology	2018	152	66	Academic rank
Turaga[188]	2013	US	Departmental websites	General surgery	2012	87	3	Publications, h-index
Valsangkar[189]	2016	US	Departmental websites	General surgery	2016	3087	928	Academic rank, publications, federal research

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
								funding, departmental chair
Villablanca[190]	2017	US	Survey	Multiple	2010	552	322	Academic rank
Vydareny[191]	2000	US	Survey	Radiology	1997	494	213	Academic rank, tenure
Wagner[192]	2007	US	Survey	Physical rehabilitation and medicine	2007	164	107	Academic rank, federal research funding, tenure
Waisbren[193]	2008	US	Database	Multiple	2003	3557	1884	Academic rank
Waljee[194]	2015	US	Survey	Multiple	2012	420	283	Academic rank
Wallis[195]	1981	US	Survey	Multiple	1978	949	112	Academic rank
Warner[196]	2017	US	Database	Multiple	2015	3010	2435	Publications, h-index
Waseem[197]	2019	Global	Departmental websites	Internal medicine	2018	221	89	Academic rank
Wasserstein[198]	2007	US	Survey	Multiple	2003	1062	370	Academic rank
Weaver[199]	2015	US	Survey	Family medicine	2010	513	263	Salary
Whitley[200]	1987	US	Survey	Radiology	1986	999	200	Academic rank, publications
Wiler[201]	2019	US	Survey	Emergency medicine	2017	1430	797	Salary
Wilkes[202]	2015	UK, Ireland	Departmental websites	Neurosurgery	2014	290	25	H-index
Wilkinson[203]	1986	US	Database	Anesthesia	1984	1388	374	Academic rank
Willett[204]	2015	US	Survey	Internal medicine	2012	169	72	Academic rank
Wilson[205]	2013	US	Departmental websites	Radiation oncology	2012	66	21	Academic rank
Winch[206]	2017	UK	Survey	Pediatrics	2015	60	26	Academic rank

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Author	Publication year	Region	Data source	Specialty	Data collection year	Sample size (men)	Sample size (women)	Outcomes
Wise[207]	2004	Canada	Survey	Obstetrics and gynecology	2002	187	112	Academic rank
Woods[208]	2018	US	Database	Multiple	2017	1163	988	Academic rank
Woodward[209]	2017	US	Departmental websites	Gastroenterology	2015	118	27	Academic rank
Wright[210]	2007	US	Database	Multiple	2000	283	110	Academic rank
Wu[211]	2019	North America, Europe, Oceania	Departmental websites	General surgery	2017	220	48	Academic rank
Xierali[212]	2020	US	Database	Family medicine	2017	2887	2865	Tenure
Yang[213]	2019	US, Canada	Departmental websites	Physical medicine and rehabilitation	2017	653	392	Academic rank
Yasukawa[214]	2014	Japan	Survey	Multiple	2011	195	381	Academic rank
Yu[215]	2013	US	Database	Multiple	2008	82491	42804	Academic rank
Yue[216]	2020	Canada	Departmental websites	Orthopedic surgery	2019	618	95	Academic rank
Zelnik[217]	2018	Israel	Survey	Internal medicine	2016	58	8	Academic rank
Zivin[218]	2020	US	Survey	Multiple	2017	683	627	Academic rank

* AAMC (American Association of Medical Colleges), US (United States), UK (United Kingdom)

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eTable 2b. Number of included studies by country of publication

Country	Number of included studies
Austria	1
Canada	11
Croatia	1
France	2
Germany	5
International	8
Iran	1
Israel	1
Italy	2
Japan	4
North America	14
Pakistan	1
Saudi Arabia	1
Sudan	1
Sweden	1
Switzerland	1
Turkey	1
UK	5
US	157

*UK (United Kingdom), US (United States)

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eTable 3. Demographic data reported by 90 studies

Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Aliano 2020	Men	-	-	-	-	-	71.5%	82.2%	21.3%
	Women	-	-	-	-	-	62.2%	63.0%	14.5%
Ash 2004	Men	47.0	80.0%	-	58.0	28.0%	-	-	-
	Women	45.0	82.0%	-	56.0	29.0%	-	-	-
Awad 2017	Men	-	-	9.8	-	-	-	-	-
	Women	-	-	9.5	-	-	-	-	-
Baptiste 2017	Men	-	-	-	-	-	92.3%	94.1%	-
	Women	-	-	-	-	-	71.4%	90.0%	-
Beeler 2020	Men	-	65.9%	-	-	21.5%	-	-	-
	Women	-	63.6%	-	-	12.7%	-	-	-
Bernardi 2020	Men	-	-	-	-	-	91.0%	77.0%	-
	Women	-	-	-	-	-	71.0%	71.0%	-
Buckley 2000	Men	48.0	-	12.0	-	-	-	-	-
	Women	41.0	-	7.0	-	-	-	-	-
Burns 2019	Men	45.4	-	12.9	-	-	89.1%	57.1%	-
	Women	43.6	-	11.1	-	-	76.4%	43.8%	-
Camacci 2020	Men	-	-	-	-	-	-	-	7.0%
	Women	-	-	-	-	-	-	-	15.0%

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Carr 1992	Men	40.0	97.0%	-	58.0	18.0%	-	-	-
	Women	37.0	93.0%	-	55.0	14.0%	-	-	-
Carr 1998	Men	47.0	87.0%	13.0	59.0	28.0%	88.0%	84.0%	-
	Women	45.0	91.0%	11.0	55.0	28.1%	74.0%	69.0%	-
Chaiyachati 2019	Men	-	91.0%	-	-	-	-	-	-
	Women	-	89.0%	-	-	-	-	-	-
Chauvin 2019	Men	-	-	20.0	-	-	-	-	-
	Women	-	-	15.3	-	-	-	-	-
Colletti 2000	Men	-	-	-	-	-	100.0%	99.0%	11.0%
	Women	-	-	-	-	-	89.0%	44.0%	0.0%
Cydulka 2000	Men	40.7	86.0%	-	-	-	-	-	-
	Women	38.5	84.0%	-	-	-	-	-	-
Deipolyi 2020	Men	-	-	20.0	-	-	-	-	-
	Women	-	-	16.0	-	-	-	-	-
Dennis 1990	Men	-	-	-	-	-	65.0%	-	-
	Women	-	-	-	-	-	86.0%	-	-
Dermody 2019	Men	-	-	24.6	-	-	-	-	-
	Women	-	-	18.4	-	-	-	-	-
Dorismond 2020	Men	-	-	-	-	-	-	-	23.4%
	Women	-	-	-	-	-	-	-	26.1%

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Dossani 2019	Men	-	-	-	-	-	-	-	13.9%
	Women	-	-	-	-	-	-	-	18.6%
Epperson 2020	Men	-	-	20.3	-	-	-	-	-
	Women	-	-	13.9	-	-	-	-	-
Fassiotto 2018	Men	-	63.6%	-	-	-	-	-	-
	Women	-	54.2%	-	-	-	-	-	-
Florence 2011	Men	48.7	-	15.8	-	-	88.0%	90.5%	-
	Women	43.7	-	6.8	-	-	88.0%	64.7%	-
Fritz 1991	Men	-	-	-	52.0	25.0%	95.0%	-	-
	Women	-	-	-	48.0	15.0%	70.0%	-	-
Gauthier 2013	Men	45.6	-	-	-	-	68.6%	65.2%	-
	Women	36.6	-	-	-	-	77.4%	35.5%	-
Gawad 2020	Men	-	-	18.5	-	-	-	-	9.5%
	Women	-	-	13.5	-	-	-	-	12.6%
Geltzeiler 2017	Men	-	-	18.0	-	-	-	-	-
	Women	-	-	11.0	-	-	-	-	-
Gerety 2013	Men	53.6	-	-	-	-	-	-	-
	Women	45.1	-	-	-	-	-	-	-
Gerson 2007	Men	46.0	66.0%	8.6	-	-	-	-	-
	Women	42.0	62.0%	6.5	-	-	-	-	-

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Girod 2011	Men	49.9	96.6%	21.3	-	-	-	-	-
	Women	47.2	92.9%	16.8	-	-	-	-	-
Guss 2019	Men	-	-	12.0	-	-	-	-	26.5%
	Women	-	-	10.0	-	-	-	-	18.5%
Harris 2019	Men	-	-	23.0	-	-	-	-	21.9%
	Women	-	-	17.0	-	-	-	-	36.1%
Higgins 2020	Men	-	-	-	-	-	92.7%	94.4%	-
	Women	-	-	-	-	-	68.9%	71.1%	-
Higgins 1979	Men	-	-	-	-	-	-	-	7.8%
	Women	-	-	-	-	-	-	-	3.9%
Hill 2018	Men	-	89.0%	-	-	-	93.0%	91.0%	-
	Women	-	78.0%	-	-	-	87.0%	72.0%	-
Holliday 2015	Men	40.0	70.4%	-	-	-	92.2%	82.0%	30.4%
	Women	40.5	70.6%	-	-	-	87.4%	76.6%	15.5%
Holliday 2014	Men	-	-	17.0	-	-	-	-	17.2%
	Women	-	-	12.0	-	-	-	-	9.2%
Houtrow 2020	Men	-	-	-	-	-	-	-	23.9%
	Women	-	-	-	-	-	-	-	14.6%
Ibn 2019	Men	42.6	-	9.1	-	-	86.9%	-	10.2%
	Women	38.8	-	6.2	-	-	64.2%	-	11.3%

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Inam 2020	Men	-	-	-	-	-	74.6%	-	-
	Women	-	-	-	-	-	43.8%	-	-
Jagsi 2013	Men	40.5	68.8%	4.6	59.4	65.7%	92.9%	83.0%	34.4%
	Women	39.9	68.8%	4.7	54.0	65.4%	89.0%	79.5%	21.0%
Jagsi 2012	Men	45.0	75.1%	-	63.2	58.0%	91.0%	89.3%	49.2%
	Women	45.3	76.5%	-	58.1	57.0%	87.0%	84.2%	48.6%
Jena 2016	Men	51.5	-	19.4	-	-	-	-	-
	Women	46.4	-	14.6	-	-	-	-	-
Jena 2015	Men	52.3	-	20.8	-	-	-	-	-
	Women	46.8	-	15.1	-	-	-	-	-
Jolly 2014	Men	40.6	68.3%	-	59.0	-	92.5%	83.2%	-
	Women	40.0	66.4%	-	54.0	-	88.8%	79.2%	-
Kaderli 2015	Men	50.0	-	-	-	-	83.3%	84.8%	-
	Women	43.0	-	-	-	-	76.6%	42.2%	-
Kenevan 2018	Men	52.0	-	-	-	-	-	-	-
	Women	52.5	-	-	-	-	-	-	-
Khan 2019	Men	-	-	24.0	-	-	-	-	8.6%
	Women	-	-	21.0	-	-	-	-	7.9%
Klingensmith 2006	Men	49.9	-	-	-	-	-	-	-
	Women	43.5	-	-	-	-	-	-	-

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Komlenac 2019	Men	-	-	-	54.0	-	-	64.0%	-
	Women	-	-	-	43.0	-	-	39.0%	-
Long 2010	Men	53.0	-	-	-	-	-	-	-
	Women	51.0	-	-	-	-	-	-	-
Madsen 2017	Men	-	77.8%	12.0	-	-	-	-	-
	Women	-	77.8%	8.4	-	-	-	-	-
Maxwell 2020	Men	-	-	26.3	-	-	-	-	-
	Women	-	-	18.1	-	-	-	-	-
Mayer 2017	Men	-	-	22.0	-	-	-	-	2.8%
	Women	-	-	13.0	-	-	-	-	0.3%
McAlister 2014	Men	-	-	-	-	-	-	90.0%	-
	Women	-	-	-	-	-	-	81.0%	-
Modena 1999	Men	-	-	-	39.0	-	87.0%	-	-
	Women	-	-	-	37.7	-	66.0%	-	-
Monroe 2015	Men	49.0	-	-	-	-	-	-	-
	Women	43.0	-	-	-	-	-	-	-
Mueller 2017	Men	-	-	14.3	-	-	-	-	-
	Women	-	-	12.6	-	-	-	-	-
Muffly 2016	Men	50.9	-	-	-	-	-	-	-
	Women	47.5	-	-	-	-	-	-	-

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Ngaage 2019	Men	-	-	10.0	-	-	-	-	-
	Women	-	-	4.0	-	-	-	-	-
Pagel 2015	Men	-	-	20.0	-	-	-	-	-
	Women	-	-	16.0	-	-	-	-	-
Paulus 2016	Men	51.0	-	19.0	-	-	-	-	-
	Women	45.0	-	14.0	-	-	-	-	-
Reed 2011	Men	-	-	26.6	-	-	-	-	-
	Women	-	-	26.6	-	-	-	-	-
Riaz 2020	Men	-	-	20.0	-	-	-	-	-
	Women	-	-	14.0	-	-	-	-	-
Robboy 2017	Men	-	71.4%	-	-	-	-	-	-
	Women	-	87.5%	-	-	-	-	-	-
Sadeghpour 2012	Men	49.9	-	-	52.9	-	-	-	-
	Women	42.6	-	-	50.2	-	-	-	-
Santosa 2019	Men	-	-	-	-	-	-	-	11.6%
	Women	-	-	-	-	-	-	-	29.3%
Sasor 2018	Men	-	-	16.4	-	-	-	-	4.3%
	Women	-	-	9.9	-	-	-	-	4.4%
Schnapp 2003	Men	47.0	-	-	-	-	-	-	-
	Women	42.0	-	-	-	-	-	-	-

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Schroen 2004	Men	48.1	-	15.5	-	-	96.0%	90.0%	-
	Women	42.7	-	9.3	-	-	77.0%	62.0%	-
Schwarz 2016	Men	47.1	-	-	65.5	9.1%	93.0%	86.2%	-
	Women	47.5	-	-	59.7	15.1%	74.6%	77.4%	-
Shaikh 2019	Men	-	-	-	-	-	-	-	20.4%
	Women	-	-	-	-	-	-	-	17.6%
Shollen 2009	Men	-	-	-	57.0	42.0%	-	-	-
	Women	-	-	-	57.0	42.0%	-	-	-
Singh 2008	Men	45.0	-	-	58.0	-	95.0%	97.0%	-
	Women	46.0	-	-	53.0	-	92.0%	87.0%	-
Smith 2019	Men	51.6	75.7%	14.0	-	-	-	-	4.3%
	Women	44.6	73.4%	9.0	-	-	-	-	3.8%
Speck 2012	Men	37.3	71.9%	-	-	-	54.4%	-	15.7%
	Women	36.9	66.0%	-	-	-	44.5%	-	7.2%
Spencer 2016	Men	50.0	-	19.0	60.0	-	-	-	-
	Women	42.0	-	10.0	55.0	-	-	-	-
Stadler 2017	Men	-	-	9.3	-	-	-	-	-
	Women	-	-	9.2	-	-	-	-	-
Stewart 2020	Men	-	70.2%	22.4	-	-	-	-	27.0%
	Women	-	61.7%	16.1	-	-	-	-	22.8%

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Author		Age (years)	Race (white)	Years in practice	Hours worked per week	% time conducting research	Married	Have children	Additional PhD
Tracy 2019	Men	-	-	17.2	-	-	-	-	-
	Women	-	-	10.8	-	-	-	-	-
Tsukada 2009	Men	-	-	18.1	-	-	85.4%	69.2%	-
	Women	-	-	10.8	-	-	66.7%	50.0%	-
Tulunay-Ugur 2019	Men	-	72.0%	-	-	-	96.8%	-	-
	Women	-	64.2%	-	-	-	75.0%	-	-
Valsangkar 2016	Men	-	-	-	-	-	-	-	4.9%
	Women	-	-	-	-	-	-	-	11.3%
Wagner 2007	Men	47.3	81.3%	15.2	59.8	26.1%	-	-	9.8%
	Women	45.1	79.0%	12.4	54.3	22.2%	-	-	0.9%
Warner 2017	Men	-	71.7%	-	-	-	-	-	9.3%
	Women	-	71.8%	-	-	-	-	-	4.5%
Weaver 2015	Men	45.0	-	6.0	-	-	86.0%	46.0%	-
	Women	42.0	-	5.0	-	-	80.0%	45.0%	-
Wise 2004	Men	52.8	-	-	-	-	96.0%	-	18.0%
	Women	43.4	-	-	-	-	74.0%	-	28.0%
Yasukawa 2014	Men	-	-	-	-	-	90.0%	83.0%	53.0%
	Women	-	-	-	-	-	72.0%	78.0%	40.0%
Median across studies	Men	47.4	74.0%	18.0	58.0	27.1%	91.0%	84.0%	14.8%
	Women	43.5	75.0%	12.0	54.7	25.2%	75.0%	71.0%	14.6%

A dashed line (-) represents values that were not reported, could not be calculated from reported data, and authors were not able to provide.

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eTable 4. Quality assessment of included studies

Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
AAMC 1992	1	1	1	2	0	2	0	7
Abdellatif 2019	0	1	1	2	1	2	1	8
Ahmadi 2018	1	1	1	2	2	2	1	10
Akabas 2019	0	1	0	2	0	1	1	5
Aliano 2020	1	1	0	2	1	1	1	7
Al-Tamimi 2004	1	1	0	2	0	1	1	6
Amering 2011	1	1	1	2	0	2	1	8
Ash 2004	1	1	0	2	2	1	1	8
Awad 2017	1	1	1	2	2	2	1	10
Baptiste 2017	1	1	0	2	0	1	1	6
Battaglia 2019	1	1	1	2	0	2	1	8
Beeler 2020	0	1	0	2	0	1	1	5
Bernardi 2020	0	0	0	2	0	1	0	3
Bickel 1988	1	1	1	2	0	2	1	8
Bickel 1991	1	1	1	2	0	2	0	7
Bickel 1993	1	1	1	2	0	2	0	7
Bissing 2019	1	1	1	2	0	2	1	8
Breyer 2020	1	1	1	2	0	1	1	7

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Bruggmann 2017	1	1	1	2	0	2	0	7
Buckley 2000	1	1	0	2	0	1	1	6
Burns 2019	0	1	0	2	0	1	1	5
Camacci 2020	0	1	1	2	0	2	1	7
Carnevale 2020	1	1	1	2	0	2	1	8
Carpenter 2018	0	1	1	2	0	2	1	7
Carr 1992	1	1	0	2	2	1	1	8
Carr 1998	1	1	1	2	2	1	1	9
Caturegli 2020	1	1	1	2	2	2	1	10
Chaiyachati 2019	1	1	0	2	2	1	1	8
Chandrasekar 2020	1	1	1	2	2	2	1	10
Chang 2016	1	1	1	2	0	2	1	8
Chaudhary 2020	1	1	1	2	0	2	1	8
Chauvin 2019	1	1	1	2	1	2	1	9
Cheng 2019	1	1	0	2	0	1	1	6
Cheng 2019	1	1	1	2	2	2	1	10
Chevreul 2018	1	1	1	2	2	2	1	10
Chokshi 2009	0	1	0	2	2	1	1	7
Colletti 2000	1	1	0	2	0	1	1	6

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Cuoco 2020	0	1	1	2	0	2	1	7
Cydulka 2000	1	1	0	2	2	1	1	8
Danic 2003	1	1	1	2	0	2	1	8
Day 2010	1	1	1	2	0	2	1	8
Deipolyi 2020	1	1	1	2	2	2	1	10
Dennis 1990	0	1	0	2	0	1	1	5
Dermody 2019	1	1	1	2	2	2	1	10
Desai 2018	1	1	1	2	1	2	1	9
Diehl 2008	0	1	0	2	2	1	1	7
Dorismond 2020	0	1	1	2	0	2	1	7
Dossani 2019	1	1	1	2	2	2	1	10
Dubosh 2019	1	1	1	2	0	2	1	8
Duke 2019	1	1	1	2	0	2	1	8
Eloy 2013	1	1	1	2	1	2	1	9
Eloy 2013	0	1	1	2	2	2	1	9
Epperson 2020	1	1	1	2	2	2	1	10
Fassiotto 2018	1	1	1	2	2	2	1	10
Florence 2011	0	1	0	2	0	1	1	5
Fritz 1991	1	1	0	2	0	1	1	6
Gambhir 2020	1	1	1	2	2	2	1	10

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Garstka 2019	1	1	1	2	1	2	1	9
Gauthier 2013	1	1	0	2	0	1	1	6
Gawad 2020	1	1	1	2	2	2	1	10
Geltzeiler 2017	1	1	1	2	1	2	1	9
Gerety 2013	0	1	0	2	0	1	1	5
Gerson 2007	0	1	0	2	0	1	1	5
Girod 2011	0	1	0	2	2	1	1	7
Gjerde 1994	0	1	0	2	0	1	1	5
Grimm 2016	1	1	1	2	0	2	0	7
Guss 2019	1	1	1	2	2	2	1	10
Hafeez 2019	0	1	1	2	0	2	1	7
Harrington 2018	0	1	1	2	0	2	1	7
Harris 2019	0	1	1	2	2	2	1	9
Herrick-Reynolds 2019	1	1	1	2	0	2	1	8
Herzke 2020	0	1	0	2	0	1	1	5
Higgins 1979	1	1	1	2	1	2	0	8
Higgins 2020	0	1	0	2	0	1	1	5
Hill 2018	0	1	0	2	0	1	1	5
Holliday 2014	1	1	1	2	2	2	1	10
Holliday 2015	0	1	0	2	0	1	1	5

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Hoof 2020	1	1	1	2	0	2	1	8
Hoops 2018	1	1	1	2	0	2	1	8
Hoover 2005	1	1	1	2	0	2	1	8
Houtrow 2011	1	1	0	2	0	1	1	6
Houtrow 2020	1	1	1	2	2	1	1	9
Hurley 2014	1	1	1	2	1	2	1	9
Ibn 2019	1	1	0	2	0	1	1	6
Ilin 2020	1	1	1	2	0	2	1	8
Inam 2020	1	1	0	2	0	1	1	6
Ingram 2015	1	1	1	2	0	2	1	8
Jacquet 2014	0	1	0	2	0	1	1	5
Jagsi 2012	0	1	1	2	2	1	1	8
Jagsi 2013	0	1	1	2	2	2	1	9
Jaque 2020	1	1	1	2	0	2	1	8
Jena 2015	1	1	1	2	2	2	1	10
Jena 2016	1	1	1	2	2	2	1	10
John 2016	1	1	1	2	1	2	1	9
Johnson 1992	1	1	1	2	0	2	0	7
Jolly 2014	0	1	0	2	1	1	1	6
Kaderli 2015	1	1	0	2	0	1	1	6

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Karimian 2012	1	1	0	2	0	1	1	6
Kenevan 2018	0	1	1	2	0	2	1	7
Khan 2014	1	1	1	2	1	2	1	9
Khan 2019	1	1	1	2	1	2	1	9
Khurshid 2018	1	1	1	2	0	2	1	8
Kilic 2020	1	1	1	2	0	2	1	8
Killaspy 2003	1	1	0	2	0	1	1	6
Kim 2019	1	1	1	2	2	2	1	10
Kirchner 1984	0	1	0	2	0	1	0	4
Klingensmith 2006	1	1	0	2	0	1	1	6
Kloosterboer 2020	0	1	1	2	0	2	1	7
Komlenac 2019	0	1	0	2	0	1	1	5
Larese 2019	1	1	1	2	0	2	1	8
LaRocca 2018	1	1	1	2	2	2	1	10
Levey 1990	0	1	0	2	0	1	1	5
Lewis-Stevenson 2001	1	1	0	2	0	1	1	6
Long 2010	1	1	1	2	0	2	1	8
Madsen 2017	1	1	0	2	2	1	1	8
Mascarenhas 2017	1	1	1	2	0	2	0	7

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Maxwell 2020	1	1	1	2	2	2	1	10
Mayer 2017	1	1	1	2	0	2	1	8
McAlister 2014	0	1	0	2	0	1	1	5
McDermott 2018	1	1	1	2	1	2	1	9
McDonald 2017	1	1	1	2	0	2	1	8
Mendoza 2015	0	1	0	2	0	1	1	5
Modena 1999	0	1	0	2	0	1	1	5
Moghimi 2019	1	1	1	2	0	2	1	8
Monroe 2015	1	1	0	2	0	1	1	6
Morgan 2019	0	1	0	2	0	1	1	5
Morris 2018	1	1	1	2	0	2	1	8
Mueller 2017	1	1	1	2	1	2	1	9
Muffly 2016	1	1	1	2	0	2	1	8
Nakata 2017	1	1	1	2	2	2	1	10
Ngaage 2019	1	1	1	2	2	2	1	10
Nguyen 2018	1	1	1	2	2	2	1	10
Nickerson 1990	1	1	1	2	0	2	1	8
Niu 2020	1	1	1	2	0	2	1	8
Nowling 2018	1	1	1	2	0	2	1	8

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Odell 2019	1	1	1	2	0	2	1	8
Okafor 2020	1	1	1	2	2	2	1	10
Okoshi 2014	1	1	1	2	0	2	1	8
O'Neill 2019	1	1	1	2	0	2	1	8
Orhurhu 2020	1	1	1	2	2	2	1	10
Pace 2020	1	1	1	2	2	2	1	10
Pagel 2015	0	1	1	2	1	2	1	8
Paik 2014	1	1	1	2	1	2	1	9
Paulus 2016	1	1	0	2	0	1	1	6
Penfold 1987	0	1	0	2	0	1	0	4
Pfeiffer 2016	1	1	1	2	0	2	1	8
Pololi 2013	1	1	0	2	0	1	1	6
Qamar 2018	1	1	1	2	1	2	1	9
Qamar 2020	1	1	1	2	2	2	1	10
Rao 2018	1	1	1	2	0	2	1	8
Raptis 2012	1	1	0	2	0	1	1	6
Rayburn 2012	1	1	1	2	1	2	1	9
Reed 2011	1	1	1	2	2	2	1	10
Reiser 1993	0	1	0	2	0	1	1	5
Riaz 2020	1	1	1	2	2	2	1	10

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Robboy 2017	1	1	1	2	0	2	1	8
Rose 2015	1	1	1	2	2	2	1	10
Rosso 2019	1	1	1	2	0	2	1	8
Rotbart 2012	1	1	1	2	1	2	1	9
Sadeghpour 2012	0	1	0	2	2	1	1	7
Sandhu 2007	1	1	1	2	0	1	1	7
Santosa 2019	0	1	1	2	2	2	1	9
Sasor 2018	1	1	1	2	0	2	1	8
Schnapp 2003	1	1	0	2	0	1	1	6
Schroen 2004	0	1	0	2	0	1	1	5
Schwarz 2016	1	1	0	2	0	2	1	7
Selter 2020	1	1	1	2	0	2	1	8
Shah 2018	1	1	1	2	0	2	1	8
Shaikh 2019	1	1	1	2	0	2	1	8
Sheikh 2018	1	1	1	2	0	2	1	8
Shiber 2020	1	1	0	2	2	1	1	8
Shollen 2009	1	1	0	2	0	1	1	6
Singh 2008	0	1	0	2	2	1	1	7
Smith 2019	1	1	1	2	1	2	1	9
Soothill 2010	0	1	0	2	0	1	1	5

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Speck 2012	1	1	1	2	0	2	1	8
Spencer 2016	0	1	0	2	2	1	1	7
Sperling 2019	1	1	1	2	2	2	1	10
Stadler 2017	1	1	0	2	0	1	1	6
Steger 2009	0	1	0	2	0	1	1	5
Stewart 2020	0	1	1	2	2	2	1	9
Svider 2014	0	1	1	2	2	2	1	9
Tekerek 2016	0	0	0	2	0	1	1	4
Therattil 2016	1	1	1	2	2	2	1	10
Tracy 2019	1	1	1	2	0	2	1	8
Tsukada 2009	0	1	0	2	0	1	1	5
Tulunay-Ugur 2019	0	0	0	2	0	1	1	4
Turaga 2013	0	1	1	2	1	2	1	8
Valsangkar 2016	1	1	1	2	1	2	1	9
Villablanca 2017	1	1	0	2	0	1	1	6
Vydareny 2000	0	1	0	2	0	1	1	5
Wagner 2007	0	1	0	2	2	1	1	7
Waisbren 2008	1	1	1	2	2	2	1	10
Waljee 2015	1	1	0	2	0	1	1	6

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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Wallis 1981	0	1	0	2	0	1	0	4
Warner 2017	1	1	1	2	2	2	1	10
Waseem 2019	1	1	1	2	0	1	1	7
Wasserstein 2007	1	1	0	2	0	1	1	6
Weaver 2015	0	1	0	2	2	1	1	7
Whitley 1987	0	1	0	2	0	1	1	5
Wiler 2019	1	1	0	2	2	1	1	8
Wilkes 2015	1	1	1	2	0	2	1	8
Wilkinson 1986	1	1	1	2	0	2	1	8
Willett 2015	1	1	0	2	2	1	1	8
Wilson 2013	0	1	1	2	0	2	1	7
Winch 2017	0	1	0	2	0	1	1	5
Wise 2004	1	1	0	2	0	1	1	6
Woods 2018	1	1	1	2	0	2	1	8
Woodward 2017	1	1	1	2	0	2	1	8
Wright 2007	1	1	1	2	2	2	1	10
Wu 2019	1	1	1	2	0	1	1	7
Xierali 2020	1	1	1	2	0	2	1	8
Yang 2019	1	1	1	2	0	2	1	8

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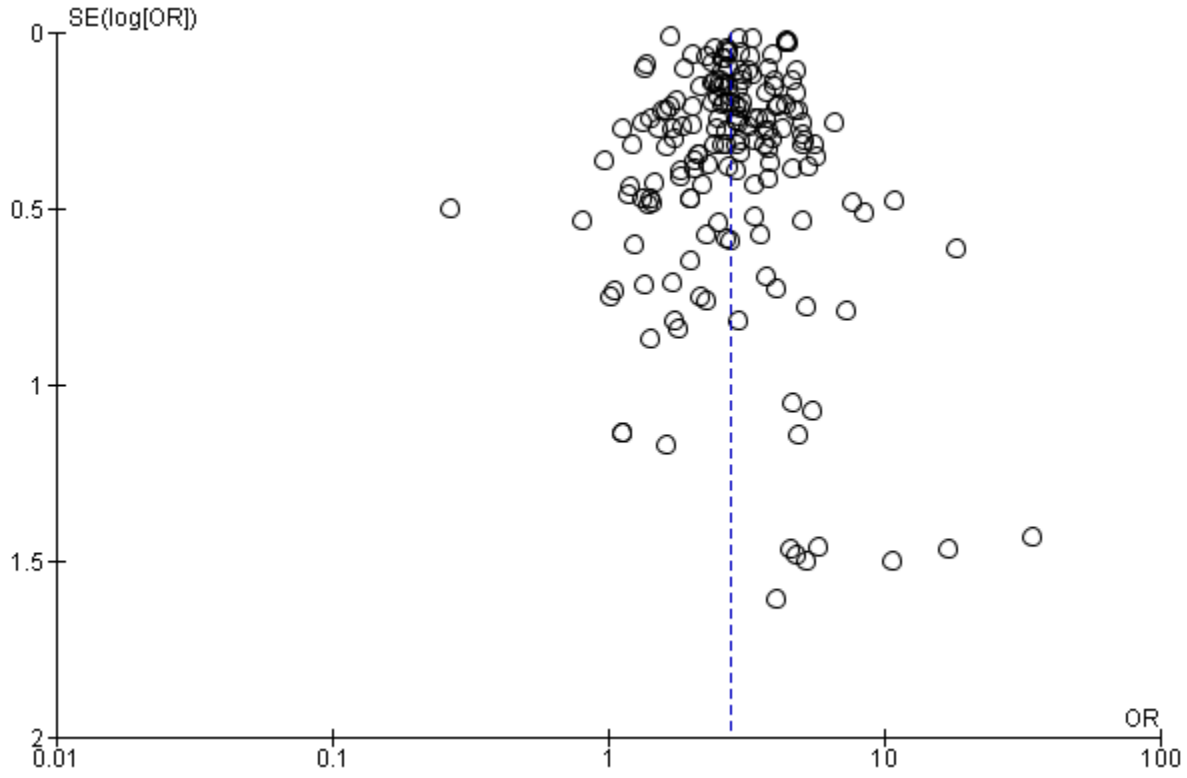
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Author	Representativeness of cohort	Adequacy of sample size	Characterization of non-respondents	Ascertainment of exposure	Comparability of groups	Outcome assessment	Statistical tests	NOS Score
Yasukawa 2014	1	1	0	2	2	1	1	8
Yu 2013	1	1	1	2	1	2	1	9
Yue 2020	1	1	1	2	0	2	1	8
Zelnik 2018	0	1	0	2	0	1	1	5
Zivin 2020	1	1	0	2	2	1	1	8

*Quality assessment was performed using Newcastle-Ottawa Scale (NOS) for cross-sectional studies. A maximum of one point is awarded for representativeness of cohort, adequacy of sample size, characterization of non-respondents, and statistical tests. A maximum of two points is awarded for characterization of non-respondents, ascertainment of exposure, and comparability of groups. An overall maximum score of 10 can be awarded per study. NOS score ≥ 8 is low risk, 6 – 7 is moderate risk, and ≤ 5 is high risk.

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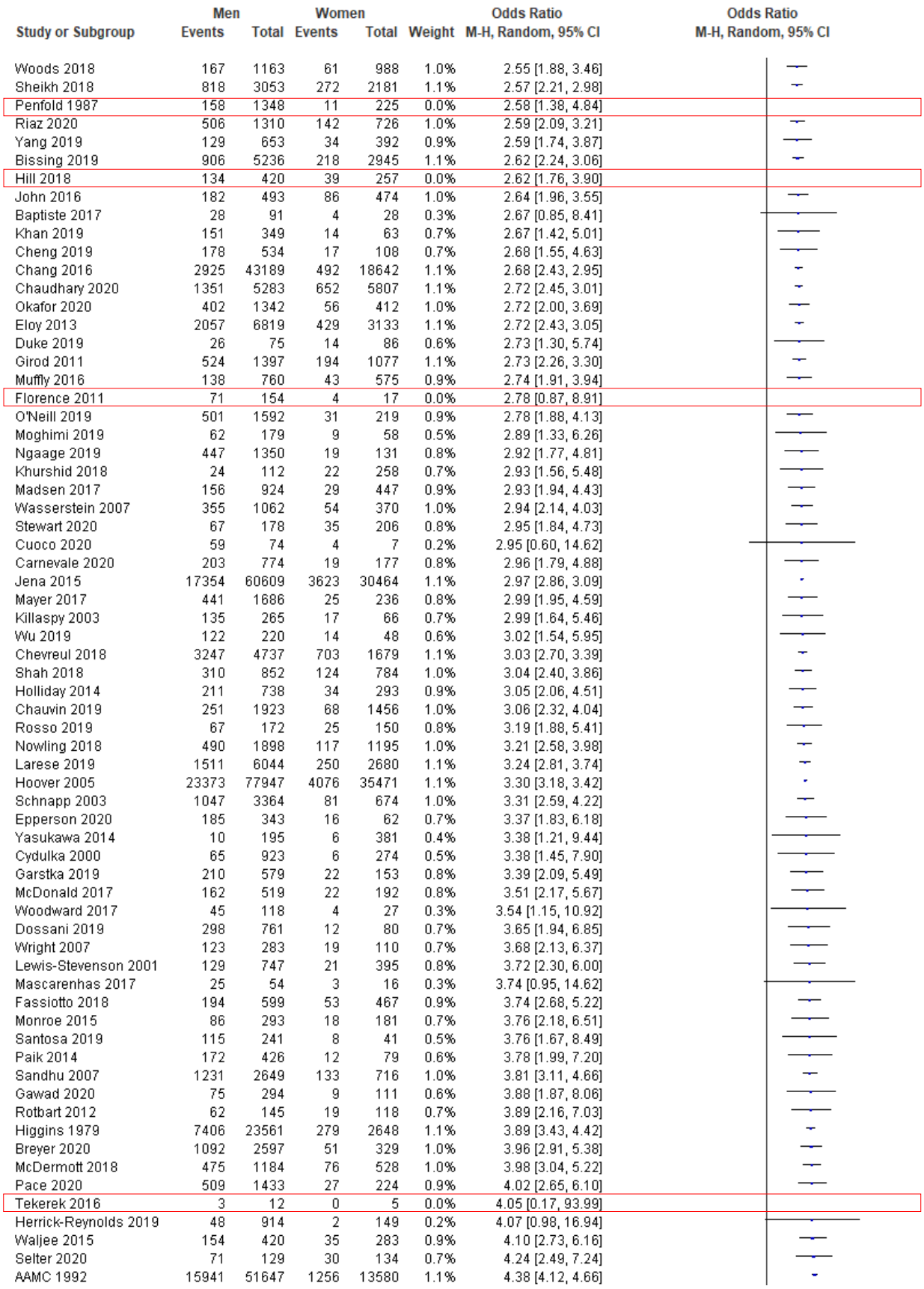
eFigure 1. Funnel plot of full professorship for men vs. women. SE (standard error), OR (odds ratio).

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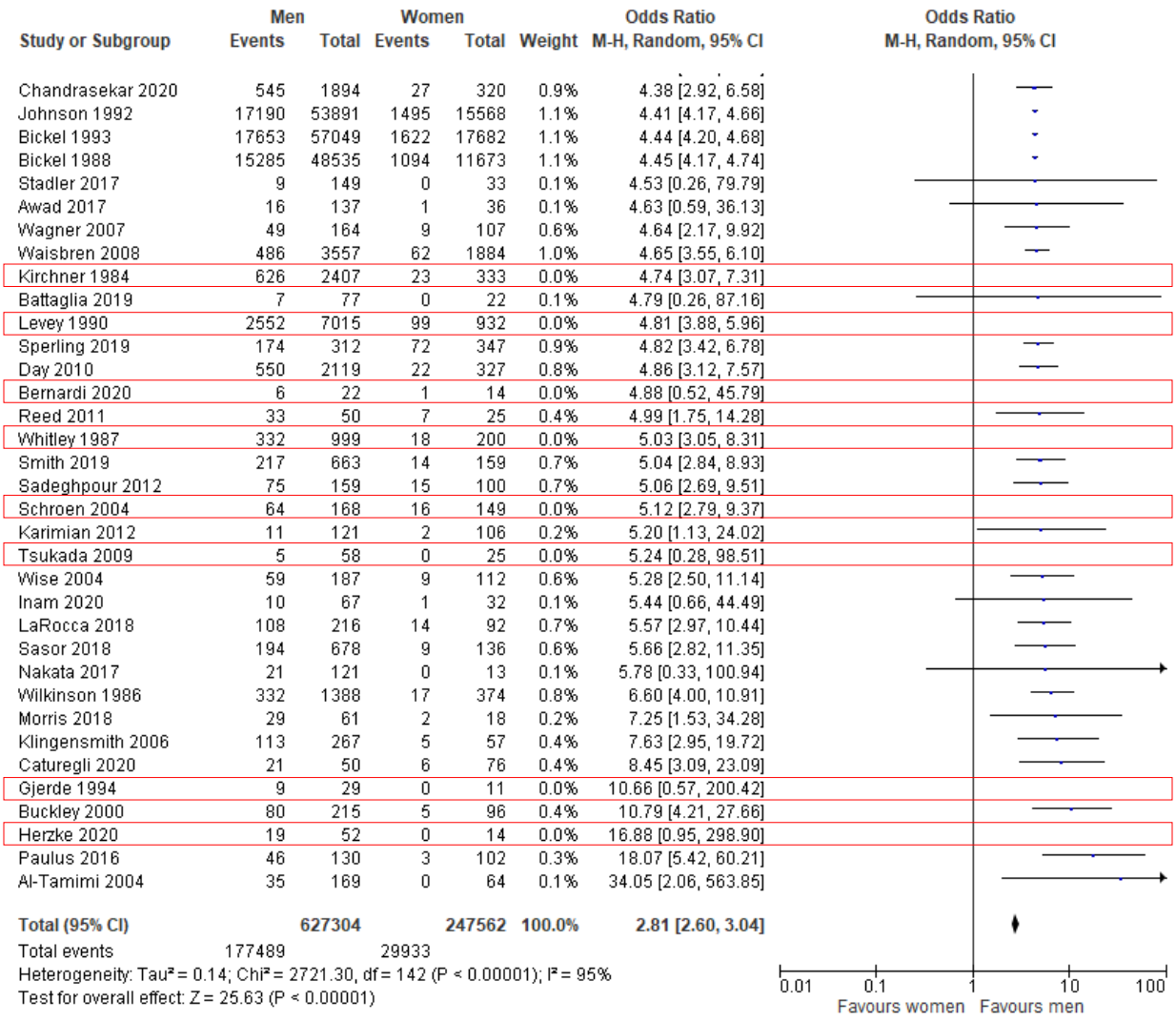
Study or Subgroup	Men Events	Men Total	Women Events	Women Total	Weight	Odds Ratio M-H, Random, 95% CI
Schwarz 2016	7	200	11	93	0.4%	0.27 [0.10, 0.72]
Wilson 2013	19	66	7	21	0.4%	0.81 [0.28, 2.32]
Shaikh 2019	162	294	19	34	0.6%	0.97 [0.47, 1.98]
Hoops 2018	8	42	3	16	0.2%	1.02 [0.23, 4.45]
Gerety 2013	36	104	3	9	0.0%	1.06 [0.25, 4.49]
Zelnik Yovel 2018	8	58	1	8	0.0%	1.12 [0.12, 10.36]
Zivin 2020	8	58	1	8	0.1%	1.12 [0.12, 10.36]
Ahmadi 2018	77	358	21	107	0.7%	1.12 [0.65, 1.92]
Kenevan 2018	20	101	8	46	0.5%	1.17 [0.47, 2.90]
Geltzeiler 2017	30	281	7	77	0.5%	1.20 [0.50, 2.84]
Beeler 2020	73	135	27	55	0.0%	1.22 [0.65, 2.29]
Dorismond 2020	12	47	5	23	0.3%	1.23 [0.38, 4.05]
Winch 2017	32	60	12	26	0.0%	1.33 [0.53, 3.36]
Higgins 2020	108	287	28	90	0.0%	1.34 [0.81, 2.22]
Aliano 2020	272	654	334	965	1.0%	1.35 [1.10, 1.65]
Morgan 2019	3	25	8	87	0.0%	1.35 [0.33, 5.51]
Pololi 2013	331	1046	295	1172	1.1%	1.38 [1.14, 1.66]
Jacquet 2014	21	81	7	35	0.0%	1.40 [0.53, 3.68]
Jagsi 2012	73	553	24	247	0.8%	1.41 [0.87, 2.30]
McAlister 2014	23	283	6	102	0.0%	1.42 [0.56, 3.58]
Jagsi 2013	4	593	2	419	0.2%	1.42 [0.26, 7.77]
Burns 2019	22	192	6	73	0.0%	1.45 [0.56, 3.72]
Long 2010	33	94	10	37	0.5%	1.46 [0.63, 3.38]
Carr 1998	35	1010	22	953	0.7%	1.52 [0.88, 2.61]
Harris 2019	275	426	55	102	0.8%	1.56 [1.01, 2.41]
Odell 2019	46	281	14	129	0.6%	1.61 [0.85, 3.04]
Ibn Auf 2019	3	100	1	53	0.1%	1.61 [0.16, 15.85]
Nickerson 1990	106	663	29	278	0.8%	1.63 [1.06, 2.53]
Yu 2013	22245	82491	7737	42804	1.1%	1.67 [1.63, 1.72]
Tracy 2019	181	1086	29	274	0.9%	1.69 [1.11, 2.56]
Orhurhu 2020	84	520	18	176	0.7%	1.69 [0.98, 2.90]
Holliday 2015	6	685	3	582	0.0%	1.71 [0.42, 6.85]
Guss 2019	60	166	20	81	0.7%	1.73 [0.95, 3.13]
Dubosh 2019	8	77	2	32	0.2%	1.74 [0.35, 8.68]
Danic 2003	165	333	60	168	0.9%	1.77 [1.21, 2.59]
Jolly 2014	5	612	2	437	0.2%	1.79 [0.35, 9.28]
Harrington 2018	33	144	10	71	0.5%	1.81 [0.84, 3.93]
Deipolyi 2020	82	734	7	108	0.5%	1.81 [0.82, 4.04]
Shollen 2009	80	244	23	110	0.7%	1.85 [1.08, 3.14]
Ash 2004	322	941	190	873	1.0%	1.87 [1.52, 2.31]
Carpenter 2018	45	117	7	29	0.4%	1.96 [0.78, 4.97]
Houtrow 2020	5	68	5	130	0.3%	1.98 [0.55, 7.11]
Dermody 2019	35	161	6	49	0.4%	1.99 [0.78, 5.06]
Gerson 2007	39	195	29	262	0.0%	2.01 [1.19, 3.38]
Niu 2020	1497	5795	332	2256	1.1%	2.02 [1.77, 2.30]
Morrison 2014	116	398	36	213	0.9%	2.02 [1.33, 3.07]
Yue 2020	98	618	8	95	0.6%	2.05 [0.96, 4.36]
Qamar 2018	45	190	11	84	0.6%	2.06 [1.01, 4.22]
Gambhir 2020	65	120	18	50	0.6%	2.10 [1.06, 4.15]
Willett 2015	54	169	13	72	0.6%	2.13 [1.08, 4.22]
Okoshi 2014	33	326	2	40	0.2%	2.14 [0.49, 9.28]
Hoof 2020	852	3889	50	434	1.0%	2.15 [1.59, 2.92]
Soothill 2010	42	102	9	37	0.0%	2.18 [0.93, 5.09]
Kilic 2020	1005	2436	415	1755	1.1%	2.27 [1.98, 2.60]
Ilin 2020	55	164	4	22	0.3%	2.27 [0.73, 7.04]
Fritz 1991	4	22	4	45	0.2%	2.28 [0.51, 10.13]
Wallis 1981	142	949	8	112	0.0%	2.29 [1.09, 4.80]
Valsangkar 2016	1272	3087	211	928	1.1%	2.38 [2.01, 2.82]
Villablanca 2017	313	552	114	322	1.0%	2.39 [1.80, 3.17]
Vydareny 2000	176	494	40	213	0.0%	2.39 [1.62, 3.54]
Tulunay-Udur 2019	72	152	18	66	0.0%	2.40 [1.28, 4.50]
Rao 2018	316	1029	70	452	1.0%	2.42 [1.81, 3.22]
Jena 2016	2543	6692	717	3549	1.1%	2.42 [2.20, 2.66]
Waseem 2019	175	221	54	89	0.7%	2.47 [1.44, 4.21]
Ingram 2015	17	50	6	35	0.4%	2.49 [0.87, 7.16]
Qamar 2020	89	597	22	335	0.8%	2.49 [1.53, 4.06]
Maxwell 2020	275	1543	36	450	0.9%	2.49 [1.73, 3.59]
Bickel 1991	642	2918	67	669	1.0%	2.53 [1.94, 3.31]

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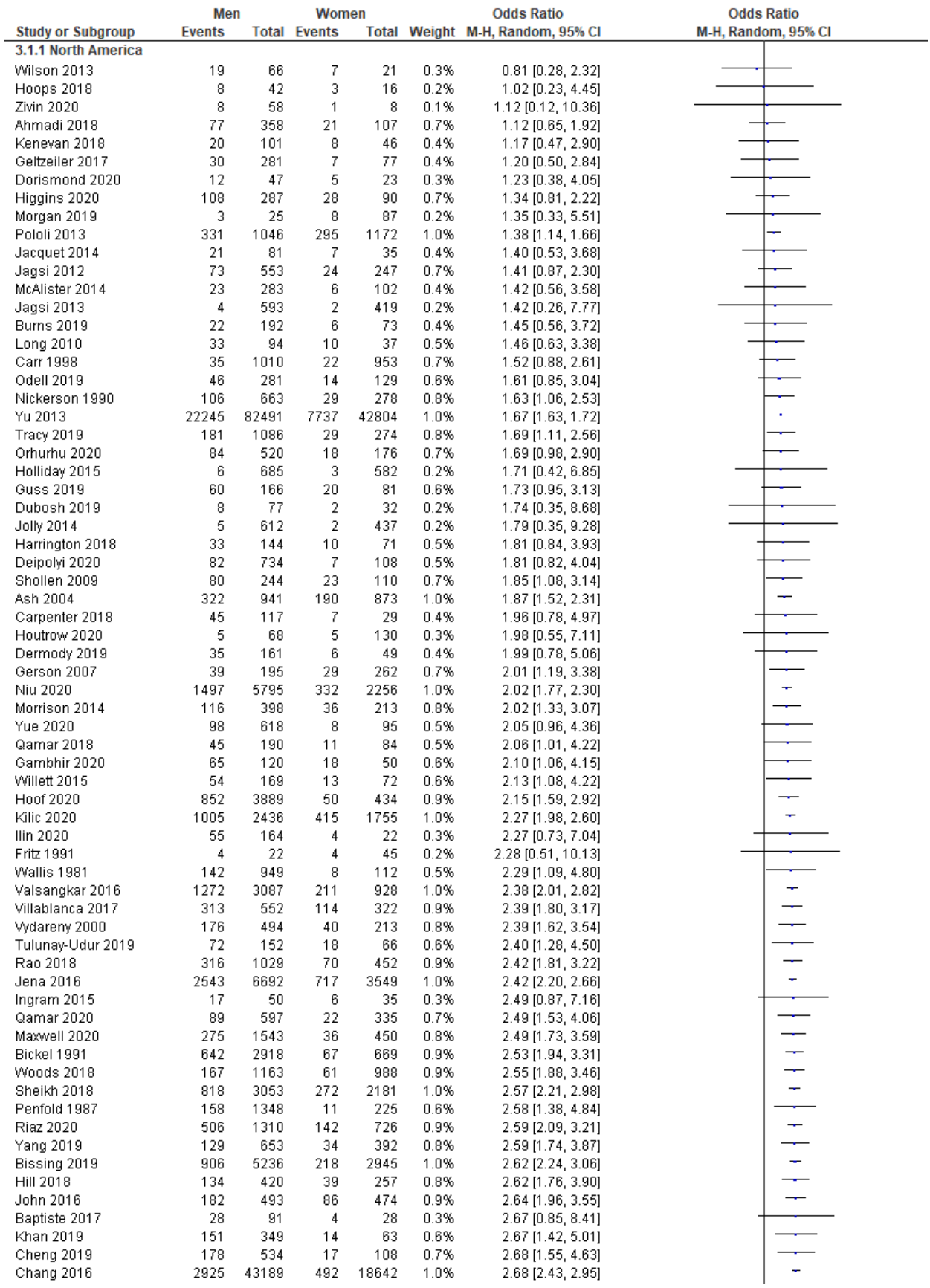
Study or Subgroup	Men Events	Men Total	Women Events	Women Total	Weight	Odds Ratio M-H, Random, 95% CI
Woods 2018	167	1163	61	988	1.0%	2.55 [1.88, 3.46]
Sheikh 2018	818	3053	272	2181	1.1%	2.57 [2.21, 2.98]
Penfold 1987	158	1348	11	225	0.0%	2.58 [1.38, 4.84]
Riaz 2020	506	1310	142	726	1.0%	2.59 [2.09, 3.21]
Yang 2019	129	653	34	392	0.9%	2.59 [1.74, 3.87]
Bissing 2019	906	5236	218	2945	1.1%	2.62 [2.24, 3.06]
Hill 2018	134	420	39	257	0.0%	2.62 [1.76, 3.90]
John 2016	182	493	86	474	1.0%	2.64 [1.96, 3.55]
Baptiste 2017	28	91	4	28	0.3%	2.67 [0.85, 8.41]
Khan 2019	151	349	14	63	0.7%	2.67 [1.42, 5.01]
Cheng 2019	178	534	17	108	0.7%	2.68 [1.55, 4.63]
Chang 2016	2925	43189	492	18642	1.1%	2.68 [2.43, 2.95]
Chaudhary 2020	1351	5283	652	5807	1.1%	2.72 [2.45, 3.01]
Okafor 2020	402	1342	56	412	1.0%	2.72 [2.00, 3.69]
Eloy 2013	2057	6819	429	3133	1.1%	2.72 [2.43, 3.05]
Duke 2019	26	75	14	86	0.6%	2.73 [1.30, 5.74]
Girod 2011	524	1397	194	1077	1.1%	2.73 [2.26, 3.30]
Muffly 2016	138	760	43	575	0.9%	2.74 [1.91, 3.94]
Florence 2011	71	154	4	17	0.0%	2.78 [0.87, 8.91]
O'Neill 2019	501	1592	31	219	0.9%	2.78 [1.88, 4.13]
Moghimi 2019	62	179	9	58	0.5%	2.89 [1.33, 6.26]
Ngaage 2019	447	1350	19	131	0.8%	2.92 [1.77, 4.81]
Khurshid 2018	24	112	22	258	0.7%	2.93 [1.56, 5.48]
Madsen 2017	156	924	29	447	0.9%	2.93 [1.94, 4.43]
Wasserstein 2007	355	1062	54	370	1.0%	2.94 [2.14, 4.03]
Stewart 2020	67	178	35	206	0.8%	2.95 [1.84, 4.73]
Cuoco 2020	59	74	4	7	0.2%	2.95 [0.60, 14.62]
Carnevale 2020	203	774	19	177	0.8%	2.96 [1.79, 4.88]
Jena 2015	17354	60609	3623	30464	1.1%	2.97 [2.86, 3.09]
Mayer 2017	441	1686	25	236	0.8%	2.99 [1.95, 4.59]
Killaspy 2003	135	265	17	66	0.7%	2.99 [1.64, 5.46]
Wu 2019	122	220	14	48	0.6%	3.02 [1.54, 5.95]
Chevreul 2018	3247	4737	703	1679	1.1%	3.03 [2.70, 3.39]
Shah 2018	310	852	124	784	1.0%	3.04 [2.40, 3.86]
Holliday 2014	211	738	34	293	0.9%	3.05 [2.06, 4.51]
Chauvin 2019	251	1923	68	1456	1.0%	3.06 [2.32, 4.04]
Rosso 2019	67	172	25	150	0.8%	3.19 [1.88, 5.41]
Nowling 2018	490	1898	117	1195	1.0%	3.21 [2.58, 3.98]
Larese 2019	1511	6044	250	2680	1.1%	3.24 [2.81, 3.74]
Hoover 2005	23373	77947	4076	35471	1.1%	3.30 [3.18, 3.42]
Schnapp 2003	1047	3364	81	674	1.0%	3.31 [2.59, 4.22]
Epperson 2020	185	343	16	62	0.7%	3.37 [1.83, 6.18]
Yasukawa 2014	10	195	6	381	0.4%	3.38 [1.21, 9.44]
Cydulka 2000	65	923	6	274	0.5%	3.38 [1.45, 7.90]
Garstka 2019	210	579	22	153	0.8%	3.39 [2.09, 5.49]
McDonald 2017	162	519	22	192	0.8%	3.51 [2.17, 5.67]
Woodward 2017	45	118	4	27	0.3%	3.54 [1.15, 10.92]
Dossani 2019	298	761	12	80	0.7%	3.65 [1.94, 6.85]
Wright 2007	123	283	19	110	0.7%	3.68 [2.13, 6.37]
Lewis-Stevenson 2001	129	747	21	395	0.8%	3.72 [2.30, 6.00]
Mascarenhas 2017	25	54	3	16	0.3%	3.74 [0.95, 14.62]
Fassiotto 2018	194	599	53	467	0.9%	3.74 [2.68, 5.22]
Monroe 2015	86	293	18	181	0.7%	3.76 [2.18, 6.51]
Santosa 2019	115	241	8	41	0.5%	3.76 [1.67, 8.49]
Paik 2014	172	426	12	79	0.6%	3.78 [1.99, 7.20]
Sandhu 2007	1231	2649	133	716	1.0%	3.81 [3.11, 4.66]
Gawad 2020	75	294	9	111	0.6%	3.88 [1.87, 8.06]
Rotbart 2012	62	145	19	118	0.7%	3.89 [2.16, 7.03]
Higgins 1979	7406	23561	279	2648	1.1%	3.89 [3.43, 4.42]
Breyer 2020	1092	2597	51	329	1.0%	3.96 [2.91, 5.38]
McDermott 2018	475	1184	76	528	1.0%	3.98 [3.04, 5.22]
Pace 2020	509	1433	27	224	0.9%	4.02 [2.65, 6.10]
Tekerek 2016	3	12	0	5	0.0%	4.05 [0.17, 93.99]
Herrick-Reynolds 2019	48	914	2	149	0.2%	4.07 [0.98, 16.94]
Waljee 2015	154	420	35	283	0.9%	4.10 [2.73, 6.16]
Selter 2020	71	129	30	134	0.7%	4.24 [2.49, 7.24]
AAMC 1992	15941	51647	1256	13580	1.1%	4.38 [4.12, 4.66]

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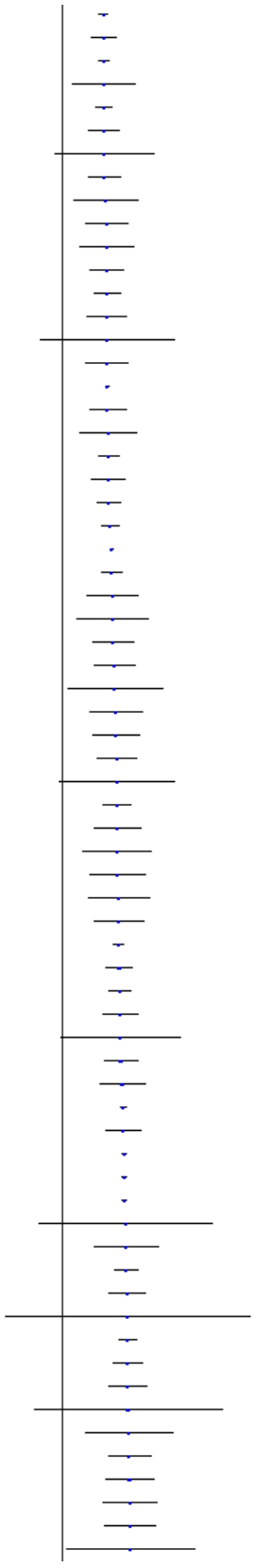
eFigure 2. Sensitivity analysis of full professorship removing 27 poor quality studies (red box, NOS score ≤ 5). M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

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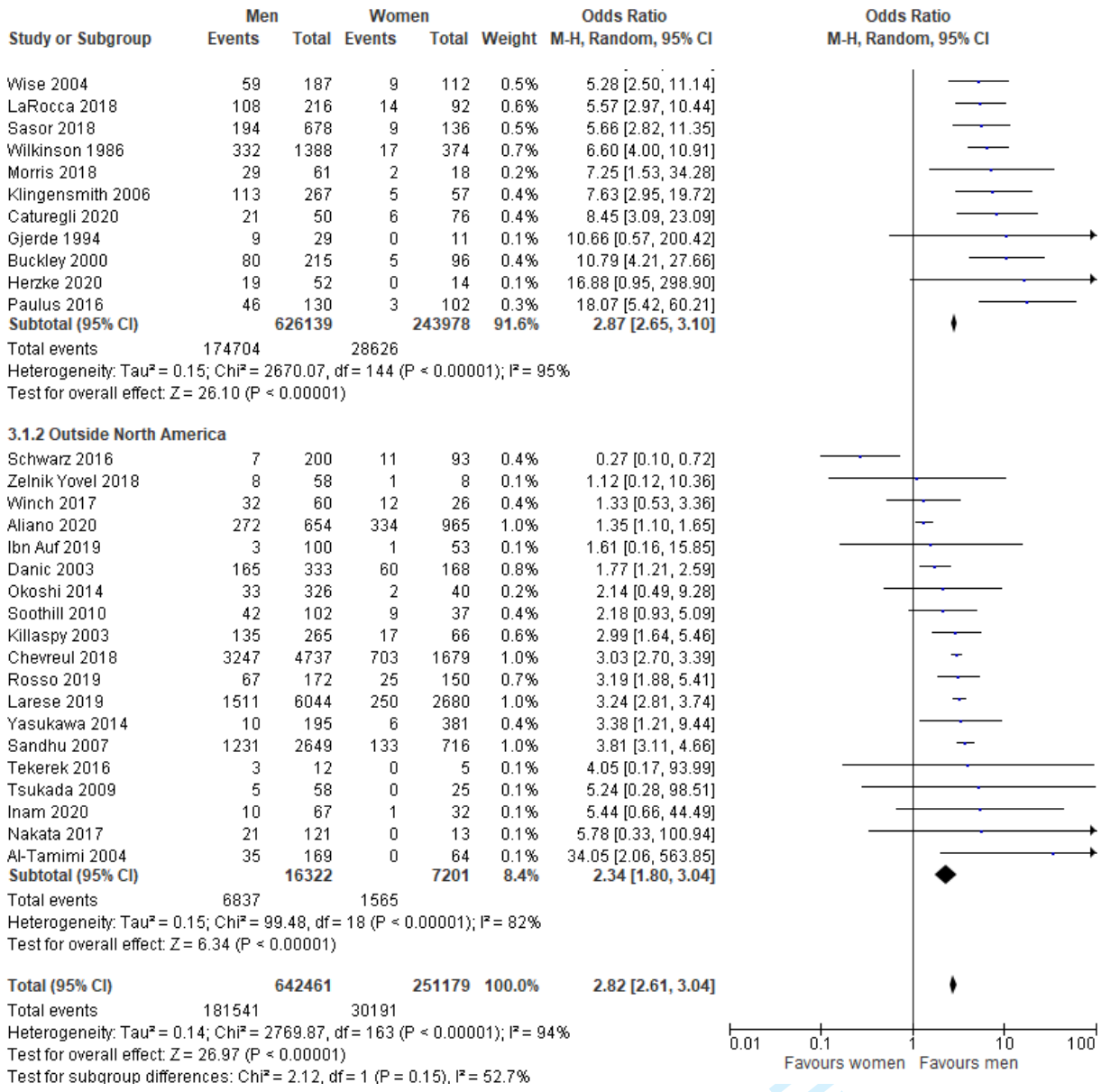


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Study or Subgroup	Men Events	Men Total	Women Events	Women Total	Weight	Odds Ratio M-H, Random, 95% CI
Chaudhary 2020	1351	5283	652	5807	1.0%	2.72 [2.45, 3.01]
Okafor 2020	402	1342	56	412	0.9%	2.72 [2.00, 3.69]
Eloy 2013	2057	6819	429	3133	1.0%	2.72 [2.43, 3.05]
Duke 2019	26	75	14	86	0.5%	2.73 [1.30, 5.74]
Girod 2011	524	1397	194	1077	1.0%	2.73 [2.26, 3.30]
Muffly 2016	138	760	43	575	0.8%	2.74 [1.91, 3.94]
Florence 2011	71	154	4	17	0.3%	2.78 [0.87, 8.91]
O'Neill 2019	501	1592	31	219	0.8%	2.78 [1.88, 4.13]
Moghimi 2019	62	179	9	58	0.5%	2.89 [1.33, 6.26]
Ngaage 2019	447	1350	19	131	0.7%	2.92 [1.77, 4.81]
Khurshid 2018	24	112	22	258	0.6%	2.93 [1.56, 5.48]
Madsen 2017	156	924	29	447	0.8%	2.93 [1.94, 4.43]
Wasserstein 2007	355	1062	54	370	0.9%	2.94 [2.14, 4.03]
Stewart 2020	67	178	35	206	0.7%	2.95 [1.84, 4.73]
Cuoco 2020	59	74	4	7	0.2%	2.95 [0.60, 14.62]
Carnevale 2020	203	774	19	177	0.7%	2.96 [1.79, 4.88]
Jena 2015	17354	60609	3623	30464	1.0%	2.97 [2.86, 3.09]
Mayer 2017	441	1686	25	236	0.8%	2.99 [1.95, 4.59]
Wu 2019	122	220	14	48	0.6%	3.02 [1.54, 5.95]
Shah 2018	310	852	124	784	0.9%	3.04 [2.40, 3.86]
Holliday 2014	211	738	34	293	0.8%	3.05 [2.06, 4.51]
Chauvin 2019	251	1923	68	1456	0.9%	3.06 [2.32, 4.04]
Nowling 2018	490	1898	117	1195	0.9%	3.21 [2.58, 3.98]
Hoover 2005	23373	77947	4076	35471	1.0%	3.30 [3.18, 3.42]
Schnapp 2003	1047	3364	81	674	0.9%	3.31 [2.59, 4.22]
Epperson 2020	185	343	16	62	0.6%	3.37 [1.83, 6.18]
Cydulka 2000	65	923	6	274	0.4%	3.38 [1.45, 7.90]
Garstka 2019	210	579	22	153	0.7%	3.39 [2.09, 5.49]
McDonald 2017	162	519	22	192	0.7%	3.51 [2.17, 5.67]
Woodward 2017	45	118	4	27	0.3%	3.54 [1.15, 10.92]
Dossani 2019	298	761	12	80	0.6%	3.65 [1.94, 6.85]
Wright 2007	123	283	19	110	0.7%	3.68 [2.13, 6.37]
Lewis-Stevenson 2001	129	747	21	395	0.7%	3.72 [2.30, 6.00]
Mascarenhas 2017	25	54	3	16	0.2%	3.74 [0.95, 14.62]
Fassiotto 2018	194	599	53	467	0.9%	3.74 [2.68, 5.22]
Monroe 2015	86	293	18	181	0.7%	3.76 [2.18, 6.51]
Santosa 2019	115	241	8	41	0.5%	3.76 [1.67, 8.49]
Paik 2014	172	426	12	79	0.6%	3.78 [1.99, 7.20]
Gawad 2020	75	294	9	111	0.5%	3.88 [1.87, 8.06]
Rotbart 2012	62	145	19	118	0.6%	3.89 [2.16, 7.03]
Higgins 1979	7406	23561	279	2648	1.0%	3.89 [3.43, 4.42]
Breyer 2020	1092	2597	51	329	0.9%	3.96 [2.91, 5.38]
McDermott 2018	475	1184	76	528	0.9%	3.98 [3.04, 5.22]
Pace 2020	509	1433	27	224	0.8%	4.02 [2.65, 6.10]
Herrick-Reynolds 2019	48	914	2	149	0.2%	4.07 [0.98, 16.94]
Waljee 2015	154	420	35	283	0.8%	4.10 [2.73, 6.16]
Selter 2020	71	129	30	134	0.7%	4.24 [2.49, 7.24]
AAMC 1992	15941	51647	1256	13580	1.0%	4.38 [4.12, 4.66]
Chandrasekar 2020	545	1894	27	320	0.8%	4.38 [2.92, 6.58]
Johnson 1992	17190	53891	1495	15568	1.0%	4.41 [4.17, 4.66]
Bickel 1993	17653	57049	1622	17682	1.0%	4.44 [4.20, 4.68]
Bickel 1988	15285	48535	1094	11673	1.0%	4.45 [4.17, 4.74]
Awad 2017	16	137	1	36	0.1%	4.63 [0.59, 36.13]
Wagner 2007	49	164	9	107	0.5%	4.64 [2.17, 9.92]
Waisbren 2008	486	3557	62	1884	0.9%	4.65 [3.55, 6.10]
Kirchner 1984	626	2407	23	333	0.8%	4.74 [3.07, 7.31]
Battaglia 2019	7	77	0	22	0.1%	4.79 [0.26, 87.16]
Levey 1990	2552	7015	99	932	0.9%	4.81 [3.88, 5.96]
Sperling 2019	174	312	72	347	0.8%	4.82 [3.42, 6.78]
Day 2010	550	2119	22	327	0.8%	4.86 [3.12, 7.57]
Bernardi 2020	6	22	1	14	0.1%	4.88 [0.52, 45.79]
Reed 2011	33	50	7	25	0.3%	4.99 [1.75, 14.28]
Whitley 1987	332	999	18	200	0.7%	5.03 [3.05, 8.31]
Smith 2019	217	663	14	159	0.6%	5.04 [2.84, 8.93]
Sadeghpour 2012	75	159	15	100	0.6%	5.06 [2.69, 9.51]
Schroen 2004	64	168	16	149	0.6%	5.12 [2.79, 9.37]
Karimian 2012	11	121	2	106	0.2%	5.20 [1.13, 24.02]

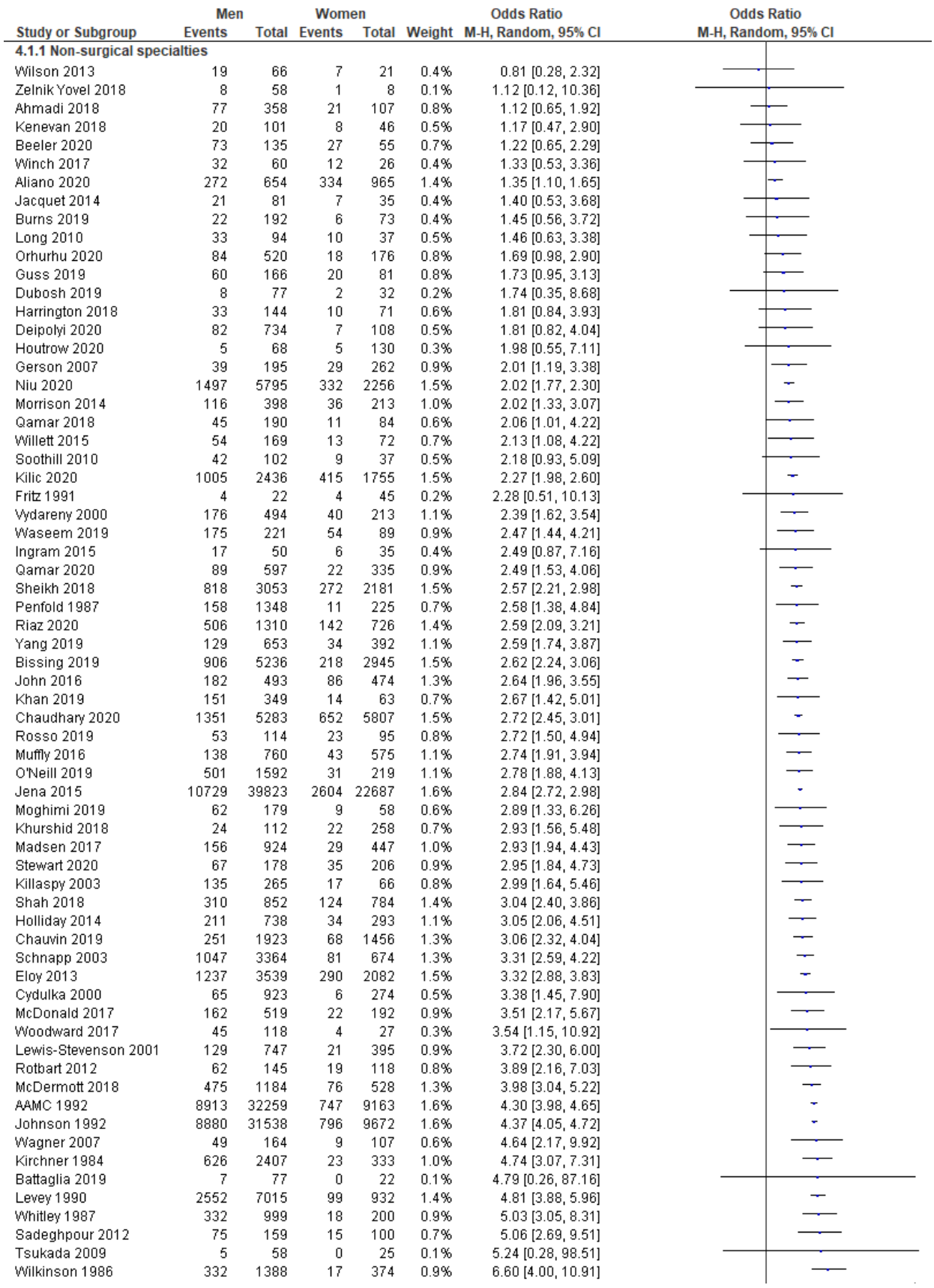


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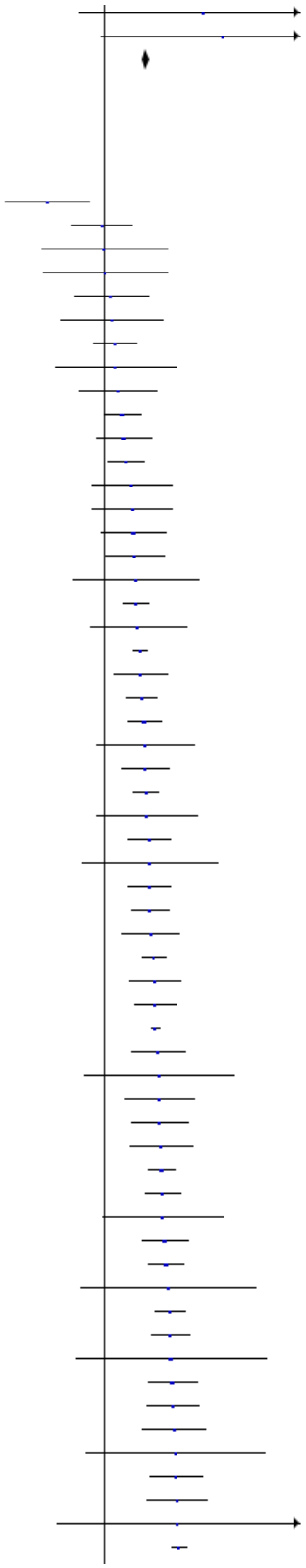
eFigure 3. Subgroup analysis of full professorship by geographic region (within North America versus outside of North America). M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

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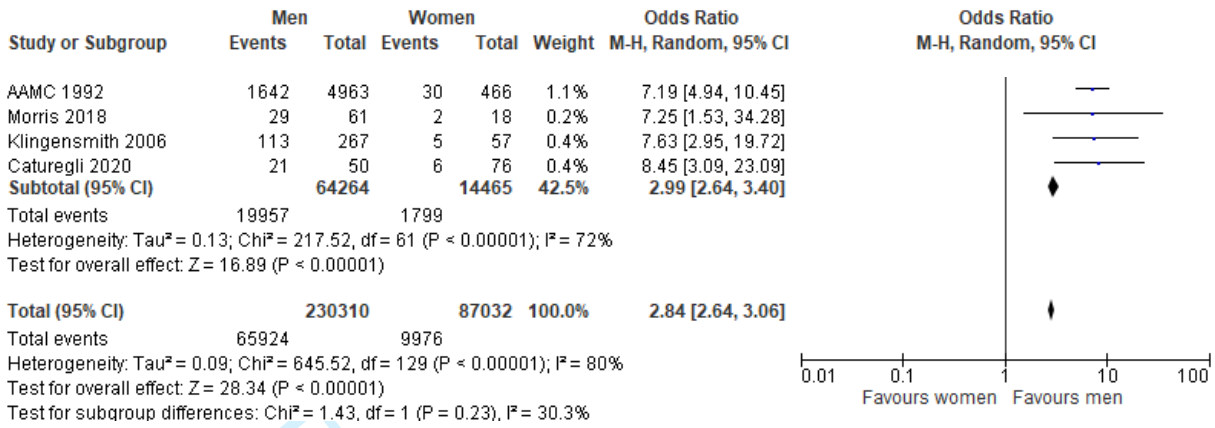


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Study or Subgroup	Men Events	Men Total	Women Events	Women Total	Weight	Odds Ratio M-H, Random, 95% CI
Gjerde 1994	9	29	0	11	0.1%	10.66 [0.57, 200.42]
Herzke 2020	19	52	0	14	0.1%	16.88 [0.95, 298.90]
Subtotal (95% CI)		**166046**		**72567**	**57.5%**	**2.72 [2.48, 2.98]**
Total events	45967		8177			
Heterogeneity: Tau² = 0.08; Chi² = 423.11, df = 67 (P < 0.00001); I² = 84%						
Test for overall effect: Z = 21.33 (P < 0.00001)						
4.1.2 Surgical specialties						
Schwarz 2016	7	200	11	93	0.4%	0.27 [0.10, 0.72]
Shaikh 2019	162	294	19	34	0.6%	0.97 [0.47, 1.98]
Hoops 2018	8	42	3	16	0.2%	1.02 [0.23, 4.45]
Gerety 2013	36	104	3	9	0.2%	1.06 [0.25, 4.49]
Geltzeiler 2017	30	281	7	77	0.5%	1.20 [0.50, 2.84]
Dorismond 2020	12	47	5	23	0.3%	1.23 [0.38, 4.05]
Higgins 2020	108	287	28	90	0.9%	1.34 [0.81, 2.22]
Morgan 2019	3	25	8	87	0.2%	1.35 [0.33, 5.51]
McAlister 2014	23	283	6	102	0.4%	1.42 [0.56, 3.58]
Harris 2019	275	426	55	102	1.0%	1.56 [1.01, 2.41]
Odell 2019	46	281	14	129	0.7%	1.61 [0.85, 3.04]
Tracy 2019	181	1086	29	274	1.0%	1.69 [1.11, 2.56]
Carpenter 2018	45	117	7	29	0.4%	1.96 [0.78, 4.97]
Dermody 2019	35	161	6	49	0.4%	1.99 [0.78, 5.06]
Yue 2020	98	618	8	95	0.6%	2.05 [0.96, 4.36]
Gambhir 2020	65	120	18	50	0.7%	2.10 [1.06, 4.15]
Okoshi 2014	33	326	2	40	0.2%	2.14 [0.49, 9.28]
Hoof 2020	852	3889	50	434	1.2%	2.15 [1.59, 2.92]
Ilin 2020	55	164	4	22	0.3%	2.27 [0.73, 7.04]
Valsangkar 2016	1272	3087	211	928	1.5%	2.38 [2.01, 2.82]
Tulunay-Udur 2019	72	152	18	66	0.7%	2.40 [1.28, 4.50]
Maxwell 2020	275	1543	36	450	1.1%	2.49 [1.73, 3.59]
Hill 2018	134	420	39	257	1.1%	2.62 [1.76, 3.90]
Baptiste 2017	28	91	4	28	0.3%	2.67 [0.85, 8.41]
Cheng 2019	178	534	17	108	0.8%	2.68 [1.55, 4.63]
Okafor 2020	402	1342	56	412	1.2%	2.72 [2.00, 3.69]
Florence 2011	71	154	4	17	0.3%	2.78 [0.87, 8.91]
Ngaage 2019	447	1350	19	131	0.9%	2.92 [1.77, 4.81]
Cuoco 2020	59	74	4	7	0.2%	2.95 [0.60, 14.62]
Carnevale 2020	203	774	19	177	0.9%	2.96 [1.79, 4.88]
Mayer 2017	441	1686	25	236	1.0%	2.99 [1.95, 4.59]
Wu 2019	122	220	14	48	0.7%	3.02 [1.54, 5.95]
Eloy 2013	553	1740	66	534	1.3%	3.30 [2.50, 4.36]
Epperson 2020	185	343	16	62	0.7%	3.37 [1.83, 6.18]
Garstka 2019	210	579	22	153	0.9%	3.39 [2.09, 5.49]
Jena 2015	3925	13010	436	3873	1.5%	3.41 [3.06, 3.79]
Dossani 2019	298	761	12	80	0.7%	3.65 [1.94, 6.85]
Rosso 2019	10	22	2	11	0.2%	3.75 [0.65, 21.51]
Santosa 2019	115	241	8	41	0.5%	3.76 [1.67, 8.49]
Paik 2014	172	426	12	79	0.7%	3.78 [1.99, 7.20]
Gawad 2020	75	294	9	111	0.6%	3.88 [1.87, 8.06]
Breyer 2020	1092	2597	51	329	1.2%	3.96 [2.91, 5.38]
Pace 2020	509	1433	27	224	1.0%	4.02 [2.65, 6.10]
Herrick-Reynolds 2019	48	914	2	149	0.2%	4.07 [0.98, 16.94]
Selter 2020	71	129	30	134	0.9%	4.24 [2.49, 7.24]
Chandrasekar 2020	545	1894	27	320	1.1%	4.38 [2.92, 6.58]
Awad 2017	16	137	1	36	0.1%	4.63 [0.59, 36.13]
Sperling 2019	174	312	72	347	1.2%	4.82 [3.42, 6.78]
Day 2010	550	2119	22	327	1.0%	4.86 [3.12, 7.57]
Bernardi 2020	6	22	1	14	0.1%	4.88 [0.52, 45.79]
Smith 2019	217	663	14	159	0.8%	5.04 [2.84, 8.93]
Schroen 2004	64	168	16	149	0.8%	5.12 [2.79, 9.37]
Wise 2004	59	187	9	112	0.6%	5.28 [2.50, 11.14]
Inam 2020	10	67	1	32	0.1%	5.44 [0.66, 44.49]
LaRocca 2018	108	216	14	92	0.7%	5.57 [2.97, 10.44]
Sasor 2018	194	678	9	136	0.6%	5.66 [2.82, 11.35]
Nakata 2017	21	121	0	13	0.1%	5.78 [0.33, 100.94]
Johnson 1992	3147	9672	128	1711	1.4%	5.96 [4.96, 7.18]



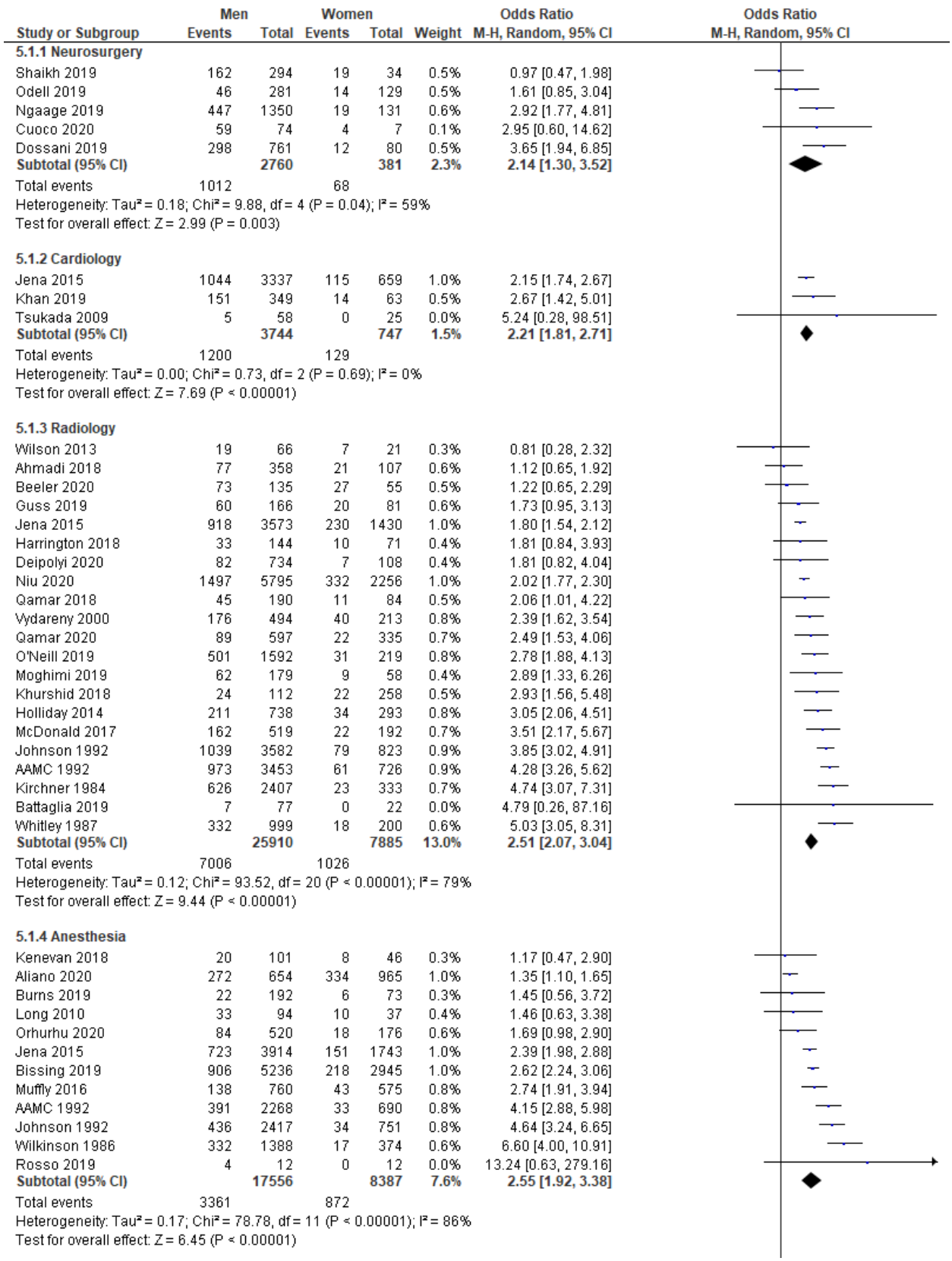
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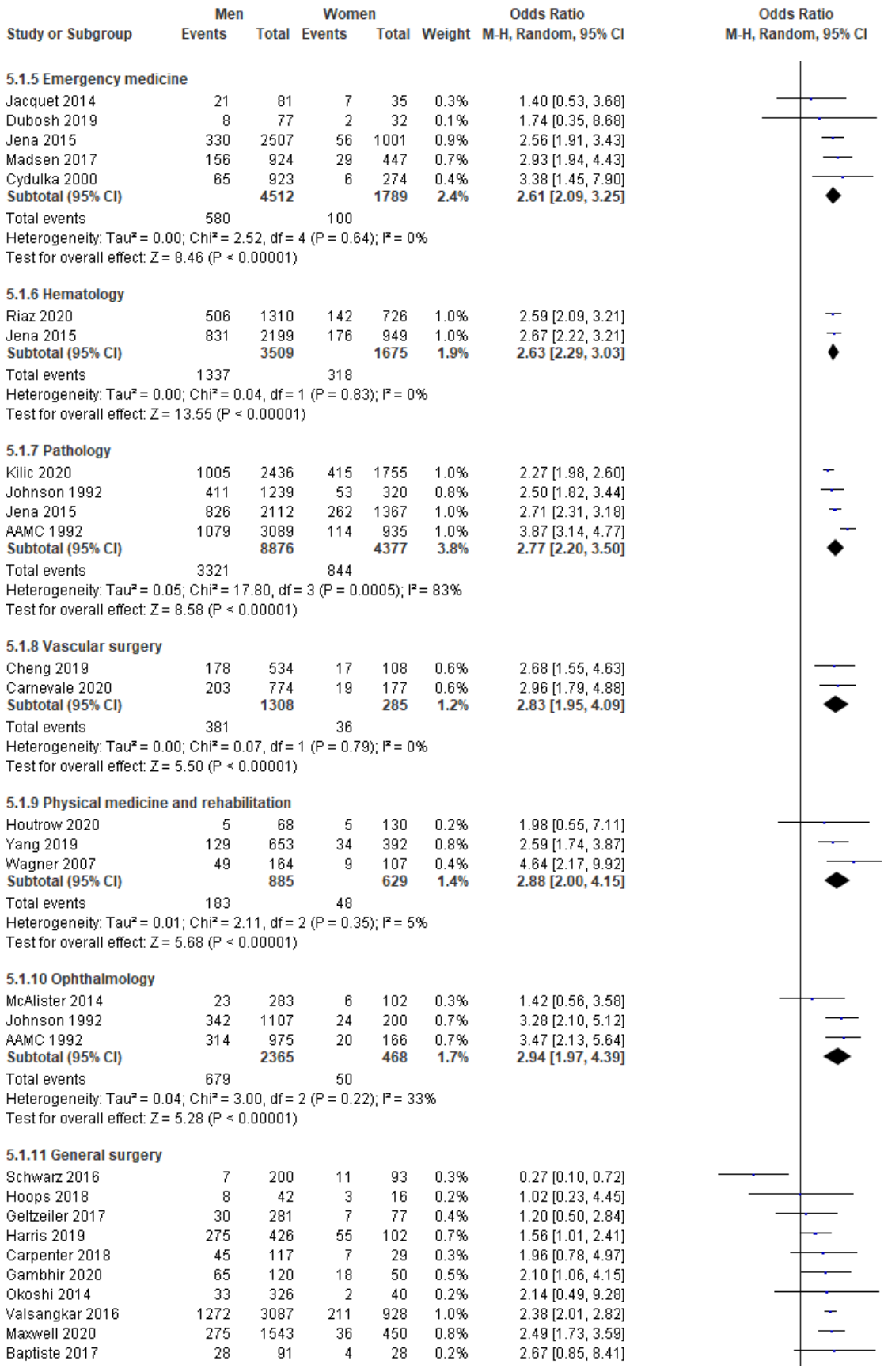
eFigure 4. Subgroup analysis of full professorship for surgical versus non-surgical specialties. M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

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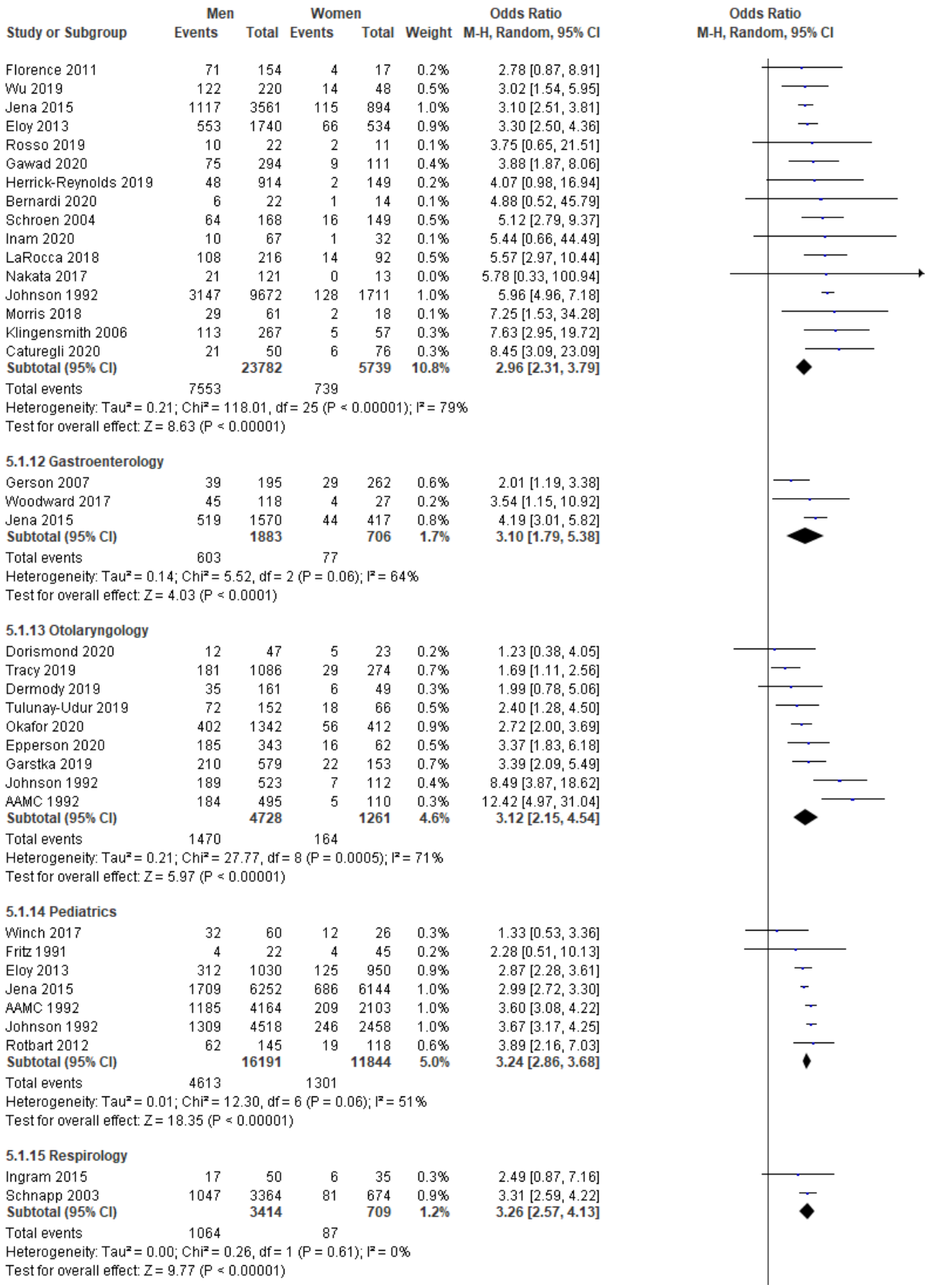
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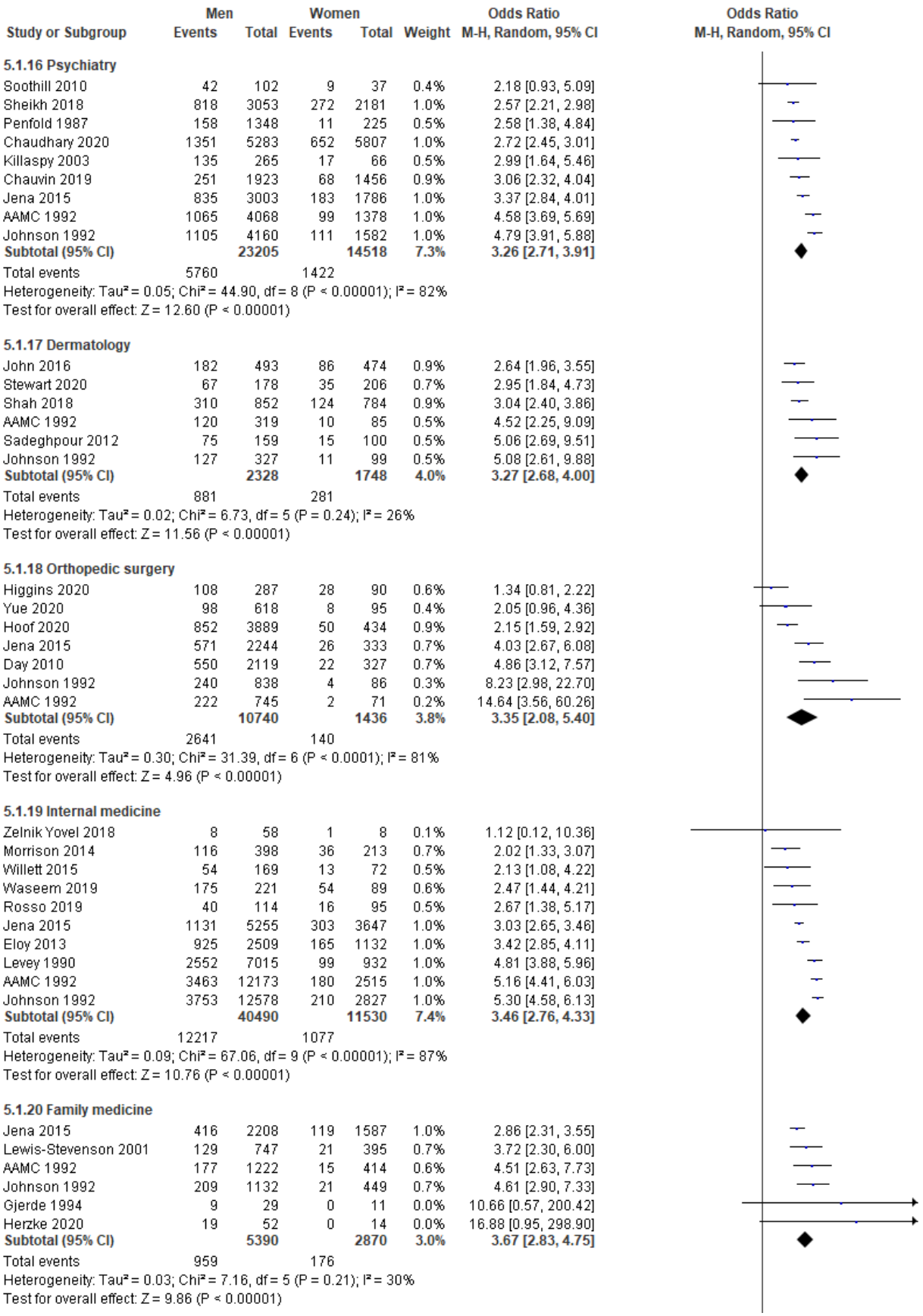
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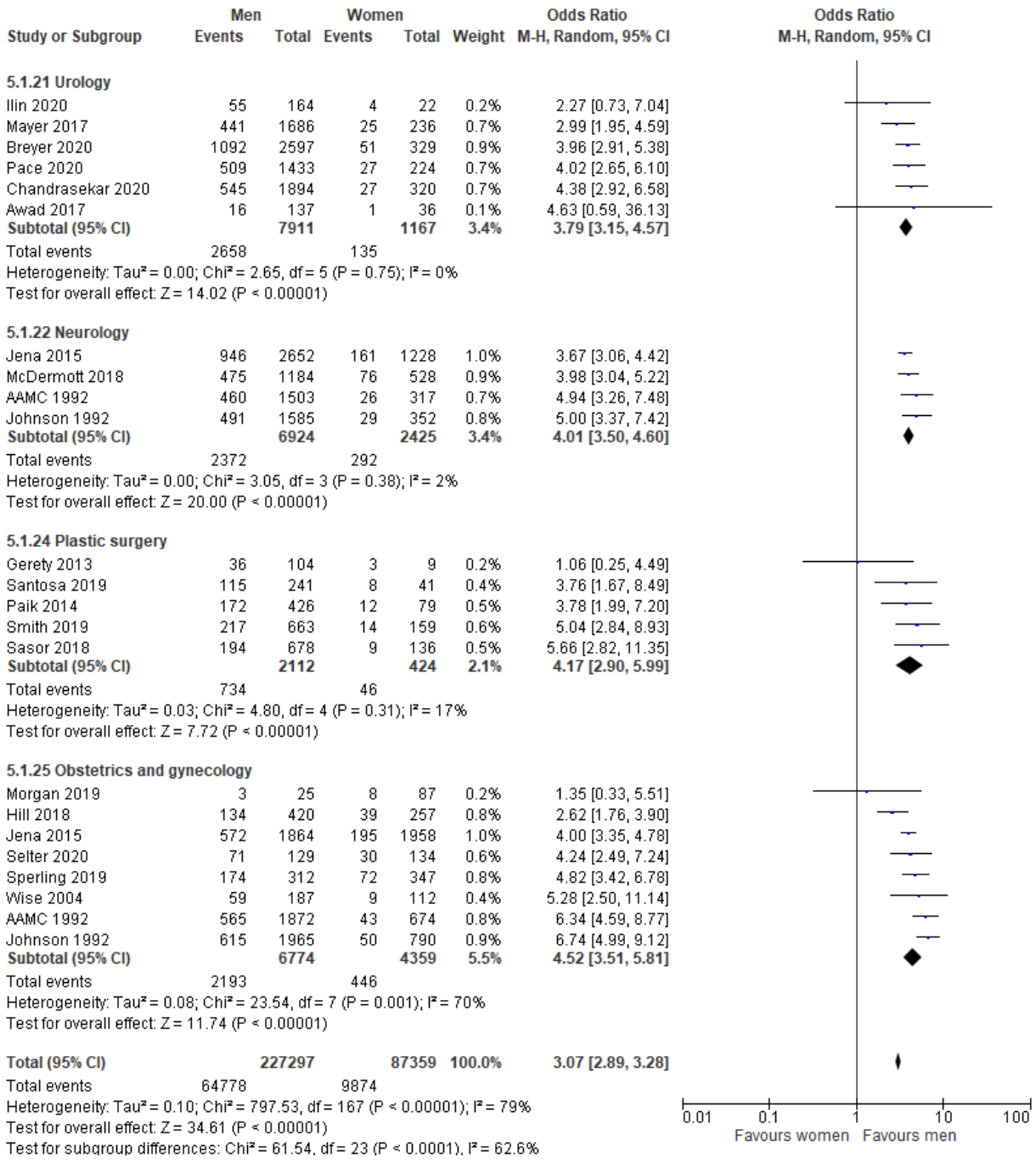
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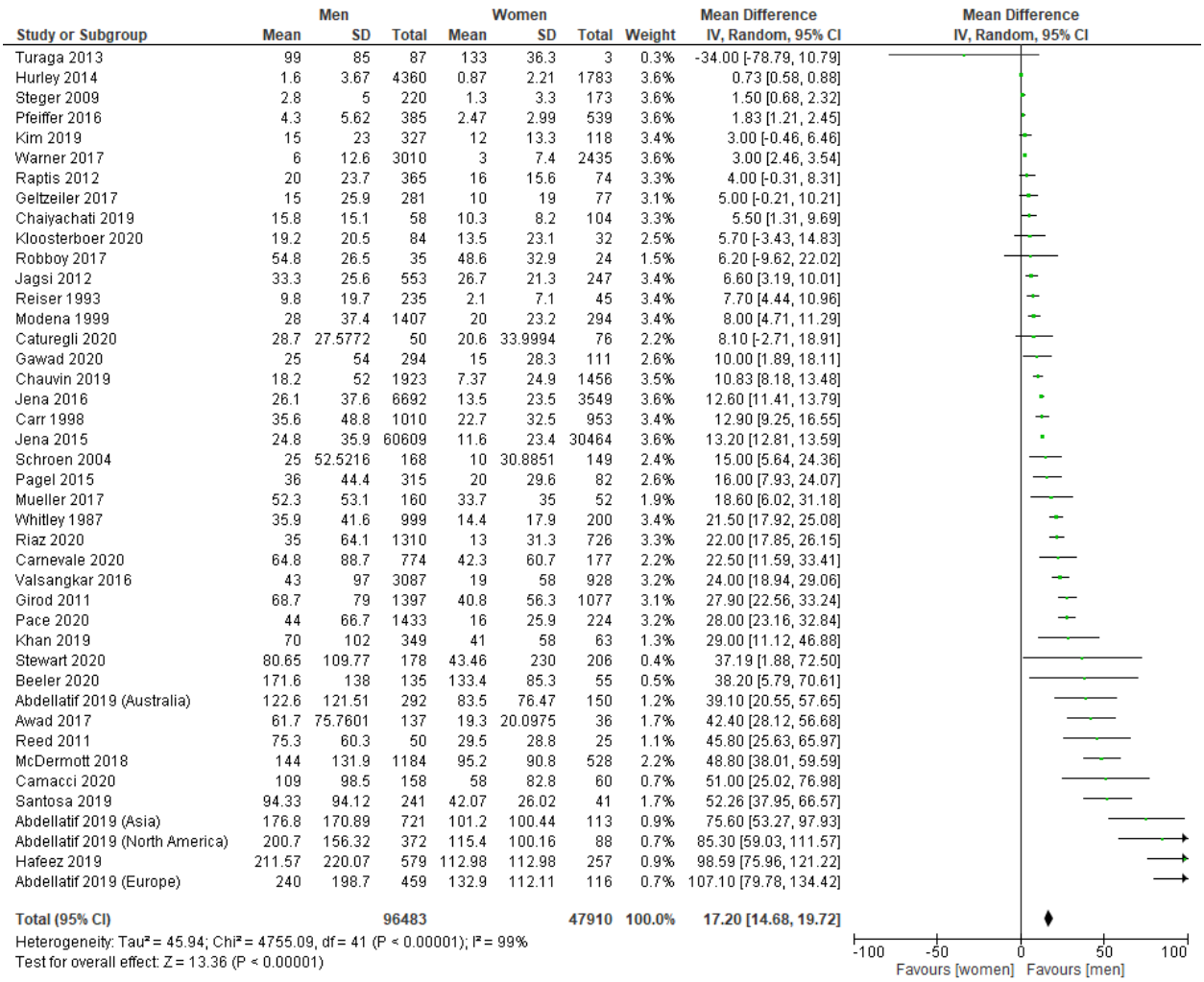


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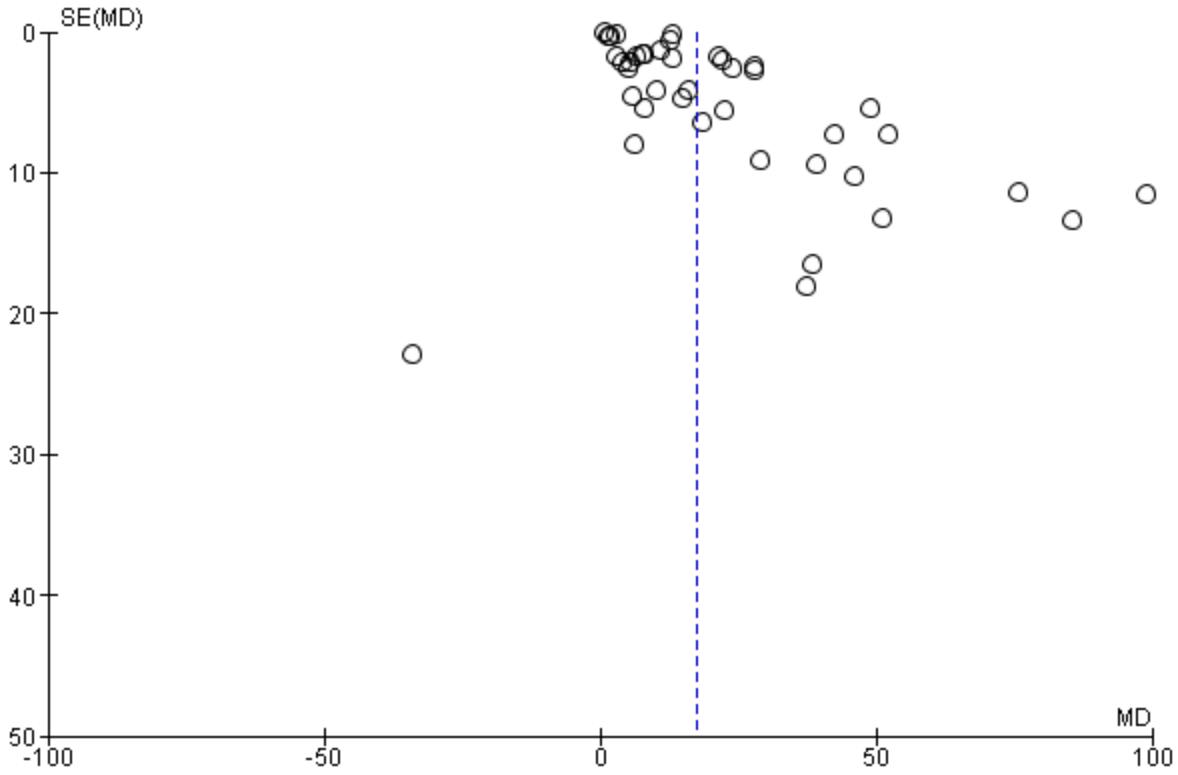
eFigure 5. Subgroup analysis of full professorship by specialty. M-H (Mantel-Haenszel), Random (random effects model), CI (confidence interval).

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eFigure 6. Forest plot of total publications for men vs. women. SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).

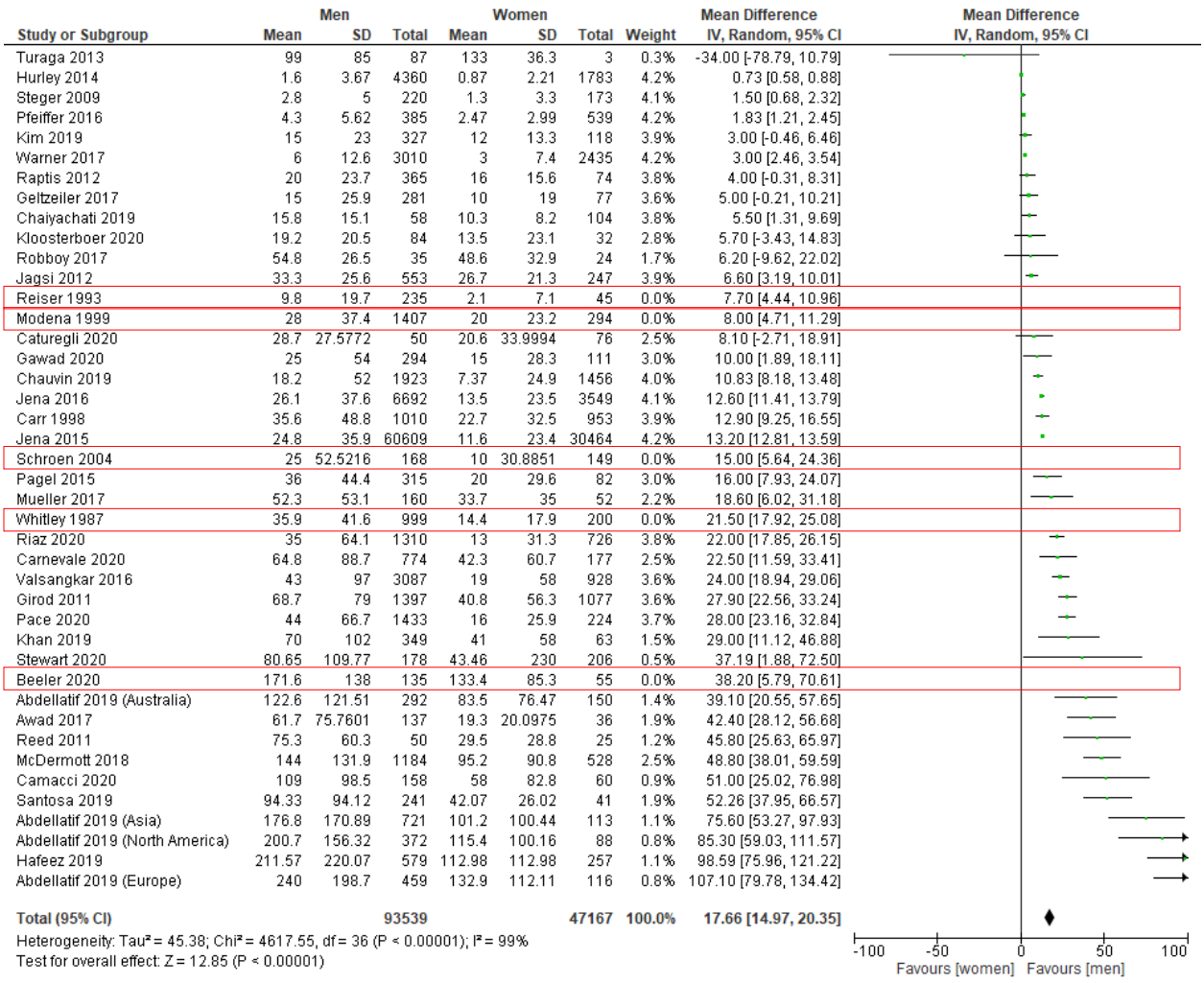
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eFigure 7. Funnel plot of total publications for men vs. women. SE (standard error), MD (mean difference).

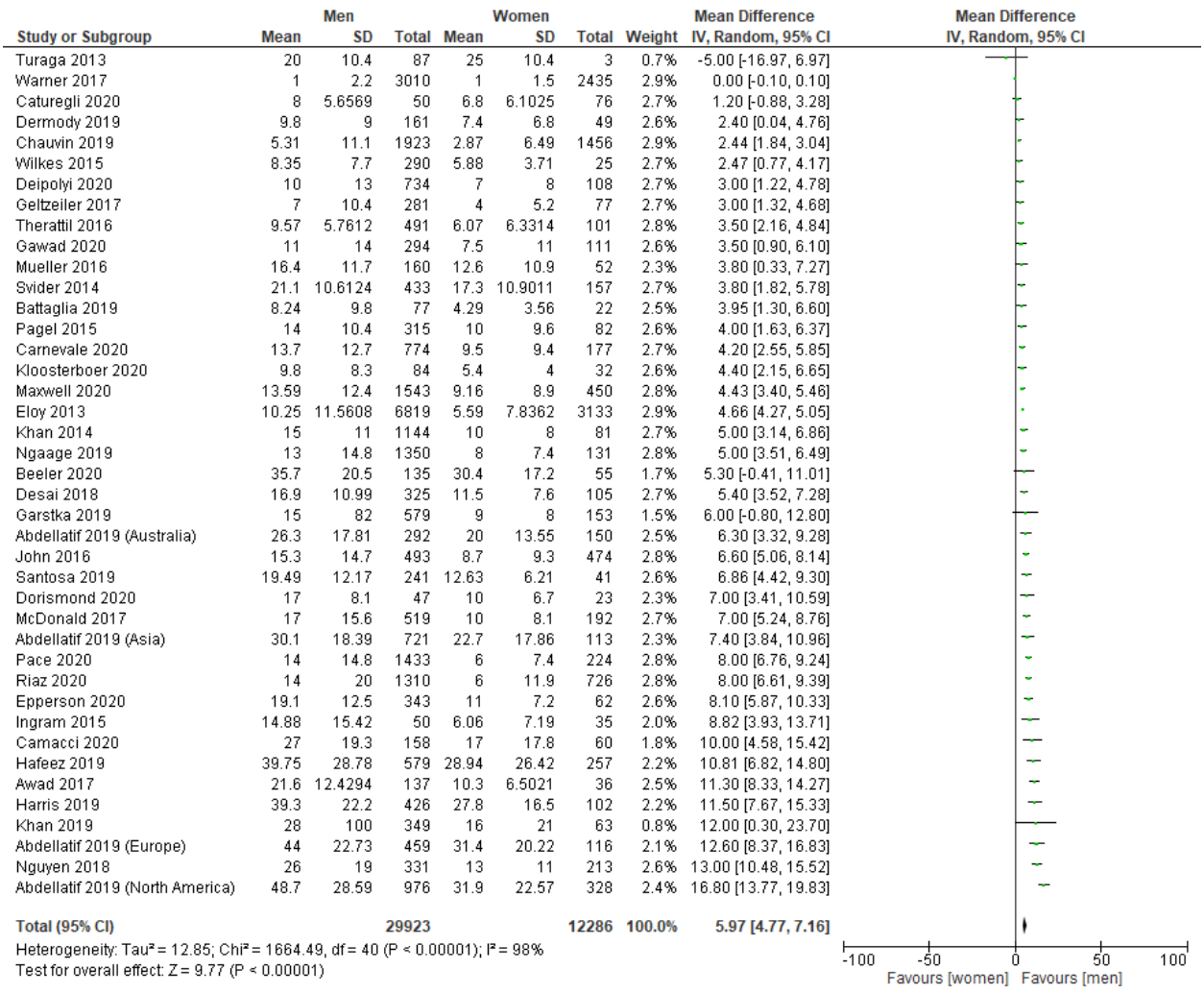
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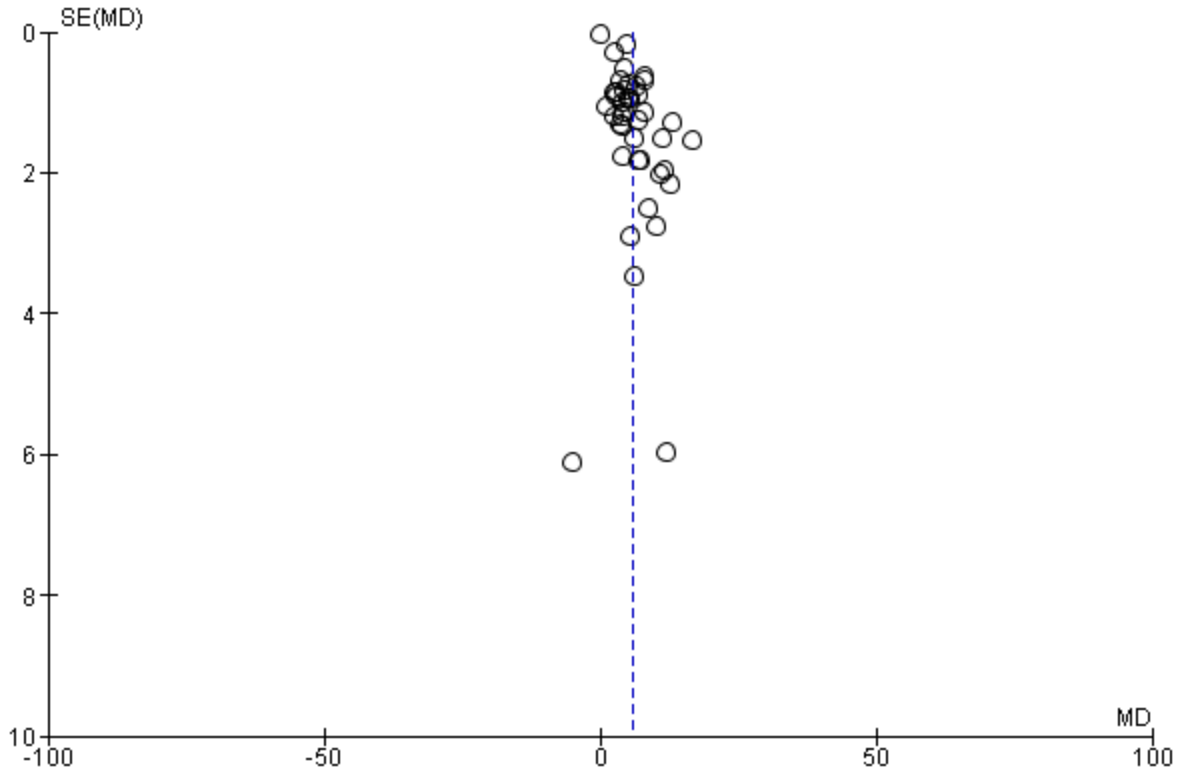
eFigure 8. Sensitivity analysis of total publications removing 5 poor quality studies (red box, NOS score ≤ 5). SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).

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eFigure 9. Forest plot of h-index for men vs. women. SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).

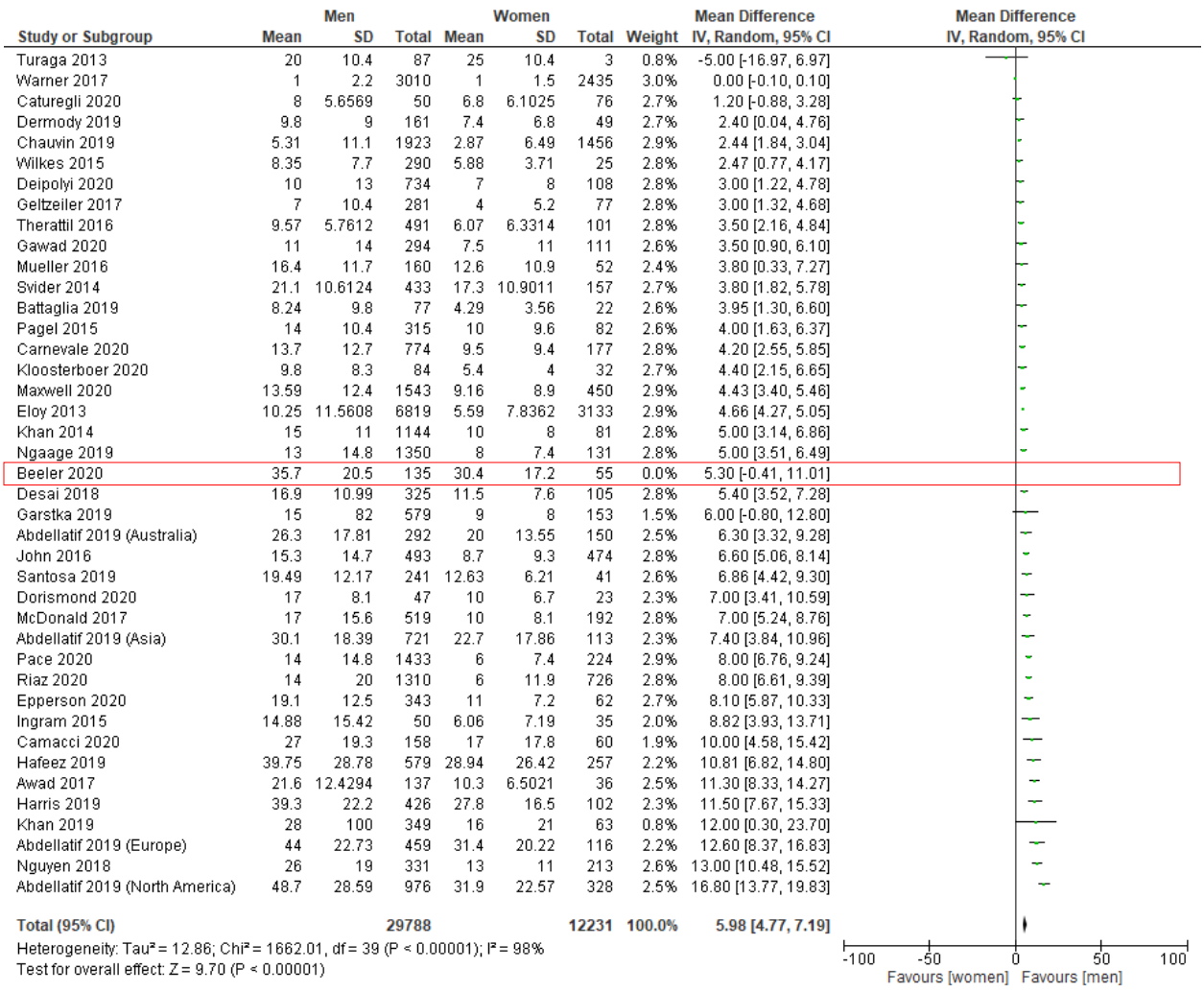
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eFigure 10. Funnel plot of h-index for men vs. women. SE (standard error), MD (mean difference).

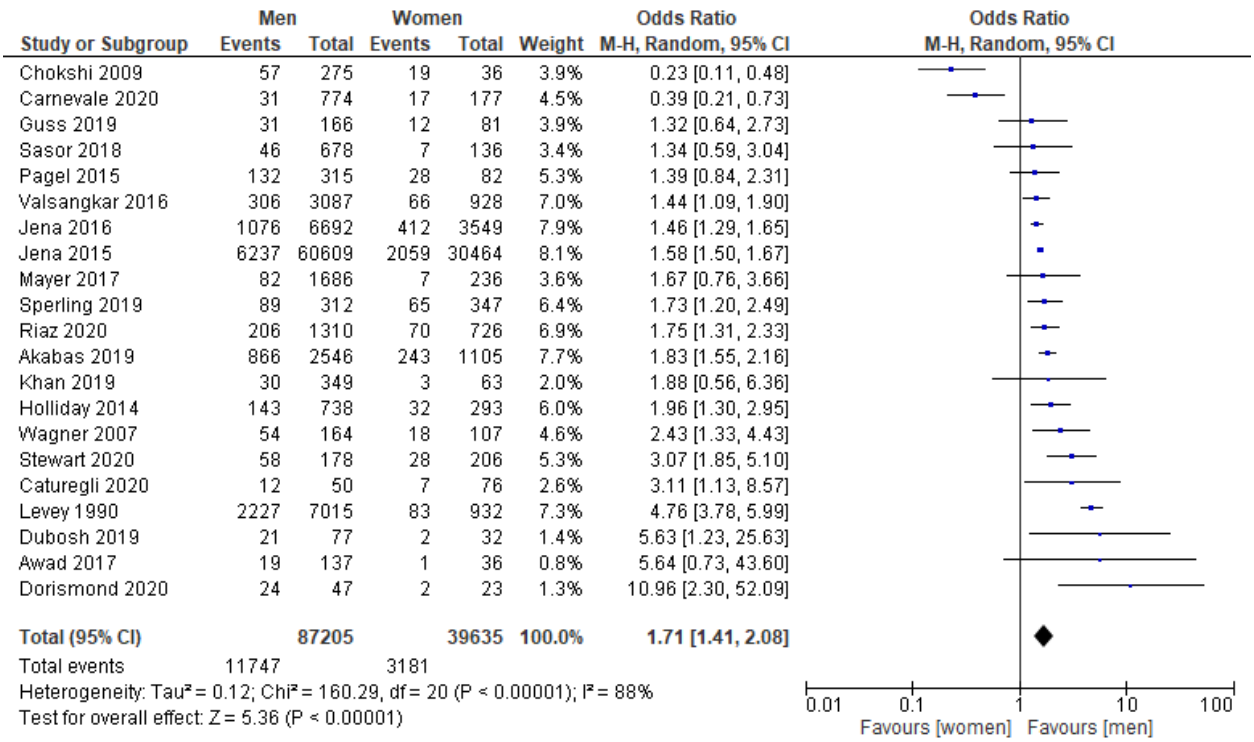
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eFigure 11. Sensitivity analysis of h-index removing 1 poor quality study (red box, NOS score ≤ 5). SD (standard deviation), Total (sample size), IV (inverse variance), Random (random effects model), CI (confidence interval).

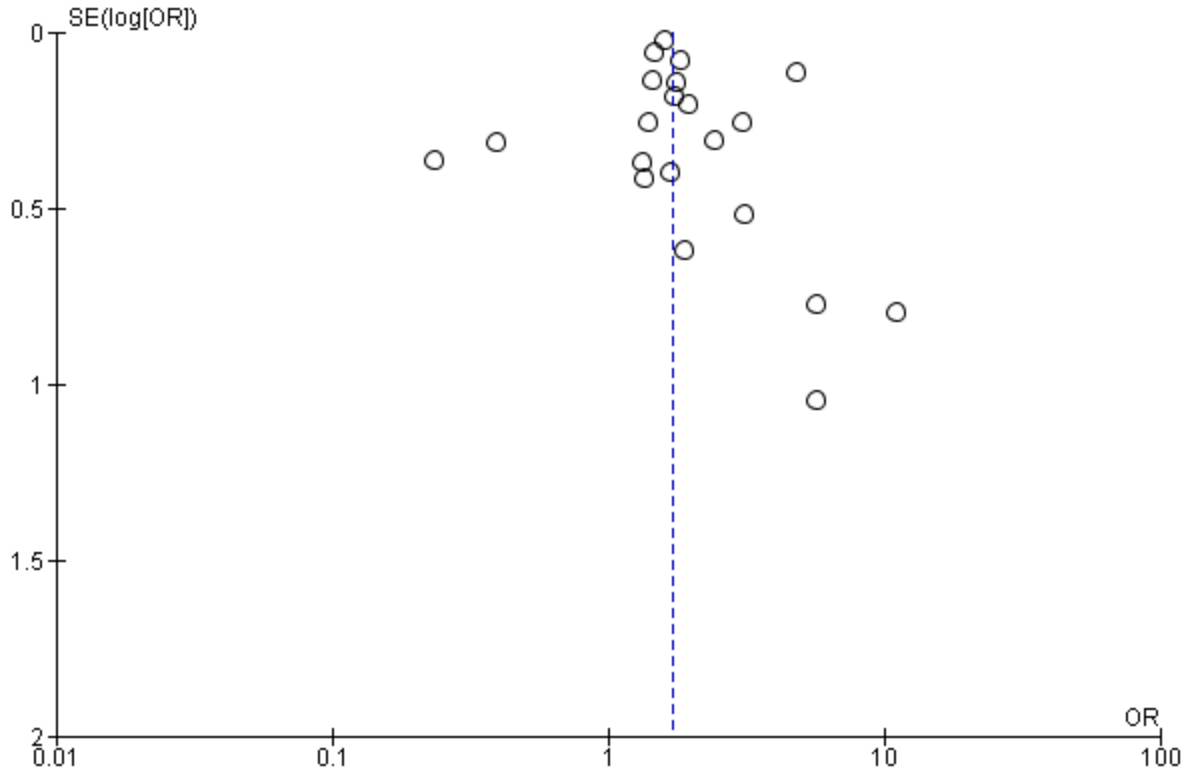
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eFigure 12. Forest plot of federal research funding for men vs. women. M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).

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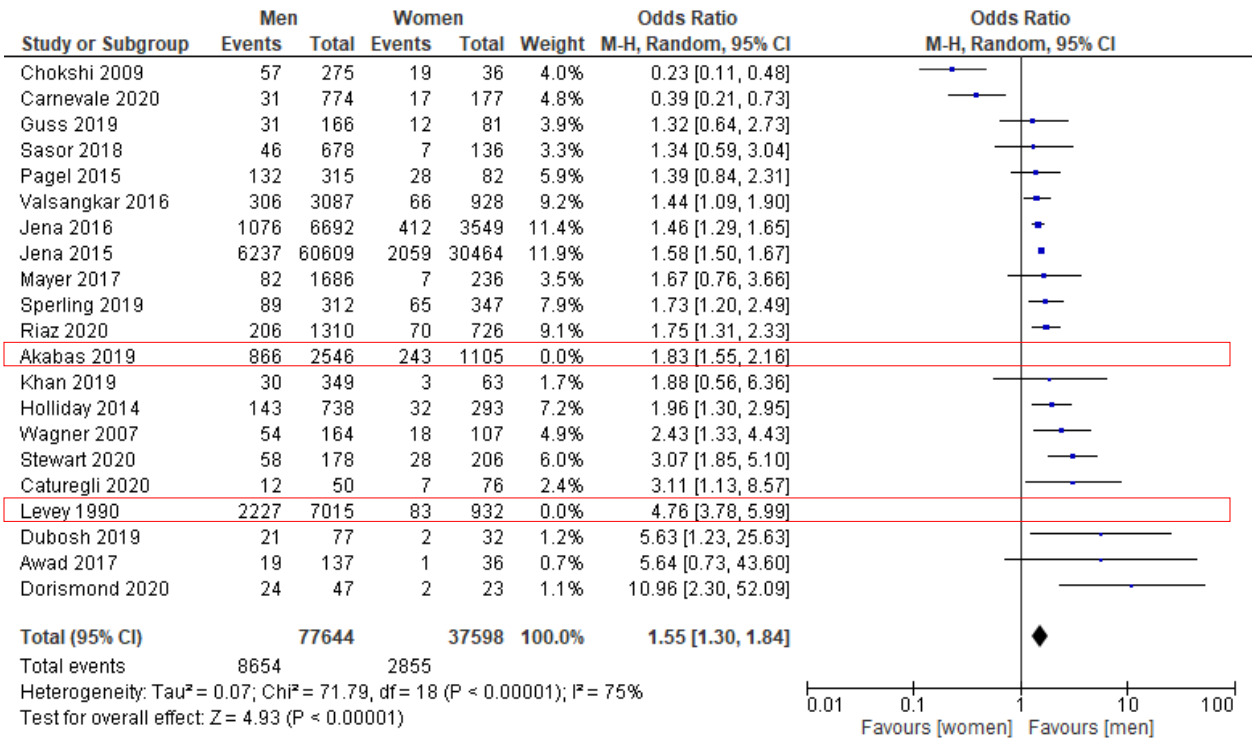
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eFigure 13. Funnel plot of federal research funding for men vs. women. SE (standard error), OR (odds ratio).

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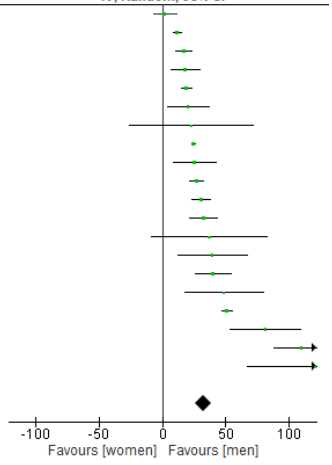
eFigure 14. Sensitivity analysis of federal research funding removing 2 poor quality studies (red box, NOS score ≤ 5). M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).

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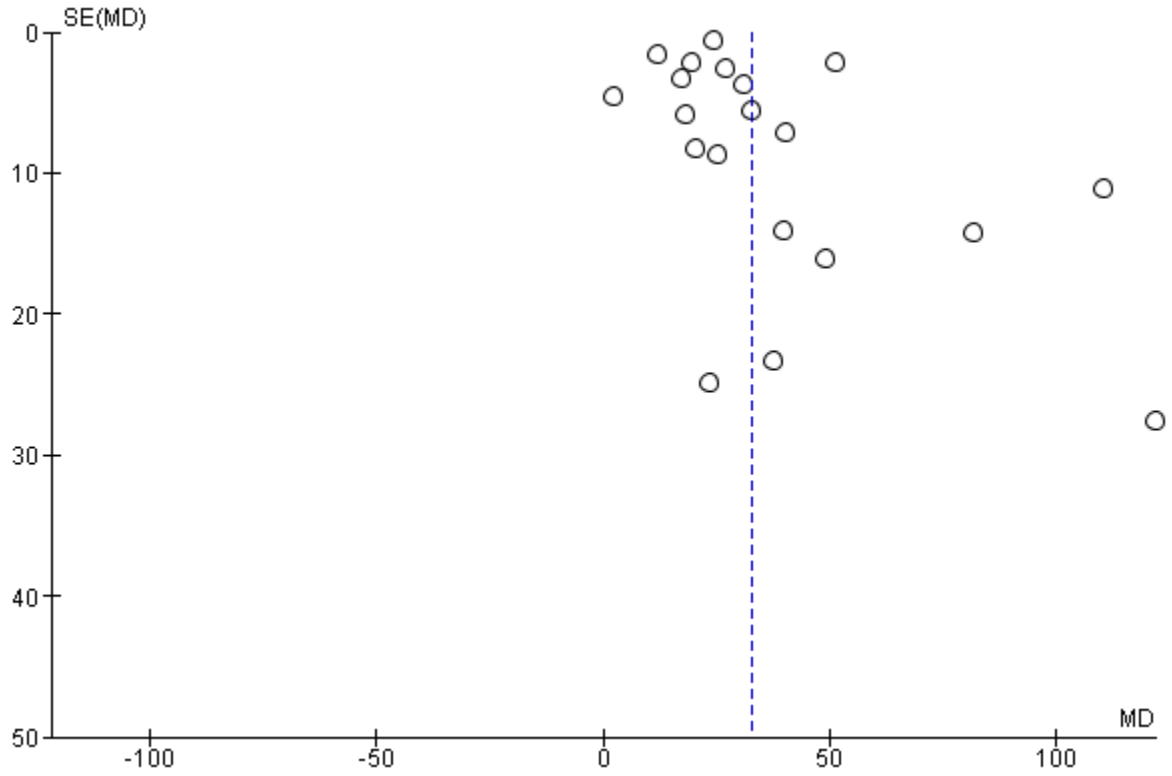
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Study or Subgroup	Men Mean [$ (1000's)]	Men SD [$ (1000's)]	Men Total	Women Mean [$ (1000's)]	Women SD [$ (1000's)]	Women Total	Weight	Mean Difference IV, Random, 95% CI
Dermody 2019	266.707	31.624	161	264.674	27.027	49	6.4%	2.03 [-6.97, 11.04]
Wiler 2019	252	39.365	1430	240	36.127	797	7.1%	12.00 [8.77, 15.23]
Weaver 2015	202	44.444	513	185	44.444	263	6.7%	17.00 [10.39, 23.61]
Madsen 2017	342.795	100.342	924	324.713	101.307	447	6.0%	18.08 [6.68, 29.49]
Maxwell 2020	287.717	45.379	1543	268.426	41.339	450	7.0%	19.29 [14.85, 23.73]
Houtrow 2020	244.798	52.906	68	224.497	60.756	130	5.0%	20.30 [3.95, 36.65]
Cheng 2019	259.649	126.724	37	236.514	80.887	35	1.5%	23.13 [-25.71, 71.98]
Dennis 1990	74.6	3.392	85	50.294	4.206	51	7.2%	24.31 [22.94, 25.67]
Shiber 2020	269.981	49.388	41	244.661	36.582	88	4.9%	25.32 [8.38, 42.26]
Ash 2004	125	66	941	98	45	873	6.9%	27.00 [21.83, 32.17]
Jagsi 2013	172.164	59.60251	593	141.325	59.54479	419	6.6%	30.84 [23.39, 38.29]
Jagsi 2012	200.433	74.03391	553	167.669	73.82341	247	6.0%	32.76 [21.68, 43.85]
Sadeghpour 2012	274.367	94.849	159	237	220.776	100	1.7%	37.37 [-8.35, 83.08]
Houtrow 2011	219.4	64.2	25	179.9	44.6	61	3.3%	39.50 [11.96, 67.04]
Diehl 2008	172	72	276	132	40	49	5.5%	40.00 [25.94, 54.06]
Guss 2019	390.147	144.043	166	341.173	103.429	81	2.8%	48.97 [17.55, 80.40]
Jena 2016	257.957	137.202	6692	206.641	88.238	3549	7.0%	51.32 [46.93, 55.70]
Spencer 2016	400	148.148	660	318.422	111.111	73	3.2%	81.58 [53.70, 109.46]
Singh 2008	280	80	168	170	46	25	4.1%	110.00 [88.29, 131.71]
Gambhir 2020	468.959	225.454	120	347.381	128.975	50	1.3%	121.58 [67.68, 175.48]
Total (95% CI)			**15155**			**7837**	**100.0%**	**32.52 [25.79, 39.26]**

Heterogeneity: Tau² = 164.35; Chi² = 339.12, df = 19 (P < 0.00001); I² = 94%
Test for overall effect: Z = 9.47 (P < 0.00001)

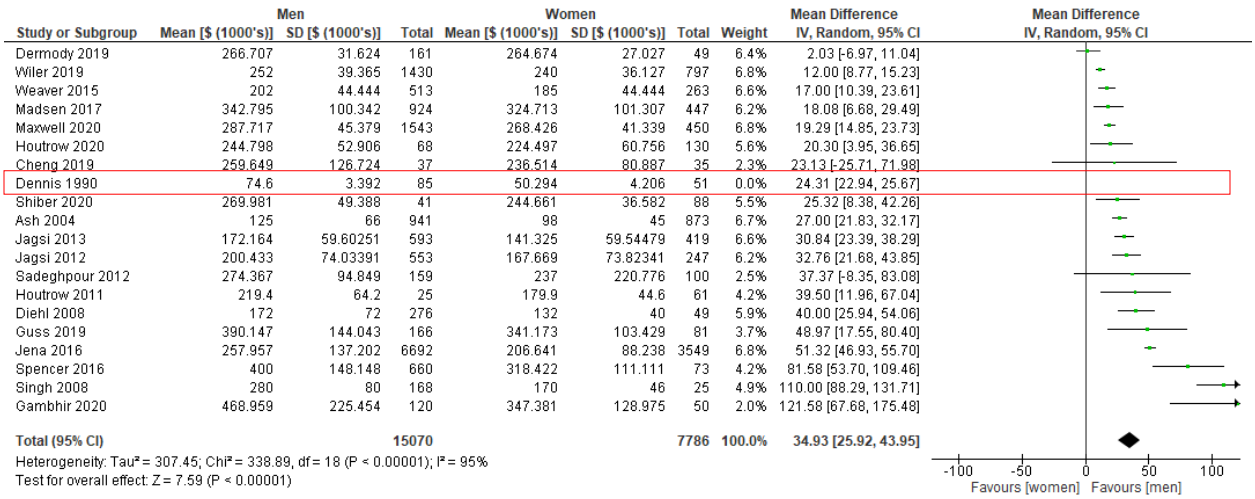


eFigure 15. Forest plot of salary for men vs. women. Means and standard deviations (SD) are presented in thousands of dollars. Total (sample size), IV (inverse variance), Random (random effects), CI (confidence interval).



eFigure 16. Funnel plot of salary for men vs. women. SE (standard error), MD (mean difference).

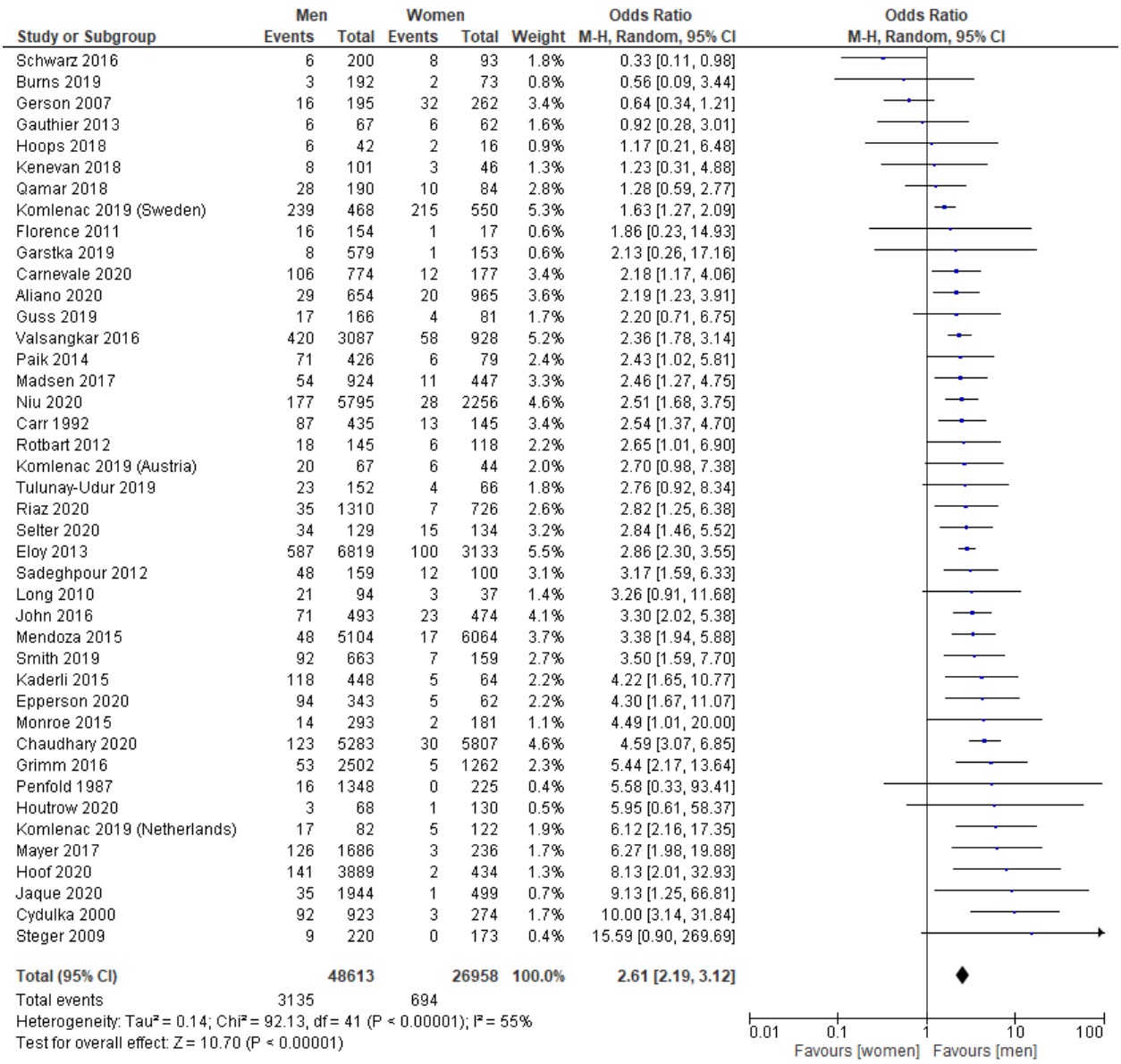
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eFigure 17. Sensitivity analysis of salary removing 1 poor quality study (red box, NOS score ≤ 5). Means and standard deviations (SD) are presented in thousands of dollars. Total (sample size), IV (inverse variance), Random (random effects), CI (confidence interval).

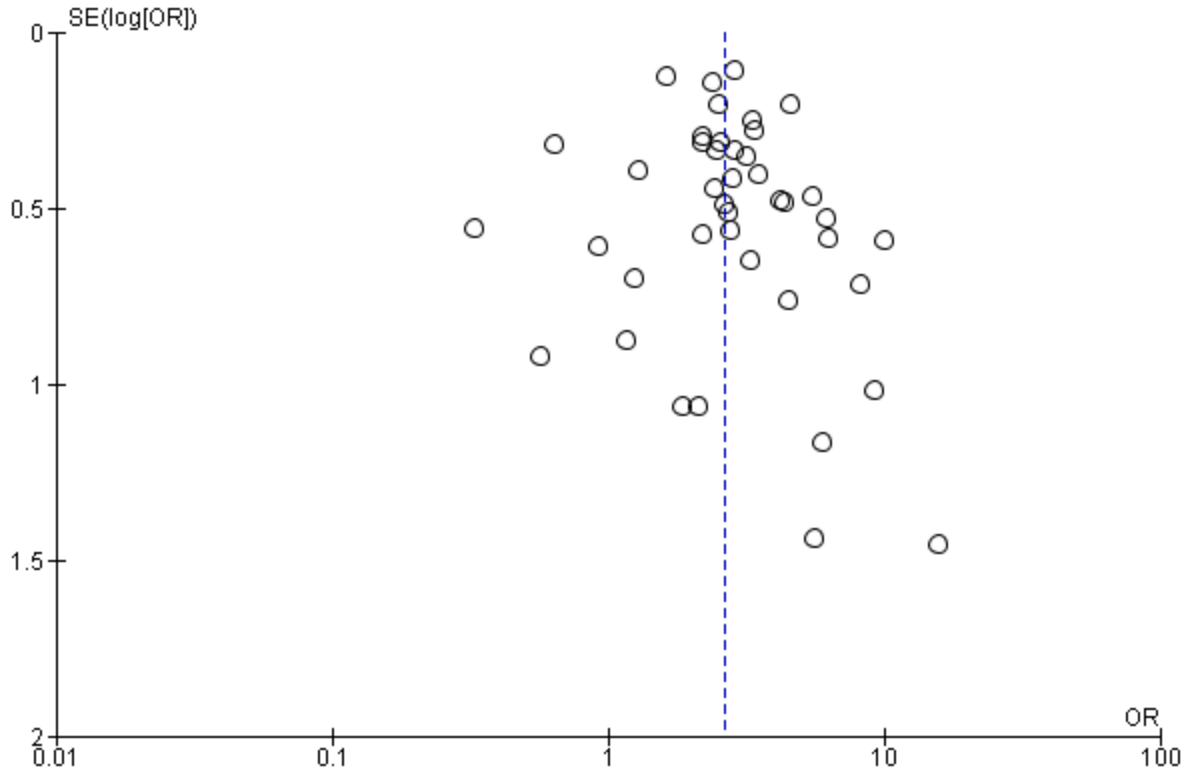
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eFigure 18. Forest plot of leadership (departmental chair) for men vs. women. M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).

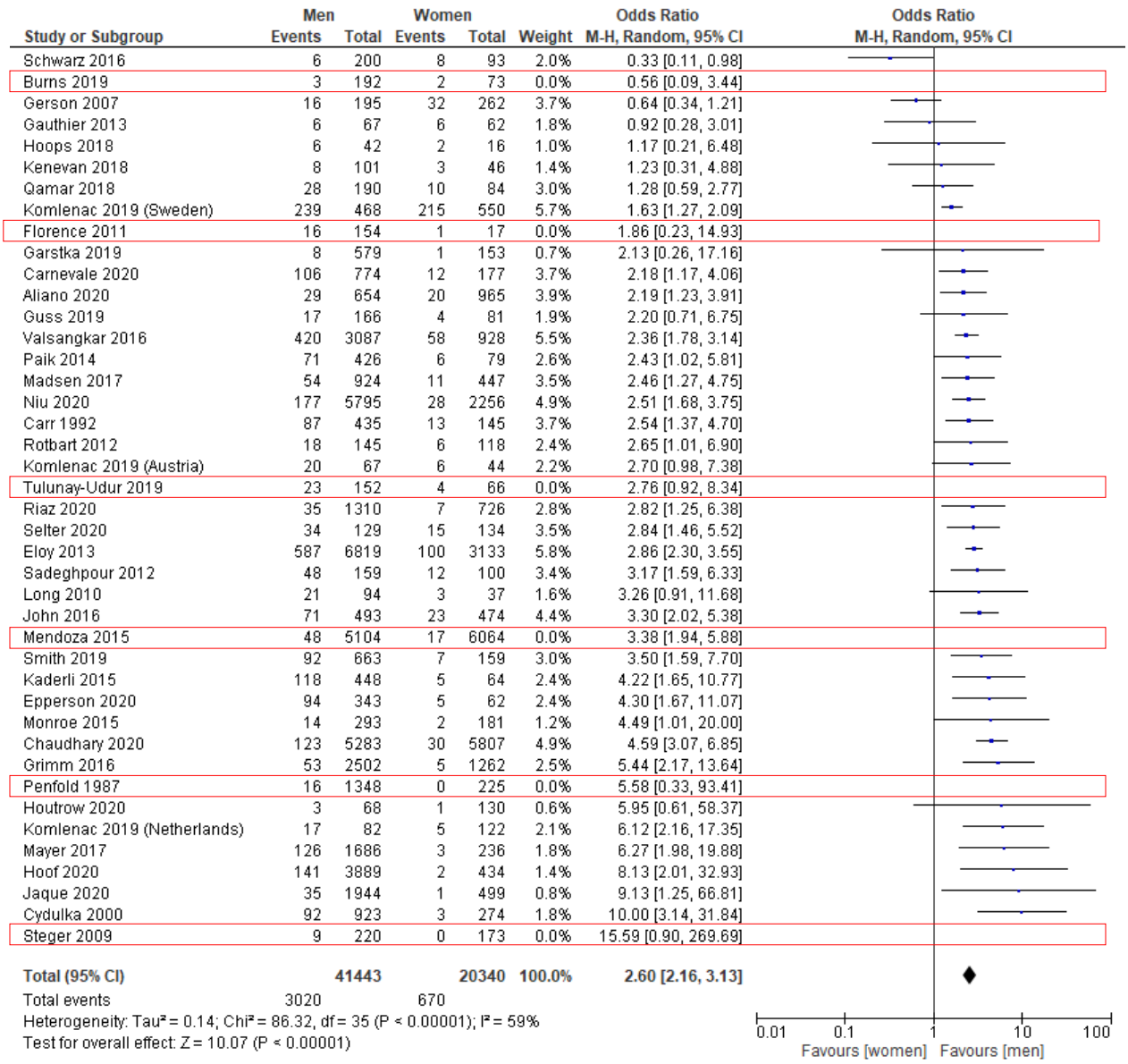
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eFigure 19. Funnel plot of leadership (departmental chair) for men vs. women. M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).

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eFigure 20. Sensitivity analysis of leadership (departmental chair) removing 6 poor quality studies (red box, NOS score ≤ 5). M-H (Mantel-Haenszel), Random (random effects), CI (confidence interval).

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	eTable 1 (Supplement pages 2-5)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6-7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6-7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7-9

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Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7-9

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6-9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8-9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10, eTable 2a (Supplement pages 6-19)
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	11, eTable 4 (Supplement pages 30-41)
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	11-15
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	11-15
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11-15
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	11-15
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	15-19
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	18-19
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	19-20

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FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	21

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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